Algo One 5.2.0.6

Integrated Market and Credit Risk
Mark-to-Future Aggregator Finance Functions

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Chapter 1 Conventions and Notation

This document defines the finance functions available in the Integrated Market and Credit Risk (IMCR) solution.

The following notation and conventions are used in this document:

Symbol	Definition
I_k	Instrument in a portfolio *P*
P_l	Sub-portfolios in a portfolio *P*
$V_{I_k}^{ij}$	Instrument attribute `Value` (default is `THEO/Value`) at simulation time t_i under scenario s_j, obtained from the MtF table (the value has been scaled by multiplying the number of positions)
$d(t_{ref}, t_{ref}, t_i)$	Discount factor in which the rates are taken from the discount curve as indicated by the Instrument attribute `Discount Curve` under scenario s_j
t_i or t_k	Scenario time step (trigger time)
t_0	Session time
K	Number of time steps (trigger times)
n	Number of scenarios

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Chapter 2 Expression Definitions and Instances

The main role of Integrated Market and Credit Risk (IMCR) calculation is to calculate statistical measures and scenario-based results with inputs from the following three types of data:

- Hierarchy data, which includes the expressions
- Mark-to-Future® (MtF) cubes that hold position, instrument, and portfolio data
- Real-time deal values

You need to follow the three step process to perform the desired evaluation using the IMCR solution:

1. Create a statistical function and a collateral model in a stand-alone application (`$MAG_HOME/bin/defineFunctions`) that runs on the server where the IMCR solution is installed. For details of the available statistical functions and collateral models, refer to Chapter 6 Functions on page 75 and Chapter 4 MAG Collateral Models on page 7.
2. Create an expression in `CreditExpressions.csv` and refer the attributes `FunctionId` and `CollateralStyle` to the statistical function and collateral model created in step 1.
3. Create an expression instance in `ExpressionInstances.csv` and refer the attribute `ExpressionId` to the expression created in step 2.

Note

Steps 2 and 3 can be completed either using a command line interface or via the RTCE graphical interface. For more information about `CreditExpressions.csv` and `ExpressionInstances.csv` files, see the *Integrated Market and Credit Risk Data Dictionary*.

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Chapter 3 Netting Rule

The value of a portfolio on a node is determined by the netting feature specified by `NettingRule` (i.e. one of the attributes defined in the expression `CreditExpressions.csv`), `CloseoutNetting` (i.e. one of the attributes defined in `MasterAgreement.csv`), `AgreementEnforceable` (i.e. one of the attributes defined in `Counterparty.csv`), as well as the perspective direction of the expression.

The attribute `NettingRule` can take one of the following values: NONET, AGREEMENT, LEGAL_OPINION.

3

Netting Rule

Let $V^X(t_i, s_j)$ be equal to the portfolio value denominated in `SimulationCurrency` on node X at time t_i under scenario s_j.

Let $MtM^X(t_i, s_j)$ be equal to the netted sum of the mark-to-market of all positions denominated in `SimulationCurrency` on node X at time t_i under scenario s_j. That is $MtM^X(t_i, s_j) = \sum_{k=1}^{K} v_k(t_i, s_j)$, where $v_k(t_i, s_j)$ is the mark-to market value of position k at time t_i under scenario s_j.

Let $P^X(t_i, s_j)$ be equal to the sum of the mark-to-market of all positions denominated in `SimulationCurrency` that are positive (positive positions are those positions which are in-the-money for the user) on node X at time t_i under scenario s_j. That is $P^X(t_i, s_j) = \sum_{k=1}^{K} \max\left(v_k(t_i, s_j), 0\right)$.

Let $N^X(t_i, s_j)$ be equal to the sum of the mark-to-market of all positions denominated in `SimulationCurrency` that are negative (negative positions are those positions which are in-the-money for the counterparty) on node X at time t_i under scenario s_j. That is $N^X(t_i, s_j) = \sum_{k=1}^{K} \min\left(v_k(t_i, s_j), 0\right)$.

Portfolio Value From The Perspective Of the User (Forward View)

Case 1.1. NettingRule = NONET

$V^X(t_i, s_j) = P^X(t_i, s_j)$

Case 1.2. NettingRule = AGREEMENT

If CloseoutNetting = True

$V^X(t_i, s_j) = MtM^X(t_i, s_j)$

Else

$V^X(t_i, s_j) = P^X(t_i, s_j)$

Case 1.3. NettingRule = LEGAL_OPINION

If Closeout Netting Pools (CNPs) are present under the MA

$V^X(t_i, s_j) = \sum_{l=1}^{L} \max\left(MtM^X_{CNP_l}(t_i, s_j), 0\right) + P^X_{N\_CNP}(t_i, s_j)$

where CNP_l (l = 1, ..., L) are the closeout netting pools on node X;

$MtM^X_{CNP_l}(t_i, s_j)$ is the netted sum of the mark-to-market of all positions within CNP_l on node X at time t_i under scenario s_j;

and $P^X_{N\_CNP}(t_i, s_j)$ is the sum of the mark-to-market of all positive positions on node X that don't belong to any closeout netting pools on node X.

Else

If AgreementEnforceable = True

If CloseoutNetting = True

$V^X(t_i, s_j) = MtM^X(t_i, s_j)$

```
                 Else
                         V^X(t_i, s_j) = P^X(t_i, s_j)
        Else V^X(t_i, s_j) = P^X(t_i, s_j)
```

Portfolio Value From The Perspective Of the Counterparty (Reverse View)

Case 2.1. NettingRule = NONET

$V^X(t_i, s_j) = -N^X(t_i, s_j)$

Case 2.2. NettingRule = AGREEMENT or LEGAL_OPINION

```
If CloseoutNetting = True
        V^X(t_i, s_j) = -MtM^X(t_i, s_j)
Else V^X(t_i, s_j) = -N^X(t_i, s_j)
```

3.1 Non-simulated values

Details on how the non-simulated values should be aggregated on nodes will be available at a later stage.

Chapter 4 MAG Collateral Models

- **Introduction**

Two types of collateral are considered in calculating collateralized exposure:

- Initial margin, and
- Variation margin (also referred to as Collateral Balance as it includes Independent Amounts as defined in the ISDA CSA agreement: http://assets.isda.org/media/e0f39375/b61bd039-pdf)

A MAG collateral model defines the methodology to be used for the variation margin calculation, as well as whether to include variation margin, initial margin or both when performing the collateralized exposure calculation.

For a counterparty for which a collateral agreement is in place, calculating the exposure over time requires knowing the amount of collateral that will be posted in the future, i.e. future variation margin. This amount is dependent on the terms of the agreement and the value of the transactions under the agreement. There are 2 main collateral models in MAG to estimate future variation margin: the *Path Dependent Model* and the *Settlement Adjusted Path Dependent Model*. The *Shortcut Empirical Model*, is built on a simplified version of the Path Dependent model.

4.1 Variation Margin Models

4.1.1 The Path Dependent Collateral Model – Collateral Balance and Margin Call

The collateral balance and margin call computation performed by MAG in the path dependent collateral model depends on the value assigned to attributes `CPPostStyle` and `UserPostStyle`. `CPPostStyle` and `UserPostStyle` can be set to one of the following values:

- Net: Collateral requirement is calculated by treating all trades in the portfolio on a net basis.
- Gross: Collateral requirement is calculated by treating all trades in the portfolio on a gross basis.
- None: Collateral requirement is always zero as no variation margin is posted in this case.

By default, `CPPostStyle` = `Net` and `UserPostStyle` = `Net`, which means collateral balance and margin call are computed by treating all trades in the trade portfolio on a net basis. The default setting represents the traditional way to model collateral by assuming netting within a CSA (i.e. bilateral netting variation margin computation).

When either `CPPostStyle` or `UserPostStyle` is set to `Gross`, the path dependent model produces two separate collateral balances and margin calls for a given time step and scenario, with one collateral balance and margin call for user and the other collateral balance and margin call for counterparty. The two separate collateral balances are reflected in the collateralized exposure computation.

The collateral model attribute `Use Physical Collateral` is used to specify whether or not the collateral posted in either direction at the future time steps is modeled in the form of credit mitigant portfolio (CMP).

- `Use Physical Collateral` = `True`. Two CMPs, CMP_{user} and CMP_{cpty} are created and used as representative sets of model collateral posted by user and counterparty.

- `Use Physical Collateral` = `False`. Collateral posted in either direction is assumed to be in the form of cash.

4.1.1.1 Model Inputs Overview

MAG's collateral models accept three types of inputs:

- The simulated, netted mark-to-market of all the transactions under the collateral agreement in the currency of the agreement, *CSAcy*, over scenarios and time steps (simulation dates). A positive mark-to-market indicates the portfolio is in-the-money for the user and a negative mark-to-market indicates that it is in-the-money for the counterparty. (When the risk of over-collateralization due to maturing trades is excluded from the exposure computation, the netted mark-to-market (in *CSAcy*) of all transactions that have not matured by the next simulation date will also be an input.)
- The RTCE inputs which include the terms of the agreement – CSA parameters.
- The model parameters' values.

For a full description of the latter two inputs, see the section 4.1.1.7 Model Inputs and Notation on page 27.

4.1.1.2 Call Dates

The amount of collateral posted and margin call made at a reporting date requires knowledge of margin calls made at previous call dates. Since it is not computationally feasible (and sometimes not necessary) to compute the margin call at all call dates, the collateral model uses a subset of these. This subset, which is referred to as the set of relevant call dates, depends on:

- The margin period of risk, represented by the counterparty/user *lag* + *curePeriod* (RTCE inputs).
- The call schedule, obtained from the *callPeriod* (RTCE input).
- The model parameter `Additional Call Dates` that is used to ensure that the risk arising from minimum transfer amounts is captured adequately.

■ The set of all Call Dates

In RTCE, the set of *all* call dates is determined by the frequency (call period), *p*, of the margin calls.

If $t_0, t_1, t_2, \ldots$ denote the set of simulation dates in ascending order, that is, the dates for which the MtM of the portfolio has been simulated, then the set, $\bar{\Gamma}$, is defined by

$\bar{\Gamma}:=\{t_0, t_1, t_2, \ldots\}$ if the call period *p* is zero and

$\bar{\Gamma}:=\{\mathrm{c}_k\colon \mathrm{c}_k = t_0 + k \times p, k\in\aleph\}$, if the call period *p* is greater than zero, where $\aleph$ is the set of natural numbers.

In both cases, $c_0, c_1, c_2, \ldots$ denote the elements of $\bar{\Gamma}$ in ascending order.

Since we need to know how much collateral was called before $c_0 = t_0$, we need to refer to the call date immediately before, denoted by c_{-1}, and assume that the collateral called up to and including this date is the initial collateral balance, *ICB*, which is a RTCE input.

The set of call dates is then defined by

$\Gamma = \{c_{-1}, c_0, c_1, c_2, \ldots\}$

The set of Relevant Call Dates

We now define the subset of call dates used in the collateral model. This subset is determined by the set of reporting dates[*1] $\mathrm{R}{:=}\{r_0, r_1, r_2, ...\}$ which is a subset of $\{t_0, t_1, t_2, ...\}$.

We use the following notation to specify the counterparty and user margin period of risk used in the model:

lag_{cpty}:= cLag + cCurePeriod and lag_{user}:= uLag + uCurePeriod

```
IF (CPPostStyle = Net && UserPostStyle = Net)
    One set of relevant call dates is generated in the collateral model with:
    lag_max = max(lag_cpty, lag_user)
    lag_min = min(lag_cpty, lag_user)
ELSE
    Two sets of relevant call dates are generated, one for counterparty side and o
ne for user side.
    IF (collateral balance is calculated for counterparty side)
        lag_max = lag_min = lag_cp
    ELSE
        lag_max = lag_min = lag_user
```

(see 4.1.1.7 Model Inputs and Notation on page 27 section for the definitions of *cLag*, *cCurePeriod*, *uLag* and *uCurePeriod*).

For each reporting date r_i, we first define the minimum set of call dates that are used in the model to compute the collateral balance on r_i:

- All collateral called up to and including the first call date on or before r_i-lag_{max} is available by r_i (if neither party has defaulted on or before the call date). We denote this call date by c_{n_i} . (If there are no call dates before r_i-lag_{max}, set c_{n_i}:= c_{-1} .) The model estimates the amount of collateral called up to this date.
- If the lags are not equal, collateral calls between r_i-lag_{max} and r_i-lag_{min} are available by r_i (provided there is no default or or before r_i-lag_{min}) if they are called by the party with the longer lag. That is, if they are posted by the party with the shorter lag. Let $c_{n_i+1}, c_{n_i+2}, ..., c_{n_i+m_i} \in \Gamma$ be the call dates between r_i-lag_{max} (exclusive) and r_i-lag_{min} (inclusive), if any (note that *m* might be zero, that is, there might not be any such call dates). In the model we compute the margin calls (transfers) on these dates.
- To adequately capture the risk arising from minimum transfer amounts, the amount of collateral called on some dates before c_{n_i} are needed. The model parameter `Additional Call Dates` specifies the number of additional call dates that should be used. If *l* :=Additional Call Dates, we denote the *l* call dates immediately before c_{n_i} by $c_{n_i-l}, ..., c_{n_i-1}$

  Note that there might be less than *l* such call dates and possibly none, for example if c_{n_i} is close to the beginning of the session date t_0.

The set of all these dates are denoted by

$\Gamma_{cb}{}^{(i)} = \left\{c_{n_i-l}, ..., c_{n_i-1}, c_{n_i}, c_{n_i+1}, c_{n_i+2}, ..., c_{n_i+m_i}\right\}$

If $r_i = c_{q_i}$ for some $c_{q_i} \in \bar{\Gamma}$, that is, if r_i is a call date, we now define the minimum set of dates that are used in the model to compute the margin call on r_i.

- To determine the margin call on r_i, the amount of collateral called up to and including the previous call date $c_{q_i\text{-}1}$ is needed. Note that $c_{q_i\text{-}1}$ might be equal to c_{-1}.
- To more accurately capture the impact of minimum transfer amounts on the magnitude of margin calls, the model parameter `Additional Call Dates`, *l*, is used. When $l \geq 2$, the additional *l*-1 dates $c_{q_i - l}, ..., c_{q_i - 2}$

  Note that there might be less than *l* -1 such call dates and possibly none, for example if r_i is close to the beginning of the session date t_0. are used.

Define

$\Gamma_{mc}{}^{(i)} := \left\{ c_{q_i - \max\{1, l\}} ..., c_{q_i - 1}, c_{q_i} \right\}$

The set of relevant Call Dates is defined by

$\Gamma_{\text{relevant}} := \left(\bigcup_{i \in \aleph} \left(\Gamma_{cb}{}^{(i)} \cup \Gamma_{mc}{}^{(i)} \right) \right)$

To ensure consistency between two different reporting dates calculations and between the collateral balance and margin call calculations (although this consistency is not necessary, it makes validation easier.):

- the set of dates used to compute the collateral balance on r_i, $\Gamma_{\text{relevant} - cb}{}^{(i)}$, is
  - the largest subset of consecutive relevant call dates less than or equal to $c_{n_i + m_i}$ if `Link Relevant Call Dates` = `FALSE`.
  - all relevant call dates less than or equal to $c_{n_i + m_i}$ if `Link Relevant Call Dates` = `TRUE`.

  Note that this set contains $\Gamma_{cb}{}^{(i)} = \left\{ c_{n_i - l}, ..., c_{n_i - 1}, c_{n_i}, c_{n_i + 1}, c_{n_i + 2}, c_{n_i + m_i} \right\}$
- the set of dates used to compute the margin call on r_i, $\Gamma_{\text{relevant} - mc}{}^{(i)}$, is
  - the largest subset of consecutive relevant call dates less than or equal to $r_i = c_{q_i}$ if `Link Relevant Call Dates` = `FALSE`.
  - all relevant call dates less than or equal to $r_i = c_{q_i}$ if `Link Relevant Call Dates` = `TRUE`.

  Note that this set contains $\Gamma_{mc}{}^{(i)} := \left\{ c_{q_i - \max\{1, l\}}, ..., c_{q_i - 1}, c_{q_i} \right\}$

*1 This is the set of dates on which risk measures are reported. At present, the set of reporting dates is equal to the set of simulation dates.

4.1.1.3 Collateral Balance and Margin Call Algorithm when CPPostStyle = Net and UserPostStyle = Net

We now describe the computation of collateral balance and margin call for a given scenario *s* and a reporting date r_i when both `CPPostStyle` and `UserPostStyle` are set to `Net`.

Step 1: Determine the necessary call dates

When computing the collateral balance, we use the set of call dates $\Gamma^{(i)}_{relevant} := \Gamma^{(i)}_{relevant\_cb}$, and when computing the margin call, we use $\Gamma^{(i)}_{relevant} := \Gamma^{(i)}_{relevant\_mc}$.

Step 2: Determine the initial collateral post direction and create the credit mitigation portfolio (CMP)

Step 2.1: Compute the initial collateral balance (ICB) for the CSA node, i.e. the total collateral called up to and including t_0

ICB is computed as the sum of the values in the Collateral Delivery Currency of all variation margin mitigants assigned to the CSA whose `Applicability` is set to `INITIAL_BALANCE`. This computation uses the following mitigants' parameters, which are defined in 4.1.1.7 Model Inputs and Notation on page 27:

- `MTM`
- `MTMCcy`
- `PostDirection`
- `DeliveryDate`
- `ReturnDate`
- `Applicability`

If the CSA parameter `Collateral Mode` = `Cash only`, proceed to step 3.

If the CSA parameter `Collateral Mode` = `Physical collateral only in ICB post direction` or `Physical collateral in either direction`, the following Step 2.2 and Step 2.3 describe the algorithm used to determine the initial collateral post direction thus create CMP and adjust position unit at t_0.

Step 2.2: Determine the initial collateral post direction at t_0 and create the CMP

- If $ICB > 0$, *Initial collateral post direction = counterparty posts*

  Create a CMP that includes all cash and non-cash mitigants with `PostDirection` = `To User` at t_0.

  All non-cash mitigants with `PostDirection` = `To Counterparty` at t_0 are converted to equivalent amount of cash and combined with the cash mitigants with `PostDirection` = `To Counterparty` at t_0 before being used in CMP position unit adjustment.

- If $ICB < 0$, *Initial collateral post direction = user posts*

  Create a CMP that includes all cash and non-cash mitigants with `PostDirection` = `To Counterparty` at t_0.

  All non-cash mitigants with `PostDirection` = `To User` at t_0 are converted to equivalent amount of cash and combined with the cash mitigants with `PostDirection` = `To User` at t_0 before being used in CMP position unit adjustment.

- If $ICB = 0$, *Initial collateral post direction = none* OR *Initial collateral post direction = counterparty posts*

  If the zero ICB is due to the unavailability of information of collateral called up to and including t_0, it is presumed that the collateral posted by user and counterparty in the future time steps is in cash. In this case, *Initial collateral post direction = none* and no CMP needs to be created at t_0.

  Note

  When *Initial collateral post direction = none*, `Collateral Mode` is defaulted to `Cash only` and user provided input of this attribute will be ignored.

  If the zero ICB is caused by the exact offset of total user posted collateral and total counterparty posted collateral at t_0, *Initial collateral post direction = counterparty posts* and a CMP containing all cash and non-cash mitigants with `PostDirection` = `To User` at t_0 is created. All non-cash mitigants with `PostDirection` = `To Counterparty` at t_0 are converted to equivalent amount of cash and combined with cash mitigants with `PostDirection` = `To Counterparty` at t_0 before being used in CMP position unit adjustment.

Step 2.3: Compute the position unit (PU) of CMP at t_0

Once a CMP is created, the PU of the CMP at t_0 is computed to tie back to ICB:

The simulated mark-to-market of CMP at t_0 is computed as:

$$CMtM^{sim}(t_0) = \sum_{m=1}^{M} PU_m \times CMtM_m^{sim}(t_0)$$

where, PU_m is the position unit of mitigant m at t_0 obtained from source system;

$CMtM_m^{sim}(t_0)$ is the simulated mark-to-market of mitigant m at t_0; and

M is the total number of mitigants in CMP.

PU of CMP at t_0 is:

$$PU(t_0) = ICB/CMtM^{sim}(t_0)$$

Step 3: Obtain the mark-to-market of trade portfolio and credit mitigation portfolio at each call date $c \in \Gamma_{relevant}^{(i)} \backslash \{c_{-1}\}$ *2

At each call date *c* and scenario *s*, obtain the trade portfolio mark-to-market (*MtM(c,s)*), and credit mitigation portfolio mark-to-market after applying all relevant margin call haircuts*3 (*CMtMHC(c,s)*).

Note

When `Collateral Mode` = `Cash Only` or *Initial Collateral Post Direction = none*, no CMP is created at t_0 thus only *MtM(c,s)* needs to be obtained at each call date *c* and scenario *s*.

If *c* is not a simulation date, the *MtM(c,s)* and *CMtMHC(c,s)* are modeled using a Brownian Bridge (see the section 4.1.1.6 MtM on Non-Simulated Dates on page 24).

Step 4: Compute the amount of collateral required at each call date $c \in \Gamma_{relevant}^{(i)} \backslash \{c_{-1}\}$

Define the amount of collateral required by

$$\mathrm{CR}(c,s) = \begin{cases} \mathrm{MtM}(c,s) + cIA + uIA - cT \text{if} CPosts = \text{True and } \mathrm{MtM}(c,s) + cIA + uIA > cT \\ \mathrm{MtM}(c,s) + cIA + uIA - uT \text{if} UPosts = \text{True and } \mathrm{MtM}(c,s) + cIA + uIA < uT \\ 0 \qquad \text{otherwise} \end{cases}$$

where cIA is the counterparty independent amount, uIA is the user independent amount, cT is the counterpart threshold, uT is the user threshold and *cPosts* and *uPosts* are as defined in section 4.1.1.7 Model Inputs and Notation on page 27.

Step 5: Compute the cash collateral and physical collateral called as well as margin call on each call date $c \in \Gamma_{relevant}^{(i)} \backslash \{c_{-1}\}$

Let $\Gamma_{\text{relevant}}{}^{(i)} = \{c_j, c_{j+1}, \ldots, c_{j+k}\}$

1. Collateral Mode = Cash only

First, we approximate the amount of cash collateral called up to and including c_j by

$$CashCollateral(c_j,s) = \begin{cases} ICB \ if \ c_j = c_{-1} \\ CR(c_j,s) \ if \ c_j \neq c_{-1} \end{cases}$$

Then recursively define the margin calls on the other call dates in $\Gamma_{\text{relevant}}^{(i)} = \{c_j, c_{j+1}, ..., c_{j+k}\}$ as follows.

For $1 \le l \le k$,

$$MC(c_{j+l}, s) = \begin{cases} 0 \ \ if \ \ uMTA < CR(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s) < cMTA \\ round\left[CR(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s), RA\right] \ , \ otherwise \end{cases}$$

where function *round(x,m)* rounds *x* to the nearest integer multiple of *m*; *cMTA* is the counterparty minimum transfer amount; *uMTA* is the user minimum transfer amount, and *RA* is the round-off amount. All these parameters are defined in section 4.1.1.7 Model Inputs and Notation on page 27.

The total amount of cash collateral called at c_{j+l} under scenario *s* is:

$$CashCollateral(c_{j+l}, s) = CashCollateral(c_{j+l-1}, s) + MC(c_{j+l}, s)$$

The incremental amount of cash collateral called at c_{j+l} under scenario *s* is:

$$\Delta CashCollateral(c_{j+l}, s) = CashCollateral(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s) = MC(c_{j+l}, s)$$

2. Collateral Mode = Physical collateral only in ICB post direction

First, decide the collateral post direction at each call date *c* and scenario *s* as:

- If *CR*(*c*,*s*) > 0, *Collateral post direction* = *counterparty posts*
- If *CR*(*c*,*s*) < 0, *Collateral post direction* = *user posts*
- If *CR*(*c*,*s*) = 0, *Collateral post direction* = *no post*

Note

If *Initial collateral post direction* = *none*, no CMP is created at t_0, the calculation of cash collateral called and margin call is the same as in the case `Collateral Mode` = `Cash only`.

At c_j, if *Collateral post direction* = *Initial collateral post direction*, it is assumed that all collateral called is in the form of physical collateral represented by credit mitigation portfolio.

The position unit of credit mitigation portfolio at c_j is calculated as:

$$PU(c_j, s) = \begin{cases} PU(t_0) \ \ if \ \ c_j = c_{-1} \\ \dfrac{\mathrm{CR}(c_j, s)}{CMtMHC(c_j, s)} \ \ if \ \ c_j \neq c_{-1} \end{cases}$$

The amount of cash collateral at c_j is 0, that is $CashCollateral(c_j, s) = 0$.

At c_j, if *Collateral post direction* ≠ *Initial collateral post direction*, it is assumed that all collateral called is in the form of cash.

The amount of cash collateral at c_j is calculated as:

$$CashCollateral(c_j, s) = \begin{cases} ICB \ \ if \ \ c_j = c_{-1} \\ \mathrm{CR}(c_j, s) \ \ if \ \ c_j \neq c_{-1} \end{cases}$$

The position unit of credit mitigation portfolio at c_j is 0, that is $PU(c_j, s) = 0$

Then recursively define the margin calls on the other call dates in $\Gamma_{\text{relevant}}^{(i)} = \{c_j, c_{j+1}, ..., c_{j+k}\}$ as follows:

$$MC(c_{j+l}, s) = \begin{cases} 0 \ \ if \ \ uMTA < CR(c_{j+l}, s) - PU(c_{j+l-1}, s) \times CMtMHC(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s) < cMTA \\ round\left[CR(c_{j+l}, s) - PU(c_{j+l-1}, s) \times CMtMHC(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s), RA\right] \ , \ otherwise \end{cases}$$

,

where $1 \leq l \leq k$.

For each call date c_{j+l} under scenario *s*, the following is calculated:

- The total position unit of CMP: $PU(c_{j+l}, s)$
- The incremental position unit of CMP: $\Delta PU(c_{j+l}, s)$
- The total amount of cash collateral: $CashCollateral(c_{j+l}, s)$
- The incremental amount of cash collateral: $\Delta CashCollateral(c_{j+l}, s)$

If *Initial collateral post direction = counterparty posts*, on each call date c_{j+l}, the total position unit of CMP is floored at 0 (i.e. $PU(c_{j+l}, s) \geq 0$) and the total amount of cash collateral is capped at 0 (i.e. $CashCollateral(c_{j+l}, s) \leq 0$). Therefore,

$$PU(c_{j+l}, s) = Max\left[PU(c_{j+l-1}, s) + \left(MC(c_{j+l}, s) + CashCollateral(c_{j+l-1}, s)\right)/CMtMHC(c_{j+l}, s), 0\right]$$

$$\Delta PU(c_{j+l}, s) = PU(c_{j+l}, s) - PU(c_{j+l-1}, s)$$

$$CashCollateral(c_{j+l}, s) = Min\left[CashCollateral(c_{j+l-1}, s) + MC(c_{j+l}, s) + PU(c_{j+l-1}, s) \times CMtMHC(c_{j+l}, s), 0\right]$$

$$\Delta CashCollateral(c_{j+l}, s) = CashCollateral(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s)$$

If *Initial collateral post direction = user posts*, on each call date c_{j+l}, the total position unit of CMP and the total amount of cash collateral are floored at 0 (i.e. $PU(c_{j+l}, s) \geq 0$ $CashCollateral(c_{j+l}, s) \geq 0$). Therefore,

$$PU(c_{j+l}, s) = Max\left[PU(c_{j+l-1}, s) + \left(MC(c_{j+l}, s) + CashCollateral(c_{j+l-1}, s)\right)/CMtMHC(c_{j+l}, s), 0\right]$$

$$\Delta PU(c_{j+l}, s) = PU(c_{j+l}, s) - PU(c_{j+l-1}, s)$$

$$CashCollateral(c_{j+l}, s) = Max\left[CashCollateral(c_{j+l-1}, s) + MC(c_{j+l}, s) + PU(c_{j+l-1}, s) \times CMtMHC(c_{j+l}, s), 0\right]$$

$$\Delta CashCollateral(c_{j+l}, s) = CashCollateral(c_{j+l}, s) - CashCollateral(c_{j+l-1}, s)$$

3. Collateral Mode = Physical collateral in either direction

First, we approximate the position unit of CMP called up to and including c_j by

$$PU\left(c_j, s\right) = \begin{cases} PU(t_0) \ if \ \ c_j = c_{-1} \\ \dfrac{\mathrm{CR}\left(c_j, s\right)}{CMtMHC(c_j, s)} \ if \ \ c_j \neq c_{-1} \end{cases}$$

Then recursively define the margin calls on the other call dates in $\Gamma_{\mathrm{relevant}}{}^{(i)} = \left\{c_j, c_{j+1}, ..., c_{j+k}\right\}$ as follows:

$$MC(c_{j+l}, s) = \begin{cases} 0 \ if \ \ uMTA < CR(c_{j+l}, s) - PU(c_{j+l-1}, s) \times CMtMHC(c_{j+l}, s) < cMTA \\ round\left[CR(c_{j+l}, s) - PU(c_{j+l-1}, s) \times CMtMHC(c_{j+l}, s), RA\right] \ , \ otherwise \end{cases},$$

where, $1 \leq l \leq k$.

For each call date c_{j+l} under scenario *s*, the following is calculated:

- The total position unit of CMP: $PU(c_{j+l}, s) = PU(c_{j+l-1}, s) + MC(c_{j+l}, s)/CMtMHC(c_{j+l}, s)$
- The incremental postition unit of CMP: $\Delta PU(c_{j+l}, s) = PU(c_{j+l}, s) - PU(c_{j+l-1}, s)$

Step 6: Compute cashflows during the margin period of risk preceding the reporting date r_i

This step is only needed if the model parameter [**Isolate Collateral Spikes**] is set to True. In this case, the counterparty and user lags must be equal. That is $lag_{\max} = lag_{cpty} = lag_{user}$.

Let $lag := lag_{\max}$. Let c denote the first call date on or before $r_i - lag$. Let t_{k-1} and t_k denote the simulation dates before and on or after c respectively. We approximate the magnitude of the cashflows between c and t_k when $c \neq t_k$, $CF([c, t_k], s)$, as follows:

1. Assuming cashflows occur gradually between t_{k-1} and t_k, that is, if the model parameter [**Gradual Cashflows**] is set to True:.
   $CF([c, t_k], s) := \frac{(t_k - c)}{(t_k - t_{k-1})} \times [MtM(t_{k-1}, s) - MtM_{nmf}(t_{k-1}, s)]$
2. Assuming all cashflows between t_{k-1} and t_k occur after c, that is, if the model parameter [**Gradual Cashflows**] is set to False:
   $CF([c, t_k], s) := [MtM(t_{k-1}, s) - MtM_{nmf}(t_{k-1}, s)]$

Here $MtM_{nmf}(t, s)$ denotes the netted mark-to-market on simulation time step t and scenario s of all those trades that have not matured by the following simulation date.

When $c = t_k$, $CF([c, t_k], s) = 0$.

Define the cashflows during the margin period of risk preceding r_i, $S(r_i, s)$ by

$$S(r_i, s) := CF([c, t_k], s) + \sum_{t_k \leq t < r_i} [MtM(t, s) - MtM_{nmf}(t, s)]$$

where the ts are simulation dates.

Step 7: Compute the amount of collateral available (collateral balance) at r_i

Define

$$\overline{CB}(r_i, s) = [PU(c_{n_i}, s) \times CMtM_{default}(r_i, s) + CashCollateral(c_{n_i}, s)] +$$

$$\begin{cases} \sum_{l=1}^{m_i} \frac{\max(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} \times \left[\Delta PU(c_{n_i+l}, s) \times CMtM_{default}(r_i, s) + \Delta CashCollateral(c_{n_i+l}, s)\right] & if\ \ lag_{cp} < lag_{user} \\ \sum_{l=1}^{m_i} \frac{\min(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} \times \left[\Delta PU(c_{n_i+l}, s) \times CMtM_{default}(r_i, s) + \Delta CashCollateral(c_{n_i+l}, s)\right] & if\ \ lag_{cp} > lag_{user} \\ 0 & otherwise \end{cases}$$

where, $CMtM_{default}(r_i, s)$ is the mark-to-market of CMP after applying the haircuts at default[*1] specifically for exposure calculation; and the summations are zero if $m_i = 0$.

Then the collateral balance, $CB(r_i, s)$, is defined by

$$CB(r_i, s) = \begin{cases} \overline{CB}(r_i, s)\ \ if\ \ IsolateCollateralSpikes = F \\ \overline{CB}(r_i, s) - S(r_i, s)\ \ if\ \ IsolateCollateralSpikes = T \end{cases}$$

Note

Risk Measures can be defined on any of the Collateral Balance, Margin Call and Cashflow planes.

*1 The haircuts at default applied to the mark-to-market of CMP in the collateral balance computation are different from the ones used in the margin call computation. Applying haircuts at default in the collateral balance computation will produce more conservative exposure results.

*2 $\Gamma^{(i)}_{relevant}\backslash\{c_{-1}\}$ is the set of relevant call dates excluding c_{-1} .

*3 The margin call haircut can be applied to both cash and non-cash mitigants in the CMP.

4.1.1.4 Collateral Balance and Margin Call Algorithm when CPPostStyle or UserPostStyle is Gross

We now describe the computation of collateral balance and margin call for a given scenario *s* and a reporting date r_i when either `CPPostStyle` or `UserPostStyle` is set to `Gross`.

Step 1: Determine the necessary call dates

When computing the collateral balance, we use the set of call dates $\Gamma^{(i)}_{relevant} := \Gamma^{(i)}_{relevant\_cb}$, and when computing the margin call, we use $\Gamma^{(i)}_{relevant} := \Gamma^{(i)}_{relevant\_mc}$.

Step 2: Compute the initial collateral balance (ICB) and create the credit mitigation portfolio (CMP)

Step 2.1: Compute the initial collateral balance (ICB) for the CSA node, i.e. the total collateral called up to and including t_0

When `CPPostStyle` = `Gross`, the initial collateral balance for user (ICB_{cpty}) is computed by summing up the values in the Collateral Delivery Currency of all mitigants assigned to the CSA with

- `MarginType` = `Variation`
- `Applicability` = `INITIAL_BALANCE` or `BOTH`, AND
- `PostDirection` = `To User`

When `UserPostStyle` = `Gross`, the initial collateral balance for counterparty (ICB_{user}) is computed by summing up the values in the Collateral Delivery Currency of all mitigants assigned to the CSA with

- `MarginType` = `Variation`
- `Applicability` = `INITIAL_BALANCE` or `BOTH`, AND
- `PostDirection` = `To Counterparty`

ICB_{cpty} and ICB_{user} are computed using the following mitigant's parameters, which are defined in 4.1.1.7 Model Inputs and Notation on page 27:

- `MTM`
- `MTMCcy`
- `PostDirection`
- `DeliveryDate`
- `ReturnDate`
- `Applicability`

If the collateral model parameter `Use Physical Collateral` = `False`, proceed to step 3.

If the collateral model parameter `Use Physical Collateral` = `True`, proceed to Step 2.2 and 2.3.

Step 2.2: Create the CMP

When `CPPostStyle` = `Gross`, at t_0, create a CMP_{cpty} that includes all mitigants with:

- `Applicability` = `MITIGANT_POOL` or `BOTH`, AND
- `PostDirection` = `To User`

When `UserPostStyle` = `Gross`, at t_0, create a CMP_{user} that includes all mitigants with:

- `Applicability` = `MITIGANT_POOL` or `BOTH`, AND
- `PostDirection` = `To Counterparty`

It is assumed that the gross collateral posted by either side in the future time steps will be made up with the same mitigants in the same quantity ratio as is in the CMP at t_0. In the case where there is no mitigant that satisfies the required attributes settings described above to be included in the CMP_{cpty} or/and CMP_{user}, no corresponding CMP(s) is/are created, assuming that the gross collateral posted by the corresponding side(s) in the future time steps will all be in cash.

Step 2.3: Compute the position unit (PU) of CMP at t_0

The simulated mark-to-market of CMP at t_0 is computed as:

$CMtM_A^{sim}(t_0) = \sum_{m=1}^{M} PU_m \times CMtM_{A,m}^{sim}(t_0)$

where, $A = \begin{cases} counterparty\ if\ CPPostStyle = Gross \\ user\ if\ UserPostStyle = Gross \end{cases}$;

PU_m is the position unit of mitigant m at t_0 obtained from source system;

$CMtM_m^{sim}(t_0)$ is the simulated mark-to-market of mitigant m at t_0; *1 and

M is the total number of mitigants in CMP.

PU of CMP at t_0 is then computed as:

When `CPPostStyle` = `Gross`, $PU_{cpty}(t_0) = \begin{cases} \frac{ICB_{cpty}}{CMtM_{cpty}^{sim}(t_0)}\ if\ CMtM_{cpty}^{sim}(t_0) \neq 0 \\ 0\ if\ CMtM_{cpty}^{sim}(t_0) = 0 \end{cases}$

When `UserPostStyle` = `Gross`, $PU_{user}(t_0) = \begin{cases} \frac{ICB_{user}}{CMtM_{user}^{sim}(t_0)}\ if\ CMtM_{user}^{sim}(t_0) \neq 0 \\ 0\ if\ CMtM_{user}^{sim}(t_0) = 0 \end{cases}$

Note

when $CMtM_{cpty}^{sim}(t_0) = 0$, the future collateral posted by counterparty is assumed to be in cash. The same applies to user side.

Step 3: Obtain the mark-to-market of trade portfolio and credit mitigation portfolio at each call date $c \in \Gamma_{relevant}^{(i)} \backslash \{c_{-1}\}$ *3

Let $P(t_i, s_j)$ be equal to the sum of the mark-to-market of all positions in the trade portfolio denominated in `SimulationCurrency` that are positive at time t_i under scenario s_j. That is $P(t_i, s_j) = \sum_{k=1}^{K} \max\left(v_k(t_i, s_j), 0\right)$, where $v_k(t_i, s_j)$ is the mark-to market value of position k in the trade portfolio at time t_i under scenario s_j.

Let $N(t_i, s_j)$ be equal to the sum of the mark-to-market of all positions in the trade portfolio denominated in `SimulationCurrency` that are negative at time t_i under scenario s_j. That is $N(t_i, s_j) = \sum_{k=1}^{K} \min\left(v_k(t_i, s_j), 0\right)$.

At each call date c under scenario s, obtain the following:

- trade portfolio mark-to-market (*P(c,s)* when `CPPostStyle = Gross`; *N(c,s)* when `UserPostStyle = Gross`)
- credit mitigation portfolio mark-to-market ($CMtM_{cpty}(c,s)$ when `CPPostStyle = Gross`; $CMtM_{user}(c,s)$ when `UserPostStyle = Gross`)
- credit mitigation portfolio mark-to-market after applying all relevant haircuts ($CMtMHC_{cpty}(c,s)$ when `CPPostStyle = Gross`; $CMtMHC_{user}(c,s)$ when `UserPostStyle = Gross`)

If a relevant call date *c* is not a simulation date, the trade portfolio mark-to-market and credit mitigation portfolio mark-to-market after applying all relevant haircuts are modeled using a Brownian Bridge (see the section 4.1.1.6 MtM on Non-Simulated Dates on page 24).

Step 4: Compute the amount of collateral required at each call date $c \in \Gamma^{(i)}_{relevant} \setminus \{c_{-1}\}$

When `CPPostStyle = Gross`, define the amount of collateral required for counterparty as

$$\mathrm{CR}_{cpty}(c,s) = \begin{cases} P(c,s) + cIA + uIA - cT \ \ if \ \ CPosts = True \& P(c,s) + cIA + uIA > cT \\ 0 \ \ otherwise \end{cases}$$

where cIA is the counterparty independent amount, uIA is the user independent amount, cT is the counterpart threshold, and *cPosts* is as defined in section 4.1.1.7 Model Inputs and Notation on page 27.

When `UserPostStyle = Gross`, define the amount of collateral required for user as

$$\mathrm{CR}_{user}(c,s) = \begin{cases} N(c,s) + cIA + uIA - uT \ \ if \ \ UPosts = True \& N(c,s) + cIA + uIA < uT \\ 0 \ \ otherwise \end{cases}$$

where uT is the user threshold, and *uPosts* is as defined in section 4.1.1.7 Model Inputs and Notation on page 27.

Note

The attribute `CPosts` should always be set to True in the above CR_{cpty}(c,s) computation; and the attribute `UPosts` should always be set to True in the above CR_{user}(c,s) computation;

Step 5: Compute the collateral called and margin call on each call date $c \in \Gamma^{(i)}_{relevant} \setminus \{c_{-1}\}$

Let $\Gamma_{\mathrm{relevant}}{}^{(i)} = \{c_j, c_{j+1}, \dots, c_{j+k}\}$

1. Computation if no CMP is created at t_0

In this case, all collateral is in the form of cash.

- `CPPostStyle = Gross`

First, we approximate the amount of cash collateral called up to and including c_j by

$$CashCollateral_{cpty}(c_j, s) = \begin{cases} ICB_{cpty} \ \ if \ \ c_j = c_{-1} \\ CR_{cpty}(c_j, s) \ \ if \ \ c_j \neq c_{-1} \end{cases}$$

Then recursively define the margin calls on the other call dates in $\Gamma_{\mathrm{relevant}}{}^{(i)} = \{c_j, c_{j+1}, \dots, c_{j+k}\}$ as follows.

For $1 \leq l \leq k$,

$$MC_{cpty}(c_{j+l}, s) = \begin{cases} 0 \ \ if \ \ uMTA < CR_{cpty}(c_{j+l}, s) - CashCollateral_{cpty}(c_{j+l-1}, s) < cMTA \\ round\left[CR_{cpty}(c_{j+l}, s) - CashCollateral_{cpty}(c_{j+l-1}, s), RA\right] \ , \ otherwise \end{cases}$$

where function *round(x,m)* rounds *x* to the nearest integer multiple of *m*; *cMTA* is the counterparty minimum transfer amount; *uMTA* is the user minimum transfer amount, and *RA* is the round-off amount. All these parameters are defined in section 4.1.1.7 Model Inputs and Notation on page 27.

The total amount of cash collateral called at c_{j+l} under scenario *s* is:

$$CashCollateral_{cpty}(c_{j+l}, s) = \max\left[CashCollateral_{cpty}(c_{j+l-1}, s) + MC_{cpty}(c_{j+l}, s), 0\right]$$

Note

From user perspective, the amount of $CashCollateral_{cpty}$ should always be non-negative.

- `UserPostStyle = Gross`

First, we approximate the amount of cash collateral called up to and including c_j by

$$CashCollateral_{user}(c_j, s) = \begin{cases} ICB_{user} \ \ if \ \ c_j = c_{-1} \\ CR_{user}(c_j, s) \ \ if \ \ c_j \neq c_{-1} \end{cases}$$

Then recursively define the margin calls on the other call dates in $\Gamma_{\text{relevant}}{}^{(i)} = \{c_j, c_{j+1}, \dots, c_{j+k}\}$ as follows.

For $1 \leq l \leq k$,

$$MC_{user}(c_{j+l}, s) = \begin{cases} 0 \ \ if \ \ uMTA < CR_{user}(c_{j+l}, s) - CashCollateral_{user}(c_{j+l-1}, s) < cMTA \\ round\left[CR_{user}(c_{j+l}, s) - CashCollateral_{user}(c_{j+l-1}, s), RA\right] \ , \ otherwise \end{cases}$$

The total amount of cash collateral called at c_{j+l} under scenario *s* is:

$$CashCollateral_{user}(c_{j+l}, s) = \min\left[CashCollateral_{user}(c_{j+l-1}, s) + MC_{user}(c_{j+l}, s), 0\right]$$

Note

From user perspective, the amount of $CashCollateral_{user}$ should always be non-positive.

2. Computation if CMP is created at t_0

- `CPPostStyle = Gross`

First, we approximate the position unit of CMP_{cpty} called up to and including c_j by

$$PU_{cpty}(c_j, s) = \begin{cases} PU_{cpty}(t_0) \ \ if \ \ c_j = c_{-1} \\ \dfrac{\text{CR}_{cpty}(c_j, s)}{CMtMHC_{cpty}(c_j, s)} \ \ if \ \ c_j \neq c_{-1} \end{cases}$$

Then recursively define the margin calls on the other call dates in $\Gamma_{\text{relevant}}{}^{(i)} = \{c_j, c_{j+1}, \dots, c_{j+k}\}$ as follows:

$$MC_{cpty}(c_{j+l}, s) = \begin{cases} 0 \ \ if \ \ uMTA < CR_{cpty}(c_{j+l}, s) - PU_{cpty}(c_{j+l-1}, s) \times CMtMHC_{cpty}(c_{j+l}, s) < cMTA \\ round\left[CR_{cpty}(c_{j+l}, s) - PU_{cpty}(c_{j+l-1}, s) \times CMtMHC_{cpty}(c_{j+l}, s), RA\right] \ , \ otherwise \end{cases},$$

where, $1 \leq l \leq k$.

For each call date c_{j+l} under scenario *s*, compute the total position unit of CMP_{cpty} as:

$$PU_{cpty}(c_{j+l}, s) = \max\left[PU_{cpty}(c_{j+l-1}, s) + MC_{cpty}(c_{j+l}, s) / CMtMHC_{cpty}(c_{j+l}, s), 0\right] .$$

If the CMP_{cpty} is created at t_0, but at one time step $\text{CMtM}_{\text{cpty}}$ becomes 0 or negative under certain scenario, the collateral that counterparty posts thereafter is assumed to all be in cash. The collateral computation will switch from **Computation if the CMP is created at t0**to **Computation if no CMP is created at t0** at the time when $\text{CMtM}_{\text{cpty}}$ becomes 0 or negative.

- `UserPostStyle = Gross`

First, we approximate the position unit of user CMP called up to and including c_j by

$$PU_{user}(c_j, s) = \begin{cases} PU_{user}(t_0)\ if\ \ c_j = c_{-1} \\ \dfrac{\mathrm{CR}_{user}(c_j, s)}{CMtMHC_{user}(c_j, s)}\ if\ \ c_j \neq c_{-1} \end{cases}$$

Then recursively define the margin calls on the other call dates in $\Gamma_{\text{relevant}}{}^{(i)} = \{c_j, c_{j+1}, ..., c_{j+k}\}$ as follows:

$$MC_{user}(c_{j+l}, s) = \begin{cases} 0\ if\ \ uMTA < CR_{user}(c_{j+l}, s) - PU_{user}(c_{j+l-1}, s) \times CMtMHC_{user}(c_{j+l}, s) < cMTA \\ round\left[CR_{user}(c_{j+l}, s) - PU_{user}(c_{j+l-1}, s) \times CMtMHC_{user}(c_{j+l}, s), RA\right]\ ,\ otherwise \end{cases},$$

where, $1 \leq l \leq k$.

For each call date c_{j+l} under scenario s, compute the total position unit of counterparty CMP as:

$$PU_{user}(c_{j+l}, s) = \max\left[PU_{user}(c_{j+l-1}, s) + MC_{user}(c_{j+l}, s)/CMtMHC_{user}(c_{j+l}, s), 0\right] .$$

If the CMP_{user} is created at t_0, but at one time step $\text{CMtM}_{\text{user}}$ becomes 0 or positive under certain scenario, the collateral user posts thereafter is assumed to all be in cash. The collateral computation will switch from **Computation if the CMP is created at t0**to **Computation if no CMP is created at t0** at the time $\text{CMtM}_{\text{user}}$ becomes 0 or positive.

If we are computing the gross margin call, this is the final step; if we are computing the collateral balance, proceed to step 6.

Step 6: Compute cashflows during the margin period of risk preceding the reporting date r_i

This step is only needed if the model parameter [**Isolate Collateral Spikes**] is set to True. When `CPPostStyle = Gross` or `UserPostStyle = Gross`, the counterparty and user lags do not have to be equal. That is, when `CPPostStyle = Gross`, $lag = lag_{cpty}$; when `UserPostStyle = Gross`, $lag = lag_{user}$

Let c denote the first call date on or before $r_i - lag$. Let t_{k-1} and $t_k$denote the simulation dates before and on or after c respectively. We approximate the magnitude of the cashflows between c and t_k when $c \neq t_k$, $CF([c, t_k], s)$, as follows:

- `CPPostStyle = Gross`

1. Assuming cashflows occur gradually between t_{k-1} and t_k, that is, if the model parameter [**Gradual Cashflows**] is set to True:

   $$CF_{cpty}([c, t_k], s) := \frac{(t_k - c)}{(t_k - t_{k-1})} \times [P(t_{k-1}, s) - P_{nmf}(t_{k-1}, s)]$$

2. Assuming all cashflows between t_{k-1} and $t_k$occur after c, that is, if the model parameter [**Gradual Cashflows**] is set to False:

   $$CF_{cpty}([c, t_k], s) := [P(t_{k-1}, s) - P_{nmf}(t_{k-1}, s)]$$

Here $P_{nmf}(t, s)$ denotes the sum of the positions with positive mark-to-market value on simulation time step t under scenario s and have not matured by the following simulation date in the trade portfolio.

When $c = t_k$, $CF_{cpty}([c, t_k], s) = 0$.

Define the cashflows during the margin period of risk preceding r_i, $S(r_i, s)$ by

$$S_{cpty}(r_i, s) := CF_{cpty}([c, t_k], s) + \sum_{t_k \le t < r_i} [P(t, s) - P_{nmf}(t, s)]$$

where the ts are simulation dates.

- `UserPostStyle` = `Gross`

1. Assuming cashflows occur gradually between t_{k-1} and t_k, that is, if the model parameter [**Gradual Cashflows**] is set to True:

   $$CF_{user}([c, t_k], s) := \frac{(t_k - c)}{(t_k - t_{k-1})} \times [N(t_{k-1}, s) - N_{nmf}(t_{k-1}, s)]$$

2. Assuming all cashflows between t_{k-1} and $t_k$occur after c, that is, if the model parameter [**Gradual Cashflows**] is set to False:

   $$CF_{user}([c, t_k], s) := [N(t_{k-1}, s) - N_{nmf}(t_{k-1}, s)]$$

Here $N_{nmf}(t, s)$ denotes the sum of the positions with negative mark-to-market value on simulation time step t under scenario s and have not matured by the following simulation date in the trade portfolio.

When $c = t_k$, $CF_{user}([c, t_k], s) = 0$.

Define the cashflows during the margin period of risk preceding r_i, $S(r_i, s)$ by

$$S_{user}(r_i, s) := CF_{user}([c, t_k], s) + \sum_{t_k \le t < r_i} [N(t, s) - N_{nmf}(t, s)]$$

where the ts are simulation dates.

Step 7: Compute the amount of collateral available (collateral balance) at r_i

Define

$$\overline{CB_A}(r_i, s) = PU_A(c_{n_i}, s) \times CMtM_{A,default}(r_i, s) + CashCollateral_A(c_{n_i}, s)$$

where $A = \begin{cases} counterparty \ \ if \ \ CPPostStyle = Gross \\ user \ \ if \ \ UserPostStyle = Gross \end{cases}$, and

$CMtM_{A,default}(r_i, s)$ is the mark-to-market of CMP_A after applying the haircuts at default*2 specifically for exposure calculation.

Then the collateral balance, $CB_A(r_i, s)$, is defined by

$$CB_A(r_i, s) = \begin{cases} \overline{CB_A}(r_i, s) \ \ if \ \ IsolateCollateralSpikes = F \\ \overline{CB_A}(r_i, s) - S_A(r_i, s) \ \ if \ \ IsolateCollateralSpikes = T \end{cases}$$

*1 It is assumed that post direction is reflected in the simulated MtM of CMP at t_0, therefore $CMtM^{sim}_{cpty}(t_0) \ge 0$ and $CMtM^{sim}_{user}(t_0) \le 0$

*2 The haircuts at default applied to the mark-to-market of CMP_A in the collateral balance computation are different from the ones used in the margin call computation. Applying haircuts at default in the collateral balance computation will produce more conservative exposure results.

*3 $\Gamma^{(i)}_{relevant} \backslash \{c_{-1}\}$ is the set of relevant call dates excluding c_{-1} .

4.1.1.5 Collateralized Exposure

Let us define the (simulated) *Collateralized exposure* (this excludes the addons part of the exposure), $Exp(t_i, s_j)$, at time t_i under scenario s_j by:

$$Exp(t_i, s_j) = \max(CV_{Exp}(t_i, s_j), 0)$$

where the intermediate (simulated) collateralized value, $CV_{Exp}(t_i, s_j)$ is defined below.

The collateralized exposure is used for most credit expressions. However, FVA expressions are not based on $Exp(t_i, s_j)$ nor $CV_{Exp}(t_i, s_j)$. For the details of how the collateralized value for FVA $CV_{FVA}(t_i, s_j)$ is defined, see FVA fucntion document 6.20 FVA/FCA/FBA/MVA on page 93.

Definitions

$V^X(t_i, s_j)$ - portfolio value on the node *X*. For how this portfolio value is determined depending on the netting specified by the *NettingRule, CloseoutNetting, and AgreementEnforceable* values as well as the perspective direction of the expression, see Chapter 3 Netting Rule on page 5.

Important

We will assume that all the variation and initial margins defined here are positive.

$$VM^X_{c,\ seg}(t_i, s_j) = \begin{cases} \text{segregated variation margin posted by C on node } X \text{ if } X \text{ is a CSA} \\ \text{segregated variation margin posted by C on all children of node } X \text{ if } X \text{ is an MA} \\ 0 \text{ otherwise} \end{cases}$$

$$VM^X_{c,\ unseg}(t_i, s_j) = \begin{cases} \text{unsegregated variation margin posted by C on node } X \text{ if } X \text{ is a CSA} \\ \text{unsegregated variation margin posted by C on all children of node } X \text{ if } X \text{ is an MA} \\ 0 \text{ otherwise} \end{cases}$$

The segregation of variation margin is specified by the collateral agreement parameters `CpVariationMarginSegregated` and `UserVariationMarginSegregated`. (Here *C* denotes either the counterparty or the user.)

$IM^X_{c,\ seg}(t_i, s_j)$ - The segregated initial margin posted by *C* at the node *X* and at all the children of the node, as well as the initial margin posted by *C* to any trades of the node.

$IM^X_{c,\ unseg}(t_i, s_j)$ - The unsegregated initial margin posted by *C* at the node *X* and at all the children of the node, as well as the initial margin posted by *C* to any trades of the node.

For each counterparty and user, both segregated initial margin and non-segregated initial margin can be specified for each node of type CSA, NC, NN or MA if there is no CSA under it. Each initial margin profile starts at `DeliveryDate` and ends at `ReturnDate`. For the determination of initial margin amount, see 4.1.1.7 Model Inputs and Notation on page 27.

$$C^X_{c,\ seg}(t_i, s_j) := \begin{cases} VM^X_{c,\ seg}(t_i, s_j) + IM^X_{c,\ seg}(t_i, s_j) & \text{if } \quad \text{CT} = \text{ALL} \\ VM^X_{c,\ seg}(t_i, s_j) & \text{if } \text{CT} = \text{VARIATION\_MARGIN} \\ IM^X_{c,\ seg}(t_i, s_j) & \text{if } \quad \text{CT} = \text{INITIAL\_MARGIN} \end{cases}$$

where CT denotes the collateral model parameter COLLATERAL TYPE.

$$C^X_{c,\ unseg}(t_i, s_j) := \begin{cases} VM^X_{c,\ unseg}(t_i, s_j) + IM^X_{c,\ unseg}(t_i, s_j) & \text{if } \quad \text{CT} = \text{ALL} \\ VM^X_{c,\ unseg}(t_i, s_j) & \text{if } \text{CT} = \text{VARIATION\_MARGIN} \\ IM^X_{c,\ unseg}(t_i, s_j) & \text{if } \quad \text{CT} = \text{INITIAL\_MARGIN} \end{cases}$$

$$C^X_c(t_i, s_j) := C^X_{c,\ seg}(t_i, s_j) + C^X_{c,\ unseg}(t_i, s_j)$$

- **Collateralized Value (for Collateralized Exposure)**

Collateralized Value From The Perspective of User (Forward View)

Case 1:

When the following conditions are satisfied:

- *NettingRule = LEGAL_OPINION*
- there is a complex structure specified or *AgreementEnforceable* is False

the collateralized value is calculated as:

```
IF (User Collateral Offset = Netted Pools)
```

$$CV(t_i, s_j)^X_{Exp} = V^X(t_i, s_j) + \max(C^X_{user,\ unseg}(t_i, s_j) + N^X_{CNP}(t_i, s_j), 0)$$

$$-\begin{cases} VM^X_{cpty}(t_i, s_j) + IM^X_{cpty}(t_i, s_j) & \text{if} \quad \text{AM} = \text{BOTH} \\ VM^X_{cpty}(t_i, s_j) & \text{if} \quad \text{AM} = \text{VARIATION MARGIN} \\ IM^X_{cpty}(t_i, s_j) & \text{if} \quad \text{AM} = \text{INITIAL MARGIN} \\ 0 & \text{if} \quad \text{AM} = \text{NONE} \end{cases}$$

where $N^X_{CNP}(t_i, s_j) = \sum_{l=1}^{L} \min\left(MtM^X_{CNP_l}(t_i, s_j), 0\right)$, $CNP_l(l = 1, \ldots, L)$ are the closeout netting pools on Node X, and $MtM^X_{CNP_l}(t_i, s_j)$ is the netted sum of the MtM of all positions within CNP_l on Node X at time t_i under scenario s_j.

```
ELSE
```

$$CV(t_i, s_j)^X_{Exp} = V^X(t_i, s_j) + C^X_{user,\ unseg}(t_i, s_j) -$$

$$\begin{cases} VM^X_{cpty}(t_i, s_j) + IM^X_{cpty}(t_i, s_j) & \text{if} \quad \text{AM} = \text{BOTH} \\ VM^X_{cpty}(t_i, s_j) & \text{if} \quad \text{AM} = \text{VARIATION MARGIN} \\ IM^X_{cpty}(t_i, s_j) & \text{if} \quad \text{AM} = \text{INITIAL MARGIN} \\ 0 & \text{if} \quad \text{AM} = \text{NONE} \end{cases}$$

where AM is determined by the values of *Collateral Type* and *Apply with Legal Opinion* as in the following table:

	Collateral Type		
Apply with Legal Opinion	**ALL**	**VARIATION_MARGIN**	**INITIAL_MARGIN**
ALL	BOTH	VARIATION MARGIN	INITIAL MARGIN
VARIATION_MARGIN	VARIATION MARGIN	VARIATION MARGIN	NONE
INITIAL_MARGIN	INITIAL MARGIN	NONE	INITIAL MARGIN
NONE	NONE	NONE	NONE

Case 2:

When the conditions specified in the above Case 1 are not satisfied, the collateralized value is calculated as:

$$CV(t_i, s_j)^X_{Exp} = V^X(t_i, s_j) - C^X_{cpty}(t_i, s_j) + C^X_{user,\ unseg}(t_i, s_j)$$

Collateralized Value From The Perspective of Counterparty (Reverse View)

$$CV(t_i, s_j)^X_{Exp} = V^X(t_i, s_j) - C^X_{user}(t_i, s_j) + C^X_{\text{cpty},\, unseg}(t_i, s_j)$$

4.1.1.6 MtM on Non-Simulated Dates

In the Path Dependent Collateral Model, the trade portfolio mark-to-market (MtM) and the credit mitigation portfolio mark-to-market after applying all relevant haircuts (CMtMHC) are needed for some non-simulated call dates (see step 3 in section 4.1.1 The Path Dependent Collateral Model – Collateral Balance and Margin Call on page 7). The missing information for such a call date is approximated as follows:

1. Trade Portfolio

- (`Isolate Collateral Spikes` = F) A Brownian Bridge is built between the simulation dates immediately before and immediately after the call date using the MtM of all positions in the CSA portfolio.
- (`Isolate Collateral Spikes` = T) First the Brownian Bridge is built between the simulation dates immediately before and immediately after the call date using the MtM of all positions in the CSA portfolio that have not matured by the later of the two dates. Second an approximation of the MtM on the call date of the trades that mature between the two simulation dates is obtained by either linearly interpolating between the two simulation dates (if `Gradual Cashflows` = T) or by taking the MtM of the maturing trades on the earlier of the two simulation dates (if `Gradual Cashflows` = F). The sum of these is defined to be the MtM on the call date.

2. Credit Mitigation Portfolio

- A Brownian Bridge is built between the simulation dates immediately before and immediately after the call date using the CMtMHC of all variant mitigants in CMP assigned to the CSA node.

Note

In the following cases, no credit mitigation portfolio is generated. Thus, the calculation for credit mitigation portfolio described below is not necessary.

- `CPPostStyle` = `Net` and `UserPostStyle` = `Net` and `Collateral Mode` = `Cash only` OR
- `User Physical Collateral` = `False`

The following model parameters are used in the construction of the Brownian Bridge:

- The model parameter `Isolate Collateral Spikes`. This is used to exclude the risk of over-collateralization due to maturing trades during the margin period of risk from exposure and to include it in a cashflows measure. This parameter is only applicable to the approximation for trade portfolio.
- The model parameter `Gradual Cashflows`. This determines how the mark-to-market of trades that mature between two simulation dates is approximated when the parameter `Gradual Cashflows` = T. This parameter is only applicable to the approximation for trade portfolio.
- The model parameter `Intra Period` indicates if the volatility for the Brownian Bridge is estimated using the right end point of the bridge or both the left and right end points
- The model parameters
  - `IQR Variance Estimator` indicates if the volatility is estimated using a generalized inter-quartile range and
  - `IQR Alpha` defines the quantile to be used

Notation

c - The call date which has not been simulated

t_i - The simulation date immediately before c

t_{i+1}- The simulation date immediately after c

$MtM_P(t_{i+1},s)$- The mark-to-market of the portfolio at t_{i+1} and scenario s

$MtM_P(t_i,s)$- If `Isolate Collateral Spikes`= False, the mark-to-market of the portfolio at t_i and scenario s , and if `Isolate Collateral Spikes` = True, the netted mark-to-market (at t_i and scenario s) of all positions in the portfolio which have not matured before t_{i+1}

$CMtMHC(t_{i+1},s)$- The mark-to-market of the credit mitigation portfolio after applying all relevant haircuts at t_{i+1} and scenario s

$CMtMHC(t_i,s)$- The mark-to-market of the credit mitigation portfolio after applying all relevant haircuts at t_i and scenario s

N– The number of scenarios

$\{s_1, s_2, ..., s_N\}$ - The set of all scenarios

The above notations are straightforward for the case of `CPPostStyle` = `Net` and `UserPostStyle` = `Net`. For the simplicity of illustration, these notations are used in the remaining of this document. However, reader should bear in mind that the following substitution happens:

```
IF (CPPostStyle = Gross)
     P substitutes MtM
     CMtMHC_cpty substitutes CMtMHC
ELSE IF (UserPostStyle = Gross)
     N substitutes MtM
     CMtMHC_user substitutes CMtMHC
```

where, P is the sum of the mark-to-market of all positions in the trade portfolio that are positive; N is the sum of the mark-to-market of all positions in the trade portfolio that are negative.

Estimating the Variance Covariance Matrix

We now estimate the variance covariance matrix of the trade portfolio and credit mitigation portfolio which is used in the Brownian Bridge computation.

If `Intra Period` = False, $x_{i,s}$ is defined as $\mathrm{MtM}_P(t_{i+1}, s)$ and $y_{i,s}$ is defined as $CMtMHC(t_{i+1}, s)$.

If `Intra Period`= True, $x_{i,s}$ is defined as $\mathrm{MtM}_P(t_{i+1}, s) - \mathrm{MtM}_P(t_i, s)$ and $y_{i,s}$ is defined as $CMtMHC(t_{i+1}, s) - CMtMHC(t_i, s)$.

Case 1 – IQR Variance Estimator = False

$$V_p(t_i) = \frac{1}{\mathrm{N}-1}\sum_{j=1}^{N}\left(x_{i,s_j} - \overline{x_i}\right)^2, \quad \text{where} \quad \overline{x_i} = \frac{1}{N}\sum_{j=1}^{N} x_{i,s_j}$$

$$V_{cmp}(t_i) = \frac{1}{\mathrm{N}-1}\sum_{j=1}^{N}\left(y_{i,s_j} - \overline{y_i}\right)^2, \quad \text{where} \quad \overline{y_i} = \frac{1}{N}\sum_{j=1}^{N} y_{i,s_j}$$

$$COV_{p,cmp}(t_i) = \frac{1}{\mathrm{N}-1}\sum_{j=1}^{N}\left(x_{i,s_j} - \overline{x_i}\right)\left(y_{i,s_j} - \overline{y_i}\right)$$

Case 2 - IQR Variance Estimator = True

Let α denote the model parameter `IQR Alpha`, and define:

$k_1 := \lceil \alpha \times N \rceil$, that is the smallest integer that is not less than $\alpha \times N$;

$k_2 := \lceil (1-\alpha) \times N \rceil$, that is the smallest integer that is not less than $(1-\alpha) \times N$;

$Q(\alpha)$ as the k_1th smallest element of $\left\{x_{i,s_j}:1 \le j \le N\right\}$ or $\left\{y_{i,s_j}:1 \le j \le N\right\}$;

$Q(1-\alpha)$ as the k_2th smallest element of $\left\{x_{i,s_j}:1 \le j \le N\right\}$ or $\left\{y_{i,s_j}:1 \le j \le N\right\}$; and

$N^{-1}(\alpha)$ as the inverse of the normal cumulative distribution function.

Then the generalized interquartile range is defined by:

$$\mathrm{IQR}(\alpha) = \left(\frac{Q(\alpha) - Q(1-\alpha)}{2 \times N^{-1}(\alpha)}\right)$$

The trade portfolio variance is estimated by $V_p = \mathrm{IQR}(\alpha)^2$ and the variance of the credit mitigation portfolio is estimated by $V_{cmp} = \mathrm{IQR}(\alpha)^2$.

The correlation between trade portfolio and credit mitigation portfolio is estimated by:

$$Corr_{p,cmp} = \sin(\tfrac{\pi}{2} \times \tau)$$

where τ denotes Kendall's Tau and is calculated as:

$$\tau = \frac{N_c - N_d}{N(N-1)/2} ,$$

where N_c and N_d denote the number of concordant and discordant pairs respectively; that is:

N_c is the number of pairs in $\{(j,k): j \ne k, 1 \le j,k \le N\}$ such that ($x_{i,s_j} < x_{i,s_k}$ and $y_{i,s_j} < y_{i,s_k}$) or ($x_{i,s_j} > x_{i,s_k}$ and $y_{i,s_j} > y_{i,s_k}$); and

N_d is the number of pairs in $\{(j,k): j \ne k, 1 \le j,k \le N\}$ such that ($x_{i,s_j} < x_{i,s_k}$ and $y_{i,s_j} > y_{i,s_k}$) or ($x_{i,s_j} > x_{i,s_k}$ and $y_{i,s_j} < y_{i,s_k}$).

The covariance between trade portfolio and credit mitigation portfolio is then calculated as:

$$COV_{p,cmp} = Corr_{p,cmp} \times \sqrt{V_p} \times \sqrt{V_{cmp}}$$

■ Computing the Brownian Bridge

1. Trade portfolio

The Brownian Bridge of the trade portfolio if `Isolate Collateral Spikes` = F or the mark-to-market of all positions which have not matured before t_{i+1} if `Isolate Collateral Spikes` = T at the call date c and scenario s is defined by:

$$MtM_p(c,s) = MtM_p(t_i,s) + \frac{MtM_p(t_{i+1},s) - MtM_p(t_i,s)}{t_{i+1} - t_i}(c - t_i) + \sqrt{\frac{(c-t_i)(t_{i+1}-c)}{(t_{i+1}-t_i)\Delta t}}\zeta_{p,i,s}$$

where $\Delta t = \begin{cases} t_{i+1} - t_i \ \ if \ \ IntraPeriod = True \\ t_{i+1} - t_0 \ \ if \ \ IntraPeriod = False \end{cases}$

and $\zeta_{p,i,s}$ is a random normal draw with zero mean and covariance matrix $VCV_{p,cmp}$.

2. Credit Mitigation Portfolio

The Brownian Bridge of the credit mitigation portfolio after applying all relevant haircuts at the call date *c* and scenario *s* is defined by:

$$CMtMHC(c,s) = CMtMHC(t_i,s) + \frac{CMtMHC(t_{i+1},s) - CMtMHC(t_i,s)}{t_{i+1} - t_i}(c - t_i) + \sqrt{\frac{(c-t_i)(t_{i+1}-c)}{(t_{i+1}-t_i)\Delta t}}\zeta_{cmp,i,s}$$

where $\zeta_{cmp,i,s}$ is a random normal draw with zero mean and covariance matrix $VCV_{p,cmp}$.

MtM on Call Dates

1. Trade Portfolio

The mark-to-market of the trade portfolio at the call date *c* and scenario *s* is defined by:

$$MtM(c,s) = \begin{cases} MtM_p(c,s) \;\; if \;\; IsolateCollateralSpikes = F \\ MtM_p(c,s) + CF([c, t_{i+1}], s) \;\; if \;\; IsolateCollateralSpikes = T \end{cases}$$

where $CF([c,t_k],s)$ is as defined in 4.1.1 The Path Dependent Collateral Model – Collateral Balance and Margin Call on page 7.

2. Credit Mitigation Portfolio

The mark-to-market of the credit mitigation portfolio after applying all relevant haircuts at the call date *c* and scenario *s* is the Brownian Bridge *CMtMHC*(*c*,*s*).

4.1.1.7 Model Inputs and Notation

RTCE Inputs

The source files for these inputs are described in the *Integrated Market and Credit Risk Solution Data Dictionary* and are located in `$ALGO_TOP/inputs/userdata/rtce/update/counterparties` and `$ALGO_TOP/inputs/userdata/rtce/replace/counterparties`.

Table 4.1-1 RTCE Inputs

Input	Notation	Source File	Usage in Model/Comments
AgreementCurrency	CSAcy	CollateralAgreement.csv	Specifies the currency of the CSA parameters (thresholds, minimum transfer amounts, etc.).*1
DeliveryCurrency		CollateralAgreement.csv	Specifies the currency in which variation margins are calculated.
CallPeriod	*p*	CollateralAgreement.csv	Frequency of margin calls
CPLag	cLag	CollateralAgreement.csv	In the Path Dependent collateral model, the sum of *cLag* and *cCurePeriod* represents the margin period of risk of collateral posted by the counterparty; it is denoted by lag_{cp}. In the Shortcut Empirical collateral model, the maximum of lag_{cp} and lag_{user} represents the margin period of risk; it is denoted by $lag_{\max}$. When the `Without Lags` model parameter is set to True, the value of this input in the Path Dependent collateral model is set to zero.
UserLag	uLag	CollateralAgreement.csv	In the Path Dependent collateral model, the sum of *uLag* and *uCurePeriod* represents the

			margin period of risk of collateral posted by the user; it is denoted by lag_{user}. In the Shortcut Empirical collateral model, the maximum of lag_{cp} and lag_{user} represents the margin period of risk; it is denoted by $lag_{\max}$. When the `Without Lags` model parameter is set to True, the value of this input in the Path Dependent collateral model is set to zero.
CPCurePeriod	cCurePeriod	CollateralAgree-ment.csv	See comment for CPLag.
UserCurePeriod	uCurePeriod	CollateralAgree-ment.csv	See comment for UserLag.
CPIndependent	cIA	CollateralAgree-ment.csv	The treatment of this amount in the model is as defined in the CSA agreement http://assets.isda.org/media/e0f39375/b61bd039.pdf
UserIndependent	uIA	CollateralAgree-ment.csv	The treatment of this amount in the model is as defined in the CSA agreement http://assets.isda.org/media/e0f39375/b61bd039.pdf.
CPThreshold	cT	CollateralAgree-ment.csv	When either the `Minimum Transfer Risk Under-Collateralization` or `Minimum Transfer Risk Over-Collateralization` model parameter is set to true, this value is replaced. See Table 4.1-3 Thresholds, Minimum Transfer, and Rounding Inputs on page 31.
UserThreshold	uT	CollateralAgree-ment.csv	See comment for *CPThreshold.*
CPPosts	cPosts	CollateralAgree-ment.csv	If set to false, the counterparty does not post collateral. There can still be margin calls made to the counterparty to return collateral posted by the user. (Note, at present, this is only used when the Path Dependent model is used. The adjusted collateral used in the Empirical Shortcut model does not use this parameter.)
UserPosts	uPosts	CollateralAgree-ment.csv	If set to false, the user does not post collateral. There can still be margin calls made to the user to return collateral posted by the counterparty. (Same note as for cPosts.)
CPMinTransfer	cMTA	CollateralAgree-ment.csv	See comment for *CPThreshold.*
UserMinTransfer	uMTA	CollateralAgree-ment.csv	See comment for *CPThreshold.*
RoundoffAmount	RA	CollateralAgree-ment.csv	See comment for *CPThreshold.*
CpVariationMargin-Segregated		CollateralAgree-ment.csv	If set to true, the collateral balance posted by counterparty is assumed to be segregated; that is, any collateral balance which is not used to offset exposure to user is protected in case of default.

UserVariationMarginSegregated		CollateralAgreement.csv	If set to true, the collateral balance posted by user is assumed to be segregated; that is, any collateral balance which is not used to offset exposure to counterparty is protected in case of default.
Rehypothecation		CollateralAgreement.csv	If set to true, counterparty variantion margin can be re-used. This parameter is only relevant to the computation of FVA.
Collateral Mode			Specifies whether the posted collateral is in the form of cash or physical collateral. This attribute can be set as: Cash only, Physical collateral only in ICB post direction, or Physical collateral in either direction.
CPPostStyle		CollateralAgreement.csv	Specifies counterparty's variation margin post style. This attribute can be set as: Net (default value), Gross, None.
UserPostStyle		CollateralAgreement.csv	Specifies user's variation margin post style. This attribute can be set as: Net (default value), Gross, None.
marginType		standardMitigantAttributes	Specifies the type of margin. The value can be Initial or Variation.
deliveryDate		standardMitigantAttributes	Specifies when the collateral will be delivered. For Variation Margin, only those mitigants whose delivery dates fall before or on the session date are used. For Initial Margin, the profile starts at *deliveryDate* and ends at *returnDate*.
returnDate		standardMitigantAttributes	Specifies when the collateral will be returned. For Variation Margin, only those mitigants whose return dates fall after the session date are used. For Initial Margin, the profile starts at *deliveryDate* and ends at *returnDate*.
mitigantPostDirection		standardMitigantAttributes	Specifies who is receiving the collateral. The value can be To Counterparty or To User.
segregated		standardMitigantAttributes	This attribute only applies to initial margin type. If set to true, the initial margin posted is assumed to be segregated.
MTM			Specifies the collateral value posted/received in the currency defined by *MTMCurrency*. This is used to compute the current amount of Variation Margin and Initial Margin.
MTMCurrency			See comment for *MTM*.
mitigantQuantity		standardMitigantAttributes	When *MTM* and *MTMCurrency* are not specified, this attribute, together with *mitigantCurrency*, are used to compute the collateral value posted/received.
mitigantCurrency		standardMitigantAttributes	See comment for *mitigantQuantity*.
applicability		standardMitigantAttributes	For mitigant with marginType = Variation, this attribute allows the user to define an appropriate security pool. When set to INITIAL_BALANCE (default value), the variant mitigant is used to calculate the total variation margin balance as of t_0. When set to

			MITIGANT_POOL, the variant mitigant is used inside the credit mitigation portfolio.
			For mitigant with marginType = Initial, this attribute allows user to apply dynamic initial margin model. When set to INITIAL_BALANCE (default value), the static MtM value of the initial mitigant is used to calculate initial margin balance across time. When set to MITIGANT_POOL, the dynamic initial margin model is applied.
AgreementEnforceable	AgreementEnforceable	Counterparty.csv	If set to false, netting of positions is not allowed when computing the exposure of the user to the counterparty, but collateral is computed based on a netted plane.
CloseoutNetting	CloseoutNetting	MasterAgreement.csv	If set to false, netting of positions is not allowed but collateral is computed based on a netted plane.

*1 This currency needs to be converted to DeliveryCurrency, using the closest FX rate available. If there are 2 equally "closest" simulation dates from the call date for which we need the FX rate, the latter date is chosen.

Note

Since version 5.2 of the software, Collateral Agreement schema has changed. Some of the parameters listed in the above table are no longer available (please refer to 4.1.2.9 Model Inputs and Notation on page 61 under Settlement Adjusted Path Dependent Model section for the full list of 5.2 Collateral Agreement schema). If Path Dependent Model is set up in 5.2 environment, the model inputs Lag_{cpty} and Lag_{user} can be derived with 5.2 Collateral Agreement schema as :

$$Lag_{user} = curePeriod$$

$$Lag_{cpty} = Lag\_VM + curePeriod$$

Model Parameters

The values of these parameters are set in the `CollateralModels.xml` file in `$ALGO_TOP/static/mag/` instances. They can also be defined through a GUI application (see `$MAG_HOME/bin/defineFuncitons`).

Table 4.1-2 Model Parameters

Input	Default value	Usage in Model/Comments
Additional Call Dates	1	This allows for additional call dates to be used in the model to capture more precisely the risk introduced by minimum transfer amounts. Although this parameter can be set to any integer greater than or equal to zero, a large value greatly impacts the performance of the algorithm, which means it should only be set to be greater than 1 when absolutely necessary (for example if the portfolio moves within call periods is small compared to the minimum transfer amounts). Setting this value to a large number should only be used for testing the model on relatively small portfolios and should never be used in production.
Adjustment On Time 0	False	This applies to the Shortcut Empirical Model and specifies if an adjustment should be included on the first reporting date.
Scenario Selection Mechanism	With Replacement	This applies to the Shortcut Empirical Model and specifies if the adjustments are chosen with or without replacement.
Collateral Type	All	This allows user to choose what kinds of collateral to be applied.

Apply With Legal Opinion	None	When *NettingRule* = Legal_Opinion, this, together with *Collateral Type*, determines what kinds of collateral posted by counterparty are used to offset exposures to user.
Isolate Collateral Spikes	False	If set to true, it removes over-collateralization due to maturing trades from the exposure computation and includes it in a cash-flow measure. *Important:* If set to True, the counterparty and user lags must be equal.
Gradual Cashflows	False	When *Isolate Collateral Spikes* = T, it determines how the mark-to-market of maturing trades is estimated on a non-simulated date
Intra Period	False	If set to true, indicates if the volatility used in the Brownian Bridge construction is estimated using both the left and right end points of the bridge.
IQR Variance Estimator	False	Indicates if the volatility used in the Brownian Bridge construction is estimated using a generalized interquartile range.
IQR Alpha	95%	Defines the quantile to be used in the generalized IQR when *IQR Variance Estimator* is set to true.
Minimum Transfer Risk Under-Collateralization	False	This is to allow for a conservative treatment of the risk introduced by the counterparty not posting collateral due to minimum transfer amounts (and rounding); see Table 4.1-3 Thresholds, Minimum Transfer, and Rounding Inputs on page 31 to see how this parameter works in conjunction with thresholds, minimum transfer amounts and rounding inputs
Minimum Transfer Risk Over-Collateralization	False	This is to allow for a conservative treatment of the risk introduced by the counterparty not returning collateral due to minimum transfer amounts (and rounding); see Table 4.1-3 Thresholds, Minimum Transfer, and Rounding Inputs on page 31 to see how this parameter works in conjunction with thresholds, minimum transfer amounts and rounding inputs.
Without Lags	False	Forces *cLag* and *uLag* to zero.
Link Relevant Call Dates	False	When this is set to True, margin calls are computed between two non-consecutive call dates.
Use Physical Collateral	False	If set to true, credit mitigation portfolio is used to model collateral.
User Collateral Offset	None	If set to `Netted Pools`, user posted unsegregated margin is allowed to offset the negative mark-to-market from trades belonging to closeout netting pools.

The following table describes how the thresholds, minimum transfer amounts and rounding inputs from RTCE change based on the value of the parameters `Minimum Transfer Risk Under-Collateralization` and `Minimum Transfer Risk Over-Collateralization`:

Table 4.1-3 Thresholds, Minimum Transfer, and Rounding Inputs

Minimum Transfer Risk Under-Collateralization	**Minimum Transfer Risk Over-Collateralization**	**cT**	**cMTA**	**uT**	**uMTA**	**RA**
False	False	CPThreshold	CPMinTransfer	UserThreshold	UserMinTransfer	RoundoffAmount
True	False	CPThreshold + max(CPMinTransfer, RoundoffAmount)	0	UserThreshold	0	0
False	True	CPThreshold	0	UserThreshold +	0	0

				max(CPMin-Transfer ,RoundoffA-mount)		
True	True	CPThreshold + max(CPMin-Transfer, RoundoffA-mount)	0	UserThreshold + max(CPMin-Transfer, ,RoundoffA-mount)	0	0

Comments:

- Implicit in setting the *uMTA* to zero is the assumption that the counterparty is never over-collateralized and the user never under-collateralized due to the user not returning or posting collateral due to minimum transfer amounts. This is an additional conservative assumption.
- The rounding amount is included in the new threshold to allow for the risk of rounding to be captured in the case that the rounding amount is greater than the minimum transfer amount (this is not generally the case).
- *UserThreshold* + max(*CPMinTransfer*, *RoundoffAmount*) could be positive.

4.1.2 The Settlement Adjusted Path Dependent Model – Variation Margin Balance and Margin Call

The variation margin balance and margin call computation performed by MAG in the settlement adjusted path dependent model depends on the value assigned to attributes `CPVMPostStyle` and `UserVMPostStyle`. `CPVMPostStyle` and `UserVMPostStyle` can be set to one of the following values:

- Net: Variation margin requirement is calculated by treating all trades in the portfolio on a net basis.
- None: Variation margin requirement is always zero as no variation margin is posted in this case.

By default, `CPVMPostStyle` = `Net` and `UserVMPostStyle` = `Net`, which means variation margin balance and margin call are computed by treating all trades in the trade portfolio on a net basis. The default setting represents the traditional way to model variation margin by assuming netting within a CSA (i.e. bilateral netting variation margin computation).

The collateral model attribute `Use Physical Collateral` is used to specify whether or not the collateral posted in either direction at the future time steps is modeled in the form of credit mitigant portfolio (CMP).

- `Use Physical Collateral` = `True`. Two CMPs, CMP_{user} and CMP_{cpty} are created and used as representative sets of model collateral posted by user and counterparty.
- `Use Physical Collateral` = `False`. Collateral posted in either direction is assumed to be in the form of cash.

4.1.2.1 Model Inputs Overview

The settlement adjusted path dependent model accepts four types of inputs:

- The simulated, netted mark-to-market of all the transactions under the collateral agreement over scenarios and time steps (simulation dates). A positive mark-to-market indicates the portfolio is in-the-money for the user and a negative mark-to-market indicates that it is in-the-money for the counterparty.

- Trade settlement flow of all the transactions under the collateral agreement.
- The RTCE inputs which include the terms of the agreement – CSA parameters.
- The model parameters' values.

For a full description of the latter two inputs, see the section 4.1.2.9 Model Inputs and Notation on page 61.

4.1.2.2 Call Dates

The amount of variation margin posted and margin call made at a reporting date requires knowledge of margin calls made at previous call dates. Since it is not computationally feasible (and sometimes not necessary) to compute the margin call at all call dates, the collateral model uses a subset of these. This subset, which is referred to as the set of relevant call dates, depends on:

- The margin period of risk (MPOR), represented by the max(*Lag_VM, Lag_TS*) + *curePeriod* (RTCE inputs).
- The call schedule, obtained from the *callPeriod* (RTCE input) which is greater than 0.
- The model parameter `Additional Call Dates` that is used to ensure that the risk arising from minimum transfer amounts is captured adequately.

■ The set of all Call Dates

Standard Collateral Agreement

In RTCE, the set of all call dates is determined by the frequency (call period), *p*, of the margin calls.

Note

In case of Settled To Market collateral agreement, *CallPeriod* = 1.

If $t_0, t_1, t_2, \ldots$ denote the set of simulation dates in ascending order, that is, the dates for which the MtM of the portfolio has been simulated, then the set, $\bar{\Gamma}$, is defined by

$\bar{\Gamma}$:={c$_k$: c$_k = t_0 + k \times p, k \in \aleph$} , where $\aleph$ is the set of natural numbers; $c_0, c_1, c_2, \ldots$ denote the elements of $\bar{\Gamma}$ in ascending order.

Since we need to know how much variation margin was called before $c_0 = t_0$, we need to refer to the call date immediately before, denoted by c_{-1}, and assume that the variation margin called up to and including this date is the initial variation margin balance, *IVMB* (for the calculation of IVMB, see step 2 in section 4.1.2 The Settlement Adjusted Path Dependent Model – Variation Margin Balance and Margin Call on page 32).

The set of call dates is then defined by

$\Gamma = \{c_{-1}, c_0, c_1, c_2, \ldots\}$

Exotic Collateral Agreement

RTCE allows user to override the variation margin post style for one or both of the counterparties during specified periods of time. Please refer to `VMSchedules.csv` in Data Dictionary for the detailed information. In this case, the set of all call dates depends not only on the call period, but also on the parameters `CpVMPostStyle(t)` and `UserVMPostStyle(t)` which may change over time

Let t_0 be the processing date and t_N be the last simulation date.

Find all the dates d between t_0 and t_N where:

1. Variation margin starts being posted, that is, `CPVMPostStyle(d-1)` = `NONE` and `UserVMPostStyle(d-1)` = `NONE`, and `CPVMPostStyle(d)` ≠ `NONE` or `UserVMPostStyle(d)` ≠ `NONE`; OR
2. Variation margin stops being posted, that is, `CPVMPostStyle(d-1)` ≠ `NONE` or `UserVMPostStyle(d-1)` ≠ `NONE`, and `CPVMPostStyle(d)` = `NONE` and `UserVMPostStyle(d)` = `NONE`.

Denote these dates by $d_1 < d_2 < \ldots < d_{n-1}$ and define $d_0 := t_0$, $d_n := t_N + 1$.

For each interval $[d_j, d_{j+1})$ on which the counterparty or user (or both) posts collateral, define the set of call dates as:

$\bar{\Gamma} := \{c_k | c_k = d_j + k \times p, k \in \aleph \text{ such that } 0 \le k \le \min(\lceil\frac{d_{j+1} - d_j}{p}\rceil, \frac{d_{j+2} - d_j}{p})\}$.

Here, $\lceil x \rceil$ denotes the smallest integer greater than or equal to x. If $j+1 = n$, define $d_{j+2} = \infty$.

Note

We add a call date on or after d_{j+1}, that is $d_j + \lceil\frac{d_{j+1} - d_j}{p}\rceil \times p$ (unless this date is after d_{j+2}) to ensure that we make a margin call to return collateral if necessary.

The set of Relevant Call Dates

We now define the subset of call dates used in the collateral model. This subset is determined by the set of reporting dates. This is the set of dates on which risk measures are reported. If auxiliary dates are used, the set of reporting dates is a subset of simulation dates; if auxiliary dates are not used, the set of reporting dates is equal to the set of simulation dates. $\mathrm{R}{:=}\{r_0, r_1, r_2, \ldots\}$ which is a subset of simulation dates $\{t_0, t_1, t_2, \ldots\}$.

We use the following notation to specify the margin period of risk used in the model:

$MPOR = \max(Lag\_VM, Lag\_TS) + curePeriod$

(see 4.1.2.9 Model Inputs and Notation on page 61 section for the definitions of *Lag_VM*, *Lag_TS*, and *curePeriod*).

For each reporting date r_i, we first define the minimum set of call dates that are used in the model to compute the variation margin balance on r_i:

- All variation margin called up to and including the first call date on or before r_i-*MPOR* is available by r_i (if neither party has defaulted on or before the call date). We denote this call date by c_{n_i} . (If there are no call dates before r_i-*MPOR*, set $c_{n_i} := \mathrm{c}_{-1}$.) The model estimates the amount of variation margin called up to this date.
- Variation margin calls between r_i-*MPOR* and r_i-*MPOR*+*Lag_VM* are available by r_i if they are called by the defaulting party. That is, if they are posted by the non-defaulting party. Let $c_{n_i+1}, c_{n_i+2}, \ldots, c_{n_i+m_i} \in \Gamma$ be the call dates between r_i-*MPOR* (exclusive) and r_i-*MPOR*+*Lag_VM* (inclusive), if any (note that m might be zero, that is, there might not be any such call dates). In the model we compute the margin calls (transfers) on these dates.
- To adequately capture the risk arising from minimum transfer amounts, the amount of variation margin called on some dates before c_{n_i} are needed. The model parameter `Additional`

`Call Dates` specifies the number of additional call dates that should be used. If *l* :=Additional Call Dates, we denote the *l* call dates immediately before c_{n_i} by $c_{n_i - l}, ..., c_{n_i - 1}$

Note

There might be less than *l* such call dates and possibly none, for example if c_{n_i} is close to the beginning of the session date t_0.

The set of all these dates is the minimum set of relevant call dates for the reporting date r_i and is denoted by

$$\Gamma_{VM}^{(i)} = \left\{c_{n_i - l}, ..., c_{n_i - 1}, c_{n_i}, c_{n_i + 1}, c_{n_i + 2}, ..., c_{n_i + m_i}\right\}$$

To ensure consistency between the variation margin balance calculation for two different reporting dates (although this consistency is not necessary, it makes validation easier), the set of dates used to compute the variation margin balance on r_i, $\Gamma_{relevant\_VM}^{(i)}$, is

- the largest subset of consecutive relevant call dates less than or equal to $c_{n_i + m_i}$ if `Link Relevant Call Dates` = `FALSE`.
- all relevant call dates less than or equal to $c_{n_i + m_i}$ if `Link Relevant Call Dates` = `TRUE`.

Note that this set contains $\Gamma_{VM}^{(i)} = \left\{c_{n_i - l}, ..., c_{n_i - 1}, c_{n_i}, c_{n_i + 1}, c_{n_i + 2}, c_{n_i + m_i}\right\}$.

4.1.2.3 Variation Margin Balance and Margin Call Algorithm when CPVMPostStyle = Net and UserVMPostStyle = Net

Here, we describe the computation of variation margin balance and margin call for entities covered by bilateral variation margin agreement under the Settlement Adjusted Path Dependent Model.

General Assumption

Mitigants specified by user at t_0 can be used to proxy what each entity will post at future time steps. For a given time step and scenario, at most two different collateral pools can exist, one for user and the other for counterparty. The ratio (based on quantity) of mitigant in each collateral pool will remain constant across time when computing the variation margin balance at the beginning of the margin period of risk.

All Settlement Flows occur on or before `maturityDate` set in `standardTradeAttributes`, and the mark-to-market of the trade is zero after this maturity date.

Definition

Let *CSA* denote a Legacy, VM or IMVM CSA.

Define

$$mat_{node} := \max_{x \in node} \{mat_x\}$$

where

$$mat_x := \begin{cases} maturityDate, & EarlyExit = FALSE \\ nextEarlyTerminationDate, & EarlyExit = TRUE \end{cases}$$

Both *maturityDate* and *nextEarlyTerminationDate* are defined in `standardTradeAttributes` of trade *x*.

- **The Settlement Adjusted Path Dependent Model**

MAG performs bilateral netting variation margin computation in any of the following cases:

- `CPVMPostStyle` = `Net` and `UserVMPostStyle` = `Net`
- Only `CPVMPostStyle` = `Net`
- Only `UserVMPostStyle` = `Net`

Variation margin balance is calculated for all reporting dates r_i such that $mat_{CSA} \geq c_{n_i}$.

The computation of bilateral netting variation margin balance and margin call for a given scenario *s* on a reporting date r_i is specified below:

Step 1: Determine the necessary call dates

When computing the variation margin balance, we use the set of relevant call dates $\Gamma^{(i)}_{relevant\_VM}$.

Note

When the portfolio maturity falls between c_{n_i} and r_i, the call dates within this period are included in the relevant call dates list.

Step 2: Compute the initial variation margin balance (IVMB) and create the credit mitigation portfolio (CMP)

Step 2.1: Compute the IVMB

When `SettledToMarketAgreement` = `TRUE`, IVMB is calculated as the setttled amount SA_0. The settled amount is defined as:

$\mathrm{SA}_0 := \mathrm{SA}_{0,\,\mathrm{CSA}} := \sum_{i\,\in\,\mathrm{CSA}} \mathrm{MtM}_{0,\,i}$

where the summation is over all trades in the CSA and $MtM_{0,i}$ is the mark-to-market of trade *i* at t_0 from the cube.

When `SettledToMarketAgreement` = `FALSE`, at t_0, IVMB is computed using mitigants with `Applicability` = `INITIAL_BALANCE`.

$$IVMB = \begin{cases} \sum_{m\,=\,1}^{M} CMtM_m^{sys}(t_0), \; if \;\; CPVMPostStyle = Net \;\; \& UserVMPostStyle = Net \\ \sum_{c\,=\,1}^{C} CMtM_c^{sys}(t_0), \; if \qquad\qquad only \;\; CPVMPostStyle = Net \\ \sum_{u\,=\,1}^{U} CMtM_u^{sys}(t_0), \; if \qquad\qquad only \;\; UserVMPostStyle = Net \end{cases}$$

where,

$CMtM_m^{sys}(t_0)$ is the source system mark-to-market (MtM) in agreement currency for mitigant *m* (*m = 1, 2, ..., M*) , and *M* is the total number of mitigants with any `PostDirection` settings;

$CMtM_c^{sys}(t_0)$ is the source system MtM in agreement currency for mitigant *c* (*c = 1, 2, ..., C*) with `PostDirection` = `To User`, and *C* is the total number of mitigants with `PostDirection` = `To User`; and

$CMtM_u^{sys}(t_0)$ is the source system MtM in agreement currency for mitigant *u* (*u = 1, 2, ..., U*) with `PostDirection` = `To Counterparty`, and *U* is the total number of mitigants with `PostDirection` = `To Counterparty`.

Note

Post direction is reflected in $CMtM_m^{sys}(t_0)$, $CMtM_c^{sys}(t_0)$ and $CMtM_u^{sys}(t_0)$.

Note

The user can access the IVMB value using detail function Variation Margin Required.

IVMB computation uses the following mitigants' parameters, which are defined in 4.1.2.9 Model Inputs and Notation on page 61:

- `MTM`
- `MTMCcy`
- `PostDirection`
- `DeliveryDate`
- `ReturnDate`
- `Applicability`

If the collateral model parameter `Use Physical Collateral` = `False`, the IVMB computation above is the only computation that needs to be done for Step 2.

If the collateral model parameter `Use Physical Collateral` = `True`, proceed to Step 2.2 and 2.3.

Step 2.2: Create the CMP

At t_0, two CMPs, i.e. CMP_{cpty} and CMP_{user}, are created at most for mitigants with `Applicability` = `MITIGANT_POOL`:

- If `CPVMPostStyle` = `Net` and `UserVMPostStyle` = `Net`, both CMP_{cpty} and CMP_{user} might be created;
- If only `CPVMPostStyle` = `Net`, only CMP_{cpty} might be created;
- If only `UserVMPostStyle` = `Net`, only CMP_{user} might be created.

CMP_{cpty} is created by grouping all the mitigants with `PostDirection` = `To User`. If there is no mitigant with `PostDirection` = `To User`, CMP_{cpty} is not created. The variation margin posted by counterparty at future time steps is assumed to be in cash in agreement currency.

CMP_{user} is created by grouping all mitigants with `PostDirection` = `To Counterparty`. If there is no mitigant with `PostDirection` = `To Counterparty`, CMP_{user} is not created. The variation margin posted by user at future time steps is assumed to be in cash in agreement currency.

If neither CMP_{cpty} nor CMP_{user} is created, the model parameter `Use Physical Collateral` is automatically switched to `False` even if it is set to `True` originally.

Step 2.3: Compute the position unit (PU) of CMP at t_0

First, compute the simulated mark-to-market (MtM) of each CMP at t_0, $CMtM_A^{sim}(t_0)$, as:

$$CMtM_A^{sim}(t_0) = \sum_{m=1}^{M} PU_m \times CMtM_{A,\,m}^{sim}(t_0)$$

where, A is either cpty or user;

PU_m is the position unit of mitigant m at t_0 obtained from source system;

$CMtM_{A,\,m}^{sim}(t_0)$ is the simulated MtM of mitigant m in CMP_A at t_0 after accounting for the post direction; and

M is the total number of mitigants in CMP_A.

Position unit of CMP_A at t_0 is then computed as:

$$PU_A(t_0) = \begin{cases} Max(\frac{IVMB}{CMtM_A^{sim}(t_0)}, 0), \ if \quad CMtM_A^{sim}(t_0) \neq 0 \\ \qquad 0, \qquad\qquad\qquad otherwise \end{cases}$$

Note

In case where $CMtM_A^{sim}(t_0) = 0$, the future collateral posted by A is assumed to be in cash.

Step 3: Obtain the MtM at each call date $c \in \Gamma^{(i)}_{relevant\_VM} \backslash \{c_{-1}\}$ *1

Note

This step is not need for call dates $c > mat_{CSA}$.

At each call date *c* and under scenario *s*, obtain:

- trade portfolio mark-to-market ($MtM(c, s)$), and
- credit mitigation portfolio mark-to-market after applying all relevant margin call haircuts*2 ($CMtMHC_A(c, s)$) for each CMP_A*3

Note

When `User Physical Collateral` = `False`, only $MtM(c, s)$ needs to be obtained for each call date *c* under scenario *s* as no CMP is created at t_0.

If *c* is not a simulation date, the $MtM(c, s)$ and $CMtMHC_A(c, s)$ are modeled using Brownian Bridge (see the section 4.1.2.5 MtM on Non-Simulated Dates on page 50).

Step 4: Compute the amount of variation margin required at each call date $c \in \Gamma^{(i)}_{relevant\_VM} \backslash \{c_{-1}\}$

Note

This step is not need for call dates $c > mat_{CSA}$.

Define the counterparty and user independent amounts on trade *x* at time *t* by

$$cpIA(x,t) := \begin{cases} cpIAInput(x,t), & t \leq mat_x \\ 0, & t > mat_x \end{cases}$$

$$uIA(x,t) := \begin{cases} uIAInput(x,t), & t \leq mat_x \\ 0, & t > mat_x \end{cases}$$

where, *cpIAInput*(*x*, *t*) is the input value of the counterparty independent amount profile on trade *x* at time *t*, and *uIAInput*(*x*, *t*) is the input value of the user independent amount profile on trade *x* at time *t*. Please refer to 4.1.1.7 Model Inputs and Notation on page 27 for the definition.

mat_x is the maturity of trade *x*.

For a scenario *s* and time *t*, define the planes

$$cpIA(x,t,s) := FX_{ccy_{cpIA,x} \rightarrow ccy_{CSA}}(t,s) \times cpIA(x,t)$$

$$uIA(x,t,s) := FX_{ccy_{uIA,x} \rightarrow ccy_{CSA}}(t,s) \times uIA(x,t)$$

where, $ccy_{cpIA,x}$ is the currency of the counterparty independent amount profile on trade *x*,

$ccy_{uIA,x}$ is the currency of the user independent amount profile on trade *x*,

ccy_{CSA} is the `AgreementCurrency` of the CSA,

$FX_{ccy_1 \to ccy_2}(t, s)$ is the FX rate from ccy_1 to ccy_2 at time *t* and scenario *s*. If *t* is not a simulation date, take the FX rate from the simulation date closest to *t*. If the closest simulation date for a given *t* is not unique, use the FX rate from the simulation date after *t*. In particular, if *t* is before the session date, take the spot FX rate, that is, the FX rate at t_0.

For a CSA node, define the planes

$$cIA(t, s) := CPIndependent + \sum_{x \in CSA} cpIA(x, t, s)$$

$$uIA(t, s) := UserIndependent + \sum_{x \in CSA} uIA(x, t, s)$$

where, *CPIndependent* is the CSA counterparty independent amount and *UserIndependent* is the CSA user independent amount.

Note

In case of Settled To Market collateral agreement, *CPIndependent* = *UserIndependent* = 0.

If there are no variation margin caps present in the collateral agreement, the amount of variation margin required[*4] under scenario *s* is defined as

$$\mathrm{VMR_{NoCap}}(c, s) = \begin{cases} \mathrm{MtM}(c, s) + \mathrm{cIA}(c, s) + \mathrm{uIA}(c, s) - \mathrm{cVMT}(c), & \text{if } \mathrm{CPVMPostStyle}(c) = \mathrm{Net} \ \& \ \mathrm{MtM}(c, s) + \mathrm{cIA}(c, s) + \mathrm{uIA}(c, s) > \mathrm{cVMT}(c) \\ \mathrm{MtM}(c, s) + \mathrm{cIA}(c, s) + \mathrm{uIA}(c, s) - \mathrm{uVMT}(c), & \text{if } \mathrm{UserVMPostStyle}(c) = \mathrm{Net} \ \& \ \mathrm{MtM}(c, s) + \mathrm{cIA}(c, s) + \mathrm{uIA}(c, s) < \mathrm{uVMT}(c) \\ 0, & \text{otherwise} \end{cases}$$

where $cVMT$ is the counterpart variation margin threshold, $uVMT$ is the user variation margin threshold and *CPVMPostStyle* and *UserVMPostStyle* are as defined in section 4.1.2.9 Model Inputs and Notation on page 61.

Note

cVMT, *uVMT*, *CPVMPostStyle* and *UserVMPostStyle* may change over time if specified in VMScheduel.csv.

Note

In case of Settled To Market collateral agreement, *cVMT* = *uVMT* = 0.

If a variation margin cap is present in the collateral agreement, the variation margin required under scenario *s* is defined as

$$\mathrm{VMR}(c, s) = \begin{cases} \min\left[\mathrm{VMR_{NoCap}}(c, s), \mathrm{cVMCap}(c)\right], & \text{if } \mathrm{VMR_{NoCap}}(c, s) \geq 0 \\ \max\left[\mathrm{VMR_{NoCap}}(c, s), \mathrm{uVMCap}(c)\right], & \text{if } \mathrm{VMR_{NoCap}}(c, s) < 0 \end{cases}$$

where *cVMCap* is the counterparty variation margin cap and *uVMCap* is the user variation margin cap. Please refer to 4.1.1.7 Model Inputs and Notation on page 27 for the definition.

Note

cVMCap and *uVMCap* may change over time if specified in VMScheduel.csv.

Step 5: Compute the cash collateral and physical collateral called as well as margin call on each call date $c \in \Gamma^{(i)}_{relevant\_VM} \backslash \{c_{-1}\}$

Note

This step is not need for call dates $c > mat_{CSA}$.

The computation of the variation margin posted by user and counterparty for each margin call depends on the $\mathrm{CMP_A}$ created at t_0 and the evolution of $\mathrm{CMtMHC_A}$ over time.

The computation of CMP position unit and cash collateral at each relevant call date $c \in \Gamma^{(i)}_{relevant\_VM} \backslash \{c_{-1}\}$ and under scenario *s* (i.e. $PU_{cpty}(c,s)$, $PU_{user}(c,s)$, $CashCollateral_{cpty}(c,s)$ and $CashCollateral_{user}(c,s)$) varies by the following cases:

1. Both CMP_{cpty} and CMP_{user} are created at t_0

```
If CMtMHC_cpty  has been positive up to and including c in all scenarios
      compute PU_cpty(c, s)
Else
      compute CashCollateral_cpty(c, s)
If CMtMHC_user  has been negative up to and including c in all scenarios
      compute PU_user(c, s)
Else
      compute CashCollateral_user(c, s)
```

2. Only CMP_{cpty} is created at t_0

```
compute CashCollateral_user(c, s)
If CMtMHC_cpty  has been positive up to and including c in all scenarios
      compute PU_cpty(c, s)
Else
      compute CashCollateral_cpty(c, s)
```

3. Only CMP_{user} is created at t_0

```
compute CashCollateral_cpty(c, s)
If CMtMHC_user  has been negative up to and including c in all scenarios
      compute PU_user(c, s)
Else
      compute CashCollateral_user(c, s)
```

4. Neither CMP_{cpty} nor CMP_{user} are created at t_0 , compute $CashCollateral_{cpty}(c, s)$ and $CashCollateral_{user}(c, s)$.

In the computation, it is assumed that both user and counterparty will first fully return the variation margin it holds before posting variation margin to satisfy a margin call requirement.

Let $\Gamma^{(i)}_{relevant\_VM} = \left\{c_j, c_{j+1}, \ldots, c_{j+k}\right\}$

Computation at call date c_j under scenario *s*

- The position unit of CMP_{cpty} and CMP_{user} are computed as:

  $$PU_A\left(c_j, s\right) = \begin{cases} PU_A(t_0), & if \quad c_j = c_{-1} \\ Max\left(\dfrac{VMR(c_j, s)}{CMtMHC_A(c_j, s)}, 0\right), & if \quad c_j \neq c_{-1} \end{cases}$$

  where *A* is either *cpty* or *user*.

- The amount of cash collateral posted by counterparty is computed as:

  $$CashCollateral_{cpty}(c_j, s) = \begin{cases} \max(IVMB, 0), & if \quad c_j = c_{-1} \\ \max(VMR(c_j, s), 0), & if \quad c_j \neq c_{-1} \end{cases}$$

- The amount of cash collateral posted by user is computed as:

$$CashCollateral_{user}(c_j, s) = \begin{cases} \min(IVMB, 0), & if \quad c_j = c_{-1} \\ \min(VMR(c_j, s), 0), & if \quad c_j \neq c_{-1} \end{cases}$$

Margin calls computation on other call dates

For $1 \le l \le k$, recursively define the margin calls *5 on the other call dates as follows:

$$MC(c_{j+l}, s) = \begin{cases} 0, \quad if \quad uVMMTA(c) < VMR(c_{j+l}, s) - \sum_A (PU_A(c_{j+l-1}, s) \times CMtMHC_A(c_{j+l}, s) + \\ CashCollateral_A(c_{j+l-1}, s)) < cVMMTA(c) \\ round\left[VMR(c_{j+l}, s) - \sum_A (PU_A(c_{j+l-1}, s) \times CMtMHC_A(c_{j+l}, s) + CashCollateral_A(c_{j+l-1}, s)), RA\right], \\ otherwise \end{cases}$$

,

where function *round(x,m)* rounds *x* to the nearest integer multiple of *m*; *cVMMTA* is the counter-party variation margin minimum transfer amount; *uVMMTA* is the user variation margin minimum transfer amount, and *RA* is the round-off amount. All these parameters are defined in section 4.1.1.7 Model Inputs and Notation on page 27.

Note

cVMMTA and *uVMMTA* may change over time if specified in VMScheduel.csv.

Note

In case of Settled To Market collateral agreement, *cVMMTA* = *uVMMTA* = 0, *RA* = 0.

Computation at each other call date c_{j+l} under scenario *s*

- The total position unit of CMP_{cpty} is computed as:

$$PU_{cpty}(c_{j+l}, s) = Max\left[PU_{cpty}(c_{j+l-1}, s) + \frac{MC(c_{j+l}, s) + PU_{user}(c_{j+l-1}, s) \times CMtMHC_{user}(c_{j+l}, s) + CashCollateral_{user}(c_{j+l-1}, s)}{CMtMHC_{cpty}(c_{j+l}, s)}, 0\right]$$

- The incremental position unit of CMP_{cpty} is computed as:

$$\Delta PU_{cpty}(c_{j+l}, s) = PU_{cpty}(c_{j+l}, s) - PU_{cpty}(c_{j+l-1}, s)$$

- The total position unit of CMP_{user} is computed as:

$$PU_{user}(c_{j+l}, s) = Max\left[PU_{user}(c_{j+l-1}, s) + \frac{MC(c_{j+l}, s) + PU_{cpty}(c_{j+l-1}, s) \times CMtMHC_{cpty}(c_{j+l}, s) + CashCollateral_{cpty}(c_{j+l-1}, s)}{CMtMHC_{user}(c_{j+l}, s)}, 0\right]$$

- The incremental position unit of CMP_{user} is computed as:

$$\Delta PU_{user}(c_{j+l}, s) = PU_{user}(c_{j+l}, s) - PU_{user}(c_{j+l-1}, s)$$

- The total amount of cash collateral posted by user is computed as:

$$CashCollateral_{user}(c_{j+l}, s) = Min\left[\sum_A CashCollateral_A(c_{j+l-1}, s) + MC(c_{j+l}, s) + PU_{cpty}(c_{j+l-1}, s) \times CMtMHC_{cpty}(c_{j+l}, s), 0\right]$$

- The incremental amount of cash collateral posted by user is computed as:

$$\Delta CashCollateral_{user}(c_{j+l}, s) = CashCollateral_{user}(c_{j+l}, s) - CashCollateral_{user}(c_{j+l-1}, s)$$

- The total amount of cash collateral posted by counterparty is computed as:

$$CashCollateral_{cpty}(c_{j+l}, s) = Max\left[\sum_A CashCollateral_A(c_{j+l-1}, s) + MC(c_{j+l}, s) + PU_{user}(c_{j+l-1}, s) \times CMtMHC_{user}(c_{j+l}, s), 0\right]$$

- The incremental amount of cash collateral posted by counterparty is computed as:

  $\Delta CashCollateral_{cpty}(c_{j+l}, s) = CashCollateral_{cpty}(c_{j+l}, s) - CashCollateral_{cpty}(c_{j+l-1}, s)$

Step 6: Compute the amount of variation margin available (variation margin balance) at r_i

Define the variation margin balance used for $Exp_{smooth}(r_i, s)$ computation by:

$$VM_{smooth}(r_i, s) = \sum_A \left[PU_A(c_{n_i}, s) \times CMtM_{A,default}(r_i, s) + CashCollateral_A(c_{n_i}, s)\right]$$

where, $CMtM_{A,default}(r_i, s)$ is the mark-to-market of CMP_A after applying the haircuts at default*6 specifically for exposure calculation.

Define the variation margin balance used for $Exp_{full}(r_i, s)$ computation by:

$$VM_{full}(r_i, s) = \sum_A \left[PU_A(c_{n_i+m_i}, s) \times CMtM_{A,default}(r_i, s) + CashCollateral_A(c_{n_i+m_i}, s)\right]$$

where,

- If $c_{n_i+m_i} \leq mat_{CSA}$

$$PU_A(c_{n_i+m_i}, s) = PU_A(c_{n_i}, s) + \begin{cases} \sum_{l=1}^{m_i} \frac{\min\left(0, MC(c_{n_i+l}, s)\right)}{MC(c_{n_i+l}, s)} \times \Delta PU_A(c_{n_i+l}, s) & for \quad forward \quad view \quad computation \\ \sum_{l=1}^{m_i} \frac{\max\left(0, MC(c_{n_i+l}, s)\right)}{MC(c_{n_i+l}, s)} \times \Delta PU_A(c_{n_i+l}, s) & for \quad reverse \quad view \quad computation \end{cases}$$

$$CashCollateral_A(c_{n_i+m_i}, s) = CashCollateral_A(c_{n_i}, s) +$$

$$\begin{cases} \sum_{l=1}^{m_i} \frac{\min\left(0, MC(c_{n_i+l}, s)\right)}{MC(c_{n_i+l}, s)} \times \Delta CashCollateral_A(c_{n_i+l}, s) & for \quad forward \quad view \quad computation \\ \sum_{l=1}^{m_i} \frac{\max\left(0, MC(c_{n_i+l}, s)\right)}{MC(c_{n_i+l}, s)} \times \Delta CashCollateral_A(c_{n_i+l}, s) & for \quad reverse \quad view \quad computation \end{cases}$$

  If $m_i = 0$, $PU_A(c_{n_i+m_i}, s) = PU_A(c_{n_i}, s)$ and $CashCollateral_A(c_{n_i+m_i}, s) = CashCollateral_A(c_{n_i}, s)$.*7

- If $c_{n_i+m_i} > mat_{CSA}$

  Define $c_{<mat}$ the last relevant call date on or before mat_{CSA}.

  Forward View:

$$PU_A(c_{n_i+m_i}, s) := \begin{cases} PU_A(c_{<mat}, s), & A = user \\ 0, & A = cpty \end{cases}$$

$$CashCollateral_A(c_{n_i+m_i}, s) := \begin{cases} CashCollateral_A(c_{<mat}, s), & A = user \\ 0, & A = cpty \end{cases}$$

  Reverse View:

$$PU_A(c_{n_i+m_i}, s) := \begin{cases} PU_A(c_{<mat}, s), & A = cpty \\ 0, & A = user \end{cases}$$

$$CashCollateral_A(c_{n_i+m_i}, s) := \begin{cases} CashCollateral_A(c_{<mat}, s), & A = cpty \\ 0, & A = user \end{cases}$$

Forward view computation means collateralized value is computed from the perspective of user. Reverse view computation means collateralized value is computed from the perspective of counterparty.

When the collateral model parameter `Use Physical Collateral` = `FALSE`, user has the option to apply the haircut at default to the cash collateral by defining the CSA level custom attributes `CPHaircutAtDefault` and `UserHaircutAtDefault` and the collateral balance is then calculated as:

$$VM_{\text{smooth}}(r_i, s) = \begin{cases} \text{CashCollateral}(c_{n_i}, s) \times (1 - \text{HC}_{\text{cpty, default}}), & \text{if CashCollateral}(c_{n_i}, s) \geq 0 \\ \text{CashCollateral}(c_{n_i}, s) \times (1 + \text{HC}_{\text{user, default}}), & \text{if CashCollateral}(c_{n_i}, s) < 0 \end{cases}$$

and

$$VM_{\text{full}}(r_i, s) = \begin{cases} \text{CashCollateral}(c_{n_i + m_i}, s) \times (1 - \text{HC}_{\text{cpty, default}}), & \text{if CashCollateral}(c_{n_i + m_i}, s) \geq 0 \\ \text{CashCollateral}(c_{n_i + m_i}, s) \times (1 + \text{HC}_{\text{user, default}}), & \text{if CashCollateral}(c_{n_i + m_i}, s) < 0 \end{cases}$$

where, $HC_{cpty,\ default}$ is read from `CPHaircutAtDefault` and $HC_{user,\ default}$ is read from `UserHaircutAtDefault`. Both are expressed as a percentage.

*1 $\Gamma^{(i)}_{relevant\_VM} \backslash \{c_{-1}\}$ is the set of relevant call dates excluding c_{-1} .

*2 The margin call haircut can be applied to both cash and non-cash mitigants in CMP_A.

*3 It is assumed that post direction is reflected in $CMtM_A(c, s)$ and $CMtMHC_A(c, s)$. Therefore, $CMtM_{cpty}(c, s) \geq 0$ and $CMtMHC_{cpty}(c, s) \geq 0$; $CMtM_{user}(c, s) \leq 0$ and $CMtMHC_{user}(c, s) \leq 0$.

*4 User can access the amount of variation margin required (VMR) on the first call date of a set of consecutive relevant call dates under scenario s using detail function Variation Margin Required.

*5 User can access the margin call amount at each call date c_{j+l} for $1 \leq l \leq k$ and scenario s using detail function Variation Margin Call.

*6 The haircuts at default applied to the mark-to-market of CMP_A in the variation margin balance computation are different from the ones used in the margin call computation. Applying haircuts at default in the variation margin balance computation will produce more conservative $Exp_{smooth}(r_i, s)$ and $Exp_{full}(r_i, s)$ results. Note, haircut at default can be applied to both cash and non-cash mitigants in CMP_A.

*7 For c < mat_{CSA}, user can access the amount of $\sum_{l=1}^{m_i} \frac{\min(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} \times \left[\Delta PU_A(c_{n_i+l}, s) + \Delta CashCollateral_A(c_{n_i+l}, s) \right]$ for forward view computation and $\sum_{l=1}^{m_i} \frac{\max(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} \times \left[\Delta PU_A(c_{n_i+l}, s) + \Delta CashCollateral_A(c_{n_i+l}, s) \right]$ for reverse view computation using detail function MPOR Settled Payments. The function option should be set to VM. MAG doesn't compute this amount after portfolio maturity.

4.1.2.4 Collateralized Exposure

Let us define the (simulated) collateralized full exposure, $Exp_{full}(r_i, s_j)$, and collateralized smooth exposure, $Exp_{smooth}(r_i, s_j)$, at time r_i under scenario s_j by:

$$Exp_{full}(r_i, s_j) = \max(CV_{Exp\_full}(r_i, s_j), 0)$$

$$Exp_{smooth}(r_i, s_j) = \max(CV_{Exp\_smooth}(r_i, s_j), 0)$$

where,

Exp_{smooth} assumes that neither the defaulting party nor the non-defaulting party makes margin (variation margin and initial margin) or trade settlement flow payments after the beginning of the margin period of risk (MPOR); and

Exp_{full} assumes that the defaulting party does not make margin or trade settlement flow payments after the beginning of the margin period of risk (MPOR), while the non-defaulting party continues making variation margin and trade settlement flow payments according to its default management process.

Exposure is calculated for all reporting dates r_i such that $mat_{CSA} \geq c_{n_i}$.

The intermediate (simulated) collateralized value, $CV_{Exp\_full}(r_i, s_j)$ and $CV_{Exp\_smooth}(r_i, s_j)$ are defined below.

The collateralized exposure is used for most credit expressions. However, FVA expressions are not supported with Settlement Adjusted Path Dependent Model.

Definitions

- **Portfolio Value**

$V^X(r_i, s_j)$ - portfolio value on the node *X*. For how this portfolio value is determined depending on the netting specified by the *NettingRule, CloseoutNetting, and AgreementEnforceable* values as well as the perspective direction of the expression, see Chapter 3 Netting Rule on page 5.

Important

We will assume that all the variation and initial margins defined here are positive.

- **Variation Margins**

$VM^X_{smooth,\ A,\ seg}(r_i, s_j)$ is defined as:

```
If X is a CSA node
       VM^X_{smooth, A, seg}(r_i, s_j)  is segregated VM_smooth posted by A on node X
Else if X is an MA node
       VM^X_{smooth, A, seg}(r_i, s_j)  is segregated VM_smooth posted by A  on all children of node X
Else
       VM^X_{smooth, A, seg}(r_i, s_j)  = 0
```

$VM^X_{smooth,\ A,\ unseg}(r_i, s_j)$ is defined as:

```
If X is a CSA node
       VM^X_{smooth, A, unseg}(r_i, s_j)  is unsegregated VM_smooth posted by A on node X
Else if X is an MA node
       VM^X_{smooth, A, unseg}(r_i, s_j)  is unsegregated VM_smooth posted by A  on all children of node X
Else
       VM^X_{smooth, A, unseg}(r_i, s_j)  = 0
```

$VM^X_{full,\ A,\ seg}(r_i, s_j)$ is defined as:

```
If X is a CSA node
       VM^X_{full, A, seg}(r_i, s_j)  is segregated VM_full posted by A on node X
Else if X is an MA node
       VM^X_{full, A, seg}(r_i, s_j)  is segregated VM_full posted by A  on all children of node X
Else
       VM^X_{full, A, seg}(r_i, s_j)  = 0
```

$VM^X_{full,\ A,\ unseg}(r_i, s_j)$

```
If X is a CSA node
       VM^X_{full, A, unseg}(r_i, s_j)  is unsegregated VM_full posted by A on node X
Else if X is an MA node
       VM^X_{full, A, unseg}(r_i, s_j)  is unsegregated VM_full posted by A  on all children of node
```

```
X
Else
```

$$VM^X_{full,\ A,\ unseg}(r_i, s_j) \quad = 0$$

The segregation of variation margin is specified by the collateral agreement parameters `CPVMSegregated` and `UserVMSegregated`. (Here *A* denotes either the counterparty or the user.)

- **Initial Margins**

For each counterparty and user, both segregated initial margin and non-segregated initial margin can be specified for each node of type CSA, NC, NN or MA. Define

$cpIM_X(t_i, s_j, ccy)$ as the node-level counterparty initial margin posted on node *X* (coverted to currency *ccy*);

$uIM_X(t_i, s_j, ccy)$ as the node-level user initial margin posted on node *X* (converted to currency *ccy*);

$cpIM_{X,unseg}(t_i, s_j, ccy)$ as the node-level counterparty unsegregated initial margin posted on node *X* (converted to currency *ccy*); and

$uIM_{X,unseg}(t_i, s_j, ccy)$ as the node-level user unsegregated initial margin posted on node *X* (converted to currency *ccy*).

Note

If initial margin is specified through the initial margin profiles, the segregation of initial margin is specified by standard mitigant attribute `segregated`. Each initial margin profile starts at `DeliveryDate` and ends at `ReturnDate`.

Note

If initial margin is specified through the initial margin sheets, the segregation of initial margin is specified by the collateral agreement parameters `CPIMSegregated` and `UserIMSegregated`.

Note

If initial margin is calculated by Dynamic Initial Margin model, the segregation of initial margin is specified by the collateral agreement parameters `CPIMSegregated` and `UserIMSegregated`.

User can also specify initial margin for individual trade. Define the counterparty and user initial margins on trade *x* at time *t* by

$$cpIM(x,t) := \begin{cases} cpIMInput(x,t), & t \le ReturnIM_x \text{ and } mat_x \ge t_0 \\ 0, & otherwise \end{cases}$$

$$uIM(x,t) := \begin{cases} uIMInput(x,t), & t \le ReturnIM_x \text{ and } mat_x \ge t_0 \\ 0, & otherwise \end{cases}$$

where, *cpIMInput*(*x*, *t*) is the input value of the counterparty initial margins on trade *x* at time *t*, and *uIMInput*(*x*, *t*) is the input value of the user initial margins on trade *x* at time *t*. Please refer to 4.1.1.7 Model Inputs and Notation on page 27 for the definition.

$$ReturnIM_x := \begin{cases} mat_x, & \text{if } returnStyle = MATURITY \\ mat_x + IMLag, & \text{if } returnStyle = EXTENDED\_MATURITY \end{cases}$$

$$IMLag := \begin{cases} MPOR, & \text{if } x \text{ belongs to a variation margin CSA} \\ 0, & otherwise \end{cases}$$

Variation margin CSA includes VMCSA, IMVMCSA and Legacy CSA.

For a scenario s_j, timestep t_i and currency *ccy*, define

$$cpIM(x, t_i, s_j, ccy) := FX_{ccy_{cpIM,x} \to ccy}(t_i, s_j) \times cpIM(x, t_i)$$

$$uIM(x, t_i, s_j, ccy) := FX_{ccy_{uIM,x} \to ccy}(t_i, s_j) \times uIM(x, t_i)$$

$$cpIM_{unseg}(x, t_i, s_j, ccy) := \begin{cases} cpIM(x, t_i, s_j, ccy), & \text{if } cpIM(x, t_i) \text{ is unsegregated} \\ 0, & otherwise \end{cases}$$

$$uIM_{unseg}(x, t_i, s_j, ccy) := \begin{cases} uIM(x, t_i, s_j, ccy), & \text{if } uIM(x, t_i) \text{ is unsegregated} \\ 0, & otherwise \end{cases}$$

where, $ccy_{cpIM,x}$ is the currency of the counterparty initial margins on trade *x*,

$ccy_{uIM,x}$ is the currency of the user initial margins on trade *x*,

ccy_{CSA} is the `AgreementCurrency` of the CSA,

$FX_{ccy_1 \to ccy_2}(t, s)$ is the FX rate from ccy_1 to ccy_2 at time *t* and scenario *s*. If *t* is not a simulation date, take the FX rate from the simulation date closest to *t*. If the closest simulation date for a given *t* is not unique, use the FX rate from the simulation date after *t*. In particular, if *t* is before the session date, take the spot FX rate, that is, the FX rate at t_0.

For a node *X*, define

$$IM^X_{cpty}(t_i, s_j) := cpIM_X(t_i, s_j, ccy) + \sum_{x \in X} cpIM(x, t_i, s_j, ccy)$$

$$IM^X_{user}(t_i, s_j) := uIM_X(t_i, s_j, ccy) + \sum_{x \in X} uIM(x, t_i, s_j, ccy)$$

$$IM^X_{cpty,unseg}(t_i, s_j) := cpIM_{X,unseg}(t_i, s_j, ccy) + \sum_{x \in X} cpIM_{unseg}(x, t_i, s_j, ccy)$$

$$IM^X_{user,unseg}(t_i, s_j) := uIM_{X,unseg}(t_i, s_j, ccy) + \sum_{x \in X} uIM_{unseg}(x, t_i, s_j, ccy)$$

- **Total Margins**

$$C^X_{smooth,\ A,\ seg}(r_i, s_j) = \begin{cases} VM^X_{smooth,\ A,\ seg}(r_i, s_j) + IM^X_{A,\ seg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = YES \\ VM^X_{smooth,\ A,\ seg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = NO \\ IM^X_{A,\ seg}(r_i, s_j) & \text{if } EVM = NO \ \&\ EIM = YES \end{cases}$$

where *EVM* is model parameter `Enable Variation Margin` and *EIM* is model parameter `Enable Initial Margin`.

$$C^X_{smooth,\ A,\ unseg}(r_i, s_j) = \begin{cases} VM^X_{smooth,\ A,\ unseg}(r_i, s_j) + IM^X_{A,\ unseg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = YES \\ VM^X_{smooth,\ A,\ unseg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = NO \\ IM^X_{A,\ unseg}(r_i, s_j) & \text{if } EVM = NO \ \&\ EIM = YES \end{cases}$$

$$C^X_{smooth,\ A}(r_i, s_j) = C^X_{smooth,\ A,\ seg}(r_i, s_j) + C^X_{smooth,\ A,\ unseg}(r_i, s_j)$$

$$C^X_{full,\ A,\ seg}(r_i, s_j) = \begin{cases} VM^X_{full,\ A,\ seg}(r_i, s_j) + IM^X_{A,\ seg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = YES \\ VM^X_{full,\ A,\ seg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = NO \\ IM^X_{A,\ seg}(r_i, s_j) & \text{if } EVM = NO \ \&\ EIM = YES \end{cases}$$

$$C^X_{full,\ A,\ unseg}(r_i, s_j) = \begin{cases} VM^X_{full,\ A,\ unseg}(r_i, s_j) + IM^X_{A,\ unseg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EVM = YES \\ VM^X_{full,\ A,\ unseg}(r_i, s_j) & \text{if } EVM = YES \ \&\ EIM = NO \\ IM^X_{A,\ unseg}(r_i, s_j) & \text{if } EVM = NO \ \&\ EIM = YES \end{cases}$$

$$C^X_{full,\ A}(r_i, s_j) = C^X_{full,\ A,\ seg}(r_i, s_j) + C^X_{full,\ A,\ unseg}(r_i, s_j)$$

Note

As the IM value used in the computation is a user input, we don't know the underlying assumption regarding when the non-defaulting party stops paying IM. Therefore, it is up to user to decide where to account for IM in the Exp_{smooth} and Exp_{full} computation.

Note

By including $IM^X_{A,\ unseg}(r_i, s_j)$ and $IM^X_{A,\ seg}(r_i, s_j)$ in the $C^X_{smooth,\ A}(r_i, s_j)$ calculation, it is assumed that both user and counterparty stops making IM payment after the beginning of the MPOR.

Define $F_{\mathrm{Lag\_TS}}(r_i, s_j)$ as follows:

```
If UseSettlementFlows = TRUE
```

$$F_{\mathrm{Lag\_TS}}(r_i, s_j) = \mathrm{TF}_{\mathrm{Lag\_TS}}(r_i, s_j)$$

```
Else if Isolate Collateral Spikes = TRUE
```

$$\begin{aligned} F_{\mathrm{Lag\_TS}}(r_i, s_j) &= -\mathrm{CF}_{\mathrm{nd}}([r_i - MPOR, r_i - MPOR + \mathrm{Lag\_TS}), s_j) \\ &= \mathrm{CF}_{\mathrm{nd}}([r_i - MPOR, r_i), s_j) - \mathrm{CF}_{\mathrm{nd}}([r_i - MPOR + \mathrm{Lag\_TS}, r_i), s_j) \end{aligned}$$

```
Else
```

$$F_{\mathrm{Lag\_TS}}(r_i, s_j) = 0$$

Note

By the above defintion, $F_{Lag\_TS}(r_i, s_j) \le 0$ in forward view and $F_{Lag\_TS}(r_i, s_j) \ge 0$ in reverse view.

$F^X_{\mathrm{Lag\_TS}}(r_i, s_j)$ is defined as:

If X is a CSA node
    $F^X_{\mathrm{Lag\_TS}}(r_i, s_j)$ is the total trade settlement flow paid by the non-defaulting party within *Lag_TS* on node X
Else if X is an MA node
    $F^X_{\mathrm{Lag\_TS}}(r_i, s_j)$ is the total trade settlement flow paid by the non-defaulting party within *Lag_TS* on all children of node X
Else
    $F^X_{\mathrm{Lag\_TS}}(r_i, s_j)$ = 0

Define $\mathrm{NetF}_{\mathrm{MPOR}}(r_i, s_j)$ as follows:

```
If UseSettlementFlows = TRUE
```

$$\mathrm{NetF}_{\mathrm{MPOR}}(r_i, s_j) = \mathrm{NetTF}_{\mathrm{MPOR}}(r_i, s_j)$$

```
Else if Isolate Collateral Spikes = TRUE
```

$$\mathrm{NetF}_{\mathrm{MPOR}}(r_i, s_j) = \mathrm{NetCF}([r_i - MPOR, r_i), s_j)$$

```
Else
```

$$\mathrm{NetF}_{\mathrm{MPOR}}(r_i, s_j) = 0$$

$\mathrm{NetF}^{X}_{\mathrm{MPOR}}(r_i, s_j)$ is defined as:

```
If X is a CSA node
     NetF^X_MPOR(r_i, s_j) is net trade settlement flow within the MPOR on node X
Else if X is an MA node
     NetF^X_MPOR(r_i, s_j) is net trade settlement flow within the MPOR on all children of no
de X
Else
     NetF^X_MPOR(r_i, s_j) = 0
```

- **Collateralized Value And Settlement Spike**

Collateralized Value From The Perspective Of User (Forward View)

Case 1:

When the following conditions are satisfied:

- `NettingRule` = LEGAL_OPINION
- there is a complex structure specified or `AgreementEnforceable` is False

the $CV_{Exp\_smooth}(r_i, s_j)$ and $CV_{Exp\_full}(r_i, s_j)$ are computed as:

```
If User Collateral Offset = Netted Pools
```

$$CV^{X}_{Exp\_smooth}(r_i, s_j) = V^{X}(r_i, s_j) + max\{C^{X}_{smooth,\,user,\,unseg}(r_i, s_j) + N^{X}_{CNP}(r_i, s_j), 0\} - \begin{cases} VM^{X}_{smooth,\ cpty}(r_i, s_j) + IM^{X}_{cpty}(r_i, s_j) & \text{if} \quad \mathrm{AM} = \mathrm{BOTH} \\ VM^{X}_{smooth,\ cpty}(r_i, s_j) & \text{if} \quad \mathrm{AM} = \mathrm{VARIATION\ MARGIN} \\ IM^{X}_{cpty}(r_i, s_j) & \text{if} \quad \mathrm{AM} = \mathrm{INITIAL\ MARGIN} \\ 0 & \text{if} \quad \mathrm{AM} = \mathrm{NONE} \end{cases} + NetF^{X}_{MPOR}(r_i, s_j)$$

$$CV^{X}_{Exp\_full}(r_i, s_j) = V^{X}(r_i, s_j) + max\{C^{X}_{full,\,user,\,unseg}(r_i, s_j) + N^{X}_{CNP}(r_i, s_j), 0\} - \begin{cases} VM^{X}_{full,\ cpty}(r_i, s_j) + IM^{X}_{cpty}(r_i, s_j) & \text{if} \quad \mathrm{AM} = \mathrm{BOTH} \\ VM^{X}_{full,\ cpty}(r_i, s_j) & \text{if} \quad \mathrm{AM} = \mathrm{VARIATION\ MARGIN} \\ IM^{X}_{cpty}(r_i, s_j) & \text{if} \quad \mathrm{AM} = \mathrm{INITIAL\ MARGIN} \\ 0 & \text{if} \quad \mathrm{AM} = \mathrm{NONE} \end{cases} + NetF^{X}_{MPOR}(r_i, s_j) - F^{X}_{Lag\_TS}(r_i, s_j)$$

where $N^{X}_{CNP}(r_i, s_j) = \sum_{l=1}^{L} \min\left(MtM^{X}_{CNP_l}(r_i, s_j), 0\right)$, $CNP_l(l = 1, \ldots, L)$ are the closeout netting pools on Node X, and $MtM^{X}_{CNP_l}(r_i, s_j)$ is the netted sum of the MtM of all positions within *CNP$_l$* on Node X at reporting time r_i under scenario s_j.

Note

User can access the amount of $-max\{C^{X}_{smooth,\,user,\,unseg}(r_i, s_j) + N^{X}_{CNP}(r_i, s_j), 0\} + VM^{X}_{smooth,\,cpty}(r_i, s_j)$ or $-max\{C^{X}_{full,\,user,\,unseg}(r_i, s_j) + N^{X}_{CNP}(r_i, s_j), 0\} + VM^{X}_{full,\,cpty}(r_i, s_j)$ using detail function Variation Margin Balance.

```
Else
```

$$CV^X_{Exp\_smooth}(r_i, s_j) = V^X(r_i, s_j) + C^X_{smooth,\ user,\ unseg}(r_i, s_j) -$$
$$\left\{ \begin{array}{lll} VM^X_{smooth,\ cpty}(r_i, s_j) + IM^X_{cpty}(r_i, s_j) & \text{if} & AM = \text{BOTH} \\ VM^X_{smooth,\ cpty}(r_i, s_j) & \text{if} & AM = \text{VARIATION MARGIN} \\ IM^X_{cpty}(r_i, s_j) & \text{if} & AM = \text{INITIAL MARGIN} \\ 0 & \text{if} & AM = \text{NONE} \end{array} \right. + NetF^X_{MPOR}(r_i, s_j)$$

$$CV^X_{Exp\_full}(r_i, s_j) = V^X(r_i, s_j) + C^X_{full,\ user,\ unseg}(r_i, s_j) -$$
$$\left\{ \begin{array}{lll} VM^X_{full,\ cpty}(r_i, s_j) + IM^X_{cpty}(r_i, s_j) & \text{if} & AM = \text{BOTH} \\ VM^X_{full,\ cpty}(r_i, s_j) & \text{if} & AM = \text{VARIATION MARGIN} \\ IM^X_{cpty}(r_i, s_j) & \text{if} & AM = \text{INITIAL MARGIN} \\ 0 & \text{if} & AM = \text{NONE} \end{array} \right. + NetF^X_{MPOR}(r_i, s_j) - F^X_{Lag\_TS}(r_i, s_j)$$

where AM is determined by the values of Collateral Sub-Model parameters as in the following table:

Collateral Sub-Model	**Variation Margin**	
Initial Margin	**Enable Variation Margin = YES & Apply with Legal Opinion = YES**	**Apply with Legal Opinion = NO**
Enable Initial Margin = YES & Apply with Legal Opinion = YES	BOTH	INITIAL MARGIN
Apply with Legal Opinion = NO	VARIATION MARGIN	NONE

Note

User can access the amount of $-VM^X_{smooth, user, unseg}(r_i, s_j) + VM^X_{smooth, cpty}(r_i, s_j)$ or $-VM^X_{full, user, unseg}(r_i, s_j) + VM^X_{ful, cpty}(r_i, s_j)$ using detail function Variation Margin Balance.

Case 2:

When the conditions specified in the above Case 1 are not satisfied, the collateralized value is calculated as:

$$CV^X_{Exp\_smooth}(r_i, s_j) = V^X(r_i, s_j) - C^X_{smooth,\ cpty}(r_i, s_j) + C^X_{smooth,\ user,\ unseg}(r_i, s_j) + NetF^X_{MPOR}(r_i, s_j)$$

$$CV^X_{Exp\_full}(r_i, s_j) = V^X(r_i, s_j) - C^X_{full,\ cpty}(r_i, s_j) + C^X_{full,\ user,\ unseg}(r_i, s_j) + NetF^X_{MPOR}(r_i, s_j) - F^X_{Lag\_TS}(r_i, s_j)$$

Note

User can access the amount of $-VM^X_{smooth, user, unseg}(r_i, s_j) + VM^X_{smooth, cpty}(r_i, s_j)$ or $-VM^X_{full, user, unseg}(r_i, s_j) + VM^X_{ful, cpty}(r_i, s_j)$ using detail function Variation Margin Balance.

Collateralized Value From The Perspective Of Counterparty (Reverse View)

$$CV^X_{Exp\_smooth}(r_i, s_j) = V^X(r_i, s_j) - C^X_{smooth,\ user}(r_i, s_j) + C^X_{smooth,\ \mathrm{cpty},\ unseg}(r_i, s_j) - NetF^X_{MPOR}(r_i, s_j)$$

$$CV^X_{Exp\_full}(r_i, s_j) = V^X(r_i, s_j) - C^X_{full,\ user}(r_i, s_j) + C^X_{full,\ \mathrm{cpty},\ unseg}(r_i, s_j) - NetF^X_{MPOR}(r_i, s_j) + F^X_{Lag\_TS}(r_i, s_j)$$

Settlement Spike Computation

The settlement spike*1 caused by non-defaulting party's margin and trade settlement flow payments after the beginning of the MPOR is computed as:

$$SettlementSpike^X_{All}(r_i, s_j) = Exp_{full}(r_i, s_j) - Exp_{smooth}(r_i, s_j)$$

The settlement spike[*2] caused only by non-defaulting party's trade settlement flow payments after the beginning of the MPOR is computed as:

$$SettlementSpike^X_{TF}(r_i, s_j) = F^X_{Lag\_TS}(r_i, s_j)$$

Note

In the case when `UseSettlementFlows` = FALSE and `Isolate Collateral Spikes` = TRUE, $F_{\mathrm{Lag\_TS}}(r_i, s_j)$ is an estimate of the non-defaulting party settlement flows during the Lag_TS, but it only considers flows due to maturing trades.

*1 The user can access the amount of $SettlementSpike^X_{All}(r_i, s_j)$ using the detail function MPOR Settled Payments. The function option should be set to All.

*2 User can access the amount of $SettlementSpike^X_{TF}(r_i, s_j)$ using detail function MPOR Settled Payments. The function option should be set to Settlement.

4.1.2.5 MtM on Non-Simulated Dates

In the Settlement Adjusted Path Dependent Model, the trade portfolio mark-to-market (MtM) and the credit mitigation portfolio MtM after applying all relevant haircuts are needed for some non-simulated call dates (see step 3 in section 4.1.2.3 Variation Margin Balance and Margin Call Algorithm when CPVMPostStyle = Net and UserVMPostStyle = Net on page 35). The missing information for such call dates is approximated using Brownian Bridge.

Let,

- t_i and t_{i+1} be the two adjacent simulation dates
- c_k, c_{k+1}, ..., c_{k+n} be the relevant call dates between t_i and t_{i+1}
- $MtM(t, s)$ be the net mark-to-market of all trades
- $MtM_{nmf}(t, s)$ be the net mark-to-market of trades that have not matured before the following simulation date

The MtM on each non-simulated call date is computed using Brownian Bridge constructed between the date immediately before and immediately after the call date with known trade portfolio MtM and credit mitigation portfolio MtM after applying all relevant haircuts.

Here we start computing MtM from call date c_k and finish on call date c_{k+n} in ascending order. Therefore, for $0 \le l \le n$, the MtM on c_{k+l} is computed using Brownian Bridge constructed between c_{k+l-1} and t_{i+1}, where c_{k-1} is set to be t_i

■ Portfolio MtM and Settlement Flow Adjustment

Trade Portfolio

The trade portfolio MtM used as end points for Brownian Bridge computation differs by the variation margin post style. To compute the trade portfolio MtM on call date c_{k+l} under a given scenario *s*,

- `CPVMPostStyle` = Net and/or `UserVMPostStyle` = Net

  The trade portfolio MtM used as end points are the MtM of the CSA portfolio with all the trades, $MtM^{BB}(c_{k+l-1}, s)$ and $MtM(t_{i+1}, s)$, except that when computing the trade portfolio MtM on call date c_k, the left end point is

$$\text{MtM}(t_i, s) - \begin{cases} \text{TFAdj}_p(t_i, s), & \text{if UseSettlementFlows=TRUE} \\ \text{CF}([t_i, t_{i+1}), s), & \text{if UseSettlementFlows = FALSE \& Isolate Collateral Spikes = TRUE} \end{cases}'$$

where $TFAdj_p(t_i, s)$ is an adjustment applied at t_i to capture the trade settlement flows happened between t_i and t_{i+1}; and $\text{CF}([t_i, t_{i+1}), s) := \text{MtM}(t_i, s) - \text{MtM}_{nmf}(t_i, s)$.

$TFAdj_p(t_i, s)$ is computed by converting all the trade level settlement flows from settlement date d with $t_i \leq d < t_{i+1}$ under scenario s into the agreement currency and taking the sum. The currency conversion is based on the FX rate from the simulation date that is closest to d. If the closest simulation date for a given d is not unique, use the FX rate from the simulation date after d.

Credit Mitigation Portfolio

To compute the counterparty credit mitigation portfolio MtM after applying all relevant haircuts on call date c_{k+l} under a given scenario s, $CMtMHC^{BB}_{cpty}(c_{k+l}, s)$:

- If the counterparty credit mitigation portfolio (CMP_{cpty}) has been created at t_0, and $\text{CMtMHC}_{\text{cpty}}$ has been positive from t_0 up to and including t_{i+1} across all scenarios, the end points for Brownian Bridge computation are $CMtMHC^{BB}_{cpty}(c_{k+l-1}, s)$ and $CMtMHC_{cpty}(t_{i+1}, s)$, except that when computing $CMtMHC^{BB}_{cpty}(c_k, s)$, the left end point is $CMtMHC_{cpty}(t_i, s)$.
- If CMP_{cpty} has not been created at t_0 OR CMP_{cpty} has been created at t_0, but $\text{CMtMHC}_{\text{cpty}}$ has become zero or negative on or before t_{i+1} under any scenario, there is no computation for $CMtMHC^{BB}_{cpty}(c_{k+l}, s)$.

To compute the user credit mitigation portfolio MtM after applying all relevant haircuts on call date c_{k+l} under a given scenario s, $CMtMHC^{BB}_{user}(c_{k+l}, s)$:

- If the user credit mitigation portfolio (CMP_{user}) has been created at t_0, and $\text{CMtMHC}_{\text{user}}$ has been negative from t_0 up to and including t_{i+1} across all scenarios, the end points for Brownian Bridge computation are $CMtMHC^{BB}_{user}(c_{k+l-1}, s)$ and $CMtMHC_{user}(t_{i+1}, s)$, except that when computing $CMtMHC^{BB}_{user}(c_k, s)$, the left end point is $CMtMHC_{user}(t_i, s)$.
- If CMP_{user} has not been created at t_0 OR CMP_{user} has been created at t_0, but $\text{CMtMHC}_{\text{user}}$ has become zero or negative on or before t_{i+1} under any scenario, there is no computation for $CMtMHC^{BB}_{user}(c_{k+l}, s)$

■ Estimating the Variance Covariance (VCV) Matrix

Note

A user can access the estimated VCV matrix for each time step using detail function VCV Matrix.

The following model parameters are used in the VCV matrix estimation:

- **Intra Period** indicates if the volatility for the Brownian Bridge is estimated using the value from t_{i+1} or both the value from t_{i+1} and the value from t_i;
- **IQR Variance Estimator** indicates if the volatility is estimated using a generalized interquartile range; and
- **IQR Alpha** defines the quantile to be used.

On a given non-simulated call date, maximum three MtM values need to be computed [*1]:

1. Trade portfolio MtM

2. $\mathrm{CMtMHC_{cpty}}$

3. $\mathrm{CMtMHC_{user}}$

In this case, a 3x3 VCV matrix needs to be estimated for Brownian Bridege computation.

If only trade portfolio MtM and $\mathrm{CMtMHC_{cpty}}$ or trade portfolio MtM and $\mathrm{CMtMHC_{user}}$ need to be computed, the VCV matrix reduces to the 2x2 dimension.

If only trade portfolio MtM needs to be computed, the VCV matrix estimation reduces to the variance estimation for trade portfolio MtM.

Below we specify the estimation of 3x3 VCV matrix used to compute MtM values on non-simulated call dates $c_k, c_{k+1}, \ldots, c_{k+n}$ that are between adjacent simulation dates t_i and t_{i+1}. The lower dimension VCV matrix can be estimated by removing variable(s) in the below computation.

If `Intra Period` = `False`,

- $x_{i,s}$ is defined as $MtM(t_{i+1}, s)$;
- $y_{i,s}$ is defined as $CMtMHC_{cpty}(t_{i+1}, s)$; and
- $z_{i,s}$ is defined as $CMtMHC_{user}(t_{i+1}, s)$.

If `Intra Period` = `True`,

- $x_{i,s}$ is defined as $MtM(t_{i+1}, s) - \left(MtM(t_i, s) - TFAdj_p(t_i, s)\right)$;
- $y_{i,s}$ is defined as $CMtMHC_{cpty}(t_{i+1}, s) - CMtMHC_{cpty}(t_i, s)$; and
- $z_{i,s}$ is defined as $CMtMHC_{user}(t_{i+1}, s) - CMtMHC_{user}(t_i, s)$.

Case 1 – IQR Variance Estimator = False

$$V_p(t_i) = \frac{1}{\mathrm{N-1}} \sum_{j=1}^{N} \left(x_{i,s_j} - \overline{x_i}\right)^2, \quad \text{where} \quad \overline{x_i} = \frac{1}{N} \sum_{j=1}^{N} x_{i,s_j}$$

$$V_{cmp\_cpty}(t_i) = \frac{1}{\mathrm{N-1}} \sum_{j=1}^{N} \left(y_{i,s_j} - \overline{y_i}\right)^2, \quad \text{where} \quad \overline{y_i} = \frac{1}{N} \sum_{j=1}^{N} y_{i,s_j}$$

$$V_{cmp\_user}(t_i) = \frac{1}{\mathrm{N-1}} \sum_{j=1}^{N} \left(z_{i,s_j} - \overline{z_i}\right)^2, \quad \text{where} \quad \overline{z_i} = \frac{1}{N} \sum_{j=1}^{N} z_{i,s_j}$$

$$COV_{p,cmp\_cpty}(t_i) = \frac{1}{N-1} \sum_{j=1}^{N} (x_{i,s_j} - \overline{x}_i)(y_{i,s_j} - \overline{y}_i)$$

$$COV_{p,cmp\_user}(t_i) = \frac{1}{N-1} \sum_{j=1}^{N} (x_{i,s_j} - \overline{x}_i)(z_{i,s_j} - \overline{z}_i)$$

$$COV_{cmp\_cpty,cmp\_user}(t_i) = \frac{1}{N-1} \sum_{j=1}^{N} (y_{i,s_j} - \overline{y}_i)(z_{i,s_j} - \overline{z}_i)$$

Case 2 - IQR Variance Estimator = True

Let α denote the model parameter `IQR Alpha`, and define:

$k_1 := \lceil \alpha \times N \rceil$, that is the smallest integer that is not less than $\alpha \times N$;

$k_2 := \lceil (1-\alpha) \times N \rceil$, that is the smallest integer that is not less than $(1-\alpha) \times N$;

$Q(\alpha)$ as the k_1th smallest element of $\left\{x_{i,s_j}: 1 \le j \le N\right\}$ or $\left\{y_{i,s_j}: 1 \le j \le N\right\}$ or $\left\{z_{i,s_j}: 1 \le j \le N\right\}$;

Q(1-α) as the k_2th smallest element of $\left\{x_{i,s_j}:1 \le j \le N\right\}$ or $\left\{y_{i,s_j}:1 \le j \le N\right\}$ or $\left\{z_{i,s_j}:1 \le j \le N\right\}$; and

N^{-1}(α) as the inverse of the normal cumulative distribution function.

Then the generalized interquartile range is defined by:

$$\mathrm{IQR}(\alpha) = \left(\frac{Q(\alpha) - Q(1-\alpha)}{2 \times N^{-1}(\alpha)}\right)$$

The trade portfolio variance is estimated by $V_p = \mathrm{IQR}(\alpha)^2$.

The counterparty credit mitigation portfolio variance is estimated by $V_{cmp\_cpty} = \mathrm{IQR}(\alpha)^2$.

The user credit mitigation portfolio variance is estimated by $V_{cmp\_user} = \mathrm{IQR}(\alpha)^2$.

The correlation between the portfolio and counterparty credit mitigation portfolio is estimated by $Corr_{p,\ cmp\_cpty} = \sin(\frac{\pi}{2} \times \tau)$.

The correlation between the portfolio and user credit mitigation portfolio is estimated by $Corr_{p,\ cmp\_user} = \sin(\frac{\pi}{2} \times \tau)$.

The correlation between the counterparty and user credit mitigation portfolio is estimated by $Corr_{cmp\_cpty,\ cmp\_user} = \sin(\frac{\pi}{2} \times \tau)$.

Here, τ denotes Kendall's Tau and is calculated as:

$$\tau = \frac{N_c - N_d}{N(N-1)/2},$$

where N_c and N_d denote the number of concordant and discordant pairs respectively; that is:

N_c is the number of pairs in $\{(j,k): j \ne k, 1 \le j,k \le N\}$ such that:

($x_{i,s_j} < x_{i,s_k}$ and $y_{i,s_j} < y_{i,s_k}$) or ($x_{i,s_j} > x_{i,s_k}$ and $y_{i,s_j} > y_{i,s_k}$) for $Corr_{p,cmp\_cpty}$ computation;

($x_{i,s_j} < x_{i,s_k}$ and $z_{i,s_j} < z_{i,s_k}$) or ($x_{i,s_j} > x_{i,s_k}$ and $z_{i,s_j} > z_{i,s_k}$) for $Corr_{p,cmp\_user}$ computation;

($y_{i,s_j} < y_{i,s_k}$ and $z_{i,s_j} < z_{i,s_k}$) or ($y_{i,s_j} > y_{i,s_k}$ and $z_{i,s_j} > z_{i,s_k}$) for $Corr_{cmp\_cpty,cmp\_user}$ computation.

N_d is the number of pairs in $\{(j,k): j \ne k, 1 \le j,k \le N\}$ such that:

($x_{i,s_j} < x_{i,s_k}$ and $y_{i,s_j} > y_{i,s_k}$) or ($x_{i,s_j} > x_{i,s_k}$ and $y_{i,s_j} < y_{i,s_k}$) for $Corr_{p,cmp\_cpty}$ computation;

($x_{i,s_j} < x_{i,s_k}$ and $z_{i,s_j} > z_{i,s_k}$) or ($x_{i,s_j} > x_{i,s_k}$ and $z_{i,s_j} < z_{i,s_k}$) for $Corr_{p,cmp\_user}$ computation;

($y_{i,s_j} < y_{i,s_k}$ and $z_{i,s_j} > z_{i,s_k}$) or ($y_{i,s_j} > y_{i,s_k}$ and $z_{i,s_j} < z_{i,s_k}$) for $Corr_{cmp\_cpty,cmp\_user}$ computation.

The covariance is then calculated as:

$$COV_{p,cmp\_cpty} = Corr_{p,cmp\_cpty} \times \sqrt{V_p} \times \sqrt{V_{cmp\_cpty}}$$

$$COV_{p,cmp\_user} = Corr_{p,cmp\_user} \times \sqrt{V_p} \times \sqrt{V_{cmp\_user}}$$

$$COV_{cmp\_cpty,cmp\_user} = Corr_{cmp\_cpty,cmp\_user} \times \sqrt{V_{cmp\_cpty}} \times \sqrt{V_{cmp\_user}}$$

■ Creating the Random Normal Vector

Denote the 3×3 VCV matrix computed in the last step as Σ . The random normal vector $\zeta \sim N(0, \Sigma)$ used in Brownian Bridge computation is created as:

$\zeta = zA^T$

where z is a 1×3 vector of independent random numbers following the standard normal distribution [*2]; A[*3] is the linear transformation matrix obtained via a stable version of eigenvalue decomposition of VCV matrix such that $\Sigma = AA^T$.

There is a separate random draw for each MtM^{BB}, $CMtMHC^{BB}_{cpty}$ and $CMtMHC^{BB}_{user}$ computation.

■ Computing the Brownian Bridge

1. Trade portfolio MtM

- `CPVMPostStyle` = Net and/or `UserVMPostStyle` = Net

The Brownian Bridge of the trade portfolio MtM at the call date c_{k+l} under scenario s is defined as follows.

If l=0,

```
If UseSettlementFlows = TRUE
```

$$MtM^{BB}(c_k, s) = MtM(t_i, s) - TFAdj_p(t_i, s) + \frac{MtM(t_{i+1}, s) - MtM(t_i, s) + TFAdj_p(t_i, s)}{t_{i+1} - t_i}(c_k - t_i) + \sqrt{\frac{(c_k - t_i)(t_{i+1} - c_k)}{(t_{i+1} - t_i)\Delta t}}\zeta_{k,\ s}$$

```
Else if Isolate Collateral Spikes = TRUE
```

$$\mathrm{MtM}^{BB}(c_k, s) = \mathrm{MtM}(t_i, s) - \mathrm{CF}([t_i, t_{i+1}), s) + \frac{\mathrm{MtM}(t_{i+1}, s) - \mathrm{MtM}(t_i, s) + \mathrm{CF}([t_i, t_{i+1}), s)}{t_{i+1} - t_i} \times (c_k - t_i) + \sqrt{\frac{(c_k - t_i)(t_{i+1} - c_k)}{(t_{i+1} - t_i)\Delta t}}\zeta_{k,s}$$

```
Else
```

$$\mathrm{MtM}^{BB}(c_k, s) = \mathrm{MtM}(t_i, s) + \frac{\mathrm{MtM}(t_{i+1}, s) - \mathrm{MtM}(t_i, s)}{t_{i+1} - t_i} \times (c_k - t_i) + \sqrt{\frac{(c_k - t_i)(t_{i+1} - c_k)}{(t_{i+1} - t_i)\Delta t}}\zeta_{k,s}$$

If $0 < l \leq n$,

$$MtM^{BB}(c_{k+l}, s) = MtM^{BB}(c_{k+l-1}, s) + \frac{MtM(t_{i+1}, s) - MtM^{BB}(c_{k+l-1}, s)}{t_{i+1} - c_{k+l-1}}(c_{k+l} - c_{k+l-1}) + \sqrt{\frac{(c_{k+l} - c_{k+l-1})(t_{i+1} - c_{k+l})}{(t_{i+1} - c_{k+l-1})\Delta t}}\zeta_{k+l,\ s}$$

where $\Delta t = \begin{cases} t_{i+1} - t_i \;\; if \;\; IntraPeriod = True \;\; (default) \\ t_{i+1} - t_0 \qquad if \;\; IntraPeriod = False \end{cases}$

and $\zeta_{k,s}$ and $\zeta_{k+l,s}$ are obtained from the previous step ■ Creating the Random Normal Vector on page 53.

2. Counterparty credit mitigation portfolio MtM after relevant haircuts

The Brownian Bridge of the counterparty credit mitigation portfolio after applying all relevant haircuts at the call date c_{k+l} under scenario s is defined as follows.

If l=0,

$$CMtMHC^{BB}_{cpty}(c_k, s) = CMtMHC_{cpty}(t_i, s) + \frac{CMtMHC_{cpty}(t_{i+1}, s) - CMtMHC_{cpty}(t_i, s)}{t_{i+1} - t_i}(c_k - t_i) + \sqrt{\frac{(c_k - t_i)(t_{i+1} - c_k)}{(t_{i+1} - t_i)\Delta t}}\zeta_{k,\ s}$$

If $0 < l \leq n$,

$$CMtMHC_{cpty}^{BB}(c_{k+l}, s) = CMtMHC_{cpty}^{BB}(c_{k+l-1}, s) + \frac{CMtMHC_{cpty}(t_{i+1}, s) - CMtMHC_{cpty}^{BB}(c_{k+l-1}, s)}{t_{i+1} - c_{k+l-1}}(c_{k+l} - c_{k+l-1}) + \sqrt{\frac{(c_{k+l} - c_{k+l-1})(t_{i+1} - c_{k+l})}{(t_{i+1} - c_{k+l-1})\Delta t}}\zeta_{k+l,\ s}$$

where $\zeta_{k,s}$ and $\zeta_{k+l,s}$ are obtained from the previous step ■ Creating the Random Normal Vector on page 53.

3. User credit mitigation portfolio MtM after relevant haircuts

The Brownian Bridge of the user credit mitigation portfolio after applying all relevant haircuts at the call date c_{k+l} under scenario s is defined as follows.

If $l=0$,

If $0 < l \leq n$,

$$CMtMHC_{user}^{BB}(c_{k+l}, s) = CMtMHC_{user}^{BB}(c_{k+l-1}, s) + \frac{CMtMHC_{user}(t_{i+1}, s) - CMtMHC_{user}^{BB}(c_{k+l-1}, s)}{t_{i+1} - c_{k+l-1}}(c_{k+l} - c_{k+l-1}) + \sqrt{\frac{(c_{k+l} - c_{k+l-1})(t_{i+1} - c_{k+l})}{(t_{i+1} - c_{k+l-1})\Delta t}}\zeta_{k+l,\ s}$$

where $\zeta_{k,s}$ and $\zeta_{k+l,s}$ are obtained from the previous step ■ Creating the Random Normal Vector on page 53.

■ Computing MtM on non-Simulated Call Dates

1. Trade Portfolio MtM

- `CPVMPostStyle` = `Net` and/or `UserVMPostStyle` = `Net`

```
If UseSettlementFlows = TRUE
    The trade portfolio MtM at the call date c_{k+l} under scenario s is defined by:
```

$$MtM(c_{k+l}, s) = MtM^{BB}(c_{k+l}, s) + \sum_{c_{k+l} \leq d < t_{i+1}} TF_p(d, s)$$

where $TF_p(d, s)$ is computed by converting all the trade level settlement flows (including the ones that are marked as "cash" and the ones that are marked as "security") from settlement date d under scenario s into the agreement currency and taking the sum.

```
Else if Isolate Collateral Spikes = TRUE
    The trade portfolio MtM at the call date c_{k+l} under scenario s is defined by:
```

$$\mathrm{MtM}(c_{k+l}, s) = \mathrm{MtM}^{BB}(c_{k+l}, s) + \mathrm{CF}([c_{k+l}, t_{i+1}), s)$$

where $\mathrm{CF}([c_{k+l}, t_{i+1}), s)$ is the net cumulative cashflows between c_{k+l} (inclusive) and t_{i+1} (exclusive) and is defined as:

$$\mathrm{CF}([c_{k+l}, t_{i+1}), s) = \begin{cases} \frac{(t_{i+1} - c_{k+l})}{(t_{i+1} - t_i)} \times (\mathrm{MtM}(t_i, s) - \mathrm{MtM}_{\mathrm{nmf}}(t_i, s)), & \text{if Gradual Cash Flows=TRUE} \\ \mathrm{MtM}(t_i, s) - \mathrm{MtM}_{\mathrm{nmf}}(t_i, s), & \text{if Gradual Cash Flows=FALSE} \end{cases}$$

```
Else
```

$$\mathrm{MtM}(c_{k+l}, s) = \mathrm{MtM}^{BB}(c_{k+l}, s)$$

2. Counterparty credit mitigation portfolio MtM after relevant haircuts

4

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The counterparty credit mitigation portfolio MtM after relevant haircuts at the call date c_{k+l} under scenario *s* is defined by:

$$CMtMHC_{cpty}(c_{k+l}, s) = \max\left[CMtMHC^{BB}_{cpty}(c_{k+l}, s), 0\right]$$

3. User credit mitigation portfolio MtM after relevant haircuts

The user credit mitigation portfolio MtM after relevant haircuts at the call date c_{k+l} under scenario *s* is defined by:

$$CMtMHC_{user}(c_{k+l}, s) = \min\left[CMtMHC^{BB}_{user}(c_{k+l}, s), 0\right]$$

*1 This only happens when CPVMPostStyle = Net and UserVMPostStyle = Net.

*2 User can access the random draw used in the MtM^{BB}, $CMtMHC^{BB}_{cpty}$ and $CMtMHC^{BB}_{user}$ computation for each call date c_{k+l} and scenario s using detail function Random Draws, setting the function parameter to Trades, CPCMP or UserCMP respectively.

*3 User can access this transformation matrix for each time step using detail function Transformation Matrix.

4.1.2.6 Trade Settlement Flows

- **Trade Settlement Flows in the Settlement Adjusted Path Dependent Model**

Note

The description in this document is only relevant when the model parameter `UseSettlementFlows` is set to TRUE.

Trade settlement flows include the intermediate trade settlement flows and trade settlement flows upon maturity. In the settlement adjusted path dependent model, the trade settlement flows are used in the computation of

- Settlement spike
- Exp_{full} and Exp_{smooth}

Trade settlement flows are generated by RiskWatch (RW). Settlement flows can be in cash or in security (ssettlement flows in security refers to any type of non-cash settlement flows from physical settlement of trades). On a given day, if there are multiple settlement flows in cash from a trade, the cash flows are aggregated on the trade level in RW according to the settlement netting rule specified by user. No settlement netting is applied to settlement flows in security. RW computes the cash equivalent amount for each settlement flow in security. Structured deals don't have the trade level netting implemented in the aggregator. It is assumed that no settlement netting is applied to structured deals.

Settlement flows are stored in a cash flow cube. In the cash flow cube, all the trade settlement flows are stored in the form of cash, with each trade settlement flow marked as "cash" or "security" according to its original settlement flow type and associated with a date.

- **Trade Settlement Flows for Settlement Spike Computation**

Two `standardTradeAttributes` - `DVP` and `bilaterallySettledFirstLast` - determine which trade settlement flows are included in the settlement spike computation.

Trades whose flows are settled within systems that ensure simultaneous settlement of trades' flows on a given date should have the `DVP` attribute set to TRUE. For some of these trades, such as cross-currency swaps, the notional exchanges are excluded from this beneficial settlement, and are instead settled bilaterally. For these trades the attribute `bilaterallySettledFirstLast` should be set to TRUE as well.

On the CSA level, the following trade settlement flows paid by the non-defaulting party during the *Lag_TS* are relevant to the settlement spike computation at time r_i:

- All the settlement flows of trades with attribute `DVP` set to FALSE; and
- The start date flows which occurs after t_0 and the final flows of trades with both `DVP` and `bilaterallySettledFirstLast` set to TRUE.

For the definition of *Lag_TS*, see 4.1.2.9 Model Inputs and Notation on page 61.

Note

When `DVP` = TRUE and `bilaterallySettledFirstLast` = TRUE, we don't just include the notionals in the settlement spike, but all flows occurring on the start date and final flow date.

Let TF_{Lag_TS} be the settlement spikes paid by the non-defaulting party during the *Lag_TS*. MAG provides two approaches to compute TF_{Lag_TS}[*2], depending on the setting of collateral model parameter `Net Intraday Flows`. When `Net Intraday Flows` = `FALSE`, MAG applies settlement netting to the trade level settlement flows according to the CSA currency list specified by the user. There is one CSA currency list per CSA. When `Net Intraday Flows` = `TRUE`, all flows on a given date net regardless of their currencies.

1. `Net Intraday Flows` = `FALSE`

Settlement netting is only applied to trade settlement flows from the cash flow cube that are:

- marked as "cash" AND
- in settlement currencies that are on the CSA currency list [*1].

*1 To manage performance, users are recommended to only put high volume currencies in their portfolio into the currency list.

No settlement netting is applied to any trade settlement flows from the cash flow cube that are:

- marked as "security" OR
- marked as"cash" but in settlement currencies that are not on the CSA currency list.

Let,

- $CF_{k,\ m}(d, s)$ be the trade settlement flow marked as "cash" for trade *k* in settlement currency *m* at settlement date *d* under scenario *s* from the cash flow cube
- $SF_{l,\ m}(d, s)$ be the trade settlement flow marked as "security" for trade *l* in settlement currency *m* at settlement date *d* under scenario *s* from the cash flow cube
- *H* be the set of currencies on the CSA currency list
- *M* be the set of currencies with trade settlement flows at settlement date *d* under scenario *s*
- *K*|*m* be the set of trades in the CSA with trade settlement flow marked as "cash" in currency *m* at settlement date *d* under scenario *s*
- *L*|*m* be the set of trades in the CSA with trade settlement flow marked as "security" in currency *m* at settlement date *d* under scenario *s*.

On each settlement date *d* with $r_i - MPOR \le d < r_i - MPOR + Lag\_TS$, and under a given scenario *s*, compute the settlement netted cash flow in currency *m* , $SettlementNetCF_m(d, s)$ [*1], and the settlement netted cash flow in base currency *b*, $SettlementNetCF_b(d, s)$, as:

$SettlementNetCF_m(d,s) = \sum_{k \in K|m} CF_{k,\ m}(d,s)$ for each $m \in H$

$$SettlementNetCF_b(d,s) = \begin{cases} \sum_{m \in H} \min\left(SettlementNetCF^b_m(d,s), 0\right) \ \ for \ \ forward \ \ view \ \ computation \\ \sum_{m \in H} \max\left(SettlementNetCF^b_m(d,s), 0\right) \ \ for \ \ reverse \ \ view \ \ computation \end{cases}$$

where $SettlementNetCF^b_m(d,s)$ is $SettlementNetCF_m(d,s)$ in base currency *b* converted using the FX rate at r_i under scenario *s*.

The trade settlement flows that are not eligible for settlement netting are used to compute the total no-net trade settlement flow in base currency *b*, $NoNetTF_b(d,s)$:

$$NoNetTF_b(d,s) = \begin{cases} \sum_{m \notin H} \sum_{k \in K|m} \min\left(CF^b_{k,m}(d,s), 0\right) + \sum_{m \in M} \sum_{l \in L|m} \min\left(SF^b_{l,m}(d,s), 0\right) \ \ for \ \ forward \ \ view \ \ computation \\ \sum_{m \notin H} \sum_{k \in K|m} \max\left(CF^b_{k,m}(d,s), 0\right) + \sum_{m \in M} \sum_{l \in L|m} \max\left(SF^b_{l,m}(d,s), 0\right) \ \ for \ \ reverse \ \ view \ \ computation \end{cases}$$

where,

$CF^b_{k,\ m}(d,s)$ is $CF_{k,\ m}(d,s)$ in base currency *b* converted using the FX rate at r_i under scenario *s*, and

$SF^b_{l,\ m}(d,s)$ is $SF_{l,\ m}(d,s)$ in base currency *b* converted using the FX rate at r_i under scenario *s*.

The total trade settlement flow paid by the non-defaulting party within the *Lag_TS* for the settlement spike computation at time r_i under scenario *s*, $TF_{Lag\_TS}(r_i,s)$, is then computed as:

$$TF_{Lag\_TS}(r_i,s) = \sum_{r_i - MPOR \le d < r_i - MPOR + Lag\_TS} \left(SettlementNetCF_b(d,s) + NoNetTF_b(d,s)\right)$$

2. `Net Intraday Flows` = `TRUE`

Let $F_b(d,\ s)$ be the sum of all settlement flows on date *d* and scenario *s* that are converted to the agreement currency *b* using FX rate on *d*. (If *d* is not a simulation date, use the FX rate on the simulation date closest to *d*. If the closest simulation date is not unique, use the one on the first simulation date to the right of *d*.)

For a given reporting date r_i and scenario *s*, the total settlement flows due by the non-defaulting party within the *Lag_TS* is computed as:

$$TF_{Lag\_TS} = \begin{cases} \sum_{r_i - MPOR \le d < r_i - MPOR + Lag\_TS} \min\left(F_b(d,s), 0\right), & \text{for forward view computation} \\ \sum_{r_i - MPOR \le d < r_i - MPOR + Lag\_TS} \max\left(F_b(d,s), 0\right), & \text{for reverse view computation} \end{cases}$$

- ### Trade Settlement Flows for Exp_{smooth} and Exp_{full} Computation

On the CSA level, the relevant trade settlement flows for the Exp_{full} and Exp_{smooth} computation at time r_i are all the trade settlement flows happened within the margin period of risk (MPOR). It is assumed that all the trade settlement flows within the MPOR can net with each other.

For the exposure computation at r_i under given scenario *s*, first compute the net trade settlement flow within the MPOR in each settlement currency *m*, $NetTF_m(s)$, as

$$NetTF_m(r_i,s) = \sum_{r_i - MPOR \le d < r_i} \left(\sum_{k \in K|m} CF_{k,m}(d,s) + \sum_{l \in L|m} SF_{l,m}(d,s)\right)$$

The net trade settlement flow within the MPOR in base currency, $NetTF_{MPOR}(r_i,s)$ *3 is then computed as

$$NetTF_{MPOR}(r_i,s) = \sum_{m \in M} NetTF^b_m(r_i,s)$$

where $NetTF^b_m(r_i,s)$ is $NetTF_m(r_i,s)$ in base currency *b* converted using the FX rate at r_i under scenario *s*.

*1 User can access the amount of $SettlementNetCF_m(d, s)$ for each settlement currency m on the CSA currency list using detail function Currency Netted Trade Flows.

*2 User can access the amount of $TF_{Lag\_TS}(r_i, s)$ using detail function MPOR Settled Payments. The function option should be set to Settlement.

*3 User can access the amount of $NetTF_{MPOR}(r_i, s)$ using detail function MPOR Trade Flows.

4.1.2.7 Cumulative Cashflows

- **Cumulative cashflows in the Settlement Adjusted Path Dependent Model**

The calculation in this section is only relevant when model parameter `UseSettlementFlows` = FALSE and `Isolate Collateral Spikes` = TRUE.

- **Mark-to-Market Planes**

MtM(t, s) is defined as the net mark-to-market of all trades in the portfolio.

$MtM_{nmf}(t, s)$ is defined as the net mark-to-market of all trades in the portfolio that have not matured before the following simulation date.

- **Net Cumulative Cashflows between Two Dates**

Let t_i, t_{i+1}, ... , t_{i+j} be consecutive simulation time steps and c be a date such that $t_i \le c < t_{i+j}$.

Define the net cumulative cashflows between c (inclusive) and t_{i+j} (exclusive), $\mathrm{NetCF}([c, t_{i+j}), s)))$ *1, as follows:

Case 1: `Gradual Cash Flows` = TRUE

$$\mathrm{NetCF}([c, t_{i+j}), s) = \frac{(t_{i+1} - c)}{(t_{i+1} - t_i)} \times (\mathrm{MtM}(t_i, s) - \mathrm{MtM}_{\mathrm{nmf}}(t_i, s)) + \sum_{l=1}^{j-1} (\mathrm{MtM}(t_{i+l}, s) - \mathrm{MtM}_{\mathrm{nmf}}(t_{i+l}, s))$$

Case 2: `Gradual Cash Flows` = FALSE

$$\mathrm{NetCF}([c, t_{i+j}), s) = \sum_{l=0}^{j-1} (\mathrm{MtM}(t_{i+l}, s) - \mathrm{MtM}_{\mathrm{nmf}}(t_{i+l}, s))$$

In both cases, $\mathrm{NetCF}([t_i, t_i), s) := 0$.

- **Non-Defaulting Party Cumulative Cashflows between Two Dates**

Define the non-defaulting party cumulative cashflows between c (inclusive) and t_{i+j} (exclusive), $\mathrm{CF}_{\mathrm{nd}}([c, t_{i+j}), s)$ *2, as follows:

Case 1: `Gradual Cash Flows` = TRUE

$\mathrm{CF}_{nd}([c, t_{i+j}), s)$

=

$$\begin{cases} \frac{(t_{i+1}-c)}{(t_{i+1}-t_i)} \times (\mathrm{MtM}(t_i,s) - \mathrm{MtM}_{\mathrm{nmf}}(t_i,s))^- + \sum_{l=1}^{j-1}(\mathrm{MtM}(t_{i+l},s) - \mathrm{MtM}_{\mathrm{nmf}}(t_{i+l},s))^-, & \text{for forward view computation} \\ \frac{(t_{i+1}-c)}{(t_{i+1}-t_i)} \times (\mathrm{MtM}(t_i,s) - \mathrm{MtM}_{\mathrm{nmf}}(t_i,s))^+ + \sum_{l=1}^{j-1}(\mathrm{MtM}(t_{i+l},s) - \mathrm{MtM}_{\mathrm{nmf}}(t_{i+l},s))^+, & \text{for reverse view computation} \end{cases}$$

Case 2: `Gradual Cash Flows` = FALSE

$$\mathrm{CF}_{nd}([c,t_{i+j}),s) = \begin{cases} \sum_{l=0}^{j-1}(\mathrm{MtM}(t_{i+l},s) - \mathrm{MtM}_{\mathrm{nmf}}(t_{i+l},s))^-, & \text{for forward view computation} \\ \sum_{l=0}^{j-1}(\mathrm{MtM}(t_{i+l},s) - \mathrm{MtM}_{\mathrm{nmf}}(t_{i+l},s))^+, & \text{for reverse view computation} \end{cases}$$

In both cases, $\mathrm{CF}_{\mathrm{nd}}([t_i,t_i),s) := 0$.

Here, $x^- = \min(x,0)$; $x^+ = \max(x,0)$.

*1 User can access the amount of $\mathrm{NetCF}([c,t_{i+j}),s)))$ using detail function MPOR Trade Flows.

*2 User can access the amount of $\mathrm{CF}_{\mathrm{nd}}([c,t_{i+j}),s)$ using detail function MPOR Settled Payments. The function option should be set to Settlement.

4.1.2.8 Cashflow on Break Clause Date

- **Introduction**

The calculation in this section is only relevant when a trade's `nextEarlyTerminationDate` attribute in `standardTradeAttributes` section is set and the `EarlyExit` attribute of the relevant credit expression is set to TRUE.

Note

In case of a structured deal, the `nextEarlyTerminationDate` at the structured deal level is used in the adjustment described below.

It is assumed that when a break clause is exercised, a trade settles on a net basis.

- **Break Clause Date Cashflow**

For a trade *x* with break clause date d_{bc}, let t_i be the simulation date closest to d_{bc}.

Note

If the two simulation dates closest to the break clause date are equally far from it, use the simulation date greater than the break clause date.

Under a scenario *s*, approximate the trade's cashflow on d_{bc} by:

$$F(x,d_{\mathrm{bc}},s) = \begin{cases} \mathrm{MtM}(t_i,s) + \sum_{d_{\mathrm{bc}} \le d < t_i} F(d,s), & \text{if } t_i \ge d_{\mathrm{bc}} \\ \mathrm{MtM}(t_i,s) - \sum_{t_i \le d < d_{\mathrm{bc}}} F(d,s), & \text{if } t_i < d_{\mathrm{bc}} \end{cases}$$

where, $MtM(t, s)$ is the mark-to-market of the trade on t_i and under scenario *s*;

$\sum F(d,s)$ is the sum of all cashflows of the trade on date *d* converted to the currency of the trade's MtM plane (THEO/Instrument Value in the cube) using the FX rate on t_i.

Note

In the above formula, the cashflows from the counterparty to the user are assumed to be positive and the cashflows from the user to the counterparty are assumed to be negative.

- **Brownian Bridge and Settlement Flow with Early Exit**

When applying the condition Early Exit = TRUE to the Settlement Adjusted Path Dependent Model, the following adjustment to trade portfolio MtM and settlement flows is required to the trade with break clauses.

- Set the trade's MtM to 0 for all simulation dates after d_{bc};
- Ignore all original cashflows of the trade on or after d_{bc} and approximate the cashflow on d_{bc} by $F(x, d_{bc}, s)$ as defined above.

4.1.2.9 Model Inputs and Notation

RTCE Inputs

The source files for these inputs are described in the *Integrated Market and Credit Risk Solution Data Dictionary* and are located in `$ALGO_TOP/inputs/userdata/rtce/update/counterparties` and `$ALGO_TOP/inputs/userdata/rtce/replace/counterparties`.

Table 4.1-4 RTCE Inputs

Input	Notation	Source File	Usage in Model/Comments
AgreementCurrency	CSAcy	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Specifies the currency of the CSA parameters (thresholds, minimum transfer amounts, etc.).
CallPeriod	*p*	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Frequency of margin calls
MarginLag	Lag_VM	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Given a reporting date r_i, it is assumed that only the non-defaulting party makes VM payments between $r_i - MPOR$ (exclusive) and $r_i - MPOR + Lag\_VM$ (inclusive). The margin period of risk (MPOR) is represented by max(Lag_VM, Lag_TS) + CurePeriod.
SettlementLag	Lag_TS	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Given a reporting date r_i, it is assumed that only the non-defaulting party makes trade settlement flow payments between $r_i - MPOR$ (inclusive) and $r_i - MPOR + Lag\_TS$ (exclusive).
CurePeriod	CurePeriod	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	See comment for MarginLag
SettlementNetting Ccys	H	CollateralAgreementVM.csv /	When computing the total trade settlement flow paid by the non-defaulting party during

		CollateralAgreementIMVM.csv	the Lag_TS, MAG applies settlement netting to the trade level settlement flows according to the SettlementNettingCcys.
CPIndependent	CPIndependent	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv / CollateralAgreementLegacy.csv	The treatment of this amount in the model is as defined in the CSA agreement http://assets.isda.org/media/e0f39375/b61bd039.pdf
UserIndependent	UserIndependent	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv / CollateralAgreementLegacy.csv	The treatment of this amount in the model is as defined in the CSA agreement http://assets.isda.org/media/e0f39375/b61bd039.pdf.
CPVMThreshold	cVMT	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	When either the `Minimum Transfer Risk Under-Collateralization` or `Minimum Transfer Risk Over-Collateralization` model parameter is set to true, this value is replaced. See Table 4.1-6 Thresholds, Minimum Transfer, and Rounding Inputs on page 66.
UserVMThreshold	uVMT	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	See comment for *CPVMThreshold.*
CPVMMinTransfer	cVMMTA	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	See comment for *CPVMThreshold.*
UserVMMinTransfer	uVMMTA	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	See comment for *CPVMThreshold.*
RoundoffAmount	RA	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	See comment for *CPVMThreshold.*
CPVMPostStyle	CPVMPostStyle	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Specifies counterparty's variation margin post style. This attribute can be set as: Net (default value), Gross, None.
UserVMPostStyle	UserVMPostStyle	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Specifies user's variation margin post style. This attribute can be set as: Net (default value), Gross, None.
CPVMSegregated		CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	If set to true, the variation margin balance posted by counterparty is assumed to be segregated; that is, any variation margin balance which is not used to offset exposure to user is protected in case of default.
UserVMSegregated		CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	If set to true, the variation margin balance posted by user is assumed to be segregated; that is, any variation margin balance which is not used to offset exposure to counterparty is protected in case of default.

CPIMSegregated		CollateralAgreementIM.csv / CollateralAgreementIMVM.csv	If set to true, the initial margin balance posted by counterparty is assumed to be segregated; that is, any initial margin balance which is not used to offset exposure to user is protected in case of default. This attribute only applies to IM sheet or initial margin generated by Dynamic Initial Margin model.
UserIMSegregated		CollateralAgreementIM.csv / CollateralAgreementIMVM.csv	If set to true, the initial margin balance posted by user is assumed to be segregated; that is, any initial margin balance which is not used to offset exposure to counterparty is protected in case of default. This attribute only applies to IM sheet or initial margin generated by Dynamic Initial Margin model.
CPVMCap	cVMCap	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Specifies the maximum amount of variation margin posted by counterparty.
UserVMCap	uVMCap	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Specifies the maximum amount of variation margin posted by user.
SettledToMarketAgreement		CollateralAgreementIMVM.csv	If set to true, trades in the CSA are transactions for which the mark-to-market is settled daily. The default value is false.
marginType		standardMitigantAttributes	Specifies the type of margin. The value can be Initial or Variation.
deliveryDate		standardMitigantAttributes	Specifies when the collateral will be delivered. For Variation Margin, only those mitigants whose delivery dates fall before or on the session date are used. For Initial Margin, the profile starts at *deliveryDate* and ends at *returnDate*.
returnDate		standardMitigantAttributes	Specifies when the collateral will be returned. For Variation Margin, only those mitigants whose return dates fall after the session date are used. For Initial Margin, the profile starts at *deliveryDate* and ends at *returnDate*.
mitigantPostDirection		standardMitigantAttributes	Specifies who is receiving the collateral. The value can be To Counterparty or To User.
segregated		standardMitigantAttributes	This attribute only applies to initial margin type. If set to true, the initial margin posted is assumed to be segregated.
MTM			Specifies the collateral value posted/received in the currency defined by *MTMCurrency*. This is used to compute the current amount of Variation Margin and Initial Margin.
MTMCurrency			See comment for *MTM*.
mitigantQuantity		standardMitigantAttributes	When *MTM* and *MTMCurrency* are not specified, this attribute, together with *mitigantCurrency*, are used to compute the collateral value posted/received.
mitigantCurrency		standardMitigantAttributes	See comment for *mitigantQuantity*.

applicability		standardMitigantAttributes	For mitigant with marginType = Variation, this attribute allows the user to define an appropriate security pool. When set to INITIAL_BALANCE (default value), the variant mitigant is used to calculate the total variation margin balance as of t_0. When set to MITIGANT_POOL, the variant mitigant is used inside the credit mitigation portfolio. For mitigant with marginType = Initial, when set to INITIAL_BALANCE (default value), the initial mitigant is used to calculate the total initial margin balance as of t_0. When set to MITIGANT_POOL, if there is no initial mitigant with applicability = INITIAL_BALANCE and the dynamic initial margin model is applied, the initial mitigant is used as initial margin balance as of t_0 after being adjusted by the relevant threshold.
tradeMarginType		standardTradeMitigantAttributes	Specifies whether the trade mitigant is independent amount or initial margin. The value can be INITIAL or INDEPENDENT.
variableNotional	cpIAInput(x,t) / uIAInput(x,t) / cpIMInput(x,t) / uIMInput(x.t)	standardTradeMitigantAttributes	Depending on the tradeMarginType, this attribute specifies the input value of the counterparty and user independent amount profile on trade *x* at time *t*; or the counterparty and user initial margins on trade *x* at time *t*. It contains pairs of PaymentAmount and PaymentDate in the ascending order of the payment date. The value of each PaymentAmount is applied from the associated payment date (inclusive) to the next payment date (exclusive). The value of the PaymentAmount associated with the last payment date is applied to the associated date (inclusive) to the end of the simulation which depends on the returnStyle.
notionalCurrency	ccy	standardTradeMitigantAttributes	Specifies the currency of the variableNotional.
returnStyle		standardTradeMitigantAttributes	This attribute only applies to initial margin. It specifies when the margin is returned. The value can be MATURITY or EXTENDED_MATURITY.
postDirection		standardTradeMitigantAttributes	Specifies who is receiving the margin. The value can be To Counterparty or To User.
segregated		standardTradeMitigantAttributes	This attribute only applies to initial margin. If set to true, the initial margin posted is assumed to be segregated.
DVP		standardTradeAttributes	If set to true, the trade's flows are excluded from the computation of settlement spikes.
bilaterallySettledFirstLast		standardTradeAttributes	This attribute is used together with attribute DVP. When both attributes are set to true, the trade's start date flows which occurs after t_0 and final flows are included in the computation of settlement spikes.

AgreementEnforceable	AgreementEnforceable	Counterparty.csv	If set to false, netting of positions is not allowed when computing the exposure of the user to the counterparty, but collateral is computed based on a netted plane.
CloseoutNetting	CloseoutNetting	MasterAgreement.csv	If set to false, netting of positions is not allowed but collateral is computed based on a netted plane.

Model Parameters

The values of these parameters are set in the `CollateralModels.xml` file in `$ALGO_TOP/static/mag/` instances. They can also be defined through a GUI application (see `$MAG_HOME/bin/defineFuncitons`).

Table 4.1-5 Model Parameters

Settlement Flow Path Dependent Collateral Model		
Input	**Default value**	**Usage in Model/Comments**
Payment Model		This allows to pick up a pre-defined or create a new Settlement Flow Based VM Payments Collateral Sub-Model.
Initial Margin Model		This allows to pick up a pre-defined or create a new Settlement Flow IM Payments Collateral Sub-Model.
Exposure Type	Full	This can be set to Full or Smooth, depending on the assumption used in the exposure calculation whether or not the non-defaulting party makes margin or trade settlement flow payments after the beginning of the margin period of risk (MPOR).
User Collateral Offset	None	If set to Netted Pools, user posted unsegregated margin is allowed to offset the negative mark-to-market from trades belonging to closeout netting pools.
Net Intraday Flows	False	This allows to choose the approach to compute the total trade settlement flow paid by the non-defaulting party during *Lag_TS*.

Settlement Flow Based VM Payment Collateral Sub-Model		
Input	**Default value**	**Usage in Model/Comments**
Enable Variation Margin	Yes	When this is set to Yes, variation margin is applied to this collateral agreement.
Apply With Legal Opinion	None	When *NettingRule* = Legal_Opinion, this, together with *Enable Variation Margin* determines what kinds of variation margin posted by counterparty are used to offset exposures to user.
Additional Call Dates	1	This allows for additional call dates to be used in the model to capture more precisely the risk introduced by minimum transfer amounts. Although this parameter can be set to any integer greater than or equal to zero, a large value greatly impacts the performance of the algorithm, which means it should only be set to be greater than 1 when absolutely necessary (for example if the portfolio moves within call periods is small compared to the minimum transfer amounts). Setting this value to a large number should only be used for testing the model on relatively small portfolios and should never be used in production.
Intra Period	False	If this is set to True, indicates if the volatility used in the Brownian Bridge construction is estimated using both the left and right end points of the bridge.

Settlement Flow Based VM Payment Collateral Sub-Model		
IQR Variance Estimator	False	Indicates if the volatility used in the Brownian Bridge construction is estimated using a generalized interquartile range.
IQR Alpha	95%	Defines the quantile to be used in the generalized IQR when *IQR Variance Estimator* is set to true.
Minimum Transfer Risk Under-Collateralization	False	This is to allow for a conservative treatment of the risk introduced by the counterparty not posting variation margin due to minimum transfer amounts (and rounding); see Table 4.1-6 Thresholds, Minimum Transfer, and Rounding Inputs on page 66 to see how this parameter works in conjunction with thresholds, minimum transfer amounts and rounding inputs
Minimum Transfer Risk Over-Collateralization	False	This is to allow for a conservative treatment of the risk introduced by the counterparty not returning variation margin due to minimum transfer amounts (and rounding); see Table 4.1-6 Thresholds, Minimum Transfer, and Rounding Inputs on page 66 to see how this parameter works in conjunction with thresholds, minimum transfer amounts and rounding inputs.
Link Relevant Call Dates	False	When this is set to True, margin calls are computed between two non-consecutive call dates.
Use Physical Collateral	False	If this is set to True, credit mitigation portfolio is used to model collateral.
UseSettlementFlows	True	When this is set to False, no settlement cashflow is taken into consideration in the calculation of variation margin.
Isolate Collateral Spikes	False	When this is set to True, spikes in exposure caused by maturing trades are removed. This only applies when UseSettlementFlows is set to False.
Gradual Cash Flows	False	This determines how the mark-to-market of maturing trades is estimated on a non-simulated date and the cumulative cashflows on a reporting date when *Isolate Collateral Spikes* is set to True.

Settlement Flow IM Payments Collateral Sub-Models		
Input	**Default value**	**Usage in Model/Comments**
Enable Initial Margin	Yes	When this is set to Yes, initial margin is applied to this collateral agreement.
Apply With Legal Opinion	None	When *NettingRule* = Legal_Opinion, this, together with *Enable Initial Margin* determines what kinds of initial margin posted by counterparty are used to offset exposures to user.

The following table describes how the thresholds, minimum transfer amounts and rounding inputs from RTCE change based on the value of the parameters `Minimum Transfer Risk Under-Collateralization` and `Minimum Transfer Risk Over-Collateralization`:

Table 4.1-6 Thresholds, Minimum Transfer, and Rounding Inputs

Minimum Transfer Risk Under-Collateralization	Minimum Transfer Risk Over-Collateralization	cVMT	cVMMTA	uVMT	uVMMTA	RA
False	False	CPVMThreshold	CPVMMinTransfer	UserVMThreshold	UserVMMinTransfer	RoundoffAmount
True	False	CPVMThreshold + max(CPVMMin Transfer,	0	UserVMThreshold	0	0

		RoundoffAmount)				
False	True	CPVMThreshold	0	UserVMThreshold + max(CPVMMinTransfer ,RoundoffAmount)	0	0
True	True	CPVMThreshold + max(CPVMMin Transfer, RoundoffAmount)	0	UserVMThreshold + max(CPVMMinTransfer, ,RoundoffAmount)	0	0

Comments:

- Implicit in setting the *uVMMTA* to zero is the assumption that the counterparty is never over-collateralized and the user never under-collateralized due to the user not returning or posting variation margin due to minimum transfer amounts. This is an additional conservative assumption.
- The rounding amount is included in the new threshold to allow for the risk of rounding to be captured in the case that the rounding amount is greater than the minimum transfer amount (this is not generally the case).
- *UserVMThreshold* + max(*CPVMMinTransfer*, *RoundoffAmount*) could be positive.

4.1.3 Shortcut Empirical Collateral Model

The collateral balance $CB_{emp}(r_i,s_j)$ for the reporting date r_i and scenario s_j is computed as:

$$\mathrm{CB_{emp}}(r_i, s_j) = \mathrm{CB}(r_i, s_j) - \mathrm{Adj_{emp}}(r_i, s_j)$$

where,

$CB(r_i,s_j)$ is computed using the Path Dependent Collateral Model with counterparty and user lags, lag_{cp} and lag_{user}, set to zero;

$$\mathrm{Adj_{emp}}(r_i, s_j) = \begin{cases} \frac{\sqrt{\mathrm{lag_{max}}}}{\sqrt{r_i - r_0}}(\mathrm{MtM}(r_i, \sigma_i(s_j)) - \overline{\mathrm{MtM}_i}), & \text{if } r_i \neq r_0 \text{ and Full Scaling Period } = \text{TRUE} \\ \frac{\sqrt{\mathrm{lag_{max}}}}{\sqrt{r_i - r_{i-1}}}(\Delta\mathrm{MtM}(r_i, \sigma_i(s_j)) - \overline{\Delta\mathrm{MtM}_i}), & \text{if } r_i \neq r_0 \text{ and Full Scaling Period } = \text{FALSE} \end{cases}$$

$$\mathrm{Adj_{emp}}(r_0, s_j) = \begin{cases} \mathrm{Adj_{emp}}(r_1, s_j), & \text{if Adjustment On Time 0 =TRUE} \\ 0, & \text{if Adjustment On Time 0 =FALSE} \end{cases}$$

$$\mathrm{lag_{max}} = \max\left(\mathrm{lag_{cp}} + \mathrm{cure_{cp}}, \mathrm{lag_{user}} + \mathrm{cure_{user}}\right)$$

$$\overline{\mathrm{MtM}_i} = \tfrac{1}{N}\textstyle\sum_{j=1}^{N} \mathrm{MtM}(r_i, s_j)$$

$$\overline{\Delta\mathrm{MtM}_i} = \tfrac{1}{N}\textstyle\sum_{j=1}^{N} \Delta\mathrm{MtM}(r_i, s_j)$$

$$\Delta\mathrm{MtM}(r_i, s_j) = \mathrm{MtM}(r_i, s_j) - \mathrm{MtM}(r_{i-1}, s_j)$$

and σ_i is a permutation (with or without replacement)[*1] of the set of scenarios $\{s_1, s_2, ..., s_N\}$.

`Full Scaling Period` and `Adjustment On Time 0` are model parameters.

*1 The model parameter `Scenario Selection Mechanism` determines if this is done with or without replacement.

4.2 Initial Margin Model

4.2.1 Dynamic Initial Margin

The **DIM** model is used to calculate Dynamic Initial Margin.

- **Preliminary background**

The need for banks to capture time dependent initial margin for both cleared and non-cleared trades in line with contractual arrangements has increased recently by regulations. An approach to calculate initial margin within the IMCR solution is introduced here. This approach is based upon the existing risk scenario set and dynamic initial margins are forecast through time. First the notations are listed in the table below.

Table 4.2-1 Notations

Attribute	Source	Comments
Time step t_i	Input	
Call Period	CSA Attribute	
Margin Horizon	CSA Attribute	
Margin Period of Risk ($MPoR$)	Internal Variable	If Call Period is 0, $MPoR$ = Margin Horizon, otherwise, $MPoR$ = Margin Horizon + Call Period-1
MtM $MtM_{i,s}$	Input	netted mark-to-market values on scenario s and time step t_i
Scaling/Scaling Methodology	Model Parameter /CSA Attribute	CSA Attribute takes precedence. Enumeration for the model parameter: SIMPLE, EXPONENTIAL, NONE. Enumeration for the CSA Attribute: LINEAR, EXPONENTIAL, NONE. Note that the 'Linear' scaling value is referred to as 'Simple'.
Decay Factor	Model Parameter/CSA Attribute	CSA Attribute takes precedence, only used for Exponential scaling
Buckets M	Model Parameter	
User/CP IM Post Style	CSA Attribute	Three approach: PARAMETRIC, (Non-Parametric) VAR, ES
UserIMConfidenceLevel α_{User}	CSA Attribute	percentile, confidence interval
CPIMConfidenceLevel α_{CP}	CSA Attribute	percentile, confidence interval
UserIMThreshold $Threshold_{User}$	CSA Attribute	threshold amount
CPIMThreshold $Threshold_{CP}$	CSA Attribute	threshold amount
Modelled t0	Model Parameter	true or false
IM supplied by Mitigant Pool $IM^{calibration}$	Node Mitigant/MTM	applicability= MITIGANT_POOL
IM supplied by Initial Balance IM^{IB}	Node Mitigant/MTM	applicability =INITIAL MARGIN

Attribute	Source	Comments
Required IM at time step t_1 , $IM_{user}(1)$, $IM_{cp}(1)$	Internal Variable	a single calculated initial margin at the first time step
Required IM, $IM_{user}(i, m)$ $IM_{cp}(i, m)$	Internal Variable	calculated initial margin on a bucket and at a time step
Required IM $IM_{user}(t_i, s_j)$, $IM_{cp}(t_i, s_j)$	Internal Variable	calculated initial margin on a scenario and at a time step
Calculated DIM $DIM_{user}(t_i, s_j)$, $DIM_{cp}(t_i, s_j)$	Internal Variable	final reported dynamic initial margin

Next, computation process for DIM is described in following steps.

- Bucketing

  At the previous time-step $t_{i-1}, i > 1$ the mark-to-market values for the different scenarios should be sorted by value and then separated into M buckets. The number of buckets (M) should be a configuration parameter. The total scenarios are separated by the number of buckets. Let N be total number of scenarios and K be integer part of fraction $\frac{N}{M}$, a number of $(N - M \cdot K)$ buckets can contain $(\mathrm{K}{+}1)$ scenarios, the rest of buckets can have K scenarios. For example, if number of scenarios is 800 and the number of buckets is 7, the first two buckets will contain 115 scenario values and the last 5 buckets will have 114 scenario values.

  If the previous time-step $t_{i-1}, i = 1$ corresponds to the very first time-step t_0 then the bucketing should not be applied as there is only a single mark-to-market value. This means there will only be a single bucket used at time-step t_1 .

- Adjusted *MtM*

  There is an option for user to deal with the early maturity trades or the forward start trades. When the model parameter **Maturity Adjusted Deltas** is set to true, the Maturity Adjusted *MtM* sheet is used for t_{i-1} in the delta calculation. The algorithm for determining the Maturity Adjusted *MtM* sheet is as follows:

  1) Find t_i , the first time step that is not less than maturity date $t_{maturity}$.

  2) If maturity date falls exactly on a simulation time step, $t_{maturity} = t_i$, the values at all time steps $\geq t_i$ are set to zero.

  3) Otherwise, the values at all time steps $\geq t_{i-1}$ are set to zero.

  When the model parameter **Start Date Adjusted Deltas** is set to true, the Start Date Adjusted *MtM* sheet is used for t_i in the delta calculation. The algorithm for determining the Start Date Adjusted *MtM* sheet is as follows:

  1) Find t_i , the first time step that is not less than start date t_{start} (t_i is the closest simulation date on or after start date).

  2) The values at all time steps $\leq t_i$ are set to zero.

  The calculation of delta values in next step is based on the adjusted *MtM* if applied, otherwise, the original *MtM* is used.

- Computing the delta in each bucket at every time step

  For the parametric approach:

$$Delta_{i,s} = \sqrt{\frac{MPoR}{t_i - t_{i-1}}} * (MtM_{i,s} - MtM_{i-1,s})$$

For the non-parametric approach:

$$Delta_{i,s} = \sqrt{\frac{MPoR}{t_i - t_{i-1}}} * [(MtM_{i,s} - MtM_{i-1,s}) - mean((MtM_{i,s} - MtM_{i-1,s})]$$

- Computing required initial margin in parametric approach

  Within each bucket at time step t_i first the sample standard deviation of the delta values is determined by $SD = \sqrt{\frac{\sum(X_i - \bar{X})^2}{(n-1)}}$, where X_i is the delta values in this bucket and $\bar{X}$ is the mean value of X_i , then initial margin is computed as $IM_{user}(i,m) = -SD \cdot N^{-1}(\alpha)$ for the left tail distribution, and $IM_{cp}(i,m) = SD \cdot N^{-1}(\alpha)$ for the right tail distribution, where $N^{-1}(\ .\ )$ is a normal inverse function.

- Computing required initial margin in non-parametric approach

  Within each bucket at time step t_i , assuming $\widetilde{X}_i$ is sorted (in ascending order) delta values in this bucket, then $\widetilde{X}_{i^*}$ is determined as non-parametric LT VaR for $i^* = [n * (1 - \alpha)] + 1$, where [.] denotes integer part, n is total number of delta values in this bucket, $\widetilde{X}_{i^{**}}$ is determined as non-parametric RT VaR for $i^{**} = [n * (\alpha)] + 1$. The required initial margin for User is $IM_{user}(i,m) = \min(\widetilde{X}_{i^*}, 0)$, the required initial margin for CP is $IM_{cp}(i,m) = \max(\widetilde{X}_{i^{**}}, 0)$. If the risk measure is selected as Expected Shortfall, the required initial margin for User is as $IM_{user}(i,m) = \sum_{i=1}^{i^*} \min(\widetilde{X}_i, 0)/i^*$, the required initial margin for CP is $IM_{cp}(i,m) = \sum_{i=i^{**}}^{n} \max(\widetilde{X}_i, 0)/(n - i^{**} + 1)$.

Note

In above two steps $\alpha = \alpha_{User} = \alpha_{CP}$ setting is assumed, in the case $\alpha_{User} \neq \alpha_{CP}$ use $\alpha = \alpha_{User}$ to compute left tail risk measure and use $\alpha = \alpha_{CP}$ to compute right tail risk measure separately..

- Determine Initial Margin at first time step

  If mitigants with `APPLICABILITY = INITIAL_BALANCE` are provided, $IM_{t_0} = IM^{IB}$. Otherwise, if Modelled t0 is set to true and mitigants with `APPLICABILITY = MITIGANT_POOL` are provided, $IM^{calibration}$ is adjusted by the threshold before it is assigned to IM_{t0} . If neither IM^{IB} nor $IM^{calibration}$ are provided, $IM_{t_0} = 0$.

- Scaling

  There is an attribute associated with the initial margin collateral agreement that give user option to chose None, Simple or Exponential scaling method.

  None: no scaling should be applied, $SF_i = 1$.

  Simple: scaling factor is simply the ratio of the initial margin with `APPLICABILITY = MITIGANT_POOL` to the average of calculated initial margin values across all of the different buckets at time t_1 , that is $SF_i = SF = IM^{calibration}/IM(1)$. At each of future time steps t_i , the initial margin values in each of buckets should be multiplied by this scaling factor.

Exponential: scaling factors are adjusted through time, $SF_i = 1 + (SF - 1) * exp(-decayFactor * t_i/365)$. At each of future time steps t_i , the initial margin values in each of buckets should be multiplied by this scaling factor.

Note

Once the initial margins $IM_{user}(i, m)$ ($IM_{cp}(i, m)$) in each bucket at every time step (in parametric and non-parametric approach) are calculated and adjusted by supposed scaling factor SF_i in each bucket at each time step, IMCR returns initial margin value $IM_{user}(t_i, s_j)$ ($IM_{cp}(t_i, s_j)$) corresponding to each scenario at every time step.

- Final Initial Margin taking account collateral agreement threshold

  The IMCR initial margin collateral agreement model allows threshold amounts to be defined for both the internal user and external counterparty. Thus the threshold amount (if any threshold specified) must be subtracted from the calculated initial margin value before any mitigation of overall exposure is determined. The final dynamic initial margin for user at the collateral agreement level can be expressed as

  $$DIM_{user}(t_i, s_j) = \min\left(IM_{user}(t_i, s_j) - threshold_{user}, 0\right)$$

  The final dynamic initial margin for counterparty at the collateral agreement level can be expressed as

  $$DIM_{cp}(t_i, s_j) = \max\left(IM_{cp}(t_i, s_j) - threshold_{cp}, 0\right)$$

■ DIM

● Model Parameter

- **Buckets**: an integer number
- **Scaling** : none, simple, exponential
- **Decay Factor**: a number
- **Modelled t0**: true or false
- **Maturity Adjusted Deltas**: true or false
- **Start Date Adjusted Deltas**: true or false

● CSA Attribute

- **UserIMPostStyle**: PAREMETRIC, VAR, ES
- **CPIMPostStyle** : PAREMETRIC, VAR, ES
- **UserIMConfidenceLevel**: percentile
- **CPIMConfidenceLevel**: percentile
- **IMMarginHorizon**: an interger number
- **UserIMThreshold**: a number
- **CPIMThreshold**: a number

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Chapter 5 Exposure

Exposure on an MA node

Let us define the total exposure on an MA node as:

$$\text{TotalExposure}(t,s) = \text{Exp}(t,s) + \text{AddOn}(t) + \text{IP}(t)$$

where *Exp(t,s)* is the simulated collateralized exposure at time *t* and scenario *s*, *AddOn(t)* is the addon part at time *t* on the MA node, and *IP(t)* is the initial profile at time *t* defined by user.

For the calculation of simulated collateralized exposure, refer to 4.1.1.5 Collateralized Exposure on page 21 if the path dependent collateral model is applied to the expression; refer to 4.1.2.4 Collateralized Exposure on page 43 if the settlement adjusted path dependent model is applied to the expression.

In current implementation, MAG allows the protection provided by CDS instrument to mitigate the reference entity's exposure for risk management measure. The hedge effect only applies to the forward view expressions. The reverse view expressions are not affected. To enable this hedge effect, user needs to set the attribute `applyCDSHedges` in the credit expression to TRUE.

The hedged exposure at time *t* and scenario *s* is calculated as:

$$\text{Exposure}_{\text{hedged}}(t,s) = max[\text{TotalExposure}(t,s) - \text{payoff}(t,s), 0]$$

where *payoff(t,s)* is the protection amount provided by the CDS instruments. For the calculation of *payoff(t,s)*, refer to 6.1 CDS Protection Amount on page 75.

Exposure on nodes above an MA

The exposure on nodes above an MA node is the sum of the exposure of the nodes that are one level below it.

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Chapter 6 Functions

Mark-to-Future® Aggregator (MAG) simulation functions allow you to apply various risk measures to the output of evaluation functions. Currently, there are two types of simulation functions implemented in MAG:

- Statistic functions evaluate the elementary statistics on the hierarchy.
- Utility functions filter out the desired results from the MtF cube by specified criteria.
  - Discount
  - Select Scenarios
  - Select Times
  - Sort Scenarios
  - Specified Scenario Per Time

You can create composite simulation functions made of several elementary simulation functions. For example:

Mean(*TimeWeight*(*TimePeriod*))

This composite function first performs time averaging for each scenario with `@TimeWeight(TimePeriod)`, then averages the result over all scenarios with `@Mean()`.

Before evaluation of the statistics, MAG converts the cube value in scenario consistent fashion to the simulation currency. If the simulation currency is not specified, it defaults to the reporting currency. If the simulation currency differs from the reporting currency, MAG performs a spot conversion of the statistic output from the simulation currency to the reporting currency using the exchange rate at time 0. Both `SimulationCurrency` and `ReportingCurrency` are attributes defined in `ExpressionInstances.csv` file. The reporting currency is mandatory with each expression.

This section provides details of the statistic functions.

6.1 CDS Protection Amount

This function returns the amount of protection provided by CDS deals to mitigate the reference entity exposure. This function can only be used in a forward view expression.

• Definition and Inputs

Table 6.1-1 Inputs

Input	Notation	Source File	Usage in Model/Comments
mitigantedRiskType		protectionAttributes	When set to PRESETTLEMENT,the CDS trade mitigates the exposure to the reference entity.
underlyingIssuerId		protectionAttributes	Specifies the ID of the reference entity.
masterAgreement		protectionAttributes	Together with `collateralAgreement`, specify the lowest node on the legal hierarchy of the underlying entity to which the protection is applied. If left empty, it will be defaulted to OTC node.
collateralAgreement		protectionAttributes	See the comment for `masterAgreement`.

Input	Notation	Source File	Usage in Model/Comments
facilities		protectionAttributes	Specifies the facility nodes to which the protection is applied.
effectiveDate	$effDate_{CDS}$	protectionAttributes	Specifies the date on which the trade becomes effective.
recoveryRate	RR_{CDS}	protectionAttributes	Specifies the recovery rate. The default value is 40%.
variableNotional	$Notional_{CDS}(t)$	protectionAttributes	Specifies the notional. It can be time-dependent. This is a list of dates and corresponding notionals $\{d_i, N_i\}$ where $i=1, 2, \ldots, n$.
notionalCurrency		protectionAttributes	Specifies the currency of the variable notional.

Note

When `facilities` of a CDS trade is specified, the protection is automatically applied to the parent of the specified node. So it is user's responsibility to ensure that the CDS protection is not applied twice within the facility hierarchy.

Note

If `facilities` of a CDS trade is specified, the protection is automatically applied to the legal entity node.

For any time t,

$$\mathrm{Notional}_{\mathrm{CDS}}(t) := \begin{cases} N_1 & \text{if } \mathrm{effDate}_{CDS} \le t \le d_1 \\ N_{i+1} & \text{if } d_i < t \le d_{i+1} \text{ for some } 1 \le i \le n-1 \\ 0 & \text{otherwise} \end{cases}$$

where $\mathrm{effDate}_{\mathrm{CDS}} \le d_1 < d_2 \ldots < d_n =: \mathrm{Maturity}_{\mathrm{CDS}}$ and $Maturity_{CDS}$ is the maturity date of the CDS trade. If $d_n \ne \mathrm{Maturity}_{\mathrm{CDS}}$, the CDS trade is not processed.

- **CDS Protection Amount**

For a single name CDS, the payoff-at-default is calculated as:

$$\mathrm{payoff}(t,s) = \begin{cases} \mathrm{Notional}_{\mathrm{CDS}}(t) \times (1 - \mathrm{RR}_{\mathrm{CDS}}) \times \mathrm{FX}(t,s), & t < \mathrm{Maturity}_{\mathrm{CDS}} \\ 0, & \text{otherwise} \end{cases}$$

where $FX(t,s)$ is the foreign exchange rate at time t and scenario s from the CDS's `notionalCurrency` to `simulationCurrency`.

The CDS protection amount of a node is the sum of the protections applied to this node and all nodes under it.

6.2 CVA Amount

CVA Amount function has the following arguments:

- `Mode`: Counterparty (default), or Risk Taker, or Net
- `Output Perspective`: Net (default), or Counter Party, or Risk Taker, or Simple Net
- `PD Sheet Type`: Deterministic (default), or Stochastic
- `Use Stochastic PD`: True (default), or False
- `Exposure Bucketing`: Start of Interval, or Mid of Interval, or Mid of Interval Basel, or End of Interval (default)

- `Survival Probability Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `Confidence Interval`: default value = 0

Note

`Output Perspective` is only relevant when `Mode` is Net.

CVA Amount function can be applied to exposures of individual positions and to exposures of netting nodes associated with a legal entity in a hierarchy.

For nodes above legal entity, when `DiversificationStyle` associated with the function is `ALL`, the CVA Amount function aggregates the per scenario CVA amount of legal entities below that node to reach the final results of the node.

■ Preliminary Materials

Exposure Bucketing

For Credit Value Adjustment (CVA) calculation (that is, CVA Amount, CVA Amount Audit, CVA Rate, and CVA Rate Audit), the function parameter `Exposure Bucketing` defines which exposure and discount factor of the time interval (t_{j-1},t_j) to be used to calculate the discounted exposure. When `End of Interval` is selected, both the exposure and discount factor of the time interval (t_{j-1},t_j) is taken from the end of the interval. That is,

$$\text{EndofInterval} = E(t_j) \cdot d(t_0, t_j) \ ,$$

where,

$E(t_j)$ is the exposure measure for time step t_j, and

$d(t_0, t_j)$ is the discount factor from time step t_j to t_0.

Similarly,

$$\text{StartofInterval} = E(t_{j-1}) \cdot d(t_0, t_{j-1})$$

$$\text{MidofInterval} = \frac{\left(E(t_{j-1}) + E(t_j)\right) \cdot \left(d(t_0, t_{j-1}) + d(t_0, t_j)\right)}{4}$$

$$\text{MidofIntervalBasel} = \frac{E(t_{j-1}) \cdot d(t_0, t_{j-1}) + E(t_j) \cdot d(t_0, t_j)}{2}$$

For the simplicity of illustration, in the remaining part of this section, the discounted exposure expression for `End of Interval` as shown above is used in all formulas related to CVA calculation. However, remember that it can be replaced by other formats of discounted exposure as shown above depending on the setting of `Exposure Bucketing`.

Survival Probability Bucketing

The function parameter `Survival Probability Bucketing` defines which survival probability of the time interval (t_{j-1},t_j) to be used when CVA Amount function parameter `Mode` = Net. When `StartofInterval` is selected, the probability of survival of the time interval (t_{j-1},t_j) is taken from the beginning of the interval. That is,

$$\text{StartofInterval} = sp\left(m', t_0, t_{j-1}\right) \ ,$$

where m is the opposite side. For detailed explanation of m, refer to the formula in section ■ Confidence Interval ≠ 0 on page 79.

Similarly,

$$\text{EndofInterval} = sp\left(m', t_0, t_j\right)$$

$$\text{MidofInterval} = \frac{sp\left(m', t_0, t_{j-1}\right) + sp\left(m', t_0, t_j\right)}{2}$$

For the simplicity of illustration, in the remaining part of this section, the survival probability for `StartofInterval` as shown above is used in all formulas related to CVA calculation. However, remember that it can be replaced by other formats of survival probability as shown above depending on the setting of `SurvivalProbabilityBucketing`.

Refer to the section "*Discount factors (or future value factors) in CVA calculation*" in the *Integrated Market and Credit Risk Solution Guide* for the configuration of discount factor.

Note

For `UNKNOWN_CP` and `UNKNOWN_OWN`, the value of probability of default, curve type, and recovery rate can be preconfigured in `$ALGO_TOP/cfg/IMCR/mag/mag_imcr.xml`.

Confidence Interval

The function parameter `Confidence Interval` serves as a switch of CVA function. When `Confidence Interval` is set to 0, CVA Amount function returns the regular CVA amount. When `Confidence Interval` is set to a number between 0 and 1 but other than 0, CVA Amount function returns the confidence interval of the CVA amount.

■ Confidence Interval = 0

Mode = Counterparty or Risk Taker

The CVA amount at time step t_0 is

$$\text{CV}A_A(m, t_0) = \sum_{j=1}^{K} Mean\left(E(m, t_j, s) \cdot d(t_0, t_j, s) \cdot q(m, t_{j-1}, t_j, s)\right) \cdot (1 - r(m))$$

where, K is the total number of simulation time steps;

m is either Counterparty or Risk Taker depending on Mode;

$E(m, t_j, s)$ is the sum of the simulated exposure and non-simulated value at time step t_j under scenario s. For non-simulated value of the exposure, MAG aggregates the values from trade replacement profile and node adjustment profile and returns one non-simulated value at each time step as an input to CVA calculation, which is applied to all scenarios;

Mean is the mean value across scenarios; and

$d(t_0, t_j, s)$ is the discount factor from time step t_j to t_0 under scenario s;

$q(m, t_{j-1}, t_j, s)$ is the marginal probability of default of m between t_{j-1} and t_j under scenario s.

$$q\left(m, t_{j-1}, t_j, s\right) = \begin{cases} Stochastic\ PD & if\ PDSheetType = Stochastic\ and\ UseStochasticPD = True \\ Base\ Scenario\ PD & Otherwise \end{cases}.$$

$r(m)$ is the recovery rate of m.

Mode = Net

The net CVA amount at time step t_0 is

$$CVA_A(Net,t_0)=\begin{cases}\overline{CVA_A(CP,t_0)}-\overline{CVA_A(RT,t_0)} & if \quad OutputPerspective=Net\\ \overline{CVA_A(CP,t_0)} & if \quad OutputPerspective=Counterparty\\ \overline{CVA_A(RT,t_0)} & if \quad OutputPerspective=RiskTaker\\ CVA_A(CP,t_0)-CVA_A(RT,t_0) & if \quad OutputPerspective=SimpleNet\end{cases}$$

where

$$\overline{\mathrm{CVA}_A(m,t_0)}=\sum_{j=1}^{K}Mean\Big(E(m,t_j,s)\cdot d(t_0,t_j,s)\cdot q(m,t_{j-1},t_j,s)\cdot sp(m',t_0,t_{j-1},s)\Big)\cdot(1-r(m))$$

where $sp(m', t_0, t_{j-1}, s)$ is the probability of survival between t_0 and t_{j-1} for the opposite side. For example, if *m* = *Counterparty*, *m'* = *Risk Taker*. $sp(m',t_0,t_{j-1},s)=1-\sum_{l=1}^{j-1}q(m',t_{l-1},t_l,s)$.

$q(m', t_{j-1}, t_j, s)$ is the marginal probability of default of *m'* between t_{j-1} and t_j under scenario *s*.

$$q\left(m',t_{j-1},t_j,s\right)=\begin{cases}Stochastic\ PD & if\ PDSheetType=Stochastic\ and\ UseStochasticPD=True\\ Base\ Scenario\ PD & Otherwise\end{cases}.$$

Confidence Interval ≠ 0

Suppose the confidence level specified by user through function parameter `Confidence Interval` is C. For a given random variable that is drawn N times, the average is: $\overline{X}=\frac{1}{N}\sum_{s=1}^{N}X_s$. The variance of this average is:

$$VAR(\overline{X})=\frac{1}{N(N-1)}\sum_{s=1}^{N}(X_s-\overline{X})^2$$

The confidence interval of this average is thus calculated as:

$$\delta=\sqrt{VAR(\overline{X})}\times\Phi^{-1}(\tfrac{1+C}{2})$$

where, $\Phi^{-1}(z)$ is the inverse cumulative distribution function of standard normal distribution.

When calculating confidence interval for CVA amount, the regular CVA amount (i.e. the value CVA Amount function returns when `Confidence Interval` = 0) is the $\overline{X}$ used in the above formula. The following defines X_s for different setup of `Mode`.

Mode = Counterparty or Risk Taker

When `Mode` = `Counterparty` or `Risk Taker`, X_s is the CVA amount value per scenario. That is:

$$CVA_A(m,t_0,s)=\sum_{j=1}^{K}\left[E(m,t_j,s)\times d(t_0,t_j,s)\times q(m,t_{j-1},t_j,s)\times(1-r(m))\right]$$

Mode = Net

When `Mode` = `Net`, X_s is defined as:

$$CVA_A(Net,t_0,s)=\begin{cases}\overline{CVA_A(CP,t_0,s)}-\overline{CVA_A(RT,t_0,s)} & if \quad OutputPerspective=Net\\ \overline{CVA_A(CP,t_0,s)} & if \quad OutputPerspective=Counterparty\\ \overline{CVA_A(RT,t_0,s)} & if \quad OutputPerspective=RiskTaker\\ CVA_A(CP,t_0,s)-CVA_A(RT,t_0,s) & if \quad OutputPerspective=SimpleNet\end{cases}$$

where

$$\overline{\mathrm{CVA}_A(m,t_0,s)}=\sum_{j=1}^{K}\left[E(m,t_j,s)\times d(t_0,t_j,s)\times q(m,t_{j-1},t_j,s)\times sp(m',t_0,t_{j-1},s)\times(1-r(m))\right].$$

6.3 CVA Amount Audit

This function is designed for validation of CVA Amount. It is not recommended for use in production.

In addition to CVA Amount, this function generates one text file in CSV format that contain intermediate values used for CVA Amount computation. The file is located in `$ALGO_TOP/outputs/local/IMCR/<dataset_ID>/<Variant_ID>/mag/sbinstance/audit/` directory. The file name is `CVAAuditTimeLine.csv`.

In addition to all the input parameters of CVA Amount, user must set one additional parameter: `Audit Scenario` in define`Functions` tool.

For the formulae of calculating CVA Amount, refer to the 6.2 CVA Amount on page 76 section.

■ Stochastic PD

This is the case when `PD Sheet Type` is Stochastic, `Use Stochastic PD` is True, and `EnableStochastic` of the relevant legal entity that is referred to by `Mode` is True.

When `Mode` = Counterparty or Risk Taker, the **time line** file contains the values of:

- Exposure numbers of the specified `Audit Scenario` for each simulation date "as is", i.e. regardless of the `CVAExposureBucketing` setting
- Discount factors of the specified `Audit Scenario` for each simulation date
- Discounted Exposure values of the specified `Audit Scenario` depending on the `CVAExposureBucketing` setting in 6.2 CVA Amount on page 76
- Qd, the marginal probability of default of the specified `Audit Scenario` for each simulation date
- Recovery Rate
- PD Curve used
- PD Sheet Type

When `Mode` = Net, in addition to the above-mentioned information, the **time line** file also contains the values of SP, i.e. survival probability, for each simulation date.

■ Deterministic PD

This is the case when

- `PD Sheet Type` = Deterministic; or
- `PD Sheet Type` = Stochastic, and `Use Stochastic PD` = False; or
- `PD Sheet Type` = Stochastic, `Use Stochastic PD` = True, and `EnableStochastic` of relevant legal entity = False.

When `Mode` = Counterparty or Risk Taker, the **time line** file contains the values of *M*, *U* and CVA of the specified `Audit Scenario` for each simulation date as well as the Recovery Rate, PD Curve and PD Sheet Type. It allows you to validate the CVA amount.

When `Mode` = Net, in addition to the above-mentioned information, the **time line** file also contains the values of SP, i.e. survival probability, for each simulation date.

6.4 CVA Rate

CVA Rate can be applied to exposures of individual positions and to exposures of netting nodes associated with a legal entity in a hierarchy. CVA rates are not defined above the legal entity level.

CVA Rate function has the following arguments.

- `Mode`: Counterparty (default), or Risk Taker
- `PD Sheet Type`: Deterministic (default), or Stochastic
- `Use Stochastic PD`: True (default), or False
- `Exposure Bucketing`: Start of Interval, or Mid of Interval, or Mid of Interval Basel, or End of Interval (default)
- `Survival Probability Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval

Note

CVA rates are defined only for Counterparty and Risk Taker. There is no net CVA rate.

Note

The CVA rate is measured in percentage.

The CVA rate is defined as follows:

$$CVA_R(m,t_0) = \frac{CVA_A(m,t_0)}{\sum_{j=1}^{K} Mean\left[E(m,t_j,s)\cdot d(t_0,t_j,s)\right]\cdot(1-r(m))\cdot \Delta(t_{j-1},t_j)}$$

where $\Delta(t_{j-1},t_j)$ is the time factor, that is, the number of days between t_{j-1} and t_j, divided by 365. All other notations are introduced in section 6.2 CVA Amount on page 76.

6.5 CVA Rate Audit

This function is designed for validation of CVA Rate. It is not recommended for use in production. This function generates one text file in CSV format that contain intermediate values used for CVA Rate computation, The file is located in `$ALGO_TOP/outputs/local/IMCR/<dataset_ID>/<Variant_ID>/mag/sbinstance/audit/` directory. The file is `CVAAuditTimeLine.csv`.

In addition to all input parameters of CVA Rate, user must set one additional parameter: `Audit Scenario` in `defineFunctions` tool.

For the formulae of calculating CVA Rate, refer to the 6.4 CVA Rate on page 81section.

The numerator of the CVA Rate is a CVA amount, which can be validated independently as described in 6.3 CVA Amount Audit on page 80. The CVA Rate Audit function allows you to validate the denominator of the rate. The denominator is similar to CVA amount. It contains an additional time factor $\Delta(t_{j-1},t_j)$.

Note

It is recommended to run a request that contains both CVA Rate Audit and CVA Amount Audit with identical settings. The latter will provide the numerator values.

■ Stochastic PD

This is the case when `PD Sheet Type` is Stochastic, `Use Stochastic PD` is True, and `EnableStochastic` of the relevant legal entity that is referred to by `Mode` is True.

In this case, the **time line** file contains the values of:

- Delta_t for each simulation date
- Exposure numbers of the specified `Audit Scenario` for each simulation date "as is", i.e. regardless of the `CVAExposureBucketing` setting
- Discount factors of the specified `Audit Scenario` for each simulation date
- Discounted Exposure values of the specified `Audit Scenario` depending on the `CVAExposureBucketing` setting in 6.2 CVA Amount on page 76
- Recovery Rate
- PD Curve used
- PD Sheet Type

■ Deterministic PD

This is the case when

- `PD Sheet Type` = Deterministic; or
- `PD Sheet Type` = Stochastic, and `Use Stochastic PD` = False; or
- `PD Sheet Type` = Stochastic, `Use Stochastic PD` = True, and `EnableStochastic` of relevant legal entity = False.

In this case, the **time line** file contains the values of *M*, *U* and CVA of the specified `Audit Scenario` for each simulation date as well as the Recovery Rate, PD Curve and PD Sheet Type.

6.6 CVA Capital CEM

The following are CVA Capital Current Exposure Method (CEM) input parameters with their default settings:

- `Maturity Floor Term Value` = 1 (default)
- `Maturity Floor Term Unit` = Year (default), Day, Month
- `Short Term Exposure Maturity Floor Term Value` = 1 (default)
- `Short Term Exposure Maturity Floor Term Unit` = Day (default), Month, Year
- `Weighted Average Remaining Maturity` = True (default), False
- `Evaluate All Deals` = False (default), True
- `PD Sheet Type` = Deterministic (default for CVA Capital CEM), Stochastic (default for CVA Capital CEM Stock PD)

The bank must add a capital charge to cover the risk of mark-to-market losses on the expected counterparty risk to OTC derivatives. Such losses are known as credit value adjustments (CVA). The CVA capital charge will be calculated in the manner set forth depending on the bank's approved method of calculating capital charges for counterparty credit risk and specific interest rate risk. A bank is not required to include in this capital charge (i) transactions with a central counterparty (CCP) and (ii) securities financing transactions (SFT), unless their supervisor determines that the bank's CVA loss exposures arising from SFT transactions are material.

- Banks with IMM approval and Specific Interest Rate Risk VaR model approval for bonds: Advanced CVA risk capital charge
- All other banks: standardized CVA risk capital charge

When a bank does not have the required approvals as listed above to calculate a CVA capital charge for its counterparties, the bank must calculate a portfolio capital charge using the following formula. Note the formula currently implemented assumes no hedges.

$$K = 2.33\sqrt{h}\sqrt{\left(\sum_i 0.5 w_i M_i EAD_i^{total}\right)^2 + \sum_i 0.75 w_i^2 \left(M_i EAD_i^{total}\right)^2}$$

with the following parameters

- h is the one-year risk horizon (in units of years), h=1
- w_i is the weight applicable to counterparty *i*. Counterparty *i* must be mapped to one of the seven weights w_i based on its external rating, as shown in the table below. When a counterparty does not have an external rating, the bank must, subject to supervisory approval, map the internal rating of the counterparty to one of the external ratings

Table 6.6-1 Assignment of Counterparty Weight Under the Simple CVA

Internal PD (%)	**Weight w_i (%)**
0.00-0.07	0.70
>0.07-0.15	0.80
>0.15-0.40	1.00
>0.40-2.00	2.00
>2.00-6.00	3.00
>6.00	10.00

- EAD_i^{total} is the exposure at default of counterparty *i* (summed across its netting sets), including the effect of collateral as per existing IMM, SM or CEM rules as applicable to the calculation of counterparty risk capital charges for such counterparty by the bank. For non-IMM banks the exposure should be discounted by applying the factor $\frac{1 - e^{-0.05M_i}}{0.05M_i}$

  For IMM banks, no such discount should be applied as the discount factor is already included in M_i .

- M_i is the effective maturity of the transaction with counterparty *i*. For non-IMM banks, M_i is the notional weighted average maturity , see Case 2. Original Term to Maturity ≤ 1 in 6.16 Effective Maturity on page 87 except that the Original Term to Maturity condition and `Maturity Cap Term` don't apply here. There are two approaches at trade level: Foundation Approach and Advanced Approach. If Foundation Approach is selected, OTC transactions default to 2.5 years. For the Advanced Approach, the value is the remaining maturity of translation.

The function assumes an institution is non-MM bank and uses CEM methodology to calculate EAD.

For related information, see *"Current Exposure Method functions and expressions"* in the *Integrated Market and Credit Risk Solution Guide*.

6.7 Mean

Equivalent to the `@mean` simulation function in RiskWatch®, the function calculates the mean value over scenarios of a risk factor it is attached to over the scenario set. It returns mean values of a risk factor on each time step.

$$\text{Mean} = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$$

6.8 Standard Deviation

Equivalent to the `@standard deviation` simulation function in RiskWatch®.

$$StandardDeviation = \frac{\sqrt{\sum_{i=1}^{n} (x_i - m)^2}}{\sqrt{n}}$$

where m is the mean value of x_i. This function returns the standard deviation of a risk factor on each time step.

6.9 Non-Parametric Right Tail

Input parameters:

- `Confidence Interval`: default value = 0.95
- `Confidence Interval Option`: Two Sided (default) or One Sided

The definition of non-parametric right tail coincides with that of RiskWatch®. There are n scenarios for a position x: $x_1, x_2, ..., x_n$. A new variable β that depends on the confidence level α and the tail type is introduced:

$$\beta = \begin{cases} \alpha & \text{if } \textit{one} \text{ sided} \\ \frac{1+\alpha}{2} & \text{if } \textit{two} \text{ sided} \end{cases}$$

MAG finds a whole number m,

$m = \max\{k, (k-1) < \beta \cdot n\} \equiv \text{ceiling}(\beta \cdot n)$.

If $y_1, ..., y_n$ are values of $x_1, ..., x_n$ sorted in ascending order, the non-parametric right tail value is $\text{RightTail}(\alpha) = y_m$.

6.10 PD Weighted Mean

The difference between this function and the regular Mean function is that this function assigns unequal weights to different scenarios when calculating the mean. The weights are based on the stochastic PD sheet.

- **Notation**

$pd(t_i, s_j)$: the probability of default between t_{i-1} and t_i under scenario s_j

$w(t_i, s_j)$: the weight applied to scenario s_j at time t_i and is calculated as

$$w(t_i, s_j) = \begin{cases} 1, & \text{if pd is constant across scenarios} \\ \frac{n \times \text{pd}(t_i, s_j)}{\sum_{k=1}^{n} \text{pd}(t_i, s_k)}, & \text{otherwise} \end{cases}$$

, where n is the number of scenarios.

$x(t_i, s_j)$: the exposure at time t_i under scenario s_j

- **Algorithm**

The PD weighted mean value on each time step is calculated as:

$$\text{PD Weighted Mean}(t_i) = \frac{1}{n} \times \sum_{j=1}^{n} [w(t_i, s_j) \times x(t_i, s_j)]$$

MAG calculates the PD weighted mean for nodes at all levels up to legal entity. For nodes above legal entity level, MAG returns the sum of PD weighted mean at legal entity level.

Note

PD cube needs to be generated using SOBOL for credit driver scenarios. Otherwise, the results in MAG using stochastic PD can be unstable.

6.11 PD Weighted Right Tail

Input parameters:

- `Confidence Interval`: default value = 0.95
- `Confidence Interval Option`: Two Sided (default) or One Sided

The difference between this function and Non-Parametric Right Tail function is that this function assigns unequal weights to different scenarios when calculating the right tail. The weights are based on the stochastic PD sheet.

- **Notation**

$pd(t_i, s_j)$: the probability of default between t_{i-1} and t_i under scenario s_j

$w(t_i, s_j)$: the weight applied to scenario s_j at time t_i and is calculated as

$$w(t_i, s_j) = \begin{cases} 1, & \text{if pd is constant across scenarios} \\ \frac{n \times \text{pd}(t_i, s_j)}{\sum_{k=1}^{n} \text{pd}(t_i, s_k)}, & \text{otherwise} \end{cases}$$, where n is the number of scenarios.

$x_1, x_2, ..., x_n$: the exposure at a given time step across scenarios

$y_1, y_2, ..., y_n$: the exposure $x_1, x_2, ..., x_n$ sorted in ascending order

α : the `Confidence Level`

- **Algorithm**

Introduce a new variable β which is defined as:

$$\beta = \begin{cases} \alpha, & \text{if Confidence Level Option} = \text{One Sided} \\ \frac{(1+\alpha)}{2}, & \text{if Confidence Level Option} = \text{Two Sided} \end{cases}$$

MAG finds the whole number m,

$$m = \max \{k, \sum_{p=1}^{k} w_{y_p} < \beta * n + 1\}$$

where w_{y_p} is the weight of y_p and depends on the scenario y_p corresponds to.

The PD weighted right tail value is y_m.

MAG calculates the PD weighted right tail for nodes at all levels up to legal entity. For nodes above legal entity level, MAG returns the sum of PD weighted right tail at legal entity level.

Note

PD cube needs to be generated using SOBOL for credit driver scenarios. Otherwise, the results in MAG using stochastic PD can be unstable.

6.12 Non-Parametric Left Tail

Input parameters:

- `Confidence Interval`: default value = 0.95
- `Confidence Interval Option`: Two Sided (default) or One Sided

The definition of non-parametric left tail coincides with that of RiskWatch®. There are *n* scenarios for a position *x*: $x_1, x_2, \ldots, x_n$. A new variable β that depends on the confidence level α and the tail type is introduced:

$$\beta = \begin{cases} \alpha & \text{if } \ one \ \text{ sided} \\ \dfrac{1+\alpha}{2} & \text{if } \ two \ \text{ sided} \end{cases}.$$

MAG finds a whole number *m*,

$m = \max\{k, (k-1) < \beta{\cdot}n\} \equiv \text{ceiling}(\beta{\cdot}n)$

If $z_1, \ldots, z_n$ are values of $x_1, \ldots, x_n$ sorted in descending order, the non-parametric left tail value is $\text{LeftTail}(\alpha) = z_m$.

6.13 Expected Tail Exposure

Input parameters:

- `Confidence Interval`: default value = 0.95
- `Confidence Interval Option`: Two Sided (default) or One Sided

The Expected Tail Exposure (ETE) as a function of confidence level α is defined as the expected value of a variable v under condition that v exceeds a threshold value *u*(α), which is the quantile of α. In other words,

$\text{Prob}(z \geq u(\alpha)) = \alpha$

and

$\text{ETE}(\alpha) = E\{v | v \geq u(\alpha)\}.$

MAG estimates the expected tail exposure from a Mark-to-Future® cube by averaging the non-parametric right tail. If there are *n* scenarios for a position *x*: $x_1, x_2, \ldots, x_n$, a new variable β is defined as

$$\beta = \begin{cases} \alpha & \text{if } \ one \ \text{ sided} \\ \dfrac{1+\alpha}{2} & \text{if } \ two \ \text{ sided} \end{cases}$$

MAG finds a whole number *m*,

$m = \max\{k, (k-1) < \beta{\cdot}n\} \equiv \text{ceiling}(\beta{\cdot}n).$

If $y_1, \ldots, y_n$ are values of $x_1, \ldots, x_n$ sorted in ascending order, the expected tail exposure is

$$\mathrm{ETE}(\alpha) = \frac{1}{n-m+1}\sum_{q=m}^{n} y_q.$$

6.14 Expected Shortfall

Input parameters:

- `Confidence Interval`: default value = 0.95
- `Confidence Interval Option`: Two Sided (default) or One Sided

This function is similar to the Expected Tail Exposure. In the limit when the number of scenarios tends to infinity, these two measures coincide. The Expected Shortfall (ES) as a function of confidence level α is defined as the expected value of a variable v under condition that v exceeds a threshold value *u*(α), which is the quantile of α. In other words,

$$\mathrm{Prob}(z \geq u(\alpha)) = \alpha$$

and

$$\mathrm{ES}(\alpha) = E\{v|v \geq u(\alpha)\}.$$

MAG estimates the expected shortfall from a Mark-to-Future® cube using a different formula than the expected tail exposure. If there are *n* scenarios for a position *x*: $x_1, x_2, ..., x_n$, a new variable β is defined as

$$\beta = \begin{cases} \alpha & \text{if } \ one \ \text{ sided} \\ \frac{1+\alpha}{2} & \text{if } \ two \ \text{ sided} \end{cases}$$

MAG finds a whole number *p*,

$$p = \mathrm{floor}(\beta \cdot n) + 1$$

If $y_1, ..., y_n$ are values of $x_1, ..., x_n$ sorted in ascending order, the expected shortfall is

$$\mathrm{ES}(\alpha) = \frac{1}{1-\beta} \cdot \left[\frac{1}{n}\sum_{q=p+1}^{n} y_q + \left(1 - \beta - \frac{n-p}{n}\right) \cdot y_p\right]$$

6.15 Factor

Input parameter:

- `Factor Value`: default value = 0

This simulation function takes the result passed to it and multiplies it by the entered parameter x. For example, if x=1.4, and $\mathrm{Vector}_{t_{\mathrm{SIM},i}}$ =2, then the function returns 2.8. This function returns values of a risk factor for every scenario and every time step.

6.16 Effective Maturity

The function is applied to any vector of time series and calculates one output value. The function requires one input period, and nine input parameters. Effective maturity is expressed in years.

Effective Maturity can be calculated at the level of a netting set or above up to the OTC/SFT level.

Note

When Effective Maturity function is applied at CP level, MAG returns -1.

6

Functions

The following are the Effective Maturity input parameters with their default settings:

- `Input period in Days` = 365
- `Effective Maturity Reference Year` = 1
- `Maturity Cap Term Value` = 5
- `Maturity Cap Term Unit` = Year
- `Maturity Floor Term Value` = 1
- `Maturity Floor Term Unit` = Year
- `Short Term Exposure Maturity Floor Term Value` = 1
- `Short Term Exposure Maturity Floor Term Unit` = Day
- `Weighted Average Repo Cashflow` = true
- `Weighted Average Remaining Maturity` = true
- `Evaluate All Deals` = False

The notations used in this section is defined as follows:

Symbol	Explanation
a	Effective Maturity Reference Year
b	Maturity Cap Term
c	Maturity Floor Term
d	Short Term Exposure Maturity Floor Term
e	Weighted Average Repo Cashflow
f	Weighted Average Remaining Maturity

The vector of time series is $V = (v_0, ..., v_K)$, with corresponding simulation times $t_0, ..., t_K$.

- ### Effective Maturity On And Above A Netting Set

Note

For the definition of Netting Set, see 6.21 Regulatory Capital & Risk Weighted Assets on page 98.

Let *EM* denote effective maturity.

Here, we define the Original Term to Maturity of a deal as: Maturity Date - Trade Date. The Original Term to Maturity of a node is defined as the longest original term to maturity of the underlying deals. Depending on the original term to maturity of the netting set, there are two cases (the result is in unit of year):

Note

The effective maturity is always calculated in the simulation currency. Changing the reporting currency does not affect the value of effective maturity.

Case 1: Original Term to Maturity > a

$\mathrm{EM} = \max(\min(\mathrm{Mat}, b), c)$

Note

Effective Maturity is a value between Maturity Floor Term and Maturity Cap Term, i.e.: $c \leq EM \leq b$.

To estimate *Mat*, the following notation is introduced.

τ is the Input Period.

T is the remaining term to maturity of a deal or the largest remaining time to maturity on a node.

$T_{\mathrm{Span}} = \min(T, \tau)$ is the time span.

$t_0, t_1, ..., t_K$ are ordered time steps ($t_0 < t_1 < ... < t_K$).

Denote that:

$t_0 = 0$

$t_{\mathrm{IP}} = \max(t_k | t_k \le \tau), \quad IP = \max(k | t_k \le \tau)$

t_{IP+1} is the time step following t_{IP}. If there is no time step after t_{IP},

$t_{\mathrm{IP}+1} = t_{\mathrm{IP}}$

$t_m = \max(t_k | t_k \le T), \quad m = \max(k | t_k \le T)$

$\Delta(t_k) = (t_k - t_{k-1}) / T_{\mathrm{Span}}$

$\Delta(t_\tau) = (\tau - t_{\mathrm{IP}}) / T_{\mathrm{Span}}$

$\Delta(t_{\mathrm{IP}+1}) = (t_{\mathrm{IP}+1} - \tau) / T_{\mathrm{Span}}$

$\Delta(t_{mat}) = (\mathrm{T} - t_m) / T_{\mathrm{Span}}$

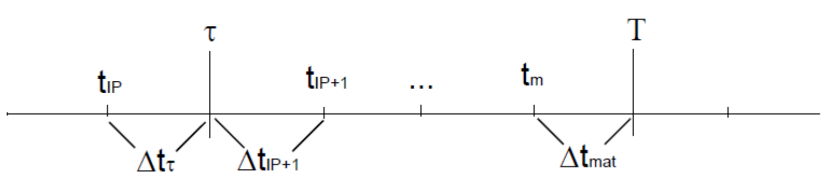
$df(t_0, t_0, t_k) \equiv df_k$ is the discount factor taken from the configured Discount Curve.

Finally:

$$\mathrm{Mat} = 1 + \frac{v_{\mathrm{IP}+1} \cdot \max(\Delta t_{\mathrm{IP}+1}, 0) \cdot df_{\mathrm{IP}+1} + \sum_{k=\mathrm{IP}+2}^{m} v_k \cdot \Delta t_k \cdot df_k + v_m \cdot \Delta t_{mat} \cdot df_T}{\sum_{k=1}^{\mathrm{IP}} \mathrm{Ratchet}(v_k) \cdot \Delta t_k \cdot df_k + \mathrm{Ratchet}(v_{\mathrm{IP}}) \cdot \Delta t_\tau \cdot df_\tau}$$

Notes

1. If the denominator is equal to 0, MAG uses the formula: *EM = max(min(1, b), c)*.
2. If $T < \tau$, MAG uses the formula: *EM = max(min(1, b), c)*.
3. Terms $V_{\mathrm{IP}+1} \cdot \max(\Delta t_{\mathrm{IP}+1}, 0) \cdot df_{\mathrm{IP}+1}$, $V_m \cdot \Delta t_{mat} \cdot df_T$ and $\mathrm{Ratchet}(V_{\mathrm{IP}}) \cdot \Delta t_\tau \cdot df_\tau$ in the Mat formula aims to address the time periods Δt_{IP} , $\Delta t_{\mathrm{IP}+1}$ and Δt_{mat} illustrated in the following figure when neither `Input Period` nor `Time to Maturity` falls on simulation time steps.



If there is no simulation time step after t_{IP}, $\Delta t_{\mathrm{IP}+1}$ is negative thus $V_{\mathrm{IP}+1} \cdot \max(\Delta t_{\mathrm{IP}+1}, 0) \cdot df_{\mathrm{IP}+1} = 0$.

If both `Input Period` and `Time to Maturity` fall on simulation time steps, the Mat formula is reduced to:

$$\mathrm{Mat} = 1 + \frac{\sum_{k=\mathrm{IP}+1}^{m} v_k \cdot \Delta t_k \cdot df_k}{\sum_{k=1}^{\mathrm{IP}} \mathrm{Ratchet}(v_k) \cdot \Delta t_k \cdot df_k}$$

The MtF cube contains the values of discount factors df_k for scenario time steps. The values df_M and df_{IP} for the maturity and estimation period are interpolated or extrapolated. Let *t* be either the maturity or the estimation period. If this time is between two time steps $t_{p-1} \leq t \leq t_p$, MAG estimates the discount factor by linearly interpolating the interest rates:

$$df_t = df_p \cdot \exp\left(-r_p \cdot \left(t - t_p\right)\right)$$

where:

$$r_p = -\frac{\ln\left(df_p\right) - \ln\left(df_{p-1}\right)}{t_p - t_{p-1}}$$

If time t is greater than the last time step t_K, MAG extrapolates the interest rates as a constant,

$$df_t = df_K \cdot \exp(-r_K \cdot (t - t_K))$$

$$r_K = -\frac{\ln(df_K) - \ln(df_{\mathrm{K}-1})}{t_K - t_{\mathrm{K}-1}}$$

Notes

1. If effective maturity is applied to another risk measure that does not depend on individual scenarios (for example, `@mean` or `@standard deviation`), the base scenario discount factors are used.

Case 2: Original Term to Maturity ≤ a

```
IF (the Netting Set is a deal)
    IF (the deal's Reg Short Term Exposure Flag*1 = 1)
        EM = max(min(μ, b), d)
    ELSE
        EM = max(min(μ, b), c)
ELSE
        EM = max(min(μ, b), FLOOR)
```

The value of *FLOOR* is determined as follows:

```
IF (FloorOverride is populated)
      FLOOR = FloorOverride
ELSE
      IF (the node is a CSA node)
            FLOOR is calculated using the table below
      ELSE IF (there are subnodes under the node)
            FLOOR = maximum of the floors of all subnodes
      ELSE
            FLOOR = 1 year
```

The *FLOOR* value at CSA level is defined as follows:

Fully Collateralized	Call Period = 1 AND Liquid Collateral	CSA Maturity FLOOR
TRUE	TRUE	1 day
TRUE	FALSE	MPOR + CallPeriod - 1
FALSE	*	1 year

Note

In the above table, Fully Collateralized = True if the `RegulatoryFullyCollateralized` attribute of a CSA node is set to True, and `CallPeriod` and `LiquidCollateral` are CSA level attributes. For the definition and calculation of MPOR, refer to 4.1.2.2 Call Dates on page 33.

Following is how μ is estimated.

```
IF (Weighted Average Remaining Maturity = true)
    μ is the notionally weighted remaining term to maturity measured in years
```

$$\mu = \frac{\sum_{\text{position}_i} \text{Notional}_i \times \mu_i}{\sum_{\text{position}_i} \text{Notional}_i}$$

where μ_i is the remaining term to maturity of deal *i*. The remaining term to maturity of a deal *i* is defined as follows:

```
        IF (Session Date ≤ Maturity Date)
            remaining term to maturity = Maturity Date - Session Date,
        ELSE
            remaining term to maturity = 0.
```

```
ELSE
    μ is the notionally weighted effective maturity measured in years
```

$$\mu = \frac{\sum_{\text{position}_i} \text{Notional}_i \times \mu_i}{\sum_{\text{position}_i} \text{Notional}_i}$$

where μ_i is the `regulatoryEffectiveMaturity` of deal *i*, which is provided by user.

Note

If user fails to provide `regulatoryEffectiveMaturity` for a deal, the remaining term to maturity of this deal will be used instead.

- **Effective Maturity of A Deal**

If Effective Maturity is required at deal level, the user can associate the function with relevant LE and set the attribute `Deal` to `True` in ExpressionInstances.csv. Then Effective Maturity will be evaluated for each deal under the LE as follows:

```
IF (Evaluate All Deals = False && the Deal is under a Netting Set)
             EM of the Deal = 0
ELSE
      EM of the Deal is calculated as Effective Maturity on A Netting Set
```

*1 The default value of `Reg Short Term Exposure Flag` is 0.

6.17 Time Weight

The input parameter of this function (`Time Buckets in days`) is one or several time periods (buckets) defined as pairs, each with start day and end day separated by comma and contained in parentheses. The function is applied to a vector of time-dependent values and returns a value for each input period. Time weight is similar to time averaging of the vector over an input period. The default time bucket is (0,365).

For a vector:

$$V = \left(V_{t_0}, \ldots, V_{t_K}\right)$$

with simulation dates $t_0, \ldots, t_K$ and for an input period with the start day t_s and the end day t_e time weight is defined as a sum.

$$\text{TimeWeight}(t_s, t_e) = \sum_{k=k_s}^{k_e} V_{t_k} \cdot \Delta t_k$$

where:

- $k_s = \min(k, t_k > \max(t_s, t_0))$
- if $t_n \geq \min(t_e, T)$, $k_e = \min(k, t_k \geq \min(t_e, T))$, else $k_e = k_n$
- T is the `Maturity Date` of a position, or the longest-dated transaction in a netting node (aggregated netting level),
- $T_P = \min(T, t_e) - \max(t_s, t_0)$ is the time span of the vector,
- $\Delta t_{k_s} = \left(t_{k_s} - \max(t_s, t_0)\right) / T_P$,
- if $t_n \geq \min(t_e, T)$, $\Delta t_{k_e} = (\min(t_e, T) - t_{k_e - 1}) / T_P$, else $\Delta t_{k_e} = (\min(t_e, T) - t_{n-1}) / T_P$
- $\Delta(t_k) = (t_k - t_{k-1}) / T_P$ for $k_s < k < k_e$
- in cases when both t_s and t_e fall within the same time step, $k_s = k_e$ and $\Delta t_k = 1$.

Notes

1. If $T \leq t_s$, *timeweight* = 0
2. If t_s, or t_e, or T do not fall on a simulation date, then take the vector value from next simulation date.
3. If there is no simulation date on or after t_e or T, then take the vector value from the preceding simulation date (that is, last simulation date)

6.18 Time Average

The function is applied to a vector of time-dependent values and returns a value. The function has no input parameters. Time Average performs time averaging of the vector over all time steps.

For a vector:

$V = \left(V_{t_0}, \ldots, V_{t_K}\right)$

with simulation dates $t_0, \ldots, t_K$ Time Average is just the mean value over all tine steps:

$$\text{TimeAverage} = \frac{1}{\text{K+1}} \sum_{k=0}^{K} V_{t_k}$$

6.19 Ratchet

This simulation function is applied to a vector of data through time and generates another vector whose values cannot decrease through time:

$ratchet_{t_{\text{SIM}, i}} = \max\left(ratchet_{t_{\text{SIM}, i-1}}, Vector_{t_{\text{SIM}, i}}\right)$

where $Vector_{t_{\text{SIM}, i}}$ is the number obtained for time $t_{SIM, i}$. For example, if the *Vector* contains exposure data across time, then $\text{Vector}_{t_{\text{SIM}, i}}$ is the exposure at time $t_{SIM, i}$.

By applying this function to the vector, we have $ratchet_{t_{\text{SIM}, i}} \geq ratchet_{t_{\text{SIM}, i-1}}$

6.20 FVA/FCA/FBA/MVA

- **Introduction**

A portfolio of derivatives that is not fully collateralized is hedged by positions with other counterparties where perfect collateral agreements [*1] are in place. Posting collateral for this hedging portfolio gives rise to a funding cost and receiving collateral a funding benefit. The function, Funding Valuation Adjustment (FVA), allows the user to calculate the funding cost/benefit at the deal, agreement level (CSA/MA), legal entity level, and funding set level.

Note

The segregation of collateral is ignored when computing FVA. Collateral is segregated to protect its owner from losing it in case the receiver of the collateral defaults. Hence the segregation of collateral impacts credit exposure measures but it does not impact funding cost measures.

A *funding set* is defined to be the set of all trades entered by an OwnSide Legal Entity. In order to compute FVA for a funding set, an aggregation node containing all Counterparties must be defined. Then the FVA of this aggregation node and an OwnSide Legal Entity will be the required FVA of the OwnSide funding set.

To calculate the FVA for a group node, when `DiversificationStyle` associated with the function is `ALL`, MAG aggregates the per scenario FVA of legal entities below that node to reach the final results of the node

- **Input**

The following inputs relevant to the calculation of FVA need to be specified in the relevant configuration and CSV files.

$\mathbf{FVF_{funding}}$

Future Value Factor (also known as FVF) converted from funding cost/benefit spread curve by RiskWatch before fed into MAG. Both of these curves are set up via custom attributes that are defined at the own legal entity level.

DF

Discount Factor converted from discount curve by RiskWatch before fed into MAG.

Simulation Currency

All curves above (DF and FVF) must be in the simulation currency.

Reporting Currency

Used to spot convert the results from simulation currency.

$VM^{X}_{cpty,\ \mathrm{reuse}}(t_i, s_j)$

All re-useable variation margin on or underneath the node *X*. The re-usability of variation margin is specified by collateral agreement parameter *Rehypothecation*.

$VM^{X}_{user}(t_i, s_j)$

All variation margin posted by the user on or underneath the node *X*.

$IM_c^X(t_i, s_j)$

All initial margin posted by *C* at the node *X* and at all the children of the node.

MAG function FVA takes the following parameters:

- `Include User Initial Margin Cost`: True, or False (default)
- `Conditional On Counterparty Survival`: True (default), or False
- `Conditional On Own Survival`: True (default), or False
- `PD Sheet Type`: Deterministic (default), or Stochastic
- `Use Stochastic PD` : True (default), or False
- `Collateralized Value Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `Survival Probability Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `Discount Factor Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `FVA Per Time`: True, or False (default)
- `Confidence Interval`: default value = 0
- `Create Audit Files`: True, or False (default)

• Preliminary Materials

Bucketing

For FVA calculation, user can choose which collateralized value, survival probability and discount factor of the time interval (t_{j-1},t_j) to be used in the calculation by setting up the 3 function parameters: `Collateralized Value Bucketing`, `Survival Probability Bucketing`, and `Discount Factor Bucketing`.

When `Start of Interval` is selected for `Survival Probability Bucketing`, the probability of survival of the time interval (t_{i-1},t_i) is taken from the beginning of the interval. That is,

$$\text{StartofInterval} = SP\left(t_0, t_{i-1}, s_j\right),$$

Similarly,

$$\text{EndofInterval} = SP\left(t_0, t_i, s_j\right)$$

$$\text{MidofInterval} = \frac{SP\left(t_0, t_{i-1}, s_j\right) + SP\left(t_0, t_i, s_j\right)}{2}$$

When `Start of Interval` is selected for `Collateralized Value Bucketing`, the collateralized value of the time interval (t_{i-1},t_i) is taken from the beginning of the interval. That is,

$$\text{StartofInterval} = CV\left(t_{i-1}, s_j\right),$$

Similarly,

$$\text{EndofInterval} = CV\left(t_i, s_j\right)$$

$$\text{MidofInterval} = \frac{CV\left(t_{i-1}, s_j\right) + CV\left(t_i, s_j\right)}{2}$$

When `Start of Interval` is selected for `Discount Factor Bucketing`, the discount factor of the time interval (t_{i-1},t_i) is taken from the beginning of the interval. That is,

$$\text{StartofInterval} = DF\left(t_0, t_{i-1}, s_j\right),$$

Similarly,

$$\text{EndofInterval} = DF(t_0, t_i, s_j)$$

$$\text{MidofInterval} = \frac{DF(t_0, t_{i-1}, s_j) + DF(t_0, t_i, s_j)}{2}$$

For the simplicity of illustration, in the remaining part of this section, it is assumed that `Survival Probability Bucketing` = `Start of Interval`, `Collateralized Value Bucketing` = `Start of Interval` and `Discount Factor Bucketing` = `End of Interval` in all formulas related to FVA calculation. However, remember that it can be replaced by other formats of survival probability, collateralized value and discount factor as shown above depending on the setting of the 3 configuration variables individually.

Simulated Collateralized Value for FVA

The simulated collateralized value when computing FVA, FCA and FBA (ignoring the addon piece) is

$$CV(t_i, s_j)^X_{sim} = MtM^X(t_i, s_j) + VM^X_{user}(t_i, s_j) - VM^X_{cpty,\ reuse}(t_i, s_j) + a\ IM^X_{user}(t_i, s_j)\ ,^{*2}$$

where $MtM^X(t_i, s_j)$ is the netted mark-to-market of all simulated trades under node *X*;

$$a = \begin{cases} 1 \ \text{if } IncludeUserIMCost = \text{True} \\ 0 \qquad\qquad \text{otherwise} \end{cases}$$

Important

We assume that all the variation and initial margins defined here are positive.

Note

In current implementation, IM doesn't change across scenarios. However, when IM currency is different from simulation currency, IM value needs to be converted to simulation currency using FX which changes over time and scenarios.

Survival Probability

The probability of survival of the counterparty/user between time t_0 and t_i under scenario s_j is calculated as:

$$SP_{cpty}(t_0, t_i, s_j) = \begin{cases} 1 - PD_{cpty}(t_0, t_i, s_j) & if \ \ ConditionalOnCptySurvival = True \\ 1 & if \ \ ConditionalOnCptySurvival = False \end{cases}$$

$$SP_{user}(t_0, t_i, s_j) = \begin{cases} 1 - PD_{user}(t_0, t_i, s_j) & if \ \ ConditionalOnOwnSurvival = True \\ 1 & if \ \ ConditionalOnOwnSurvival = False \end{cases}$$

where $PD_{cpty/user}(t_0, t_i, s_j)$ is the probability of default of the counterparty/user between time t_0 and t_i under scenario s_j and $PD_{cpty/user}(t_0, t_i, s_j) = \sum_{k=1}^{i} PD_{cpty/user}(t_{k-1}, t_k, s_j)$ where $PD_{cpty/user}(t_{k-1}, t_k, s_j)$ is the marginal probability of default between t_{k-1} and t_k under scenario s_j viewed from time t_k.

Determine PD Value

When PD value is needed in FVA/FCA/FBA calculation (that is, `Conditional On Counterparty Survival` is True or `Conditional On Own Survival` is True), MAG uses the following logic to determine which PD value to be applied.

```
If (PD sheet Type = Stochastic and Use Stochastic PD = True)
    use stochastic PD
Else use base scenario PD
```

- **Funding Valuation Adjustment Formula**

FVA Per Time = False, and, Confidence Interval = 0

The FVA of funding set is calculated as:

$$FVA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{\big[\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)\big] \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$

where

$FVF_{funding}(t_{i-1}, t_i, s_j)$ is obtained from the funding cost spread curve if $\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)$ is positive and from the funding benefit spread curve if $\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)$ is negative.

Similarly, the FVA of a node *X*, which can be a legal entity or a node under a legal entity, is calculated as:

$$FVA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{CV_{FVA}^{X}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j) \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$

where

$FVF_{funding}(t_{i-1}, t_i, s_j)$ is obtained from the funding cost spread curve if $CV_{FVA}^{X}(t_{i-1}, s_j)$ is positive and from the funding benefit spread curve if $CV_{FVA}^{X}(t_{i-1}, s_j)$ is negative.

When computing FVA on node *X*, collateralized value (i.e. $CV_{FVA}^{X}(t_{i-1}, s_j)$ in FVA formulae) is defined as: $CV_{FVA}^{X}(t_{i-1}, s_j) = CV_{sim}^{X}(t_{i-1}, s_j) + CV_{non-sim}^{X}(t_{i-1}, s_j)$ *3*4

where $CV_{non-sim}^{X}(t_{i-1}, s_j)$ = the sum of the collateralized value profiles of the non-simulated trades under node *X*.

When computing FVA at the deal level, collateral is ignored and only the mark-to-market of the deal is used if the deal is simulated or a trade or product profile is used if the deal is non-simulated.

Total funding value adjustment consists of funding cost adjustment (FCA) and funding benefit adjustment (FBA):

$$FVA = FCA + FBA$$

FVA Per Time = True, and, Confidence Interval = 0

When `FVA Per Time` is set to `True`, FVA function returns the FVA value for each simulation time step. That is:

$$FVA\_per\_Time(t_{i-1}) = \frac{1}{N}\sum_{j=1}^{N}\Big\{\big[\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)\big] \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$ for funding set, and

$$FVA\_per\_Time(t_{i-1}) = \frac{1}{N}\sum_{j=1}^{N}\Big\{CV_{FVA}^{X}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j) \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$ for a node.

FVA Per Time = False, and, Confidence Interval ≠ 0

When `Confidence Interval` is set to a number between 0 and 1 but other than 0, FVA function returns the confidence interval of FVA. Given FVA is an average over scenarios either at time 0 or for each simulation time step, the confidence interval is calculated using the following formula:

$\delta = \sqrt{VAR(\overline{X})} \times \Phi^{-1}(\frac{1+C}{2})$

where, $VAR(\overline{X}) = \frac{1}{N(N-1)}\sum_{j=1}^{N}(X_j - \overline{X})^2$;

$\overline{X}$ is the FVA value returned by FVA function when `Confidence Interval` is set to 0; and

X_j is the FVA value per scenario, i.e.

$$X_j = \sum_{i=1}^{M}\Big\{[\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)] \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$ for funding set; and

$$X_j = \sum_{i=1}^{M}\Big\{CV_{FVA}^{X}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j) \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$ for a node.

Note

`FVA Per Time` and `Confidence Interval` cannot be simultaneously set to a value other than default. If user sets `FVA Per Time` to `True` and `Confidence Interval`to a number other than 0 at the same time, MAG reports an error.

- **FCA**

FCA is borrowing by the trading desk to the funding desk.

When `FCA Per Time` is set to `False` and `Confidence Interval` = 0, FCA of a funding set is defined as:

$$FCA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{[\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)]^{+} \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$

where $x^{+} = \max(x, 0)$.

Similarly, the FCA of a node *X*, which can be a legal entity or a node under a legal entity, is calculated as:

$$FCA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{[CV_{FVA}^{X}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)]^{+} \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$

When `FCA Per Time` is set to `True` or `Confidence Interval` is other than 0, the calculation is similar to the calculation of FVA.

- **FBA**

FBA is lending by the trading desk to the funding desk.

When `FBA Per Time` is set to `False` and `Confidence Interval` = 0, FBA of a funding set is defined as:

$$FBA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{[\sum_{cpty} CV_{FVA}^{cpty}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)]^{-} \times \times Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$

where $x^{-} = \min(x, 0)$.

Similarly, the FBA of a node *X*, which can be a legal entity or a node under a legal entity, is calculated as:

$$FBA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{[CV^{X}_{FVA}(t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j)]^{-} \times$$
$$Ln[FVF_{funding}(t_{i-1}, t_i, s_j)] \times DF(t_0, t_i, s_j) \times SP_{user}(t_0, t_{i-1}, s_j)\Big\}$$

When `FBA Per Time` is set to `True` or `Confidence Interval` is other than 0, the calculation is similar to the calculation of FVA.

- **MVA**

Most of MVA functionality is the same as the above FVA functions.

MAG function MVA takes the following parameters:

- `Conditional On Counterparty Survival`: True (default), or False
- `Conditional On Own Survival`: True (default), or False
- `PD Sheet Type`: Deterministic (default), or Stochastic
- `Use Stochastic PD` : True (default), or False
- `Survival Probability Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `Discount Factor Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `Initial Margin Bucketing`: Start of Interval (default), or Mid of Interval, or End of Interval
- `Confidence Interval`: default value = 0
- `Create Audit Files`: True, or False (default)

MVA would be calculated at agreement level (CSA/MA) and legal entity level (MVA is not computed at deal level).

Segregation of collateral is ignored when computing MVA. Hence the segregation of collateral impacts credit exposure measures but it does not impact funding cost measures.

MVA is defined as:

$$MVA = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{M}\Big\{IM_{user}\big(t_{i-1}, s_j\big) \times SP_{user}(t_0, t_{i-1}, s_j) \times SP_{cpty}(t_0, t_{i-1}, s_j) \times DF(t_0, t_i, s_j)$$
$$Ln[FVF_{funding}(t_{i-1}, t_i, s_j)\Big\}$$

where Initial Margin is obtained from IM time profile and IM sheet .

*1 By a perfect collateral agreement, we mean one where there are no Thresholds, no MTAs and margin calls are daily, and so on.

*2 For the purpose of FVA, some clients want to assume the lags are zero when computing the variation margin. In the Path Dependent model, this can be achieved by setting the collateral model parameter `Without Lags` to True. The Empirical Shortcut method with lags of zero is simply the Path Dependent algorithm and hence the latter should be used instead.

*3 When computing Collateralized Value for the purpose of FVA, the setup in `AgreementsEnforceable` and `CloseoutNetting` is ignored.

*4 For the purpose of FVA, the Default Remote Upfront Amount should be set to False in the collateral model.

6.21 Regulatory Capital & Risk Weighted Assets

The Regulatory Capital function and the Risk Weighted Assets function compute Basel II IRB Regulatory Capital and its corresponding Risk Weighted Assets (RWA) respectively.

Definition of Netting Set

A netting set for the purpose of calculating capital is defined in Annex 4 of Basel II. In this implementation, a netting set is assumed to be:

- A Master Agreement for which both the AgreementEnforceable flag and the CloseoutNetting flag are set to True;
- An individual deal under a Master Agreement for which either the AgreementEnforceable flag or the CloseoutNetting flag is set to False; or
- An individual deal under a non-netting set.

If regulatory capital or risk weighted assets values are requested for node below a netting set, regulatory capital and risk weighted assets are calculated as if the node were a netting set. In this case, the AgreementEnforceable flag and the CloseoutNetting flag still apply.

Regulatory Capital (Cap) on a Netting Set

Inputs

Basel multiplier (α)

Default value is 1.4.

Probability of Default (PD)

This is the average percentage of obligors that default in a given rating band over one year. It is different from the LE-specific PD used in other functions such as CVA.

Loss Given Default (LGD)

This is obtained from the existing LE recovery rate. LGD = 1 - Recovery Rate.

Effective Maturity (EM)

This is calculated internally. For the algorithm of EM, refer to 6.16 Effective Maturity on page 87.

Exposure at Default (EAD)

This is defined to be the effective Expected Positive Exposure (eEPE) over a one year period times the Basel multiplier α , where eEPE is equal to a composition of existing MAG functions:

$eEPE = TimeWeight(0, 365)[Ratchet(ExpectedExposureOverTime)]$,

where $ExpectedExposureOverTime$ is the sum of Add-on (if any) and Simulated Expected Exposure:

- Add-on: the Expected Exposure add-on vector;
- Simulated Expected Exposure (i.e. EE_i) = $\frac{1}{N}\sum_{j=1}^{N} Max\left(Exposure(t_i, s_j), 0\right)$, where $Exposure(t_i, s_j)$ is the exposure value on the netting set at simulation time step t_i for scenario s_j and *N* is the total number of scenarios.

Formula

Regulatory Capital is calculated as:

$Cap = K \times EAD = K \times eEPE \times \alpha$

where

$$K = \frac{\left(LGD \times N(\frac{G(PD)}{\sqrt{1-R}} + \frac{R}{\sqrt{1-R}} \times G(0.999)) - PD \times LGD\right) \times \left(1 + (EM - 2.5) \times b\right)}{1 - 1.5 \times b}$$

$$R = 0.12 \times \frac{1 - e^{-50 \times PD}}{1 - e^{-50}} + 0.24 \times \left(1 - \frac{1 - e^{-50 \times PD}}{1 - e^{-50}}\right)$$

$$b = (0.11852 - 0.05478 \times \ln(PD))^2$$

N is the Standard Normal Cumulative Distribution Function and G is its inverse.

- **Regulatory Capital (Cap) above a Netting Set**

Regulatory Capital for a node that is above a netting set is defined as the sum of the Regulatory Capital of all Netting Sets below it.

- **Risk Weighted Assets (RWA)**

Inputs

Capital Adequacy Ratio (CAR)

The default value is 8%.

Regulatory Capital

Calculated as described above.

Formula

$$RWA = \frac{1}{CAR} \times Cap$$

6.22 Expected Shortfall (Market)

Input parameter:

- `Confidence Interval` α : default value = 0.95

ES (Expected Shortfall) is the average of all losses that meet or exceed the VaR. For this reason, it is also known as "conditional VaR" or "tail conditional expectation". Thus, like VaR, it is a quantile-based risk measure that requires a confidence level to be specified before it can be calculated.

The non-parametric expected shortfall at the $100(1-\alpha)\,\%$ level is defined to be

$$nES_\alpha = E[\text{loss}|\text{loss} \geq nVaR_\alpha]$$

Note

1. The loss L_i will be the difference between the reference value V_{ref} (i.e. the base scenario value at time zero) and scenario value V_i at scenario *i* $(i = 1, ..., n)$.
2. For information about how non-parametric VaR at confidence interval α ($nVaR_\alpha$) is calculated, refer to 6.23 Non-Parametric Value at Risk (VaR) on page 101.

Let p_i be the probability of scenario *i* $(i = 1, ..., n)$, and denote the size of the loss in scenario *i* by L_i (it is assumed that losses are positive while gains are negative). Let *y* be a permutation of the scenarios so that $L_{y(k)} \geq L_{y(k+1)}$ for $k = 1, 2, ..., n{-}1$ (that is, $y(1)$ and $y(n)$ are the scenarios with the largest and smallest losses, respectively). Let k^α satisfy

$$\sum_{k=1}^{k^{\alpha}-1} p_{y(k)} < (1-\alpha) \quad \text{and} \quad \sum_{k=1}^{k^{\alpha}} p_{y(k)} \geq (1-\alpha)$$

Then nES_{α} is calculated as follows:

$$nES_{\alpha} = \frac{1}{1-\alpha}\left[\left(\sum_{k=1}^{k^{\alpha}-1} p_{y(k)} L_{y(k)}\right) + \left(1-\alpha-\sum_{k=1}^{k^{\alpha}-1} p_{y(k)}\right) L_{y\left(k^{\alpha}\right)}\right]$$

6.23 Non-Parametric Value at Risk (VaR)

Input parameters:

- `Confidence Interval` α : default value = 0.95
- `Confidence Interval Option`: One Sided (default value), Two Sided
- `Relative To`: Scenario Index, Base Scenario. Mean (default value)
- `Comparison`: Difference (default value), Difference %, Ratio, Log ration
- `Marked Scenario`: default value = 0

This function calculates a non-parametric VaR estimate, denoted by $\mathrm{VaR}_{np}^{h}(\alpha, ref)$, as

$$\mathrm{VaR}_{np}^{h}(\alpha, ref) = \begin{cases} V_L[(1-\alpha)] - V_{ref} & if \;\; ConfidenceIntervalOption = OneSided \\ V_L[(1-\alpha)/2] - V_{ref} & if \;\; ConfidenceIntervalOption = TwoSided \end{cases}$$

Here, h is the number of units of the deal if the function is applied to an individual deal and, in this case, each V_i is the deal's value under scenario *i*. If `@value at risk non-parametric` is applied to a portfolio, h is the vector of the portfolio's deal units, and in this case, each V_i is the portfolio value under scenario *i*.

For any given percentile $0 < \beta < 1$, $V_L[\beta]$ is the largest value such that the sum of the normalized weights of the values less than it, is less than β. That is

$$V_L[\beta] = \max\left\{V_k;(1 \leq k \leq n), \left(\sum_{j=1}^{k-1} w_j / W\right) < \beta\right\}$$

The calculation is based on the following approach. Let *V* be the value of a position or a portfolio at a future time (for example, next day) and assume that it can take any one of the values from V_i, $i = 1, 2, ..., n$. Since scenario *i* has a weight w_i, we can assign each value V_i a probability w_i / W . This gives a probability measure Pr such that $\mathrm{Pr}\{V_i\} = w_i / W$, and for any numbers $a < b$, (they can possibly be $\pm\infty$)

$$\mathrm{Pr}\{a < V < b\} = \left(\sum_{a < V_j < b} w_j\right) / W$$

Let h be the units (holdings) of the position. Given a confidence level 0 < α < 1 and a reference value V_{ref}, the non-parametric VaR, $\mathrm{VaR}_{np}^{h}(\alpha, ref)$, is the largest number among the V_i s that satisfies

$$\mathrm{Pr}\left\{\mathrm{V} - V_{ref} < \mathrm{VaR}_{np}^{h}(\alpha, ref)\right\} < \frac{(1-\alpha)}{2}$$

which is, equivalently

$$\mathrm{Pr}\left\{V < \mathrm{VaR}_{np}^{h}(\alpha, ref) + V_{ref}\right\} < \frac{(1-\alpha)}{2}$$

Recall that if `ConfidenceIntervalStrict` = `False`, $V_L[(1-\alpha)/2]$ is the largest value among all V_is, such that the sum of probabilities of V_j s, $j < i$, is less than $(1-\alpha)/2$. $V_L[(1-\alpha)/2]$ is calculated. The definition of the probability measure $\Pr\{\}$ implies that

$$\Pr\{V < V_L[(1-\alpha)/2]\} = \left\{\sum_{j\in J} w_j\right\}/W < \frac{(1-\alpha)}{2}$$

where the index set J consists of all indices *j* such that $V_j < V_L[(1-\alpha)/2]$. From the last two inequalities, it follows that

$$\mathrm{VaR}^h_{np}(\alpha, ref) + V_{ref} = V_L[(1-\alpha)/2]$$

or

$$\mathrm{VaR}^h_{np}(\alpha, ref) = V_L[(1-\alpha)/2] - V_{ref}$$

The simulation function `@value at risk non-parametric` returns $\mathrm{VaR}_{np}(\alpha, ref)$ using the formula above. The formula can also be written as

$$\mathrm{VaR}^h_{np}(\alpha, ref) = LT_{np}(\alpha) - V_{ref}$$

where $\mathrm{LT}_{np}(\alpha)$ is the non-parametric left tail corresponding to the confidence level α .

6.24 Relative

Input parameters:

- `Relative to`: Scenario Index, Base Scenario, Mean (default value)
- `Comparison`: Difference (default value), Difference %, Ratio, Log Ratio
- `Marked Scenario`: default value = 0

This function returns a relative value according to each option of `Comparison` as follows:

- `Comparison` = Difference

  $$\mathrm{Re}\,lative = V_i - V_{ref}$$

- `Comparison` = Difference %

  $$\mathrm{Re}\,lative = \frac{V_i - V_{ref}}{|V_{ref}|}$$

- `Comparison` = Ratio

  $$\mathrm{Re}\,lative = \frac{V_i}{V_{ref}}$$

- `Comparison` = Log Ratio

  $$\mathrm{Re}\,lative = \begin{cases} \mathrm{Ln}\left(\frac{V_i}{V_{ref}}\right), & \text{if } \frac{V_i}{V_{ref}} \geq 0; \\ N/A, & otherwise \end{cases}$$

  Note

  Ln denotes the natural logarithmic function.

Here, V_i is the scenario value for scenario *i*; V_{ref} is the reference value to which the scenario value is relative. V_{ref} is determined by the setup in parameter `Relative to` as follows:

- `Relative to` = Scenario Index

$V_{ref} = V_m$, where is the value scenario *m* which is specified by parameter `Marked Scenario`.

- `Relative to` = Base Scenario

  $V_{ref} = V_{Base}$

- `Relative to` = Mean

  $V_{ref} = \sum_{i=1}^{n} V_i / n$

6.25 CEM

The function `CEM` computes the credit exposure at default for OTC transactions based on the Basel Current Exposure Method.

● Input Parameters

- OverrideNGR_NN: accepts 0 or 1. If set to 1, the net-to-gross ratio (NGR) of a no net node is forced to 1.
- CEM_Ratio: defines the beta parameter in the add-on calculation. Defaults to 0.4.
- NettingRule: accepts NO_NET, AGREEMENT and LEGAL_OPINION.
- TableColumn: A lookup key that is associated with table CEMAddonFactors.

● CEM

The Current Exposure Method (CEM) capital is given by the sum of the net mark-to-market current exposure plus an add-on based on the effective notional of the transactions. It is defined on the following nodes of legal hierarchy:

- OTC
- NN
- MA
- CSA
- NC

At OTC node:

$$CEM_{OTC} = \sum_{\text{X under OTC}} CEM_X$$

At X = NN, MA, CSA, or NC node:

$$CEM_X = CurrentExposure_X + AddOn_X$$

where:

- $CurrentExposure_X$ is defined in section ● Current Exposure on page 104.
- $AddOn_X$ is defined in section ● Add-On on page 105.
- NN stands for a no-net node, which could be:
  - an OTC node where MA does not exist;
  - an NN_OTC node;
  - an MA or CSA node under NettingRule = `NO_NET`;
  - an MA or CSA node under NettingRule = `AGREEMENT` and `CloseoutNetting` = False;

- an MA or CSA node under NettingRule = `LEGAL_OPINION`, under which closeout netting pool does not exist and `AgreementsEnforceable` or `CloseoutNetting` is set to False.

- **Netting Sets**

A netting set is:

1. An NC, CSA or MA node if all trades in the node net. This is when:
   - NettingRule = `AGREEMENT` and `CloseoutNetting` = TRUE; or
   - NettingRule = `LEGAL_OPINION` and either
     - no closeout netting pools are present and `AgreementsEnforceable` = TRUE and `CloseoutNetting` = TRUE; or
     - all trades in the node belong to a unique non-empty closeout netting pool or there is a single trade in the node; or
   - NettingRule = `NO_NET` and there is a single trade in the node.
2. When not all trade in an NC, CSA or MA node net, the node is subdivided accordingly and each subset forms a netting set.
   - NettingRule = `AGREEMENT` or `NO_NET`: each trade forms a netting set
   - NettingRule = `LEGAL_OPINION`: each closeout netting pool forms a netting set and for trades that don't belong to any closeout netting pools, each trade forms a netting set.
3. A set containing a single trade of the NN node forms a netting set.

- **Current Exposure**

Current Exposure of node X is calculated as follows:

$CurrentExposure_X = Max(MtM_X - C_X, 0)$

where:

C_X denotes the current collateral posted by the counterparty minus the unsegregated collateral posted by the user, converted to domestic currency. Only the node mitigants (IM and VM) and t0 value from the IM profiles are included.

MtM_X is calculated based on the `NettingRule`:

- NettingRule = NONET:
  $MtM_X = \sum_{trade \in X} \max(MtM_{trade}, 0)$
- NettingRule = AGREEMENT:
  - If `CloseoutNetting` = True:
    $MtM_X = \sum_{trade \in X} MtM_{trade}$
  - Else:
    $MtM_X = \sum_{trade \in X} \max(MtM_{trade}, 0)$
- NettingRule = LEGAL_OPINION
  - If Closeout Netting Pools are present under the MA:
    $MTM_X = \sum_{NS \in X} Max(\sum_{trade \in NS} MtM_{trade}, 0)$
  - Else:
    - If `AgreementEnforceable` = True and `CloseoutNetting` = True:

$MtM_X = \sum_{trade \in X} MtM_{trade}$

- Else:

  $MtM_X = \sum_{trade \in X} \max(MtM_{trade}, 0)$

where NS stands for a netting set as defined section • Netting Sets on page 104 and MtM_{trade} is the mark-to-market of trade in domestic currency (of the legal entity).

• Add-On

The trade level add-on, denoted by $AddOn_{trade}$, represents the future credit exposure of the trade. It is calculated by multiplying the effective notional of the trade by the regulatory prescribed percentage presenting the volatility of the trade.

$AddOn_{trade} = N_{trade} \cdot SF_{trade}$

where N_{trade} is the effective notional and SF_{trade} is the volatility factor looked up from table `CEMAddonFactors` using the trade's product class and remaining maturity. The product class and remaining maturity are associated with a factorLookup key.

The gross add-on for node X is defined as:

$AddOn_X^{GROSS} = \sum_{trade \in X} AddOn_{trade}$

The net-to-gross ratio is defined as:

$$NGR_X = \begin{cases} 1, & \text{if} X \in NN \text{ and OverrideNGR\_NN=TRUE} \\ \frac{\max(MtM_X, 0)}{\sum_{trade \in X} \max(MtM_{trade}, 0)}, & \text{otherwise} \end{cases}$$

where

- NN stands for a no-net node as defined in section CEM.
- `OverrideNGR_NN` is an input function parameter.
- MtM_X is defined in section • Current Exposure on page 104.

Note

$\frac{\max(MtM_X, 0)}{\sum_{trade \in X} \max(MtM_{trade}, 0)} = 0$ if $\max(MtM_X, 0)$ is zero.

Finally, the node add-on is defined as:

$AddOn_X = \beta \cdot AddOn_X^{Gross} + (1 - \beta) \cdot NGR_X \cdot AddOn_X^{Gross}$

where β is read from the input function parameter `CEM_Ratio` and defaults to 40%.

• Total Exposure

Total Exposure of node X is calculated as follows:

$TotalExposure_X = CurrentExposure_X + AddOn_X^{Gross}$

where current exposure and gross add-on is defined in sections above.

• Reporting

The function `CurrentExposure_NS` reports the current exposure of node X , $CurrentExposure_X$.

The function `PotentialExposure_NS` reports the gross add-on of node X , $AddOn_X^{GROSS}$.

The function `TotalExposure_NS` reports the total exposure of node X , $TotalExposure_X$.

6.26 CEM Indicative Regulatory Capital & Risk Weighted Assets

These 2 functions use the same algorithm and formula as those in the computation of Basel II IRB Regulatory Capital and its corresponding Risk Weighted Assets (RWA). The only difference is that these 2 functions use effective maturity and EAD of Current Exposure Method as inputs to the formula.

The following are the common input parameters of CEM Indicative Regulatory Capital function and CEM Indicative RWA function with their default settings:

- `Maturity Floor Term Value` = 1
- `Maturity Floor Term Unit` = Year (default), Month, or Day
- `Short Term Exposure Maturity Floor Term Value` = 1
- `Short Term Exposure Maturity Floor Term Unit` = Day (default), Month, or Year
- `Weighted Average Remaining Maturity` = true (default) or false
- `Evaluate All Deals` = false (default) or true
- `PD Sheet Type` = Deterministic (default) or Stochastic

The following parameter is unique to CEM Indicative RWA function:

- `Capital Adequacy Ratio` = 0.08

Note

SFT deals and nodes are not in the scope of current implementation for these 2 functions.

- **CEM Indicative Regulatory Capital (Cap) on a Netting Set**

Inputs

Basel multiplier (α)

Default value is 1.4.

Probability of Default (PD)

This is the average percentage of obligors that default in a given rating band over one year. It is read from the cube. When `PD Sheet Type` = `Deterministic`, the value is read from the curve assigned to LE `CreditDefaultCurve` for the relevant rating. When `PD Sheet Type` = `Stochastic`, the value is read from the curve assigned to LE `DefaultModel` for the relevant rating.

Loss Given Default (LGD)

This is obtained from the existing LE recovery rate. LGD = 1 - Recovery Rate.

Effective Maturity (M)

This is the same EM input to CVA Capital CEM function. Refer to 6.6 CVA Capital CEM on page 82.

Exposure at Default (EAD)

This is defined to be the Total Exposure (TE) under Current Exposure Method times the Basel multiplier α . For the information about how to calculate Total Exposure under Current Exposure Method, see *Current Exposure Method functions and expressions* in the *Integrated Market and Credit Risk Solution Guide*.

Formula

Regulatory Capital is calculated as:

$$CEMindCap = K \times EAD = K \times TE \times \alpha$$

where

$$K = \frac{\left(LGD \times N(\frac{G(PD)}{\sqrt{1-R}} + \frac{R}{\sqrt{1-R}} \times G(0.999)) - PD \times LGD\right) \times \left(1 + (M - 2.5) \times b\right)}{1 - 1.5 \times b}$$

$$R = 0.12 \times \frac{1 - e^{-50 \times PD}}{1 - e^{-50}} + 0.24 \times \left(1 - \frac{1 - e^{-50 \times PD}}{1 - e^{-50}}\right)$$

$$b = (0.11852 - 0.05478 \times \ln(PD))^2$$

N is the Standard Normal Cumulative Distribution Function and G is its inverse.

- **CEM Indicative Regulatory Capital (Cap) above a Netting Set**

CEM Indicative Regulatory Capital for a node that is above a netting set is defined as the sum of the CEM Indicative Regulatory Capital of all Netting Sets below it.

- **CEM Indicative Risk Weighted Assets (RWA)**

Inputs

Capital Adequacy Ratio (CAR)

The default value is 8%.

CEM Indicative Regulatory Capital

Calculated as described above.

Formula

$$CEMindRWA = \frac{1}{CAR} \times CEMindCap$$

6.27 SACCR Capital Exposure At Default

The function `SACCR Capital Exposure At Default` computes the Exposure at Default based on the Basel SA-CCR methodology.

- **Input Parameters**

  - Alpha1: A parameter to scale the portfolio RC, defaults to 1.4
  - Alpha2: A parameter to scale the portfolio AddOn, defaults to 1.4
  - IRTenorOffsetting: A boolean attribute, defaults to True. Setting it to False disallows offsetting benefit between maturity buckets for Interest Rate derivatives.
  - MultiplierFloor: Sets the floor for AddOn multiplier, defaults to 0.05.
  - MultiplierFactor: Sets the factor for AddOn multiplier, defaults to 2.

- MaturityFactorMultiplier: Sets the multiplier for margined maturity factor, defaults to 1.5.
- NettingRule: accepts NO_NET, AGREEMENT and LEGAL_OPINION.
- SupervisoryDurationDF: Sets the discount rate in supervisory duration calculation, defaults to 0.05.
- IRShift: accepts PER_CURRENCY or PER_TRADE. It specifies whether the IR shift amount is currency dependent or trade dependent.
- CollateralModel: Sets the collateral model for the expression: NONE, NonSim_CollateralModel, etc.
- IncludeVMInRCUnmargined: An optional collateral model parameter, defaults to 1 (True).
- TableColumn: A lookup key that is associated with the following inputs: SACCRSupervisoryFactor, SACCROptionVolatility, SACCRCorrelations, SACCRIRCorrelations, SACCRFXHaircut, SACCRCCYPFEHaircuts and VolatilityAdjustments.

• Exposure at Default

The SACCR Capital Exposure at Default (EAD) is given by the product of the multiplier *alpha* and an estimate of the effective expected positive exposure (EEPE). The EEPE estimate captures the weighted average of expected exposure over the risk horizon and consists of two terms: replacement cost (RC) and potential future exposure (PFE). RC captures the loss that would occur if a counterparty were to default and were closed out of its transactions immediately or at a future time. PFE captures the expected increase in exposure during a given risk horizon.

Let X be the node/trade at which the base calculation of SACCR EAD is performed, where X equals one of the following:

1. A CSA node (VM or VMIM or Legacy)
2. An MA node with no CSA
3. An NC node
4. An NN trade

The abbreviations are defined in section • Legal Hierarchy on page 109. If X is such a node, the EAD is calculated according to the two cases below:

- **Case 1**: X consists of a single SA-CCR netting set (see section • Netting Sets on page 109)
- **Case 2**: X consists of multiple SA-CCR netting sets

At node X ,

$$EAD = a_1 \cdot RC + a_2 \cdot PFE$$

For any nodes above X in the legal hierarchy, the EAD is the sum of the EADs of its child nodes.

Here:

- a_1 and a_2 are expression parameters Alpha1 and Alpha2.
- The replacement cost RC is defined in topic 6.28 SACCR Capital Replacement Cost on page 111.
- The potential future exposure PFE is defined in topic 6.29 SACCR Capital Potential Future Exposure on page 121.

Note

Under case 1, the EAD for a margined set is capped by the EAD of the same portfolio without a margin agreement. For a margined set, the replacement cost and potential future exposure are computed twice, with margining and without margining.

- ## Legal Hierarchy

Each trade is allocated to a node in the legal hierarchy. The node is defined by:

- Counterparty (CP)
- OwnEntity (OE)
- Container: a master agreement (MA) or a no-net node (NN). Multiple MAs may be signed with a counterparty. Trades that do not meet the filtering criteria of any MA are allocated to an NN node.
- SubContainer (optional): a credit support annex (CSA) or a non-collateralized node (NC). There may be one CSA, no CSA, or multiple CSAs appended to an MA. If CSA exists under the MA, trades that meet the MA filtering criteria but not the criteria of any CSA under that MA are allocated to a residual "NC" node under the MA.

- ## Netting Sets

The SA-CCR netting set (referred to as *netting set* in subsequent sections) is a subset of a master agreement (MA) or a no-net node (NN). It is defined as follows:

1. A netting set is an NC node under the MA, a CSA node under the MA, or an MA node with no CSAs if all trades in the NC, CSA or MA node net. This is when:
   - NettingRule = `AGREEMENT` and `CloseoutNetting` = TRUE; or
   - NettingRule = `LEGAL_OPINION` and either:
     - no closeout netting pools are present under the MA and `AgreementsEnforceable` = TRUE and `CloseoutNetting` = TRUE; or
     - all trades in the node belong to a unique non-empty closeout netting pool or there is a single trade in the node; or
   - NettingRule = `NO_NET` and there is a single trade in the node.
2. When not all trade in an NC, CSA or MA node net, the node is partitioned accordingly and each subset forms a netting set.
   - NettingRule = `AGREEMENT` or `NO_NET`: each trade forms a netting set
   - NettingRule = `LEGAL_OPINION`: each closeout netting pool forms a netting set and for trades that don't belong to any closeout netting pools, each trade forms a netting set.
3. For NN node, a set containing a single trade of the NN node forms a netting set.

- ## Margined vs. Unmargined Set

The measurement of EAD is different depending on whether node X is margined or unmargined. If the expression parameter `CollateralModel` = NONE, the netting set is considered unmargined. Otherwise, the margined nodes are those subject to a bilateral CSA agreement for which the counterparty post collateral, that is for which `CPVMPostStyle` = Gross or Net. When `CPVMPostStyle` = None, then node X is considered unmargined. When node X is not a CSA, it is considered unmargined as well.

- **Sold Options Exception**

The sold options exception may apply when X is an MA node with no CSA, an NC node, or an NN trade; that is, X is not subject to variation margin. X might be made up of one or more netting sets, $NS_{X,1}$, $NS_{X,2}$, etc. For each netting set $NS_{X,i}$, for which all of the following conditions are satisfied:

1. $NS_{X,i}$ has a negative mark-to-market
2. $NS_{X,i}$ consists of only sold options (saccr.optionparams.sold=True)
3. All premiums of the trades in $NS_{X,i}$ have been received upfront (saccr.optionparams.remainingPremiums = False)

the PFE of $NS_{X,i}$ is set to zero and the mark-to-market of $NS_{X,i}$ in the replacement cost is set to zero.

- **Trade-level IM and IA**

Counterparty and User Independent Amounts and Initial Margins can be linked to a deal i . These profiles are specified in `standardTradeMitigantAttributes` as:

- TradeMarginType: Initial or Independent
- PaymentDate and PaymentAmount pair:

Date	Value	Comment
D_1	V_1	V_1 applies from D_1 (inclusive) to D_2 (exclusive)
D_2	V_2	V_2 applies from D_2 (inclusive) to D_3 (exclusive)
...	...	...
D_n	V_n	V_n applies from D_n (inclusive) to the end of simulation

Note

V_i should be entered as a positive value.

- PostDirection: To user or To counterparty
- Currency: The currency of the PaymentAmount, can be an acceptable currency code.
- Segregated: only used for IM, True or False.

- **Settled-to-Market Agreement**

Settled-to-Market (STM) trades are transactions for which the mark-to-market is settled daily, effectively making the transaction net present value zero. These trades can belong to a master agreement which might also have non-STM trades. Under SACCR, STM trades receive a favourable treatment as they are considered unmargined with a maturity factor of 10 business days (14 calendar days in IMCR). To support this special treatment, we will use a VMCSA (Legacy, VMCSA or IMVMCSA) node with a flag indicating that the trades in it are settled-to-market.

- Data Input and Validation
    - `SettledToMarketAgreement`: a boolean CSA-level attribute, defaults to False. The attribute of a VMCSA (Legacy, VMCSA or IMVMCSA) node must be set to True in order to indicate that the trades in it are settled-to-market.

- Some other CSA parameters must be set to specific values: `CPVMThreshold` = `UserVMThreshold` = 0, `CPVMMinTransfer` = `UserVMMinTransfer`= 0, `RoundoffAmount` = 0, `CallPeriod` = 1, `CPIndependent` = `UserIndependent` = 0, `CPVMPostStyle` = `UserVMPostStyle` = NET, `CPVMCap` = `UserVMCap` = empty, `CPVMSegregated` = `UserVMSegregated` = FALSE, `CPVMRehypothecation` = `UserVMRehypothecation` = FALSE, `RegulatoryFullyCollateralized` = TRUE, `LiquidCollateral` = TRUE.
- A master agreement with at least one VMCSA (Legacy, VMCSA or IMVMCSA) node with `SettledToMarketAgreement`= True must:
    - Not have closeout netting pools.
    - Have `CloseoutNetting` = TRUE.
    - Have `AgreementEnforceable` = TRUE at counterparty level.

Note

For the VMCSA(s) (Legacy, VMCSA or IMVMCSA) with `SettledToMarketAgreement`= True under an MA node:

1. MAG will fail batch if CSA-level `CPVMPostStyle` ≠ NET or `UserVMPostStyle` ≠ NET, or MA-level has closeout netting pools or `CloseoutNetting` = FALSE, or CP-level `AgreementEnforceable` = FALSE.
2. MAG will issue warning(s) if `CPVMThreshold`, `UserVMThreshold`, `CPVMMinTransfer`, `UserVMMinTransfer`, `CPIndependent`, `UserIndependent` aren't zeros. MAG will override these values with zeros.
3. MAG will override `CallPeriod` with 1 silently.
4. Only node-level and trade-level initial margin mitigants are processed. All other mitigants are discarded with a warning.

6.28 SACCR Capital Replacement Cost

The function `SACCR Capital Replacement Cost` computes the RC component of function `SACCR Capital Exposure At Default` based on the SA-CCR methodology.

Input Parameters

- CollateralModel: Sets the collateral model for the expression: NONE, NonSim_CollateralModel, etc.
- IncludeVMInRCUnmargined: An optional collateral model parameter, defaults to 1 (True).
- TableColumn: A lookup key that is associated with the following inputs: SACCRSupervisoryFactor, SACCROptionVolatility, SACCRCorrelations, SACCRIRCorrelations, SACCRFXHaircut, SACCRCCYPFEHaircuts and VolatilityAdjustments.

Replacement Cost

The calculation of replacement cost (`RC`) is first performed for all MAs with no CSAs, CSAs, NCs, and trades under NN. Define X as such a node:

- For a node X with `SettledtoMarketAgreement` = False,

  Case 1: X consists of a single SA-CCR netting set

  The `RC` captures the loss that would occur if a counterparty were to default and were closed out of its transactions immediately.

  $$RC^{Unmargined} = \max\left(V_{NS} - C^{Unmargined}, 0\right)$$

  $$RC^{Margined} = \max\left(V_{NS} - C^{Margined}, TH + MTA - NICA, 0\right)$$

where:

- V_{NS} is the net mark-to-market value of all derivative transactions in the netting set. See • Formula: V on page 114 for more details.
- TH is the threshold of exposure below which collateral will not be called. See • Formula: TH and MTA on page 118 for more details.
- MTA is the minimum transfer amount of the collateral applicable to the counterparty. See • Formula: TH and MTA on page 118 for details.
- $NICA$ is the net independent collateral amount that the bank may use to offset its exposure to the counterparty. See • Formula: NICA on page 116 for details.
- $C^{Unmargined}$ is the haircut adjusted value of net collateral held in an unmargined set. See • Formula: C Unmargined on page 114 for details.
- $C^{Margined}$ is the haircut adjusted value of net collateral held in a margined set. See • Formula: C Margined on page 115 for details.

Case 2: X consists of multiple SA-CCR netting sets

If a single margin agreement is applied to multiple netting sets, the collateral posted to the CSA is shared among all transactions covered in the agreement and cannot be allocated to individual netting sets. When bank is a net receiver of collateral, the net collateral received can offset the positive market-to-market value of derivatives in the margin agreement; when bank is a net poster of collateral, Basel allows the offset of net collateral posted if the overall mark-to-market value of derivative transactions is negative.

$$RC^{Unmargined} = \max\left\{\sum_{NS \in X} \max(V_{NS}, 0) - \max\left(C^{Unmargined}, 0\right), 0\right\}$$
$$+ \max\left\{\sum_{NS \in X} \min(V_{NS}, 0) - \min\left(C^{Unmargined}, 0\right), 0\right\}$$
$$RC^{Margined} = \max\left\{\sum_{NS \in X} \max(V_{NS}, 0) - \max\left(C^{Margined}, 0\right), 0\right\}$$
$$+ \max\left\{\sum_{NS \in X} \min(V_{NS}, 0) - \min\left(C^{Margined}, 0\right), 0\right\}$$

where V_{NS} , $C^{Unmargined}$ and $C^{Margined}$ are as defined in Case 1 above.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True,

$$RC^{STM} = \begin{cases} max\{-(cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy), 0\} & \text{if CollateralModel} \neq \text{NONE} \\ 0 & \text{if CollateralModel} = \text{NONE} \end{cases}$$ where:

  - $cpIM(t_0, ccy)$ is the haricut-adjusted value of the initial margin (including node-level and trade-level intial margin) posted by the counterparty on a VMCSA (Legacy, VM or IM-VMCSA) node with the exception that the haircut uses a risk horizon of 14 days ($\frac{14}{365}$ in years), see section • Formula: Initial Margin and Independent Amount on page 118 for details.
  - $uIM_{unseg}(t_0, ccy)$ is the haricut-adjusted value of the initial margin (including node-level and trade-level intial margin) posted by the user on the VMCSA (Legacy, VM or IMVMCSA) node with the exception that the haircut uses a risk horizon of 14 days ($\frac{14}{365}$ in years), see section • Formula: Initial Margin and Independent Amount on page 118 for details.

• Valuation Input

The table summarizes the variables used in replacement cost calculation.

Description	Symbol	Input
Reporting date	t	Internal variable

Description	Symbol	Input
Domestic currency	ccy_d	<counterparty input>DomesticCurrency of node's Own Entity.
Threshold	TH_{raw}	<collateral agreement input>CPVMThreshold
Minimum transfer amount	MTA_{raw}	<collateral agreement input>CPVMMinTransfer
Counterparty independent amount	IA^{CSA}_{ctpy}	<collateral agreement input>CPIndependent
User independent amount	IA^{CSA}_{user}	<collateral agreement input>UserIndependent
Settlement lag	$SettleLag_{CSA}$	<collateral agreement input>SettlementLag
Margin lag	$MarginLag_{CSA}$	<collateral agreement input>MarginLag
Frequency of margin calls	$CallPeriod_{CSA}$	<collateral agreement input>CallPeriod
Cure period or liquidation period in the event of default	$CurePeriod_{CSA}$	<collateral agreement input>CurePeriod
Collateral agreement currency	ccy_{CSA}	<collateral agreement input>AgreementCurrency
Deal mark-to-market value	V_i	SACCR attributes
Deal currency	ccy_i	SACCR attributes
Deal maturity	mat_i	maximum of standard trade attribute `maturity-date` and SACCR attribute `maturity`
FX conversion factor from currency1 to domestic currency	FX_{ccy_1, ccy_d}	Internal variable
FX conversion factor from currency1 to collateral agreement currency	$FX_{ccy_1, ccy_{CSA}}$	Internal variable
Input value of the counterparty Independent Amount on deal i at time t_0	$cpIAInput(i, t_0)$	Input from `standardTradeMitigantAttributes`
Input value of the user Independent Amount on deal i at time t_0	$uIAInput(i, t_0)$	Input from `standardTradeMitigantAttributes`
Input value of the counterparty Initial Margin on deal i at time t_0	$cpIMInput(i, t_0)$	Input from `standardTradeMitigantAttributes`
Input value of the user Initial Margin on deal i at time t_0	$uIMInput(i, t_0)$	Input from `standardTradeMitigantAttributes`
Currency of the counterparty Independent Amount on deal i	$ccy_{cpIA, i}$	Input from `standardTradeMitigantAttributes`
Currency of the user Independent Amount on deal i	$ccy_{uIA, i}$	Input from `standardTradeMitigantAttributes`
Currency of the counterparty Initial Margin on deal i	$ccy_{cpIM, i}$	Input from `standardTradeMitigantAttributes`
Currency of the user Initial Margin on deal i	$ccy_{uIM, i}$	Input from `standardTradeMitigantAttributes`
Counterparty Independent Amount on deal i at time t_0	$cpIA(i, t_0)$	Internal Value

Description	Symbol	Input
User Independent Amount on deal i at time t_0	$uIA(i, t_0)$	Internal Value
Counterparty Initial Margin on deal i at time t_0	$cpIM(i, t_0)$	Internal Value
User Initial Margin on deal i at time t_0	$uIM(i, t_0)$	Internal Value

- **Mitigants**

Types of mitigants are summarized as follows:

Table 6.28-1 Mitigant Types

Mitigant Type	Applicable nodes	Haircut	Mitigant Input
Node Level Initial Margin	CSA, NC and MA (i.e. no CSA)	Y	node-level cash/security mitigants
Node Level Variation Margin	CSA	Y	node-level cash/security mitigants
Initial Margin Profile	CSA, NC and MA (i.e. no CSA)	N	initial margin profile*1

*1 The first collateral is used for SACCR Capital. Interpolation is not supported.

- **Formula: V**

V_{NS} is the net mark-to-market value of all unmatured transactions from the same netting set, spot converted to `Domestic Currency`.

$V_{NS} = \sum_{i \in NS} V_i \cdot FX_{ccy_i, ccy_d}$

- **Formula: C Unmargined**

$C^{Unmargined}$ is the haircut adjusted value of net collateral received under an unmargined set. If `CollateralModel` = NONE, $C^{Unmargined} = 0$. If `CollateralModel` ≠ NONE:

Case 1: X consists of a single SA-CCR netting set:

$C^{Unmargined} = cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy) + VM$

Case 2: X consists of multiple SA-CCR netting sets:

$C^{Unmargined} = cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy) - VM_{user, unseg}$

where:

- $cpIM(t_0, ccy)$: the haircut adjusted total value of segregated and unsegregated initial margin posted by the counterparty. See section • Formula: Initial Margin and Independent Amount on page 118 for details.
- $uIM_{unseg}(t_0, ccy)$: the haircut adjusted total value of unsegregated initial margin posted by the user. See section • Formula: Initial Margin and Independent Amount on page 118 for details.
- $VM_{user,unseg}$: the haircut adjusted value of unsegregated variation margin posted by the user. The user variation margins are ignored when `<collateral agreement input>UserVMPostStyle` = None.
- VM is defined as:

- $VM = \begin{cases} -VM_{user,unseg} & \text{if IncludeVMinRCUnmargined=True} \\ \delta_{ctpy} \cdot cpIA(t_0) - \delta_{user} \cdot uIA(t_0) & \text{if IncludeVMinRCUnmargined=False} \end{cases}$
- δ_{ctpy} = 1 if `<collateral agreement input>CPVMPostStyle` $\neq$ None; 0 otherwise.
- δ_{user} = 1 if `<collateral agreement input>UserVMPostStyle` $\neq$ None and `<collateral agreement input>UserVMSegregated` = False; 0 otherwise.

Here the initial/variation margin type is read from `<mitigant input>MarginType`; post direction is read from `<mitigant input>PostDirection`; Segregated/unsegregated flag is read from `<collateral agreement input>UserVMSegregated` for variation margin posted by the user and `<mitigant input>Segregated` otherwise.

The haircut adjusted mark-to-market captures the potential unfavorable change in mitigant value during a risk horizon of one year. The counterparty (user) mitigant value is decreased (increased) by a haircut amount; more specifically:

- the MTM of a node mitigant posted by the counterparty is: $M_i \cdot \max(0, 1 - H_i) \cdot \max\left(0, 1 - H_{default}\right)$
- the MTM of a node mitigant posted by the user is: $M_i \cdot (1 + H_i) \cdot \left(1 + H_{default}\right)$

where:

- M_i : the current mitigant value before haircut, converted to domestic currency. See section ● Formula: M on page 117 for details.
- H_i : the annualized haircut factor for the mitigant. See section ● Formula: H on page 117 for details.
- $H_{default}$: the haircut at default for the mitigant, expressed in absolute value. It is applicable to variation margin only, any haircut at default applied to initial margin will be ignored.

● Formula: C Margined

$C^{Margined}$ is the haircut adjusted value of net collateral received for a margined set.

$C^{Margined} = cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy) + VM_{ctpy} - VM_{user,\,unseg}$

where:

- $cpIM(t_0, ccy)$: the haircut adjusted total value of segregated and unsegregated initial margin posted by the counterparty. See section ● Formula: Initial Margin and Independent Amount on page 118 for details.
- $uIM_{unseg}(t_0, ccy)$: the haircut adjusted total value of unsegregated initial margin posted by the user. See section ● Formula: Initial Margin and Independent Amount on page 118 for details.
- VM_{ctpy} : the total haircut adjusted value of segregated and unsegregated variation margin posted by the counterparty. The counterparty variation margins are ignored when `<collateral agreement input>CPVMPostStyle` = None.
- $VM_{user,unseg}$: the total haircut adjusted value of unsegregated variation margin posted by the user. The user variation margins are ignored when `<collateral agreement input>UserVMPostStyle` = None.

For margined set, the risk horizon to be considered for the potential unfavorable change in collateral value is during the ● Formula: MPOR on page 116:

- the MTM of a node mitigant posted by the counterparty: $M_i \cdot \max\left(0, 1 - H_i \cdot \sqrt{\frac{MPOR_{CSA}}{365}}\right) \cdot \max(0, 1 - H_{default})$
- the MTM of a node mitigant posted by the user: $M_i \cdot \left(1 + H_i \cdot \sqrt{\frac{MPOR_{CSA}}{365}}\right) \cdot (1 + H_{default})$

where:

- $MPOR_{CSA}$: the margin period of risk. See section ● Formula: MPOR on page 116 for details.
- M_i : the current value of the collateral before haircut, converted to domestic currency. See section ● Formula: M on page 117 for details.
- H_i : the annualized haircut factor. See section ● Formula: H on page 117 for details.
- $H_{default}$: the haircut at default for the mitigant, expressed in absolute value. It is applicable to variation margin only, any haircut at default applied to initial margin will be ignored.

● Formula: NICA

The net independent collateral amount (NICA) is the collateral that bank sets aside to offset its exposure to the counterparty. It includes the initial margin and the independent amount. Mathematically:

$NICA = \left(cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy)\right) + (cpIA(t_0) - \delta_{user} \cdot uIA(t_0))$

where:

- $cpIA(t_0)$: the total independent amount posted by the counterparty for a CSA node, expressed in absolute value. See section ● Formula: Initial Margin and Independent Amount on page 118 for details.
- $uIA(t_0)$: the total independent amount posted by the user for a CSA node, expressed in absolute value. See section ● Formula: Initial Margin and Independent Amount on page 118 for details.
- δ_{user} = 1 if `<collateral agreement input>UserVMPostStyle` ≠ None and `<collateral agreement input>UserVMSegregated` = False; 0 otherwise.
- $cpIM(t_0, ccy)$: the total haircut adjusted mark-to-market value of segregated and unsegregated initial margin posted by the counterparty as defined in ● Formula: Initial Margin and Independent Amount on page 118.
- $uIM_{unseg}(t_0, ccy)$: the total haircut adjusted mark-to-market value of unsegregated initial margin posted by the user as defined in ● Formula: Initial Margin and Independent Amount on page 118.

● Formula: MPOR

Margin Period of Risk (MPOR) is the time period from the last exchange of collateral covering a netting set of transactions with a defaulting counterparty until that counterparty is closed out and the resulting market risk is re-hedged. In MAG it is calculated as follows for each CSA:

$MPOR_{CSA} = \max(MarginLag_{CSA}, SettleLag_{CSA}) + CallPeriod_{CSA} + CurePeriod_{CSA} - 1$

where:

- $MarginLag_{CSA}$: the number of days that the user continues making margin payments after the beginning of the MPOR.

- $SettleLag_{CSA}$: the number of days that the user continues making trade settlement payments after the beginning of the MPOR.
- $CallPeriod_{CSA}$: the frequency of margin calls.
- $CurePeriod_{CSA}$: the number of days between the time that the user stops making any payments (margin and trade settlement) and the time the portfolio is hedged and/or liquidated.

- **Formula: M**

M_i is the current value of individual collateral before haircut, converted to domestic currency. Denote $M_{raw,i}$ as the raw collateral value before FX conversion, and ccy_m_i as the collateral currency:

$M_i = M_{i,raw} \cdot FX_{ccy\_m_i, ccy_d}$

where the raw value and collateral currency for security mitigants are read from `<mitigant input>MTM` and `<mitigant input>MTMCcy`. For cash mitigants:

- If `<mitigant input>MTM` is populated:
  - $M_{raw,i}$ is read from `<mitigant file>MTM`
  - ccy_m_i is read from `<mitigant file>MTMCcy`
- If `<mitigant input>MTM` is empty:
  - $M_{raw,i}$ is read from `<mitigant file>NumberOfUnits`
  - ccy_m_i is read from `<mitigant file>Currency`

- **Formula: H**

The annualized haircut factor (H_i) captures the potential unfavorable change in value of the collateral over one year:

$H_i = HC_i + HC_{FX}$

where

- HC_i is the annualized mitigant haircut, that is, the input haircut factor multiplied by $\sqrt{1/LH_{HC}}$, where LH_{HC} is the liquidity horizon (in years) associated with the mitigant haircut. For cash mitigants, the mitigant haircut is taken from table `SACCRCCYPFEHaircuts` based on mitigant MTM currency; for security mitigants, the mitigant haircut is read from table `VolatilityAdjustments` based on the security class.
- HC_{FX} is the annualized FX haircut, that is, the input haircut factor multiplied by $\sqrt{1/LH_{FX}}$, where LH_{FX} is the liquidity horizon (in years) associated with the FX haircut. The FX haircut, read from table `SACCRFXHaircut`, is applied only if the currency of the collateral is different from the settlement currency. For CSA netting sets, use the CSA `Agreement Currency` as settlement currency; for MA or NC netting set, use the MA `Termination Currency`; and for trades in an NN node, use the currency of the trade.

  Note

  In the case where the collateral is assigned to an IMCSA ($\subset$ VMCSA), the `Agreement Currency` of the VMCSA is used as settlement currency.

- **Formula: TH and MTA**

The threshold (TH) is the threshold of exposure below which collateral will not be called. The minimum transfer amount (MTA) is the minimum block of collateral that can be called for at a time. TH plus MTA represents the maximum uncollateralized exposure under a margined agreement.

For each CSA, TH and MTA are read from `<collateral agreement input>` and converted to `Domestic Currency`:

$TH = TH_{raw} \cdot FX_{ccy_{CSA}, ccy_d}$

$MTA = MTA_{raw} \cdot FX_{ccy_{CSA}, ccy_d}$

- **Formula: Initial Margin and Independent Amount**

  - Trade-level Independent Amount and Initial Margin

    Define the trade-level counterparty and user independent amounts on deal i at time t_0 by

$$cpIA(i, t_0)_{raw} = \begin{cases} cpIAInput(i, t_0) & if\ t_0 \le mat_i \\ 0 & if\ t_0 > mat_i \end{cases}$$

$$uIA(i, t_0)_{raw} = \begin{cases} uIAInput(i, t_0) & if\ t_0 \le mat_i \\ 0 & if\ t_0 > mat_i \end{cases}$$

    where:

    - $cpIA(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t_0 posted by the counterparty.
    - $uIA(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t_0 posted by the user.

    Define the counterparty and user initial margins on deal i at time t_0 by

$$cpIM(i, t_0)_{raw} = \begin{cases} cpIMInput(i, t_0) & if\ t_0 \le mat_i \\ 0 & if\ t_0 > mat_i \end{cases}$$

$$uIM(i, t_0)_{raw} = \begin{cases} uIMInput(i, t_0) & if\ t_0 \le mat_i \\ 0 & if\ t_0 > mat_i \end{cases}$$

$$uIM_{unseg}(i, t_0)_{raw} = \begin{cases} uIM(i, t_0)_{raw} & if\ uIM(i, t_0) \text{ is unsegregated} \\ 0 & \text{otherwise} \end{cases}$$

    where:

    - $cpIM(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level segregated and unsegregated initial margin on deal i at time t_0 posted by the counterparty.
    - $uIM(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level initial margin on deal i at time t_0 posted by the user.
    - $uIM_{unseg}(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level unsegregated initial margin on deal i at time t_0 posted by the user.

  - FX Haircut of Trade-level Initial Margins

Assuming that any trade-level initial margin (or independent amount) is posted in cash, an FX haircut is applied to the trade-level initial margin if the currency of the initial margin is different from the settlement currency of the deal.

Note

An FX haircut is not applied to trade-level independent amounts for consistency with the treatment of a CSA independent amount.

Define the settlement currency of a deal i to be:

- the CSA currency if deal i is in a CSA (VMCSA, IMVMCSA or Legacy CSA)
- the MA termination currency if deal i is in an MA and not in a CSA
- the currency of the deal if deal i is in an NN node

If the settlement currency of a deal i is different from the currency of its trade-level initial margin $IM(i, t_0)$ (where $IM(i, t_0) := cpIM(i, t_0)$ or $uIM(i, t_0)$), define

$$HC_{FX, IM(i, t_0)} = \frac{HC_{FX}}{\sqrt{LH_{FX}}} \times \sqrt{RH}$$

where:

- HC_{FX} : the FX haircut, read from the table `SACCRFXHaircut`.
- LH_{FX} : the liquidity horizon (in years) associated with the FX haircut, read from the table `SACCRFXHaircut`.
- $RH = \begin{cases} \frac{MPOR_{CSA}}{365} & \text{for a margined set} \\ 1 & \text{for an unmargined set} \end{cases}$: the risk horizon (in years) to be considered for the potential unfavorable change in collateral value, see section • Formula: MPOR on page 116 for details.

Otherwise $HC_{FX, IM(i, t_0)} = 0$.

- FX Conversion and Aggregation
  - Independent Amounts

    At time t_0 , define the trade-level independent amounts after FX conversion:

    $$cpIA(i, t_0) = FX_{ccy_{(cpIA, i)}, ccy_{CSA}} \times cpIA(i, t_0)_{raw}$$

    $$uIA(i, t_0) = FX_{ccy_{(uIA, i)}, ccy_{CSA}} \times uIA(i, t_0)_{raw}$$

    where:

    - $cpIA(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t_0 posted by the counterparty.
    - $uIA(i, t_0)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t_0 posted by the user.

    At time t_0 , for a CSA node, define the total independent amounts:

    $$cpIA(t_0) = IA_{ctpy}^{CSA} + \sum_{i \in CSA} cpIA(i, t_0)$$

    $$uIA(t_0) = IA_{user}^{CSA} + \sum_{i \in CSA} uIA(i, t_0)$$

    where:

- IA_{ctpy}^{CSA} : the csa level independent amount posted by the counterparty, expressed in absolute value.
- IA_{user}^{CSA} : the csa level independent amount posted by the user, expressed in absolute value.

- Initial Margins

  At time t_0 , for a currency ccy , define the trade-level initial margin on deal i in currency ccy :

  $$cpIM(i,t_0,ccy) = FX_{ccy_{cpIM,i},ccy} \times cpIM(i,t_0)_{raw} \times max(0, 1 - HC_{FX, cpIM(i,t_0)})$$

  $$uIM(i,t_0,ccy) = FX_{ccy_{uIM,i},ccy} \times uIM(i,t_0)_{raw} \times (1 + HC_{FX, uIM(i,t_0)})$$

  $$uIM_{unseg}(i,t_0,ccy) := \begin{cases} uIM(i,t_0,ccy) & if\ uIM(i,t_0) \text{ is unsegregated} \\ 0 & \text{otherwise} \end{cases}$$

  where:

  - $cpIM(i,t_0)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level segregated and unsegregated initial margin on deal i at time t_0 posted by the counterparty.
  - $uIM(i,t_0)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level initial margin on deal i at time t_0 posted by the user.
  - $HC_{FX, cpIM(i,t_0)}$: FX haircut of the trade-level IM as defined in • Formula: Initial Margin and Independent Amount on page 118.
  - $HC_{FX, cpIM(i,t_0)}$: FX haircut of the trade-level IM as defined in • Formula: Initial Margin and Independent Amount on page 118.

  At time t_0 , for a currency ccy , for a node X , define the total initial margin:

  $$cpIM(t_0,ccy) = IM_{ctpy}(t_0,ccy) + \sum_{i \in X} cpIM(i,t_0,ccy)$$

  $$uIM(t_0,ccy) = IM_{user}(t_0,ccy) + \sum_{i \in X} uIM(i,t_0,ccy)$$

  $$uIM_{unseg}(t_0,ccy) = IM_{user, unseg}(t_0,ccy) + \sum_{i \in X} uIM_{unseg}(i,t_0,ccy)$$

  where:

  - $IM_{ctpy}(t_0,ccy)$: the node-level haircut adjusted value of segregated and unsegregated initial margin posted by the counterparty. For margined set, the mitigant value is decreased by a haircut amount over the risk horizon of • Formula: MPOR on page 116: $M_i \cdot \max\left(0, 1 - H_i \cdot \sqrt{\frac{MPOR_{CSA}}{365}}\right)$. For unmargined set, the mitigant value is decreased by a haircut amount over the risk horizon of 1 year: $M_i \cdot \max(0, 1 - H_i)$.
  - $IM_{user}(t_0,ccy)$: the node-level haircut adjusted value of initial margin posted by the user. For margined set, the mitigant value is increased by a haircut amount over the risk horizon of • Formula: MPOR on page 116: $M_i \cdot \max\left(0, 1 + H_i \cdot \sqrt{\frac{MPOR_{CSA}}{365}}\right)$. For unmargined set, the mitigant value is increased by a haircut amount over the risk horizon of 1 year: $M_i \cdot \max(0, 1 + H_i)$.
  - $IM_{user,unseg}(t_0,ccy)$: the node-level haircut adjusted value of unsegregated initial margin posted by the user. For margined set, the mitigant value is increased by a haircut

amount over the risk horizon of • Formula: MPOR on page 116: $M_i \cdot \left(1 + H_i \cdot \sqrt{\frac{MPOR_{CSA}}{365}}\right)$. For unmargined set, the mitigant value is increased by a haircut amount over the risk horizon of 1 year: $M_i \cdot (1 + H_i)$.

- $cpIM(t_0, ccy)$: the total haircut adjusted value of segregated and unsegregated initial margin posted by the counterparty for a node X.
- $uIM(t_0, ccy)$: the total haircut adjusted value of initial margin posted by the user for a node X.
- $uIM_{unseg}(t_0, ccy)$: the total haircut adjusted value of unsegregated initial margin posted by the user for a node X

6.29 SACCR Capital Potential Future Exposure

The function `SACCR Capital Potential Future Exposure` computes the PFE component of function `SACCR Capital Exposure At Default` based on the SA-CCR methodology.

• Input Parameters

- IRTenorOffsetting: A boolean attribute, defaults to True. Setting it to False disallows offsetting benefit between maturity buckets for Interest Rate derivatives.
- MultiplierFloor: Sets the floor for AddOn multiplier, defaults to be 0.05.
- MultiplierFactor: Sets the factor for AddOn multiplier, defaults to be 2.
- MaturityFactorMultiplier: Sets the multiplier for margined maturity factor, defaults to be 1.5.
- SupervisoryDurationDF: Sets the discount rate in supervisory duration calculation, defaults to be 0.05.
- IRShift: accepts PER_CURRENCY or PER_TRADE. It specifies whether the IR shift amount is currency dependent or trade dependent.
- CollateralModel: Sets the collateral model for the expression: NONE, NonSim_CollateralModel, etc.
- TableColumn: A lookup key that is associated with the following inputs: SACCRSupervisoryFactor, SACCROptionVolatility, SACCRCorrelations, SACCRIRCorrelations, SACCRFXHaircut, SACCRCCYPFEHaircuts and VolatilityAdjustments.

• Potential Future Exposure

The `PFE` add-on consists of 1) an aggregate add-on component that captures the expected increase of the netting set mark-to-market over a given time horizon and 2) a multiplier that allows for the recognition of excess collateral or negative mark-to-market value for the transactions.

The calculation of `PFE` add-on is first performed for all MAs with no CSAs, CSAs, NCs, and trades under NN. Define X as such a node:

- If the node X has `SettledtoMarketAgreement` = False,

  Case 1: X consists of a single SA-CCR netting set
  The `PFE` add-on is the product of aggregate add-on component and a multiplier:
  $PFE^{Unmargined} = multiplier^{Unmargined} \cdot AddOn^{Unmargined}_{Aggregate}$
  $PFE^{Margined} = multiplier^{Margined} \cdot AddOn^{Margined}_{Aggregate}$

6 Functions

Case 2: X consists of multiple SA-CCR netting sets

The PFE add-on in this case is calculated individually for each netting set according to the unmargined methodology, then summed together:

$$PFE^{Unmargined} = PFE^{Margined} = \sum_{NS \in CSA} PFE_{NS}^{Unmargined}$$

where:

$$PFE_{NS}^{Unmargined} = multiplier^{Unmargined} \cdot AddOn_{Aggregate}^{Unmargined}$$

The calculation for • PFE Multiplier on page 122 and • PFE AddOn on page 123 are explained in sections below.

- If the node X is a VMCSA (Legacy, VM or IMVMCSA) with `SettledtoMarketAgreement` = True,

$$PFE^{STM} = multiplier^{STM} \cdot Addon_{Aggregate}^{STM}$$

The calculation for • PFE Multiplier on page 122 and • PFE AddOn on page 123 are explained in sections below.

• PFE Multiplier

The PFE multiplier reduces the PFE add-on when the netting set is over-collateralized or out-of-the-money.

Case 1: X consists of a single SA-CCR netting set

- If the node X has `SettledtoMarketAgreement` = False, the multiplier is calculated as follows:

$$multiplier^{Unmargined} = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{V_{NS} - C^{Unmargined}}{m\_factor \cdot \left(1 - m\_floor\right) \cdot AddOn_{Aggregate}^{Unmargined}}\right)\right)$$

$$multiplier^{Margined} = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{V_{NS} - C^{Margined}}{m\_factor \cdot \left(1 - m\_floor\right) \cdot AddOn_{Aggregate}^{Margined}}\right)\right)$$

where:

  - m_floor is read from expression parameter `MultiplierFloor`, defaults to be 5%.
  - m_factor is read from expression parameter `MultiplierFactor`, defaults to be 2.
  - V_{NS} is the net mark-to-market value of all derivative transactions in the netting set. See • Formula: V on page 114 for details.
  - $C^{Unmargined}$ is the haircut adjusted value of net collateral held in an unmargined set. See Formula: • Formula: C Unmargined on page 114 for details.
  - $C^{Margined}$ is the haircut adjusted value of net collateral held in a margined set. See • Formula: C Margined on page 115 for details.
  - $AddOn_{Aggregate}^{Unmargined}$ and $AddOn_{Aggregate}^{Margined}$ are the aggregate add-on components of all derivative transactions in the netting set and consist of add-ons from each asset class. See section • PFE AddOn on page 123 for details.

Note

The multiplier is activated in case of over-collateralization (where net collateral held exceeds net market value of transactions).

- If the node X is a VMCSA (Legacy, VM or IMVMCSA) with `SettledtoMarketAgreement` = True,

  $multiplier^{STM} =$

  $$\begin{cases} min\left\{1, m\_floor + (1 - m\_floor) \cdot exp\left(\frac{-(cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy))}{m\_factor \cdot (1 - m\_floor) \cdot Addon^{STM}_{Aggregate}}\right)\right\} & \text{if CollateralModel} \neq \text{NONE} \\ 1 & \text{if CollateralModel} = \text{NONE} \end{cases}$$

  where $cpIM(t_0, ccy)$ and $uIM_{unseg}(t_0, ccy)$ are defined in the section • Formula: Initial Margin and Independent Amount on page 118.

Case 2: X consists of multiple SA-CCR netting sets

The multiplier is calculated at netting set level individually. Since it is problematic to allocate common collateral to individual netting sets, the multiplier is activated only when the current value of the derivative transactions in a netting set is negative. The multiplier formulation recognizes that the PFE is smaller as the netting set moves further away from an at-the-money set.

$$multiplier^{Unmargined} = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{CV_{NS}}{m\_factor \cdot \left(1 - m\_floor\right) \cdot AddOn^{Unmargined}_{Aggregate}}\right)\right)$$

where:

- m_floor is read from expression parameter `MultiplierFloor`, defaults to be 5%.
- m_factor is read from expression parameter `MultiplierFactor`, defaults to be 2.
- $CV_{NS} = \begin{cases} V_{NS} - IM^{Trade}_{NS} & \text{If X has SettledtoMarket} = \text{False} \\ -IM^{Trade}_{NS} & \text{If X has SettledtoMarket} = \text{True} \end{cases}$ is the current value of all derivative transactions in the netting set. See • Formula: V on page 114 for details.
- $IM^{Trade}_{NS} = \sum_{i \in NS} cpIM(i, t_0, ccy) - \sum_{i \in NS} uIM_{unseg}(i, t_0, ccy)$ is the sum of trade-level initial margin over all trades in the netting set. See section • Formula: Initial Margin and Independent Amount on page 118 for details.
- $AddOn^{Margined}_{Aggregate}$ is the aggregate add-on components of all derivative transactions in the netting set and consist of add-ons from each asset class. See section • PFE AddOn on page 123 for details.

• PFE AddOn

Five asset classes are defined under SA-CCR: Interest Rate (IR), Foreign Exchange (FX), Credit (CR), Equity (EQ), and Commodity (CM). Trades are allocated into one of the five asset classes according to their primary risk factors. For more complex trades that may have more than one risk driver (for example, multi-asset or hybrid derivatives), they may be allocated to more than one asset class.

The aggregate add-on for the netting set is a simple sum of add-ons calculated from each asset class:

$$AddOn^{Margined}_{Aggregate} = AddOn^{(IR)}_{Margined} + AddOn^{(FX)}_{Margined} + AddOn^{(CR)}_{Margined} + AddOn^{(EQ)}_{Margined} + AddOn^{(CM)}_{Margined}$$

$$AddOn^{Unmargined}_{Aggregate} = AddOn^{(IR)}_{Unmargined} + AddOn^{(FX)}_{Unmargined} + AddOn^{(CR)}_{Unmargined} + AddOn^{(EQ)}_{Unmargined} + AddOn^{(CM)}_{Unmargined}$$

where the add-on for each asset class is calculated using asset-class-specific formulas as described in sections below.

Note

The add-on for a margined netting set and an unmargined netting set uses different time horizon assumptions. The add-on under unmargined calculation represents the potential change in value of trades over a one-year time horizon, while the add-on under margined calculation represents the potential change in exposure during the • Formula: MPOR on page 116. The difference is captured in the maturity factors calculation in the • Preliminary Materials on page 126 section.

In the following sections, the add-on's margined/unmargined superscript is omitted for simplicity.

• Aggregation Hierarchy

MAG first computes the trade level effective notionals, then aggregates across trades, hedging buckets, hedging sets, and trade types for the final asset class add-on. The hierarchical nodes are defined from top to bottom as follows:

- `Asset Class`: Five asset classes are defined: Interest Rate, Foreign Exchange, Equity, Credit, and Commodity
- `Trade Type`: Three trade types are defined: Regular, Basis, and Volatility.

  A basis transaction is a non-foreign exchange transaction in which the cash flows of both legs depend on different risk factors from the same asset class.

  A volatility transaction is one in which the reference asset depends on the volatility (historical or implied) of a risk factor.
- `Hedging Set`: A hedging set is a set of transactions within a single netting set for which partial or full offsetting is recognized. Hedging sets are defined based on asset classes and trade types.

Table 6.29-1 Hedging Set Definition where Trade Type = Regular

Asset Class	Hedging Set Definition
Interest Rate	Each currency forms its own hedging set.
Foreign Exchange	Each currency pair forms its own hedging set.*1
Credit	All transactions form a single hedging set.
Equity	All transactions form a single hedging set.
Commodity	Each broad category of commodity forms its own hedging set.

Table 6.29-2 Hedging Set Definition where Trade Type = Basis

Asset Class	Hedging Set Definition
Interest Rate	Each pair of risk factors forms its own hedging set (for example: 3-month Libor vs. 6-month Libor)
Foreign Exchange	Basis transaction is not applicable to FX asset class.
Credit	Each pair of risk factors forms its own hedging set (for example: entity1 vs entity2).
Equity	Each pair of risk factors forms its own hedging set (for example: entity1 vs entity2).
Commodity	Each pair of risk factors forms its own hedging set (for example: Brent Crude oil vs. Henry Hub gas).

Table 6.29-3 Hedging Set Definition where Trade Type = Volatility

Asset Classes and Hedging Set Definition
Interest Rate, Foreign Exchange and Commodity volatility hedging sets follow the same hedging set construction rule as regular hedging sets, except that they are placed under Volatility trade type. Credit and Equity volatility hedging sets are defined per risk driver (i.e. per reference entity).

- `Hedging Bucket`: for Interest Rate, Credit, Equity and Commodity asset classes, hedging sets are further divided into subsets (namely, hedging buckets) that allow full offsetting among trades. Hedging bucket definition is shared among Regular, Basis, and Volatility trade types except for Credit and Equity asset classes.

Asset Class	Hedging Bucket Definition
Interest Rate	Trades are divided into three maturity buckets based on the end date of the transactions: 1. less than or equal to one year 2. between one (exclusive) and five years (inclusive) 3. more than five years
Foreign Exchange	All transactions within a hedging set form a single hedging bucket.
Credit	For regular trades: each reference entity (a single entity or an index) form its own hedging bucket. For basis and volatility trade: all transactions within a hedging set form a single hedging bucket.
Equity	For regular trades: each reference entity (a single entity or an index) form its own hedging bucket. For basis and volatility trade: all transactions within a hedging set form a single hedging bucket.
Commodity	Each commodity forms its own hedging bucket.

- `Trade`: trade is the lowest node of the hierarchy, where parameters such as maturity factor, supervisory delta adjustment, supervisory factor, adjusted notional and effective notional are calculated.

*1 MAG orders the hedging set currency pair alphabetically and trades that reference the same currency pair (regardless of the long/short direction) will be mapped to the same hedging set. For example, an FX forward that pays (short) CAD and receives (long) USD is short in the hedging set hedging set (CAD - USD). Another FX forward that pays (short) USD and receives (long) CAD is long in the same hedging set (CAD-USD).

Input recommendation: For option deltas to correctly offset, the quotation direction for strike (K) and underlying forward price (F) should be consistent across all trades that reference the same currency pair.

● Valuation Inputs

The table summarizes the variables used in add-on calculation.

Table 6.29-4 Valuation Inputs

Description	Symbol	Applicable Asset Class	Input
Reporting date	t	All	Internal variable
Domestic currency	ccy_d	All	<counterparty input>DomesticCurrency of node's Own Entity.
Maturity factor of trade i calculated on a margined basis	$MF_i^{Margined}$	All	Internally computed: see *"Maturity Factors"* in ● Preliminary Materials on page 126
Maturity factor of trade i calculated on an unmargined basis	$MF_i^{Unmargined}$	All	Internally computed: see *"Maturity Factors"* in ● Preliminary Materials on page 126
Supervisory duration of trade i	SD_i	IR, CS	Internally computed: see *"Supervisory Duration"* in ● Preliminary Materials on page 126
Supervisory delta adjustment of trade i	δ_i	All	Internally computed: see *"Supervisory Delta Adjustment"* in ● Preliminary Materials on page 126
Notional of trade i	N_i	IR, CR, EQ, CM	SACCR attributes*1
Notional currency of trade i	ccy_i	IR, CR, EQ, CM	SACCR attributes
Long notional of FX trade i	$N_{i,L}$	FX	SACCR attributes

Description	Symbol	Applicable Asset Class	Input
Short notional of FX trade i	$N_{i,S}$	FX	SACCR attributes
Long notional currency of FX trade i	$ccy_{i,L}$	FX	SACCR attributes
Short notional currency of FX trade i	$ccy_{i,S}$	FX	SACCR attributes
FX conversion factor from currency1 to domestic currency	FX_{ccy_1,ccy_d}	All	Internal variable
Adjusted notional for trade i of asset class a	$d_i^{(a)}$	All	Internally computed
Effective notional for trade i of asset class a	$D_i^{(a)}$	All	Internally computed
Effective notional for hedging bucket k of hedging set j , asset class a	$D_{jk}^{(a)}$	All	Internally computed
Effective notional for hedging set j of asset class a	$EffectiveNotional_j^{(a)}$	All	Internally computed
Supervisory factor for hedging bucket k of asset class a	$SF_k^{(a)}$	All	Internally computed
Correlation factor for hedging bucket k of asset class a	$\rho_k^{(a)}$	All	Internally computed
Add-on for hedging set j of asset class a	$AddOn_j^{(a)}$	All	Internally computed
Add-on for trade type x of asset class a	$AddOn_x^{(a)}$	All	Internally computed
Add-on of asset class a	$AddOn^{(a)}$	All	Internally computed

*1 For Interest Rate, Foreign Exchange and Credit derivatives, trade notional is the notional amount of the derivative; for Equity and Commodity derivatives, the trade notional is defined as the product of the current price of one unit of stock or commodity and the number of units referenced by the trade.

- ## Preliminary Materials

This section describes the calculation of the common parameters that are shared among asset classes:

1. **Time Factors**

   The following four time factors are used in PFE add-on calculation:

Table 6.29-5 Time Factors

t_{target}	Symbol	Description	Calculation
Maturity date	M_i	The maturity of a contract is the latest date when the contract may still be active. If the derivative (for example, a swaption) will be physically exercised into underlying contracts, then maturity of the contract is the final settlement date of the underlying derivative contract. For addon calculation, maturity is read from SACCR attribute `maturity`.	$M_i = \begin{cases} \frac{\max(t_{target} - t, 14)}{365}, & \text{if} t_{target} > t \\ 0, & \text{if} t_{target} \le t \end{cases}$
Start date	S_i	Applicable to Interest Rate and Credit derivatives. It is the start date of the time period referenced by the contract. If the derivative references the value of another interest rate or credit instrument, the time period must be determined on the basis of the underlying instrument.	$S_i = Max\left(\frac{t_{target} - t}{365}, 0\right)$
End date	E_i	Applicable to Interest Rate and Credit derivatives. It is the end date of the time period referenced by the contract. If the derivative references the value of another interest rate or credit instrument, the time period must be determined on the basis of the underlying instrument.	$E_i = Max\left(\frac{t_{target} - t}{365}, 0\right)$

t_{target}	Symbol	Description	Calculation
Exercise date	T_i	It is the latest contractual exercise date referenced by the option contract.	$T_i = Max\left(\frac{t_{target} - t}{365}, 0\right)$

2. **Maturity Factors**

- For a node with `SettledtoMarketAgreement` = False, the maturity factor, MF_i , is calculated differently for margined and unmargined sets.

  If trade *i* is unmargined, a time horizon of one year is assumed:

  $MF_i^{Unmargined} = \sqrt{\min(M_i, 1)}$

  If trade *i* is margined, the appropriate time horizon is the • Formula: MPOR on page 116 of the margin agreement:

  $MF_i^{Margined} = m\_mf \cdot \sqrt{\frac{MPOR_{CSA}}{365}}$

  where is m_mf read from expression parameter `MaturityFactorMultiplier`, defaults to 1.5.

- For a VMCSA (Legacy, VM or IMVMCSA) node with `SettledtoMarketAgreement` = True, $Addon_{Aggregate}^{STM}$ is defined the same way with the exception that the maturity factor of all trades in the CSA is defined as: $MF = \sqrt{\frac{14}{365}}$.

3. **Supervisory Duration**

For Interest Rate and Credit derivatives, the trade notional is adjusted by the supervisory duration, SD_i . The supervisory duration represents the discount factors for cashflows between contract start date and end date, using a discount rate of r = 0.05 (read from `Supervisory-DurationDF`). Mathematically,

$$SD_i = \max\left(\frac{14}{365}, \frac{\exp(-0.05 \cdot S_i) - \exp(-0.05 \cdot E_i)}{r}\right)$$

where S_i and E_i are start and end dates, expressed as time factor respectively, as defined in *"time factors"* in • Preliminary Materials on page 126.

4. **Supervisory Delta Adjustment**

The supervisory delta adjustment, δ_i , is defined at the trade level and is applied to the adjusted notional amount to reflect the direction and non-linearity of the trade. If SACCR attribute `DeltaOverride` is populated, δ_i is read from delta override. Otherwise, δ_i is determined as follows:

Table 6.29-6 Supervisory delta adjustment for non-options and non-CDO-tranches

δ_i	**Long in the primary risk factor**	**Short in the primary risk factor**
Trades that are not options or CDO tranches	1	-1

Note

For regular trades, long and short flag to Interest Rate, Credit, Equity, and Commodity hedging sets are clearly defined in the SACCR attributes. For Foreign Exchange asset class, each currency pair forms a (regular) hedging set. The hedging set currency pair is ordered alphabetically, for example (CAD-USD) or (EUR-GBP). The trade's (long - short) currency pair is compared with the hedging set currency pair to determine if a FX trade is long or short the hedging set. For example, an FX forward that pays (short) CAD and receives (long) USD is short in the hedging set hedging set (CAD - USD). Another FX forward that pays (short) USD and receives (long) CAD is long in the same hedging set (CAD-USD).

Note

For basis trades, the long and short flag are determined using both the long/short indicator (as read from SACCR attributes) and the order of hedging set risk factor pair vs. trade risk factor pair. The hedging set risk factor pair is ordered alphabetically, for example 3M Libor vs. 6M Libor. The trade's risk factor pair is defined as the primary risk factor vs. ancillary risk factor. The original delta sign is determined using the long/short indicator. Then, if the trade's basis pair is the reversal of the hedging set pair, the delta sign will also get reversed. Here the long/short indicator is expected to indicate the long/short position on the primary risk factor.

Table 6.29-7 Supervisory delta adjustment for options:

δ_i	**Call Option**	**Put Option**
Options	See Figure 6.29-1 Supervisory delta adjustment for call options on page 128.	See Figure 6.29-2 Supervisory delta adjustment for put options on page 128.

$$\delta_i = ls \cdot \begin{cases} \Phi\left(\dfrac{\ln\left(\dfrac{P_i+\lambda_j}{K_i+\lambda_j}\right)+0.5\cdot\sigma_i^2\cdot T_i}{\sigma_i\cdot\sqrt{T_i}}\right) & \text{if} t < T_i \\ 1 & \text{if} T_i \le t \le M_i \text{and} P > K \\ 0 & \text{if} T_i \le t \le M_i \text{and} P \le K \end{cases}$$

Figure 6.29-1 Supervisory delta adjustment for call options

$$\delta_i = ls \cdot \begin{cases} \Phi\left(-\dfrac{\ln\left(\dfrac{P_i+\lambda_j}{K_i+\lambda_j}\right)+0.5\cdot\sigma_i^2\cdot T_i}{\sigma_i\cdot\sqrt{T_i}}\right) & \text{if} t < T_i \\ 0 & \text{if} T_i \le t \le M_i \text{and} P > K \\ 1 & \text{if} T_i \le t \le M_i \text{and} P \le K \end{cases}$$

Figure 6.29-2 Supervisory delta adjustment for put options

where for each option,

- Call/Put flag is read from SACCR attributes.
- Φ is the cumulative distribution function of a standard normal random variable.
- ls = 1 if the trade is long the hedging set, -1 otherwise. The long/short flag is determined according to the non-option case above.
- P_i is read from SACCR attributes.
- K_i is read from SACCR attributes.

- λ_j is the shift amount, applicable to Interest Rate derivatives only. For other asset classes, λ_j = 0.
  - If expression parameter `IRShift` = PER_CURRENCY, the shift amount is read from table `SACCRNegativeInterestRateLimits`, based on currency j that trade i is mapped to.
  - If expression parameter `IRShift` = PER_TRADE, $\lambda_j = \max(\text{IRFloor} - \min(P, K), 0)$, where `IRFloor` is a Mag configuration setting (`Finance.SACCR_SFT.Defaults.IRFloor`), expressed in percentage points. By default `IRFloor` = 0.1 (0.1%).
- T_i and M_i are internally calculated, see *"Time Factors"* in • Preliminary Materials on page 126 for details.
- σ_i is the supervisory option volatility of the trade, see *"Supervisory Parameters"* in • Preliminary Materials on page 126 for details.

Table 6.29-8 Supervisory delta adjustment for CDO tranches:

δ_i	**Purchased (long protection)**	**Sold (short protection)**
CDO tranches	$\frac{15}{(1+14 \cdot A_i) \cdot (1+14 \cdot D_i)}$	$-\frac{15}{(1+14 \cdot A_i) \cdot (1+14 \cdot D_i)}$

where for each CDO tranche,

- Purchased/Sold flag is read from SACCR attributes.
- A_i is the attachment point of the CDO tranche, read from SACCR attributes.
- D_i is the detachment point of the CDO tranche, read from SACCR attributes.

5. **Supervisory Parameters**

   Three supervisory parameters are used in the add-on calculation:

   - The supervisory factor, $SF_k^{(a)}$, of hedging bucket k asset class a captures the volatility of the asset class. It is looked up from table `SACCRSupervisoryFactor`.
   - The correlation factor, $\rho_k^{(a)}$, of hedging bucket k asset class a determines the degree of offsetting/hedging benefit within the asset class. It is looked up from table `SACCRIRCorrelations` for interest rate derivatives and table `SACCRCorrelations` for other asset classes.
   - The supervisory option volatility, σ_i , of trade i is used to calculate supervisory delta adjustment for options. It is looked from table `SACCROptionVolatility`.

   The supervisory factors given in the table are for `Regular` transaction hedging sets. For a `basis` transaction hedging set, the supervisory factor applicable to its relevant asset class must be multiplied by one-half. For a `volatility` transaction hedging set, the supervisory factor applicable to its relevant asset class must be multiplied by a factor of five.

• AddOn - Interest Rate

Interest Rate hedging buckets, hedging sets and trade types are defined in section • Aggregation Hierarchy on page 124. The add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $$d_i^{(IR)} = N_i \cdot SD_i \cdot FX_{ccy_i, ccy_d}$$

2. Trade-level effective notional for trade i

$D_i^{(IR)} = \delta_i \cdot d_i^{(IR)} \cdot MF_i^{(type)}$

where $(type)$ = (Margined) under margined calculation and $(type)$ = (Unmargined) under unmargined calculation.

3. Hedging-bucket-level effective notional for hedging bucket (maturity bucket) MB_k from hedging set ccy_j :

   $D_{jk}^{(IR)} = \sum_{i \in \{ccy_j, MB_k\}} D_i^{(IR)}$

4. Hedging-set-level effective notional for hedging set j :

   - If `IRTenorOffsetting` = TRUE then:

$$EffectiveNotional_j^{(IR)} = \left[\left(D_{j1}^{(IR)}\right)^2 + \left(D_{j2}^{(IR)}\right)^2 + \left(D_{j3}^{(IR)}\right)^2 + 2 \cdot \rho_{1,2} \cdot D_{j1}^{(IR)} \cdot D_{j2}^{(IR)} + 2 \cdot \rho_{2,3} \cdot D_{j2}^{(IR)} \cdot D_{j3}^{(IR)} + 2 \cdot \rho_{1,3} \cdot D_{j1}^{(IR)} \cdot D_{j3}^{(IR)}\right]^{1/2}$$

     where $\rho_{m,n}$ is looked up from table `SACCRIRCorrelations` based on `TableColumn`.

   - If `IRTenorOffsetting` = FALSE then:

     $EffectiveNotional_j^{(IR)} = \left|D_{j1}^{(IR)}\right| + \left|D_{j2}^{(IR)}\right| + \left|D_{j3}^{(IR)}\right|$

     where $k = 1, 2, 3$ corresponds to the three hedging buckets within hedging set j .

5. Hedging-set-level add-on:

   $AddOn_j^{(IR)} = SF_j^{(IR)} \cdot EffectiveNotional_j^{(IR)}$

6. Trade-type-level add-on for trade type TT_x

   $AddOn_x^{(IR)} = \sum_{j \in TT_x} AddOn_j^{(IR)}$

7. Asset-class-level add-on:

   $AddOn^{(IR)} = \sum_x AddOn_x^{(IR)}$

• AddOn - Foreign Exchange

Foreign Exchange hedging buckets, hedging sets and trade types are defined in section • Aggregation Hierarchy on page 124. The add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   - If $ccy_{i,L} \neq ccy_d$ and $ccy_{i,S} \neq ccy_d$:

     $d_i^{(FX)} = \max\left(N_{i,L} \cdot FX_{ccy_{i,L}, ccy_d}, N_{i,S} \cdot FX_{ccy_{i,S}, ccy_d}\right)$

   - else if $ccy_{i,L} = ccy_d$:

     $d_i^{(FX)} = N_{i,S} \cdot FX_{ccy_{i,S}, ccy_d}$

   - else if $ccy_{i,S} = ccy_d$:

     $d_i^{(FX)} = N_{i,L} \cdot FX_{ccy_{i,L}, ccy_d}$

2. Trade-level effective notional for trade i

   $D_i^{(FX)} = \delta_i \cdot d_i^{(FX)} \cdot MF_i^{(type)}$

   where $(type)$ = (Margined) under margined calculation and $(type)$ = (Unmargined) under unmargined calculation.

3. Hedging-bucket-level effective notional for the single hedging bucket HB_k under hedging set HS_j :

$D_{jk}^{(FX)} = \sum_{i \in \{HB_k, HS_j\}} D_i^{(FX)}$

4. Hedging-set-level effective notional for hedging set HS_j :

   $EffectiveNotional_j^{(FX)} = D_{jk}^{(FX)}$

5. Hedging-set-level add-on:

   $AddOn_j^{(FX)} = SF_j^{(FX)} \cdot \left| EffectiveNotional_j^{(FX)} \right|$

6. Trade-type-level add-on for trade type TT_x :

   $AddOn_x^{(FX)} = \sum_{j \in TT_x} AddOn_j^{(FX)}$

7. Asset-class-level add-on:

   $AddOn^{(FX)} = \sum_x AddOn_x^{(FX)}$

- **AddOn - Credit**

Credit hedging buckets, hedging sets, and trade types are defined in section • Aggregation Hierarchy on page 124. The add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $d_i^{(CR)} = N_i \cdot SD_i \cdot FX_{ccy_i, ccy_d}$

2. Trade-level effective notional for trade i

   $D_i^{(CR)} = \delta_i \cdot d_i^{(CR)} \cdot MF_i^{(type)}$

   where $(type)$ = (Margined) under margined calculation and $(type)$ = (Unmargined) under unmargined calculation.

3. Hedging-bucket-level effective notional for hedging bucket $Entity_k$ of hedging set HS_j:

   $D_{jk}^{(CR)} = \sum_{i \in \{HS_j, Entity_k\}} D_i^{(CR)}$

4. Hedging-bucket-level add-on:

   $AddOn_{jk}^{(CR)} = SF_k^{(CR)} \cdot D_{jk}^{(CR)}$

5. Hedging-set-level add-on for hedging set HS_j:

   $AddOn_j^{(CR)} = \left[\left(\sum_k \rho_k^{(CR)} \cdot AddOn_{jk}^{(CR)} \right)^2 + \sum_k \left(1 - \left(\rho_k^{(CR)} \right)^2 \right) \cdot \left(AddOn_{jk}^{(CR)} \right)^2 \right]^{1/2}$

6. Trade-type-level add-on for trade type TT_x

   $AddOn_x^{(CR)} = \sum_{j \in TT_x} AddOn_j^{(CR)}$

7. Asset-class-level add-on:

   $AddOn^{(CR)} = \sum_x AddOn_x^{(CR)}$

- **AddOn - Equity**

Equity hedging buckets, hedging sets and trade types are defined in section • Aggregation Hierarchy on page 124. The add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $d_i^{(EQ)} = N_i \cdot FX_{ccy_i, ccy_d}$

2. Trade-level effective notional for trade i

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$D_i^{(EQ)} = \delta_i \cdot d_i^{(EQ)} \cdot MF_i^{(type)}$

where $(type)$ = (Margined) under margined calculation and $(type)$ = (Unmargined) under unmargined calculation.

3. Hedging-bucket-level effective notional for hedging bucket $Entity_k$ of hedging set HS_j:

   $D_{jk}^{(EQ)} = \sum_{i \in \{HS_j, Entity_k\}} D_i^{(EQ)}$

4. Hedging-bucket-level add-on:

   $AddOn_{jk}^{(EQ)} = SF_k^{(EQ)} \cdot D_{jk}^{(EQ)}$

5. Hedging-set-level add-on for hedging set HS_j:

   $AddOn_j^{(EQ)} = \left[\left(\sum_k \rho_k^{(EQ)} \cdot AddOn_{jk}^{(EQ)}\right)^2 + \sum_k \left(1 - \left(\rho_k^{(EQ)}\right)^2\right) \cdot \left(AddOn_{jk}^{(EQ)}\right)^2\right]^{1/2}$

6. Trade-type-level add-on for trade type TT_x

   $AddOn_x^{(EQ)} = \sum_{j \in TT_x} AddOn_j^{(EQ)}$

7. Asset-class-level add-on:

   $AddOn^{(EQ)} = \sum_x AddOn_x^{(EQ)}$

- **AddOn - Commodity**

Commodity hedging buckets, hedging sets and trade types are defined in section • Aggregation Hierarchy on page 124. The add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $d_i^{(CM)} = N_i \cdot FX_{ccy_i, ccy_d}$

2. Trade-level effective notional for trade i

   $D_i^{(CM)} = \delta_i \cdot d_i^{(CM)} \cdot MF_i^{(type)}$

   where $(type)$ = (Margined) under margined calculation and $(type)$ = (Unmargined) under unmargined calculation.

3. Hedging-bucket-level effective notional for hedging bucket (commodity type) $Type_k$ of hedging set HS_j:

   $D_{jk}^{(CM)} = \sum_{i \in \{HS_j, Type_k\}} D_i^{(CM)}$

4. Hedging-bucket-level add-on:

   $AddOn_{jk}^{(CM)} = SF_k^{(CM)} \cdot D_{jk}^{(CM)}$

5. Hedging-set-level add-on for hedging set HS_j :

   $AddOn_j^{(CM)} = \left[\left(\sum_k \rho_k^{(CM)} \cdot AddOn_{jk}^{(CM)}\right)^2 + \sum_k \left(1 - \left(\rho_k^{(CM)}\right)^2\right) \cdot \left(AddOn_{jk}^{(CM)}\right)^2\right]^{1/2}$

6. Trade-type-level add-on for trade type TT_x

   $AddOn_x^{(CM)} = \sum_{j \in TT_x} AddOn_j^{(CM)}$

7. Asset-class-level add-on:

   $AddOn^{(CM)} = \sum_x AddOn_x^{(CM)}$

6.30 SACCR Risk Weighted Assets under the Standardized Approach

To calculate the bank's risk based capital requirements for Counterparty Credit Risk using the standardized approach, as described in "Calculation of RWA for credit risk", BIS. The proposed methodology also complies with the EU implementation of the Risk Weighted Assets (RWA) calculation, as outlined in "Capital Requirements Regulation (CRR): REGULATION (EU) No 575/2013" .

- **Input Parameters**
  - `CentralBankRating`: A counterparty level attribute, the enumerations should be user-defined.
  - `OrganizationType`: A counterparty level attribute. This is a configurable enumeration. Custom attributes can be used here.
  - `CreditRating`: A counterparty level attribute. This is a configurable enumeration. Custom attributes can be used here also.
  - `regulatoryException`: A boolean trade level attribute, defaults ot be False. It is used as a generic attribute to determine if alternative risk weights are available for the `OrganizationType`.
  - `DefaultRiskWeight`: A default risk weight when `OrganizationType` and `CreditRating` values of the counterparty are not well defined in the table.
  - Original Maturity (OM): Measured in days. OM = Maturity Date - Trade Date of the trade
  - Residual Maturity (RM): Measured in days. RM = Maturity Date of the trade - Processing Date
  - `SACCRRWARiskweights.csv`: A Risk Weights table needs to be defined. Ensure that all `OrganizationType`enumerations and `CreditRating` enumerations are listed in the rows and columns of the table respectively. For example:

OrganizationType	ExceptionType	ExceptionValue	AAA,AA+,AA,...,CCC,C
Central Bank	Regulatory Exception	T	0, ..., 0
Central Bank	Regulatory Exception	F	20, ..., 100
Public Sector Entity	Original Term	T	20, ..., 20
Public Sector Entity	Original Term	F	20, ..., 150
Multilateral	None		20, ..., 150
Institution Rated	Residual Term		20, ..., 100
Institution Unrated	Original Term	T	20, ..., 100
Institution Unrated	Original Term	F	20, ..., 150
Central CP	None		2, ..., 2

    For certain `OrganizationType`, the risk weight is a function of each trade's original term to maturity or the trade's residual term to maturity or an exception, specified on the table as separate rows.

  - `CentralBankWeights.csv`: A Central Bank Weights table that contains a set of Risk Weights corresponding to the Central Bank Weights. For example:

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AAA	AA+	...	C
0	20	...	150

- **Assumptions**

The bank is using SACCR to calculate their Exposure At Default (EAD) for each counterparty. The scope of the calculation will only cover bilateral derivative transactions and exposures to Central Counteparties (where the RW is already pre-calculated, i.e., the solution does not take calculate RW based on default fund contributions); RWA calculations for exposures to equity investments in funds, securitizations and those arising from unsettled transactions and failed trades are not in scope. Also, the bank calculating exposures is incorporated in a jurisdiction that allows the use of external ratings for regulatory purposes.

For the input data,

- For each counterparty, user has to provide the counterparty's Central Bank Credit Rating (or must catch exception in code)
- Credit Ratings on the look up table should be defined in decreasing quality in the columns.
- Each `OrganizationType/ExceptionType` pair (except when `ExceptionType` = NONE or Residual Term) must have a T and F row
- Each `OrganizationType` has a unique exception type.

- **Methodology**

The standardised approach assigns standardised risk weights to exposures, as described below. Risk weighted assets (RWA) are calculated as the product of the standardised risk weights (RW) and the exposure amount (EAD). EAD is calculated using the SACCR methodology at the netting set level. Please refer to topic 6.27 SACCR Capital Exposure At Default on page 107 for modelling details. Thus, the RWA at the netting set level is determined as follows:

$RWA = RW \times EAD$

The Capital calculation is thus defined as:

$Capital = 8\% \times RWA$

- **Determining the Risk Weight**

In general, the risk weight (RW) of each netting set is determined as follows:

- Step 1: Determine Counterparty's `OrganizationType` and `Creditrating` from the counterparty attributes.
- Step 2:
  - CASE 1: `OrganizationType` and `Creditrating` values of the couterparty are well defined in the table.
    - CASE 1.1: `OrganizationType` has `ExceptionType` = None, RW is read from the corresponding row of the Risk Weights table
    - CASE 1.2: `OrganizationType` has `ExceptionType` = Original Term
      - CASE 1.2.1: All trades have original maturity ≤ 90 days, RW is read from the corresponding row where `ExceptionValue` = T.
      - CASE 1.2.2: At least one trade's original maturity > 90 days
        - CASE 1.2.2.1: All trades have `regulatoryException` = True and all trades have residual maturity ≤ 90 days, RW is read from the `CentralBankWeights.csv` and the column corresponding to a less favourable credit rating than the

counterparty's `CentralBankRating` (that is, the column after the Central Bank Rating) [Pick the last column if the Central Bank Rating is already in the last column.]

      - CASE 1.2.2.2: At least one trade has `regulatoryException` = False or residual maturity > 90 days, RW is read from the corresponding row where `ExceptionValue` = F and column equal to counterparty's `CentralBankRating`.

  - CASE 1.3: `OrganizationType` has `ExceptionType` = Residual Term
    - CASE 1.3.1: All trades have `regulatoryException` = True and all trades have residual maturity ≤ 90 days, RW is read from the `CentralBankWeights.csv` and the column corresponding to a less favourable credit rating than the counterparty's `CentralBankRating` (that is, the column after the Central Bank Rating) [Pick the last column if the Central Bank Rating is already in the last column.]
    - CASE 1.3.2: At least one trade has `regulatoryException` = False and all trades have residual maturity ≤ 90 days, RW is read from the corresponding row where `ExceptionValue` = T and the column equal to the counterparty's `Creditrating`.
    - CASE 1.3.3: At least one trade has residual maturity > 90 days, RW is read from the corresponding row where `ExceptionValue` = F and the column equal to the counterparty's `Creditrating`.
  - CASE 1.4: `OrganizationType` has `ExceptionType` = Regulatory Exception
    - CASE 1.4.1: All trades have `regulatoryException` = True, RW is read from corresponding row where `ExceptionValue` = T.
    - CASE 1.4.2: At least one trade has `regulatoryException` = False, RW is read from the corresponding row where `ExceptionValue` = F.
- CASE 2: `OrganizationType` and `ExceptionType` values of the Counterparty are not well defined in the table, $RW = DefaultRiskWeight$

6.31 SACCR Profile Expected Exposure

The function `SACCR Profile Expected Exposure` computes the expected exposure of the portfolio over time based on the Basel SA-CCR methodology.

- **Input Parameters**

- ReportingTimeline: A list of reporting time steps relative to the processing date.
- IRTenorOffsetting: A boolean attribute, defaults to True. Setting it to False disallows offsetting benefit between maturity buckets for Interest Rate derivatives.
- FXOffsetting: A boolean attribute, defaults to False. Setting it to True enables offsetting between FX currency pairs for Foreign Exchange derivatives.
- Alpha1: A parameter to scale the portfolio RC, defaults to 1.4
- Alpha2: A parameter to scale the portfolio AddOn, defaults to 1.4
- MultiplierFloor: Sets the floor for AddOn multiplier, defaults to 0.05.
- MultiplierFactor: Sets the factor for AddOn multiplier, defaults to 2.
- MaturityFactorMultiplier: Sets the multiplier for margined maturity factor, defaults to 1.5.
- NettingRule: accepts NO_NET, AGREEMENT and LEGAL_OPINION.

- SupervisoryDurationDF: Sets the discount rate in supervisory duration calculation, defaults to 0.05.
- IRShift: accepts PER_CURRENCY or PER_TRADE. It specifies whether the IR shift amount is currency dependent or trade dependent.
- EarlyExit: A boolean attribute, defaults to False. When set to True, trades with nextEarlyTerminationDate before the report date are removed from the exposure profile.
- CollateralModel: Sets the collateral model for the expression: NONE, NonSim_CollateralModel, etc.
- IncludeVMInRCUnmargined: An optional collateral model parameter, defaults to 1 (True).
- TableColumn: A lookup key that is associated with the following table inputs: SACCRSupervisoryFactor, SACCROptionVolatility, SACCRCorrelations, SACCRIRCorrelations, SACCRFXCorrelations, SACCRFXHaircut, SACCRCCYPFEHaircuts and VolatilityAdjustments.

• Expected Exposure Profile

The Expected Exposure for a given reporting date t consists of two terms: the replacement cost and the add-on times a multiplier. The replacement cost captures the current exposure at the processing date or at the beginning of the margin period of risk. The add-on times the multiplier captures the expected increase in exposure up to the reporting date.

Let X be the node/trade at which the base calculation of Expected Exposure is performed, where X equals one of the following:

1. A CSA node (VM or VMIM or Legacy)
2. An MA node with no CSA
3. An NC node
4. An NN trade

The abbreviations are defined in section • Legal Hierarchy on page 137. If X is such a node, at reporting time t , the Expected Exposure is defined as follows:

- **Case 1**: If X consists of a single SA-CCR netting set (see section • Netting Sets on page 137), the Expected Exposure is:
  $EE_t = a_1 \cdot RC_t + a_2 \cdot m_t \cdot AddOn_t$
- **Case 2**: If X consists of multiple SA-CCR netting sets, the multiplier and add-on are calculated for each of the netting sets as if these sets were unmargined, then summed up to node X . The Expected Exposure is:
  $EE_t = a_1 \cdot RC_t + a_2 \cdot \sum_{NS \in X} m_{t,NS} \cdot AddOn_{t,NS}$

For any nodes above X in the legal hierarchy, the Expected Exposure is the sum of the Expected Exposures of its child nodes.

Here:

- a_1 and a_2 are expression parameters Alpha1 and Alpha2.
- The replacement cost RC_t is defined in section • Replacement Cost on page 140.
- The multiplier m_t is defined in section • Multiplier on page 151.
- The add-on $AddOn_t$ is defined in section • AddOn on page 152.

Note

Under case 1, the EE_t for a margined set is capped by the EE_t of the same portfolio without a margin agreement. For a margined set, the replacement cost, multiplier and add-on are computed twice, with margining and without margining.

Note

At time 0, both the multiplier and the add-on are zero.

- **Legal Hierarchy**

Each trade is allocated to a node in the legal hierarchy. The node is defined by:

- Counterparty (CP)
- OwnEntity (OE)
- Container: a master agreement (MA) or a no-net node (NN). Multiple MAs may be signed with a counterparty. Trades that do not meet the filtering criteria of any MA are allocated to an NN node.
- SubContainer (optional): a credit support annex (CSA) or a non-collateralized node (NC). There may be one CSA, no CSA, or multiple CSAs appended to an MA. If CSA exists under the MA, trades that meet the MA filtering criteria but not the criteria of any CSA under that MA are allocated to a residual "NC" node under the MA.

- **Netting Sets**

The SA-CCR netting set (referred to as *netting set* in subsequent sections) is a subset of a master agreement (MA) or a no-net node (NN). It is defined as follows:

1. A netting set is an NC node under the MA, a CSA node under the MA, or an MA node with no CSAs if all trades in the NC, CSA or MA node net. All trades net when:
   - NettingRule = `AGREEMENT` and `CloseoutNetting` = TRUE; or
   - NettingRule = `LEGAL_OPINION` and either:
     - no closeout netting pools are present under the MA and `AgreementsEnforceable` = TRUE and `CloseoutNetting` = TRUE; or
     - all trades in the node belong to a unique non-empty closeout netting pool or there is a single trade in the node; or
   - NettingRule = `NO_NET` and there is a single trade in the node.
2. When not all trade in an NC, CSA or MA node net, the node is partitioned accordingly and each subset forms a netting set.
   - NettingRule = `AGREEMENT` or `NO_NET`: each trade forms a netting set
   - NettingRule = `LEGAL_OPINION`: each closeout netting pool forms a netting set and for trades that don't belong to any closeout netting pools, each trade forms a netting set.
3. For NN node, a set containing a single trade of the NN node forms a netting set.

- **Margined vs. Unmargined Set**

The measurement of EE_t is different depending on whether node X is margined or unmargined.

If the expression parameter `CollateralModel` = NONE, the netting set is considered unmargined. Otherwise, the margined nodes are those under a bilateral CSA agreement for which the counterparty post collateral, that is for which `CPVMPostStyle` = Gross or Net. When

`CPVMPostStyle` = None, then node X is considered unmargined. When node X is not a CSA, it is considered unmargined as well.

- **Sold Options Exception**

The sold options exception may apply when X is an MA node with no CSA, an NC node, or an NN trade; that is, when X is not subject to variation margin. X might be made up of one or more netting sets, $NS_{X,1}$, $NS_{X,2}$, etc. For each netting set $NS_{X,i}$, for which all of the following conditions are satisfied:

1. $NS_{X,i}$ has a negative mark-to-market
2. $NS_{X,i}$ consists of only sold options (saccr.optionparams.sold=True)
3. All premiums of the trades in $NS_{X,i}$ have been received upfront (saccr.optionparams.remainingPremiums=False)

the $AddOn_t$ of $NS_{X,i}$ is set to zero and the mark-to-market of $NS_{X,i}$ in the replacement cost is set to zero.

- **Early Exit**

The following applies if expression parameter `EarlyExit` is set to True: for trades with standard attribute nextEarlyTerminationDate populated, if the nextEarlyTerminationDate is strictly before the report date t , then the trade is removed in the profile calculation for EE_t .

- **Trade-Level IM and IA**

Counterparty and User Independent Amounts and Initial Margins can be linked to a deal i . These profiles are specified in `standardTradeMitigantAttributes` as:

- TradeMarginType: Initial or Independent
- PaymentDate and PaymentAmount pair:

Date	Value	Comment
D_1	V_1	V_1 applies from D_1 (inclusive) to D_2 (exclusive)
D_2	V_2	V_2 applies from D_2 (inclusive) to D_3 (exclusive)
...	...	...
D_n	V_n	V_n applies from D_n (inclusive) to the end of simulation

Note

V_i should be entered as a positive value.

- PostDirection: To user or To counterparty
- Currency: The currency of the PaymentAmount, can be an acceptable currency code.
- Segregated: only used for IM, True or False.

- **Settled-to-Market Agreement**

The inputs are defined in topic 6.27 SACCR Capital Exposure At Default on page 107, section ● Settled-to-Market Agreement on page 110

- **Valuation Input**

The table summarizes the variables used in replacement cost calculation.

Description	Symbol	Input
Reporting date	t	Internal variable
Domestic currency	ccy_d	<counterparty input>DomesticCurrency of node's Own Entity.
Counterparty threshold	$TH_{t,ctpy}$	<collateral agreement input>CPVMThreshold
User threshold	$TH_{t,user}$	<collateral agreement input>UserVMThreshold
Counterparty minimum transfer amount	$MTA_{t,\,ctpy}$	<collateral agreement input>CPVMMinTransfer
User minimum transfer amount	$MTA_{t,\,user}$	<collateral agreement input>UserVMMinTransfer
Counterparty independent amount	$IA_{t,\,ctpy}$	<collateral agreement input>CPIndependent
User independent amount	$IA_{t,\,user}$	<collateral agreement input>UserIndependent
Settlement lag	$SettleLag_{CSA}$	<collateral agreement input>SettlementLag
Margin lag	$MarginLag_{CSA}$	<collateral agreement input>MarginLag
Frequency of margin calls	$CallPeriod_{CSA}$	<collateral agreement input>CallPeriod
Cure period or liquidation period in the event of default	$CurePeriod_{CSA}$	<collateral agreement input>CurePeriod
Collateral agreement currency	ccy_{CSA}	<collateral agreement input>AgreementCurrency
Deal mark-to-market value	V_i	SACCR attributes
Deal currency	ccy_i	SACCR attributes
Deal maturity	mat_i	maximum of standard trade attribute `maturity-date` and SACCR attribute `maturity`
FX conversion factor from curren-cy1 to domestic currency	$FX_{ccy_1,\,ccy_d}$	Internal variable
FX conversion factor from curren-cy1 to collateral agreement cur-rency	$FX_{ccy_1,\,ccy_{CSA}}$	Internal variable
Input value of the counterparty Independent Amount on deal i at time t	$cpIAInput(i,t)$	Input from `standardTradeMitigantAttributes`
Input value of the user Independ-ent Amount on deal i at time t	$uIAInput(i,t)$	Input from `standardTradeMitigantAttributes`
Input value of the counterparty Initial Margin on deal i at time t	$cpIMInput(i,t)$	Input from `standardTradeMitigantAttributes`
Input value of the user Initial Margin on deal i at time t	$uIMInput(i,t)$	Input from `standardTradeMitigantAttributes`
Currency of the counterparty In-dependent Amount on deal i	$ccy_{cpIA,\,i}$	Input from `standardTradeMitigantAttributes`
Currency of the user Independent Amount on deal i	$ccy_{uIA,\,i}$	Input from `standardTradeMitigantAttributes`

Description	Symbol	Input
Currency of the counterparty Initial Margin on deal i	$ccy_{cpIM,i}$	Input from `standardTradeMitigantAttributes`
Currency of the user Initial Margin on deal i	$ccy_{uIM,i}$	Input from `standardTradeMitigantAttributes`
Counterparty Independent Amount on deal i at time t	$cpIA(i,t)$	Internal Value
User Independent Amount on deal i at time t	$uIA(i,t)$	Internal Value
Counterparty Initial Margin on deal i at time t	$cpIM(i,t)$	Internal Value
User Initial Margin on deal i at time t	$uIM(i,t)$	Internal Value

- **Replacement Cost**

- For a node X with `SettledtoMarketAgreement` = False:

  Case 1: X consists of a single netting set

  Unmargined set:

$$RC_t = \begin{cases} \max\left(V_{t,NS} - cpIM(t,ccy) + uIM_{unseg}(t,ccy) - VM_t, 0\right) & \text{if CollateralModel} \neq \text{NONE} \\ \max\left(V_{t,NS}, 0\right) & \text{if CollateralModel} = \text{NONE} \end{cases}$$

  where

  - $V_{t,NS}$: the net mark-to-market value of all derivative transactions in the netting set. See ● Formula: V on page 143 for details.
  - $cpIM(t,ccy)$: the haircut adjusted total value of segregated and unsegregated initial margin posted by the counterparty at time t . See ● Formula: Node-level IM on page 144 and ● Formula: Trade-level IM and IA on page 148 for details.
  - $uIM_{unseg}(t,ccy)$: the haircut adjusted total value of unsegregated initial margin posted by the user at time t . See ● Formula: Node-level IM on page 144 and ● Formula: Trade-level IM and IA on page 148 for details.
  - VM_t is defined as:
    - $$VM_t = \begin{cases} -VM_{t,user\_unseg} & \text{if IncludeVMinRCUnmargined=True} \\ \delta_{ctpy} \cdot cpIA(t) - \delta_{user} \cdot uIA(t) & \text{if IncludeVMinRCUnmargined=False} \end{cases}$$
    - $VM_{t,user\_unseg}$ is the current haircut adjusted value of unsegregated variation margin posted by the user. See ● Formula: VM on page 147 for details.
    - δ_{ctpy} = 1 if `CPVMPostStyle` ≠ None; 0 otherwise.
    - δ_{user} = 1 if `UserVMPostStyle` ≠ None and `UserVMSegregated` = False; 0 otherwise.
    - $cpIA(t)$ and $uIA(t)$ are the total counterparty and user independent amounts at time t , expressed in absolute values respectively. See ● Formula: Trade-level IM and IA on page 148 for details.

  Margined set:

$RC_t =$
$$\begin{cases} \max\left(V_{0,NS} - cpIM(t_0, ccy) + uIM_{unseg}(t_0, ccy) - VM_{0,ctpy} + VM_{0,user\_unseg}, 0\right) & \text{if } t = 0 \\ \max\left(TH_{t,ctpy} + MTA_{t,ctpy} - cpIA(t) + \delta_{user} \cdot uIA(t) - cpIM(t, ccy) + uIM_{unseg}(t, ccy), 0\right) & \text{otherwise} \end{cases}$$

where

- $V_{t_0,NS}$: the current net mark-to-market value of all derivative transactions in the netting set. See • Formula: V on page 143 for details.
- $cpIM(t_0, ccy)$: the current haircut adjusted total value of segregated and unsegregated initial margin posted by the counterparty. See • Formula: Node-level IM on page 144 and • Formula: Trade-level IM and IA on page 148 for details.
- $uIM(t_0, ccy)$: the current haircut adjusted total value of unsegregated initial margin posted by the user. See • Formula: Node-level IM on page 144 and • Formula: Trade-level IM and IA on page 148 for details.
- $VM_{t_0,ctpy}$: the current haircut adjusted total value of segregated and unsegregated variation margin posted by the counterparty. See • Formula: VM on page 147 for details.
- $VM_{t_0,user\_unseg}$: the current haircut adjusted value of unsegregated initial margin posted by the user. See • Formula: VM on page 147 for details.
- $TH_{t,ctpy}$: the counterparty threshold (`CPVMThreshold`) at time t .
- $MTA_{t,ctpy}$: the counterparty minimum transfer amount (`CPVMMinTransfer`) at time t .
- δ_{user} = 1 if `UserVMPostStyle` $\neq$ None and `UserVMSegregated` = False; 0 otherwise.
- $cpIA(t)$ and $uIA(t)$ are the total counterparty and user independent amounts at time t , expressed in absolute values respectively. See • Formula: Trade-level IM and IA on page 148 for details.
- $cpIM(t, ccy)$: the haircut adjusted total value of segregated and unsegregated initial margin posted by the counterparty at time t . See • Formula: Node-level IM on page 144 and • Formula: Trade-level IM and IA on page 148 for details.
- $uIM_{unseg}(t, ccy)$: the haircut adjusted total value of unsegregated initial margin posted by the user at time t . See • Formula: Node-level IM on page 144 and • Formula: Trade-level IM and IA on page 148 for details.

Case 2: X consists of multiple netting sets

In this case, not all trades in the node X nets. If CollateralModel $\neq$ NONE, the replacement cost for X is:

$$RC_t = \max\left\{\sum_{NS \in X} \max(V_{t,NS}, 0) - \max\left(VM_t + cpIM(t, ccy) - uIM_{unseg}(t, ccy), 0\right), 0\right\}$$
$$+ \max\left\{\sum_{NS \in X} \min(V_{t,NS}, 0) - \min\left(VM_t + cpIM(t, ccy) - uIM_{unseg}(t, ccy), 0\right), 0\right\}$$

where

- $V_{t,NS}$: the net mark-to-market value of all derivative transactions in the netting set. See • Formula: V on page 143 for details.
- $cpIM(t, ccy)$ and $uIM_{unseg}(t, ccy)$: defined in Case 1. See • Formula: Node-level IM on page 144 and • Formula: Trade-level IM and IA on page 148 for details.
- VM_t : the haircut adjusted value of net variation margin received at time t . If X is not a CSA or `CPVMPostStyle` = `UserVMPostStyle` = None, then VM_t = 0; otherwise,

- If CPVMPostStyle = Gross/Net and UserVMPostStyle = Gross/Net:

$$VM_t = \begin{cases} VM_{0,ctpy} - VM_{0,user\_unseg} & \text{if } t = 0 \\ \max\left(\sum_{NS \in X} V_{t,NS} + cpIA(t) - uIA(t) - TH_{t,ctpy} - MTA_{t,ctpy}, 0\right) & \text{if } t > 0, \sum_{NS \in X} V_{t,NS} + cpIA(t) - uIA(t) \geq 0 \\ \min\left(\sum_{NS \in X} V_{t,NS} + cpIA(t) - uIA(t) + TH_{t,user} + MTA_{t,user}, 0\right) & \text{if } t > 0, \sum_{NS \in X} V_{t,NS} + cpIA(t) - uIA(t) < 0 \text{ and UserVMSegregated=False} \\ 0 & \text{otherwise} \end{cases}$$

- If CPVMPostStyle = Gross/Net and UserVMPostStyle = None:

$$VM_t = \begin{cases} VM_{0,ctpy} & \text{if } t = 0 \\ \max\left(\sum_{NS \in X} V_{t,NS} + cpIA(t) - uIA(t) - TH_{t,ctpy} - MTA_{t,ctpy}, 0\right) & \text{if } t > 0 \end{cases}$$

- If CPVMPostStyle = None and UserVMPostStyle = Gross/Net:

$$VM_t = \begin{cases} -VM_{0,user\_unseg} & \text{if } t = 0 \\ \min\left(\sum_{NS \in X} V_{t,NS} + cpIA(t) - uIA(t) + TH_{t,user} + MTA_{t,user}, 0\right) & \text{if } t > 0 \text{ and UserVMSegregated=False} \\ 0 & \text{otherwise} \end{cases}$$

Here $VM_{0,ctpy}$ and $VM_{0,user\_unseg}$ are defined in • Formula: VM on page 147; $TH_{t,user}$ is the user threshold (UserVMThreshold) at time t , expressed in absolute value; and $MTA_{t,user}$ is the user minimum transfer amount (UserVMMinTransfer) at time t , expressed in absolute value. $cpIA(t)$ and $uIA(t)$ are the total counterparty and user independent amounts at time t , expressed in absolute values respectively. See • Formula: Trade-level IM and IA on page 148 for details.

If CollateralModel = NONE:

$RC_t = \max\{\sum_{NS \in X} \max(V_{t,NS}, 0), 0\} + \max\{\sum_{NS \in X} \min(V_{t,NS}, 0), 0\}$

- For a VMCSA (Legacy, VM or IMVMCSA) node X with SettledtoMarketAgreement = True,
  $RC_t^{STM} = \begin{cases} max\{-(cpIM(t,ccy) - uIM_{unseg}(t,ccy), 0\} & \text{if CollateralModel} \neq \text{NONE} \\ 0 & \text{if CollateralModel} = \text{NONE} \end{cases}$ where:
  - $cpIM(t, ccy)$ is the haricut-adjusted value of the initial margin (including node-level and trade-level intial margin) posted by the counterparty on a VMCSA (Legacy, VM or IM-VMCSA) node with the exception that the haircut uses a risk horizon of $min\{t, \frac{14}{365}\}$ (in years), see section • Formula: Trade-level IM and IA on page 148 for details.
  - $uIM_{unseg}(t, ccy)$ is the haricut-adjusted value of the initial margin (including node-level and trade-level intial margin) posted by the user on the VMCSA (Legacy, VM or IMVMCSA) node with the exception that the haircut uses a risk horizon of $min\{t, \frac{14}{365}\}$ (in years), see section • Formula: Trade-level IM and IA on page 148 for details.

• Mitigant Haircuts

The haircut adjusted mark-to-market captures the potential unfavorable change in mitigant value during the risk horizon. The counterparty (user) mitigant value is decreased (increased) by a haircut amount; more specifically:

- the MTM of a node mitigant posted by the counterparty is multiplied by
  $\max\left(0, 1 - HC \cdot \sqrt{RH} - HC_{FX} \cdot \sqrt{RH}\right) \cdot \max\left(0, 1 - HC_{default}\right)$
- the MTM of a node mitigant posted by the user is multiplied by
  $\left(1 + HC \cdot \sqrt{RH} + HC_{FX} \cdot \sqrt{RH}\right) \cdot \left(1 + HC_{default}\right)$

where

- RH : the risk horizon. For unmargined sets, the risk horizon used in collateral haircut is t years; for margined sets, the risk horizon is $\min(t, MPOR)$ in years, where MPOR is the • Margin Period of Risk on page 143.
- HC : the annualized haircut for the mitigant, that is, the input haircut multiplied by $\sqrt{1/LH_{HC}}$, where LH_{HC} : the liquidity horizon (in years) associated with the mitigant haircut. For cash mitigants, the mitigant haircut is taken from table `SACCRCCYPFEHaircuts` based on mitigant MTM currency; for security mitigants, the mitigant haircut is read from table `VolatilityAdjustments` based on the security class.
- HC_{FX} : the annualized FX haircut, that is, the input haircut factor multiplied by $\sqrt{1/LH_{FX}}$, where LH_{FX} is the liquidity horizon (in years) associated with the FX haircut. The FX haircut, read from table `SACCRFXHaircut`, is applied only if the currency of the collateral is different from the settlement currency. For CSA netting sets, use the CSA `Agreement Currency` as settlement currency; for MA or NC netting set, use the MA `Termination Currency`; and for trades in an NN node, use the currency of the trade.

Note

In the case where the collateral is assigned to an IMCSA ($\subset$ VMCSA) , the `Agreement Currency` of the VMCSA is used as settlement currency.

- $H_{default}$: the haircut at default for the mitigant, expressed in absolute value. It is applicable to variation margin only, any haircut at default applied to initial margin will be ignored.

• Margin Period of Risk

Margin Period of Risk (MPOR) is the time period from the last exchange of collateral covering a netting set of transactions with a defaulting counterparty until that counterparty is closed out and the resulting market risk is re-hedged. It is defined as follows for each CSA:

$MPOR = \max(MarginLag, SettleLag) + CallPeriod + CurePeriod - 1$

where:

- $MarginLag$: the number of days that the user continues making margin payments after the beginning of the MPOR.
- $SettleLag$: the number of days that the user continues making trade settlement payments after the beginning of the MPOR.
- $CallPeriod$: the frequency of margin calls.
- $CurePeriod$: the number of days between the time that the user stops making any payments (margin and trade settlement) and the time the portfolio is hedged and/or liquidated.

• Formula: V

The net mark-to-market value of all derivative transactions in the netting set at time t is:

$V_{t,NS} = \sum_{i \in \tau_t} MtM_{t,i}$

where

- $MtM_{t,i} = \begin{cases} MtM_{0,i}, & \text{if AmortizedMtM} = \text{False} \\ \frac{(M_i - t)}{M_i} \cdot MtM_{0,i}, & \text{if AmortizedMtM} = \text{True} \end{cases}$
- M_i : the maturity date of trade i .

- $MtM_{0,i}$: the current mark-to-market of trade i , spot converted to the `domestic currency`. The domestic currency is associated with the node's Own Entity.
- τ_t : the set of trades that have not matured by time t .

● Formula: Node-level IM

There are 2 methods allowed in calculating node-level IM for the SACCR Profile measure, user-input method and non-simulated-Dynamic-Initial-Margin (DIM) method. A model parameter `DIM_AddOn` is used to determine which method to use. When a DIM submodel is attached to a collateral model, `DIM_AddOn` is automatically set to True. `DIM_AddOn` is False otherwise.

If `DIM_AddOn` = False, the initial margin overtime is calculated using the user-input method as described below.

- User-Input Method for SACCR Profile

  The initial margin is entered either as node mitigants or as an initial margin profile:

  - (Node-level IM) The initial margin at time t is:

    $IM_{t,ctpy/user\_unseg} = \sum_{i \in \chi_t} MtM_{t,i}$

    where

    - $MtM_{t,i}$ is the time t haircut adjusted mark-to-market of mitigant i , spot converted to the `domestic currency`. Haircut adjustment is defined in section ● Mitigant Haircuts on page 142.
    - χ_t is the set of (ctpy or user_unseg) node level initial margins with delivery date on or before `Reporting Date` and return date on or after `Reporting Date`. Here the initial/variation margin type is read from `<mitigant input>MarginType`; post direction is read from `<mitigant input>PostDirection`; and segregated/unsegregated flag is read from `<mitigant input>Segregated`.
  - (IM Profile) The initial margin associated with the netting set at time t is interpolated from the initial margin profile using methods RIGHT_CONTINUOUS (take collateral value forward to next tenor), LEFT_CONTINUOUS (take collateral value backward to previous tenor) or LINEAR (approximate collateral value between two tenors). Here the profile entered is the haircut adjusted value.

If `DIM_AddOn` = True, the initial margin overtime is calculated as described below for SACCR Profile measures. This calculation is performed on the trades of an IMCSA ($\subset$ VMCSA) or IMVMCSA, and for the purpose of the following calculation all trades are assumed to net.

Note

No IM profile should be entered if the model parameter `DIM_Addon` is True.

- Dynamic Initial Margin Method for SACCR Profile

  The model is used to calculate the current initial margin and estimate the future initial margin for use in the non-simulated SACCR Profile measures. We call this estimate the non-simulated Dynamic Initial Margin (DIM).

  - Input Parameters

Attribute	Source	Comments
DIM_Addon	Model Parameter	A boolean model parameter, defaults to True.

Attribute	Source	Comments
		When set to True, DIM is used for non-simulated SACCR profile measures. When set to False, user input IM profile (from profile or node mitigants) is used for non-simulated SACCR Profile measures.
IM Margin Horizon	CSA Attribute	
CPIMThreshold	CSA Attribute	
UserIMThreshold	CSA Attribute	
DIMSaccrFactors	CSA Attribute	Points to a column of the SACCR Tables where the SupervisoryFactors, Correlations, etc used in addons from which non-simulated DIM is obtained.
Scaling/Scaling Methodology	Model Parameter /CSA Attribute	CSA Attribute takes precedence. Enumeration for the model parameter: SIMPLE, EXPONENTIAL, NONE. Enumeration for the CSA Attribute: LINEAR, EXPONENTIAL, NONE. Note that the 'Linear' scaling value is referred to as 'Simple'.
Decay Factor	Model Parameter/CSA Attribute	CSA Attribute takes precedence, only used for Exponential Scaling
Modelled t0	Model Parameter	A boolean model parameter.

- Dynamic Initial Margin

  We now define $DIM_{t,ctpy}$ and $DIM_{t,user\_unseg}$ for the forward view. The reverse view is similar.

  If `CPIMPostStyle` = NONE, $DIM_{t,ctpy} = 0$.

  If `UserIMPostStyle`= NONE, $DIM_{t,user\_unseg} = 0$.

  If `CollateralModel` = NONE, $DIM_{t,ctpy} = DIM_{t,user\_unseg} = 0$.

  $DIM_{t,ctpy}$ ($DIM_{t,user\_unseg}$) is now defined when `CPIMPostStyle` ≠ NONE (`UserIMPostStyle` ≠NONE) and `CollateralModel` ≠ NONE.

- Current Initial Margin

  $DIM_{0,ctpy} := IM_{0,ctpy}$ and $DIM_{0,user\_unseg} := IM_{0,user\_unseg}$ where $IM_{0,ctpy}$ and $IM_{0,user\_unseg}$ are calculated using the IM mitigant records and the model parameter `Modelledt0`:

  - If there are IM posted by the counterparty (user) mitigants with `Applicability` = INITIAL_BALANCE, then $IM_{0,ctpy}$ (resp. $IM_{0,user\_unseg}$) is obtained from these mitigants.
  - If there are no IM posted by the counterparty (user) mitigants with Applicability = INITIAL_BALANCE, then:
    - If `Modelledt0` = False, $IM_{0,ctpy} = 0$ (resp. $IM_{0,user\_unseg} = 0$).
    - If `Modelledt0` = True,
      - $IM_{0,ctpy}$ is obtained from the counterparty IM mitigants with `Applicability` = MITIGANT_POOL by subtracting the `CPIMThreshold` and flooring the result by

0. Here we are assuming that the MtM of the counterparty mitigants and the `CPIMThreshold` are positive.

- If `UserIMSegregated` = True, $IM_{0,\,user\_unseg} = 0$. If `UserIMSegregated` = False, $IM_{0,\,user\_unseg}$ is obtained from the user IM mitigants with `Applicability` = MITIGANT_POOL (include all mitigants regardless of the mitigant segregation flag) by adding the `UserIMThreshold` and flooring the result by 0. Here we are assuming that the MtM of the user mitigants is positive and the `UserIMThreshold` is negative. Note that in this case, the segregation of the user IM is determined by the CSA parameter `IMUserSegregated` and not by the mitigants segregation flags.

- Future Initial Margin

  Let IM_{CSA} be an IMCSA ($\subset$ VMCSA) or IMVMCSA, and let $t > 0$:

  Define $\tau_t = \{i \,\big|\, i \in IM_{CSA} \; with \; mat_i^{adj} \geq t\}$ where

  $$mat_i^{adj} := \begin{cases} nextEarlyTerminationDate_i, & if \; EarlyExit \; = \; True \; and \; nextEarlyTerminationDate_i \; exists \\ mat_i, & otherwise \end{cases},$$

  $nextEarlyTerminationDate_i$ is the next termination of trade i and mat_i is the maturity of trade i .

  We now define $DIM_{t,\,ctpy}$ and $DIM_{t,\,user\_unseg}$:

  - For $t \geq 0$, calculate the aggregate SACCR AddOn at time t , $Addon_t^{IM}$, of the trades that belong to τ_t assuming that all trades net, and using a maturity factor equal to $m_{MF} \times \sqrt{\frac{IMHor}{365}}$ where

    $$IMHor = \begin{cases} IMMarginHorizon + CallPeriod - 1 & if \; IMCSA = IMVMCSA \; or \subseteq VMCSA \; and \; CallPeriod \neq 0 \\ IMMarginHorizon & otherwise \end{cases}$$

    , m_{MF} is the maturity factor multiplier in the margined case, and equals 1 in the unmargined case. `CallPeriod` is the call period of the IMVMCSA or the VMCSA that contains the IMCSA. The SACCR parameters (SACCRSupervisoryFactors, SACCROptionVolatility, etc.) used in the addon are specified in one of the columns of the SACCR tables via the CSA parameter `DIMSaccrFactors`.

    Note

    If we need to calculate $DIM_{t,\,ctpy}$ and $DIM_{t,\,user\_unseg}$ on a VMCSA (Legacy, VM or IMVMCSA) node with `SettledtoMarketAgreement` = True, we still use the maturity factor equal to $m_{MF} \times \sqrt{\frac{IMHor}{365}}$ where

    $$IMHor = \begin{cases} IMMarginHorizon + CallPeriod - 1 & if \; IMCSA = IMVMCSA \; or \subseteq VMCSA \; and \; CallPeriod \neq 0 \\ IMMarginHorizon & otherwise \end{cases}$$

    to calculate $Addon_t^{IM}$.

  - Calibration Step

- If `ScalingMethodology` = NONE or there are no IM cpty (user) mitigants with `Applicability` = MITIGANT_POOL, then define $IM_{t, ctpy, preThreshold} := AddOn_t^{IM}$ ($IM_{t, user, preThreshold} := AddOn_t^{IM}$).
- If `ScalingMethodology` ≠ NONE, calculate $AddOn_0^{IM}$. If $AddOn_0^{IM} = 0$, then $IM_{t, ctpy, preThreshold} := AddOn_t^{IM}$ and $IM_{t, user, preThreshold} := AddOn_t^{IM}$. Otherwise, calculate $IM_{0, ctpy}^{calibration}$ from the counterparty IM mitigants with `Applicability` = MITIGANT_POOL without applying any haircuts. Define $SF_{0, ctpy} = \frac{IM_{0, ctpy}^{calibration}}{Addon_0^{IM}}$ (If `UserIMSegregated` = False, calculate $IM_{0, user}^{calibration}$ from the user IM mitigants with `Applicability` = MITIGANT_POOL without applying any haircuts. Define $SF_{0, user} = \frac{IM_{0, user}^{calibration}}{Addon_0^{IM}}$.)

  If `ScalingMethodology` = SIMPLE, define $IM_{t, ctpy, preThreshold} := SF_{0, ctpy} \times Addon_t^{IM}$ and $IM_{t, user, preThreshold} := SF_{0, user} \times Addon_t^{IM}$.

  If `ScalingMethodology` = EXPONENTIAL, define $SF_{t, ctpy} = 1 + (SF_{0, ctpy} - 1) \times e^{-decayFactor \times (\frac{t}{365})}$ and $SF_{t, user} = 1 + (SF_{0, user} - 1) \times e^{-decayFactor \times (\frac{t}{365})}$ where t is measured in number of days from t_0 . Then $IM_{t, ctpy, preThreshold} := SF_{t, ctpy} \times Addon_t^{IM}$ and $IM_{t, user, preThreshold} := SF_{t, user} \times Addon_t^{IM}$

- Apply the counterparty/user IM threshold to obtain the counterparty/user initial margin at time t :

$$DIM_{t, ctpy} = max\{0, IM_{t, ctpy, preThreshold} - CPIMThreshold\}$$

$$DIM_{t, user} = max\{0, IM_{t, user, preThreshold} + UserIMThreshold\}$$

$$DIM_{t, user\_unseg} = \begin{cases} DIM_{t, user} & if\ UserIMSegregated = False \\ 0 & if\ UserIMSegregated = True \end{cases}$$

- **Formula: VM**

The user unsegregated variation margin at time t is:

$$VM_{t, user\_unseg} = \begin{cases} \sum i \in \chi_t MtM_{0, i} & \text{if } t = 0 \\ -\min\left(V_{t, NS} + cpIA(t) - uIA(t) + TH_{t, user} + MTA_{t, user}, 0\right) & \text{if } t > 0, \text{UserVMPostStyle} \neq \text{NONE and UserVMSegregated=False} \\ 0 & \text{otherwise} \end{cases}$$

where

- $MtM_{0, i}$ is the time zero haircut adjusted mark-to-market of mitigant i , spot converted to the `domestic currency`. Haircut adjustment is defined in section • Mitigant Haircuts on page 142.
- χ_t is the set of node level unsegregated variation margins posted by the user, with delivery date on or before `Reporting Date` and return date on or after `Reporting Date`. Here the initial/variation margin type is read from `<mitigant input>MarginType`; post direction is read from `<mitigant input>PostDirection`; and segregated/unsegregated flag is read from `<collateral agreement input>UserVMSegregated` for variation margin posted by the user. The set is empty when `<collateral agreement input>UserVMPostStyle` = None.

- $V_{t,NS}$ is defined in ● Formula: V on page 143.
- $TH_{t,user}$ is the user threshold (`UserVMThreshold`) at time t , expressed in absolute value.
- $MTA_{t,user}$ is the user minimum transfer amount (`UserVMMinTransfer`) at time t , expressed in absolute value.
- $cpIA(t)$ and $uIA(t)$ are the total counterparty and user independent amounts at time t , expressed in absolute values respectively. See ● Formula: Trade-level IM and IA on page 148 for details.

The counterparty variation margin at time 0 is:

$VM_{0,ctpy} = \sum_{i \in \chi_0} MtM_{0,i}$

where

- $MtM_{0,i}$ is defined above.
- χ_0 is the set of node level segregated and unsegregated variation margins posted by the counterparty, with delivery date on or before `Reporting Date` and return date on or after `Reporting Date`. Here the initial/variation margin type is read from `<mitigant input>MarginType`; post direction is read from `<mitigant input>PostDirection`; and segregated/unsegregated flag is read from `<mitigant input>Segregated` for variation margin posted by the counterparty. The set is empty when `<collateral agreement input>CPVMPostStyle` = None.

● Formula: Trade-level IM and IA

- Trade-level Independent Amount and Initial Margin

  Define the trade-level counterparty and user independent amounts on deal i at time t by

  $$cpIA(i,t)_{raw} = \begin{cases} cpIAInput(i,t) & if\ t \le mat_i \\ 0 & if\ t > mat_i \end{cases}$$

  $$uIA(i,t)_{raw} = \begin{cases} uIAInput(i,t) & if\ t \le mat_i \\ 0 & if\ t > mat_i \end{cases}$$

  where:

  - $cpIA(i,t)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t posted by the counterparty.
  - $uIA(i,t)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t posted by the user.

  Define the counterparty and user initial margins on deal i at time t by

  $$cpIM(i,t)_{raw} = \begin{cases} cpIMInput(i,t) & if\ t \le mat_i \\ 0 & if\ t > mat_i \end{cases}$$

  $$uIM(i,t)_{raw} = \begin{cases} uIMInput(i,t) & if\ t \le mat_i \\ 0 & if\ t > mat_i \end{cases}$$

  $$uIM_{unseg}(i,t)_{raw} = \begin{cases} uIM(i,t)_{raw} & if\ uIM(i,t)\ \text{is unsegregated} \\ 0 & \text{otherwise} \end{cases}$$

  where:

  - $cpIM(i,t)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level segregated and unsegregated initial margin on deal i at time t posted by the counterparty.

- $uIM(i,t)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level initial margin on deal i at time t posted by the user.
- $uIM_{unseg}(i,t)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level unsegregated initial margin on deal i at time t posted by the user.

- FX Haircut of Trade-Level Initial Margins

  Assuming that any trade-level initial margin (or independent amount) is posted in cash, an FX haircut is applied to the trade-level initial margin if the currency of the initial margin is different from the settlement currency of the deal.

  Note

  An FX haircut is not applied to trade-level independent amounts for consistency with the treatment of a CSA independent amount.

  Define the settlement currency of a deal i to be:

  - the CSA currency if deal i is in a CSA (VMCSA, IMVMCSA or Legacy CSA)
  - the MA termination currency if deal i is in an MA and not in a CSA
  - the currency of the deal if deal i is in an NN node

  If the settlement currency of a deal i is different from the currency of its trade-level initial margin $IM(i,t)$ (where $IM(i,t) := cpIM(i,t)$ or $uIM(i,t)$), define

  $$HC_{FX,IM(i,t)} = \frac{HC_{FX}}{\sqrt{LH_{FX}}} \times \sqrt{RH(t)}$$

  where:

  - HC_{FX} : the FX haircut, read from the table `SACCRFXHaircut`.
  - LH_{FX} : the liquidity horizon (in years) associated with the FX haircut, read from the table `SACCRFXHaircut`.
  - $RH(t) = \begin{cases} min\{t, \frac{MPOR_{CSA}}{365}\} & \text{for a margined set} \\ \text{t} & \text{for an unmargined set} \end{cases}$: the risk horizon (in years) to be considered for the potential unfavorable change in collateral value at time t , see section • Margin Period of Risk on page 143 for details.

  Otherwise $HC_{FX,IM(i,t)} = 0$.

- FX Conversion and Aggregation
  - Independent Amounts

    At time t , define the trade-level independent amount profiles after FX conversion:

    $$cpIA(i,t) = FX_{ccy_{(cpIA,\,i)},\, ccy_{CSA}} \times cpIA(i,t)_{raw}$$

    $$uIA(i,t) = FX_{ccy_{(uIA,\,i)},\, ccy_{CSA}} \times uIA(i,t)_{raw}$$

    where:

    - $cpIA(i,t)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t posted by the counterparty.
    - $uIA(i,t)_{raw}$: the raw (before FX conversion and FX haircut) trade-level independent amount on deal i at time t posted by the user.

At time t , for a CSA node, define the total independent amount:

$cpIA(t) = IA_{t,ctpy} + \sum_{i \in CSA} cpIA(i,t)$

$uIA(t) = IA_{t,user} + \sum_{i \in CSA} uIA(i,t)$

where:

- $IA_{t,\,ctpy}$: the csa level independent amount posted by the counterparty at time t , expressed in absolute value.
- $IA_{t,\,user}$: the csa level independent amount posted by the user at time t , expressed in absolute value.

- Initial Margins

  At time t , for a currency ccy , define the trade-level initial margin on deal i in currency ccy :

  $cpIM(i,t,ccy) = FX_{ccy_{cpIM,\,i},\,ccy} \times cpIM(i,t)_{raw} \times max(0, 1 - HC_{FX,\,cpIM(i,\,t)})$

  $uIM(i,t,ccy) = FX_{ccy_{uIM,\,i},\,ccy} \times uIM(i,t)_{raw} \times (1 + HC_{FX,\,uIM(i,\,t)})$

  $uIM_{unseg}(i,t,ccy) := \begin{cases} uIM(i,t,ccy) & if\ uIM(i,t) \text{ is unsegregated} \\ 0 & \text{otherwise} \end{cases}$

  where:

  - $cpIM(i,t)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level segregated and unsegregated initial margin on deal i at time t posted by the counterparty.
  - $uIM(i,t)_{raw}$: the raw (before FX conversion and FX haircut) value of trade-level initial margin on deal i at time t posted by the user.
  - $HC_{FX,\,cpIM(i,\,t)}$: FX haircut of the trade-level IM as defined in FX Haircut of Trade-level Initial Margins.
  - $HC_{FX,\,cpIM(i,\,t)}$: FX haircut of the trade-level IM as defined in FX Haircut of Trade-level Initial Margins.

  At time t , for a currency ccy , for a node X , define the total initial margin:

  $cpIM(t,ccy) = IM_{ctpy}(t,ccy) + \sum_{i \in X} cpIM(i,t,ccy)$

  $uIM(t,ccy) = IM_{user}(t,ccy) + \sum_{i \in X} uIM(i,t,ccy)$

  $uIM_{unseg}(t,ccy) = IM_{user,\,unseg}(t,ccy) + \sum_{i \in X} uIM_{unseg}(i,t,ccy)$

  where:

  - $IM_{ctpy}(t,ccy)$: the node-level haircut adjusted value of segregated and unsegregated initial margin posted by the counterparty at time t .
  - $IM_{user}(t,ccy)$: the node-level haircut adjusted value of initial margin posted by the user at time t .
  - $IM_{user,unseg}(t,ccy)$: the node-level haircut adjusted value of unsegregated initial margin posted by the user at time t .

- **Multiplier**

The multiplier reduces the add-on when the portfolio is over-collateralized or out-of-the-money.

Case 1: X consists of a single netting set

At a future time step $t > 0$, the multiplier is calculated as follows:

- If the node X has `SettledtoMarketAgreement` = False:

  Unmargined set:

  - If `CollateralModel` $\neq$ NONE:

    $$m_t = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{V_{t,NS} - cpIM(t,ccy) + uIM_{unseg}(t,ccy) - VM_t}{m\_factor \cdot (1 - m\_floor) \cdot AddOn_t}\right)\right)$$

  - If `CollateralModel` = NONE:

    $$m_t = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{V_{t,NS}}{m\_factor \cdot (1 - m\_floor) \cdot AddOn_t}\right)\right)$$

  Margined set:

  $$m_t = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{TH_{t,ctpy} + MTA_{t,ctpy} - cpIA(t) + \delta_{user} \cdot uIA(t) - cpIM(t,ccy) + uIM_{unseg}(t,ccy)}{m\_factor \cdot (1 - m\_floor) \cdot AddOn_t}\right)\right)$$

  where:

  - m_floor is read from expression parameter `MultiplierFloor`, defaults to be 5%.
  - m_factor is read from expression parameter `MultiplierFactor`, defaults to be 2.
  - $V_{t,NS}$, $cpIM(t,ccy)$, $uIM_{unseg}(t,ccy)$, VM_t , $TH_{t,ctpy}$, $MTA_{t,ctpy}$ and δ_{user} are defined in section • Replacement Cost on page 140 and • Formula: Trade-level IM and IA on page 148.
  - $AddOn_t$ is the aggregate add-on defined in section • AddOn on page 152.
  - $cpIA(t)$ and $uIA(t)$ are the total counterparty and user independent amounts at time t , expressed in absolute values respectively. See • Formula: Trade-level IM and IA on page 148 for details.

- If the node X is a VMCSA (Legacy, VM or IMVMCSA) with `SettledtoMarketAgreement` = True,

  $$m_t^{STM} = \begin{cases} min\{1, m\_floor + (1 - m\_floor) \cdot exp\left(\frac{-(cpIM(t,ccy) - uIM_{unseg}(t,ccy))}{m\_factor \cdot (1 - m\_floor) \cdot Addon_{t,Aggregrate}^{STM}}\right) & \text{if CollateralModel } \neq NONE \\ & \text{if CollateralModel } = NONE \end{cases}$$

  where $cpIM(t,ccy)$ and $uIM_{unseg}(t,ccy)$ are defined in the section • Formula: Trade-level IM and IA on page 148.

Case 2: X consists of multiple netting sets

The multiplier is calculated at netting set level individually. Since it is problematic to allocate common collateral to individual netting sets, the multiplier is activated only when the current value of the derivative transactions in a netting set is negative.

$$m_{t,NS} = \min\left(1, m\_floor + \left(1 - m\_floor\right) \cdot \exp\left(\frac{CV_{t,NS}}{m\_factor \cdot \left(1 - m\_floor\right) \cdot AddOn_{t,NS}}\right)\right)$$

where:

- m_floor is read from expression parameter `MultiplierFloor`, defaults to be 5%.
- $CV_{t,NS} = \begin{cases} V_{t,NS} - IM_{t,NS}^{Trade} & \text{If X has SettledtoMarket = False} \\ -IM_{t,NS}^{Trade} & \text{If X has SettledtoMarket = True} \end{cases}$ is the current value of all derivative transactions in the netting set. See ● Formula: V on page 143 for details.
- $IM_{t,NS}^{Trade} = \sum_{i \in NS} cpIM(i, t, ccy) - \sum_{i \in NS} uIM_{unseg}(i, t, ccy)$ is the sum of trade-level initial margin over all trades in the netting set at time t . See section ● Formula: Trade-level IM and IA on page 148 for details.
- $AddOn_{t,NS}$ is the add-on of all derivative transactions in the netting set.

● AddOn

Five asset classes are defined under SA-CCR: Interest Rate (IR), Foreign Exchange (FX), Credit (CR), Equity (EQ), and Commodity (CM). Trades are allocated into one of the five asset classes according to their primary risk factors. For more complex trades that may have more than one risk driver (for example, multi-asset or hybrid derivatives), they may be allocated to more than one asset class.

At time $t > 0$, the aggregate add-on for the netting set is a simple sum of add-ons calculated from each asset class:

$$AddOn_t = AddOn_t^{(IR)} + AddOn_t^{(FX)} + AddOn_t^{(CR)} + AddOn_t^{(EQ)} + AddOn_t^{(CM)}$$

where the add-on for each asset class is calculated using asset-class-specific formulas as described in sections below.

Note

The add-on for a margined netting set and an unmargined netting set uses different risk horizon assumptions. The add-on under unmargined calculation represents the potential change in value of trades up to the reporting date, while the add-on under margined calculation represents the potential change in exposure during the Margin Period of Risk. The difference is captured in the maturity factors calculation.

● AddOn - Aggregation Hierarchy

For each time step $t > 0$, MAG first computes the trade level effective notionals, then aggregates across trades, hedging buckets, hedging sets, and trade types for the final asset class add-on. The hierarchical nodes are defined from top to bottom as follows:

- `Asset Class`: Five asset classes are defined: Interest Rate, Foreign Exchange, Equity, Credit, and Commodity
- `Trade Type`: Three trade types are defined: Regular, Basis, and Volatility.

  A basis transaction is a non-foreign exchange transaction in which the cash flows of both legs depend on different risk factors from the same asset class.

  A volatility transaction is one in which the reference asset depends on the volatility (historical or implied) of a risk factor.
- `Hedging Set`: A hedging set is a set of transactions within a single netting set for which partial or full offsetting is recognized. Hedging sets are defined based on asset classes and trade types.

Table 6.31-1 Hedging Set Definition

Trade Type	Asset Class	Hedging Set Definition
Regular	Interest Rate	Each currency forms its own hedging set.
	Foreign Exchange	If `FXOffsetting` = False, each currency pair forms its own hedging set; otherwise, all currency pairs form a single hedging set.*1
	Credit	All transactions form a single hedging set.
	Equity	All transactions form a single hedging set.
	Commodity	Each broad category of commodity forms its own hedging set.
Basis	Interest Rate	Each pair of risk factors forms its own hedging set (for example: 3-month Libor vs. 6-month Libor).
	Foreign Exchange	Basis transaction is not applicable to FX asset class.
	Credit	Each pair of risk factors forms its own hedging set (for example: entity1 vs entity2).
	Equity	Each pair of risk factors forms its own hedging set (for example: entity1 vs entity2).
	Commodity	Each pair of risk factors forms its own hedging set (for example: Brent Crude oil vs. Henry Hub gas).
Volatility	Interest Rate	Volatility of each currency forms its own hedging set.
	Foreign Exchange	Volatility of each currency pair forms its own hedging set.
	Credit	Volatility of each reference entity forms its own hedging set.
	Equity	Volatility of each reference entity forms its own hedging set.
	Commodity	Volatility of each broad category of commodity forms its own hedging set.

- `Hedging Bucket`: for Interest Rate, Credit, Equity and Commodity asset classes, hedging sets are further divided into subsets (namely, hedging buckets) that allow full offsetting among trades. Hedging bucket definition is shared among Regular, Basis, and Volatility trade types for Interest Rate and Commodity asset classes.

Asset Class	Hedging Bucket Definition
Interest Rate	Trades are divided into three maturity buckets based on the remaining time to end date of the transactions: 1. less than or equal to one year 2. between one (exclusive) and five years (inclusive) 3. more than five years
Foreign Exchange	For regular trades: if `FXOffsetting` = False, all transactions within a hedging set form a single hedging bucket; otherwise, each currency pair forms its own hedging bucket. For volatility trades: all transactions within a hedging set form a single hedging bucket.
Credit	For regular trades: each reference entity (a single entity or an index) form its own hedging bucket. For basis and volatility trade: all transactions within a hedging set form a single hedging bucket.

Asset Class	Hedging Bucket Definition
Equity	For regular trades: each reference entity (a single entity or an index) form its own hedging bucket. For basis and volatility trade: all transactions within a hedging set form a single hedging bucket.
Commodity	Each commodity forms its own hedging bucket.

- `Trade`: trade is the lowest node of the hierarchy, where parameters such as maturity factor, supervisory delta adjustment, supervisory factor, adjusted notional and effective notional are calculated.

*1 MAG orders the hedging set currency pair alphabetically and trades that reference the same currency pair (regardless of the long/short direction) will be mapped to the same hedging set. For example, an FX forward that pays (short) CAD and receives (long) USD is short in the hedging set hedging set (CAD - USD). Another FX forward that pays (short) USD and receives (long) CAD is long in the same hedging set (CAD-USD).

Input recommendation: For option deltas to correctly offset, the quotation direction for strike (K) and underlying forward price (F) should be consistent across all trades that reference the same currency pair.

● AddOn - Preliminary Materials

This section describes the calculation of the common parameters that are shared among asset classes:

1. **Maturity Factors**

   - For a node with `SettledtoMarketAgreement` = False, the maturity factor, $MF_{t,i}$, reflects the risk horizon of the trade i and is calculated up to the maturity of the trade read from SACCR attribute `maturity`.

     For unmargined transactions:

     $$MF_{t,i} = \begin{cases} \sqrt{t} & \text{for non-resetting trades} \\ \sqrt{\frac{\text{reset period}}{365}} & \text{for resetting trades} \end{cases}$$

     For margined transactions:

     $$MF_{t,i} = m\_mf \cdot \sqrt{\min\left(t, \frac{MPOR}{365}\right)}$$

     where m_mf is read from expression parameter `MaturityFactorMultiplier`, defaults to be 1.5; t is expressed in years relative to `Processing Date` and MPOR is defined as ● Margin Period of Risk on page 143.

   - For a VMCSA (Legacy, VM or IMVMCSA) node with `SettledtoMarketAgreement` = True, $Addon^{STM}_{t,\,Aggregate}$ is defined the same way with the exception that the maturity factor of all trades in the CSA is defined as: $MF_t = \begin{cases} \sqrt{min\{t, \frac{14}{365}\}} & \text{for non-resetting trades} \\ \sqrt{min\{t, \frac{14}{365}, \frac{\text{reset period}}{365}\}} & \text{for resetting trades} \end{cases}$.

2. **Supervisory Duration**

   For Interest Rate and Credit derivatives, the trade notional is adjusted by the supervisory duration, $SD_{t,i}$. The supervisory duration represents the discount factors for cashflows between processing date (if IRDurationOverride = False) or reporting date ((if IRDurationOverride = True) and end date, using a discount rate of r = 0.05 (read from expression parameter `SupervisoryDurationDF`). Mathematically,

$$SD_{t,i} = \begin{cases} \frac{\exp(-r \cdot \max(0, S_i - t)) - \exp(-r \cdot \max(0, E_i - t))}{r} & \text{if IRDurationOverride=True} \\ \frac{\exp(-r \cdot \max(0, S_i)) - \exp(-r \cdot \max(0, E_i))}{r} & \text{if IRDurationOverride=False} \end{cases}$$

where S_i and E_i are start and end date of trade i , expressed in years respectively.

3. **Supervisory Delta Adjustment**

The supervisory delta adjustment, $\delta_{t,i}$, is defined at the trade level and is applied to the adjusted notional amount to reflect the direction and non-linearity of the trade. If trade attribute `DeltaOverride` is populated, $\delta_{t,i}$ is read from delta override. Otherwise, $\delta_{t,i}$ is determined as follows:

Table 6.31-2 Supervisory delta adjustment for non-options and non-CDO-tranches

$\delta_{t,i}$	**Long in the primary risk factor**	**Short in the primary risk factor**
Trades that are not options or CDO tranches	1	-1

6

Functions

Note

For regular trades, long and short flag to Interest Rate, Credit, Equity, and Commodity hedging sets are clearly defined in the SACCR attributes. For Foreign Exchange asset class, each currency pair forms a (regular) hedging set. The hedging set currency pair is ordered alphabetically, for example (CAD-USD) or (EUR-GBP). The trade's (long - short) currency pair is compared with the hedging set currency pair to determine if a FX trade is long or short the hedging set. For example, an FX forward that pays (short) CAD and receives (long) USD is short in the hedging set hedging set (CAD - USD). Another FX forward that pays (short) USD and receives (long) CAD is long in the same hedging set (CAD-USD).

Note

For basis trades, the long and short flag are determined using both the long/short indicator (as read from SACCR attributes) and the order of hedging set risk factor pair vs. trade risk factor pair. The hedging set risk factor pair is ordered alphabetically, for example 3M Libor vs. 6M Libor. The trade's risk factor pair is defined as the primary risk factor vs. ancillary risk factor. The original delta sign is determined using the long/short indicator. Then, if the trade’s basis pair is the reversal of the hedging set pair, the delta sign will also get reversed. Here the long/short indicator is expected to indicate the long/short position on the primary risk factor.

Table 6.31-3 Supervisory delta adjustment for options:

$\delta_{t,i}$	**Call Option**	**Put Option**
Options	See Figure 6.31-4 Supervisory delta adjustment for call option on page 155.	See Figure 6.31-5 Supervisory delta adjustment for put option on page 156.

$$ls \cdot \begin{cases} \Phi\left(\frac{\ln\left(\frac{P_i + \lambda_j}{K_i + \lambda_j}\right) + 0.5 \cdot \sigma_i^2 \cdot (T_i - t)}{\sigma_i \cdot \sqrt{T_i - t}}\right) & \text{if } t < T_i \\ 1 & \text{if } T_i \le t \le M_i \text{ and } P \ge K \\ 0 & \text{if } T_i \le t \le M_i \text{ and } P < K \end{cases}$$

Figure 6.31-4 Supervisory delta adjustment for call option

$$ls \cdot \begin{cases} \Phi\left(-\dfrac{\ln\left(\dfrac{P_i+\lambda_j}{K_i+\lambda_j}\right)+0.5\cdot\sigma_i^2\cdot(T_i-t)}{\sigma_i\cdot\sqrt{T_i-t}}\right) & \text{if } t < T_i \\ 0 & \text{if } T_i \le t \le M_i \text{ and } P > K \\ 1 & \text{if } T_i \le t \le M_i \text{ and } P \le K \end{cases}$$

Figure 6.31-5 Supervisory delta adjustment for put option

where for each option,

- Φ is the cumulative distribution function of a standard normal random variable.
- ls = 1 if the trade is long the hedging set, -1 otherwise. The long/short flag is determined according to the non-option case above.
- P_i is the underlying price.
- K_i is the strike price.
- λ_j is the shift amount, applicable to Interest Rate derivatives only. For other asset classes, λ_j = 0.
  - If expression parameter `IRShift` = PER_CURRENCY, the shift amount is read from table `SACCRNegativeInterestRateLimits`, based on currency j that trade i is mapped to.
  - If expression parameter `IRShift` = PER_TRADE, $\lambda_j = \max(\text{IRFloor} - \min(P, K), 0)$, where `IRFloor` is a Mag configuration setting (`Finance.SACCR_SFT.Defaults.IRFloor`), expressed in percentage points. By default `IRFloor` = 0.1 (0.1%).
- T_i is the exercise date of the option, expressed in years.
- M_i is the maturity date of the option, expressed in years.
- σ_i is the supervisory option volatility of the trade looked up from table `SACCROptionVolatility`.

Table 6.31-4 Supervisory delta adjustment for CDO tranches:

$\delta_{t,i}$	**Purchased (long protection)**	**Sold (short protection)**
CDO tranches	$\dfrac{15}{(1+14\cdot A_i)\cdot(1+14\cdot D_i)}$	$-\dfrac{15}{(1+14\cdot A_i)\cdot(1+14\cdot D_i)}$

where for each CDO tranche,

- A_i is the attachment point of the CDO tranche.
- D_i is the detachment point of the CDO tranche.

4. **Supervisory Parameters**

Three supervisory parameters are used in the add-on calculation:

- The supervisory factor, $SF_k^{(a)}$, of hedging bucket or hedging set k , asset class a , captures the volatility of the asset class. It is looked up from table `SACCRSupervisoryFactor`.
- The correlation factor, $\rho_k^{(a)}$, of hedging bucket k , asset class a , determines the degree of offsetting/hedging benefit within the hedging set. It is looked up from table `SACCRIRCorr`

`elations` for interest rate, table `SACCRFXCorrelations` for foreign exchange, and table `SACCRCorrelations` for other asset classes.

- The supervisory option volatility, σ_i , of trade i , is used to calculate supervisory delta adjustment for options. It is looked from table `SACCROptionVolatility`.

The supervisory factors given in the table are for `Regular` transaction hedging sets. For a `basis` transaction hedging set, the supervisory factor applicable to its relevant asset class must be multiplied by one-half. For a `volatility` transaction hedging set, the supervisory factor applicable to its relevant asset class must be multiplied by a factor of five.

• AddOn - Interest Rate

Interest Rate hedging buckets, hedging sets and trade types are defined in section • AddOn - Aggregation Hierarchy on page 152. At reporting time t , the aggregate add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $d_{t,i}^{(IR)} = N_i \cdot SD_{t,i} \cdot FX_{ccy_i, ccy_d}$

   where N_i is the notional amount of the derivative and ccy_d is the domestic currency.

2. Trade-level effective notional for trade i

   $D_{t,i}^{(IR)} = \delta_{t,i} \cdot d_{t,i}^{(IR)} \cdot MF_{t,i}$

3. Hedging-bucket-level effective notional for hedging bucket (maturity bucket) MB_k from hedging set ccy_j :

   $D_{t,jk}^{(IR)} = \sum_{i \in \{ccy_j, MB_k\}} D_{t,i}^{(IR)}$

4. Hedging-set-level effective notional for hedging set j :

   - If `IRTenorOffsetting` = TRUE then:

$$EN_{t,j}^{(IR)} = \left[\left(D_{t,j1}^{(IR)}\right)^2 + \left(D_{t,j2}^{(IR)}\right)^2 + \left(D_{t,j3}^{(IR)}\right)^2 + 2 \cdot \rho_{1,2} \cdot D_{t,j1}^{(IR)} \cdot D_{t,j2}^{(IR)} + 2 \cdot \rho_{2,3} \cdot D_{t,j2}^{(IR)} \cdot D_{t,j3}^{(IR)} + 2 \cdot \rho_{1,3} \cdot D_{t,j1}^{(IR)} \cdot D_{t,j3}^{(IR)}\right]^{1/2}$$

     where $\rho_{m,n}$ is looked up from table `SACCRIRCorrelations` based on `TableColumn`.

   - If `IRTenorOffsetting` = FALSE then:

     $EN_{t,j}^{(IR)} = \left|D_{t,j1}^{(IR)}\right| + \left|D_{t,j2}^{(IR)}\right| + \left|D_{t,j3}^{(IR)}\right|$

     where $k = 1, 2, 3$ corresponds to the three hedging buckets within hedging set j .

5. Hedging-set-level add-on:

   $AddOn_{t,j}^{(IR)} = SF_j^{(IR)} \cdot EN_{t,j}^{(IR)}$

6. Trade-type-level add-on for trade type TT_x :

   $AddOn_{t,x}^{(IR)} = \sum_{j \in TT_x} AddOn_{t,j}^{(IR)}$

7. Asset-class-level add-on at time t :

   $AddOn_t^{(IR)} = \sum_x AddOn_{t,x}^{(IR)}$

• AddOn - Foreign Exchange

Foreign Exchange hedging buckets, hedging sets and trade types are defined in section • AddOn - Aggregation Hierarchy on page 152. At reporting time t , the aggregate add-on is calculated as:

1. Trade-level adjusted notional for trade i :
   - If $ccy_{i,L} \neq ccy_d$ and $ccy_{i,S} \neq ccy_d$:
     $d_{t,i}^{(FX)} = \max\left(N_{i,L} \cdot FX_{ccy_{i,L}, ccy_d}, N_{i,S} \cdot FX_{ccy_{i,S}, ccy_d}\right)$
   - If $ccy_{i,L} = ccy_d$:
     $d_{t,i}^{(FX)} = N_{i,S} \cdot FX_{ccy_{i,S}, ccy_d}$
   - If $ccy_{i,S} = ccy_d$:
     $d_{t,i}^{(FX)} = N_{i,L} \cdot FX_{ccy_{i,L}, ccy_d}$

   where ccy_d is the domestic currency; $N_{i,L}$ and $ccy_{i,L}$ are the long notional amount and currency; and $N_{i,S}$ and $ccy_{i,S}$ are the short notional amount and currency.
2. Trade-level effective notional for trade i
   $D_{t,i}^{(FX)} = \delta_{t,i} \cdot d_{t,i}^{(FX)} \cdot MF_{t,i}$
3. Hedging-bucket-level effective notional for the hedging bucket HB_k under hedging set HS_j :
   $D_{t,jk}^{(FX)} = \sum_{i \in \{HB_k, HS_j\}} D_{t,i}^{(FX)}$
4. Hedging-set-level effective notional for hedging set HS_j :
   - If `FXOffsetting` = False:
     $EN_{t,j}^{(FX)} = SF_k^{(FX)} \cdot \left|D_{t,jk}^{(FX)}\right|$
   - If `FXOffsetting` = True:
     $EN_{t,j}^{(FX)} = \sqrt{\sum_m \sum_n \rho_{m,n} \left(SF_m^{(FX)} \cdot \left|D_{t,jm}^{(FX)}\right|\right) \cdot \left(SF_n^{(FX)} \cdot \left|D_{t,jn}^{(FX)}\right|\right)}$

     where $\rho_{m,n}$ is read from expression parameter. If m = n, then $\rho_{m,n} = 1$.
5. Hedging-set-level add-on:
   $AddOn_{t,j}^{(FX)} = EN_{t,j}^{(FX)}$
6. Trade-type-level add-on for trade type TT_x :
   $AddOn_{t,x}^{(FX)} = \sum_{j \in TT_x} AddOn_{t,j}^{(FX)}$
7. Asset-class-level add-on at time t :
   $AddOn_t^{(FX)} = \sum_x AddOn_{t,x}^{(FX)}$

• AddOn - Credit

Credit hedging buckets, hedging sets and trade types are defined in section • AddOn - Aggregation Hierarchy on page 152. At reporting time t , the aggregate add-on is calculated as:

1. Trade-level adjusted notional for trade i :
   $d_{t,i}^{(CR)} = N_i \cdot SD_{t,i} \cdot FX_{ccy_i, ccy_d}$
   where N_i is the notional amount of the derivative and ccy_d is the domestic currency.
2. Trade-level effective notional for trade i
   $D_{t,i}^{(CR)} = \delta_{t,i} \cdot d_{t,i}^{(CR)} \cdot MF_{t,i}$

3. Hedging-bucket-level effective notional for hedging bucket $Entity_k$ of hedging set HS_j :

   $D_{t,jk}^{(CR)} = \sum_{i \in \{HS_j, Entity_k\}} D_{t,i}^{(CR)}$

4. Hedging-bucket-level add-on:

   $AddOn_{t,jk}^{(CR)} = SF_k^{(CR)} \cdot D_{t,jk}^{(CR)}$

5. Hedging-set-level add-on for hedging set HS_j :

   $AddOn_{t,j}^{(CR)} = \left[\left(\sum_k \rho_k^{(CR)} \cdot AddOn_{t,jk}^{(CR)}\right)^2 + \sum_k \left(1 - \left(\rho_k^{(CR)}\right)^2\right) \cdot \left(AddOn_{t,jk}^{(CR)}\right)^2\right]^{1/2}$

6. Trade-type-level add-on for trade type TT_x

   $AddOn_{t,x}^{(CR)} = \sum_{j \in TT_x} AddOn_{t,j}^{(CR)}$

7. Asset-class-level add-on at time t :

   $AddOn_t^{(CR)} = \sum_x AddOn_{t,x}^{(CR)}$

- **AddOn - Equity**

Equity hedging buckets, hedging sets and trade types are defined in section • AddOn - Aggregation Hierarchy on page 152. At reporting time t , the aggregate add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $d_{t,i}^{(EQ)} = N_i \cdot FX_{ccy_i, ccy_d}$

   where N_i is the notional amount of the derivative (defined as the product of the current price of one unit of stock and the number of units referenced by the trade) and ccy_d is the domestic currency.

2. Trade-level effective notional for trade i

   $D_{t,i}^{(EQ)} = \delta_{t,i} \cdot d_{t,i}^{(EQ)} \cdot MF_{t,i}$

3. Hedging-bucket-level effective notional for hedging bucket $Entity_k$ of hedging set HS_j :

   $D_{t,jk}^{(EQ)} = \sum_{i \in \{HS_j, Entity_k\}} D_{t,i}^{(EQ)}$

4. Hedging-bucket-level add-on:

   $AddOn_{t,jk}^{(EQ)} = SF_k^{(EQ)} \cdot D_{t,jk}^{(EQ)}$

5. Hedging-set-level add-on for hedging set HS_j :

   $AddOn_{t,j}^{(EQ)} = \left[\left(\sum_k \rho_k^{(EQ)} \cdot AddOn_{t,jk}^{(EQ)}\right)^2 + \sum_k \left(1 - \left(\rho_k^{(EQ)}\right)^2\right) \cdot \left(AddOn_{t,jk}^{(EQ)}\right)^2\right]^{1/2}$

6. Trade-type-level add-on for trade type TT_x

   $AddOn_{t,x}^{(EQ)} = \sum_{j \in TT_x} AddOn_{t,j}^{(EQ)}$

7. Asset-class-level add-on at time t :

   $AddOn_t^{(EQ)} = \sum_x AddOn_{t,x}^{(EQ)}$

- **AddOn - Commodity**

Commodity hedging buckets, hedging sets and trade types are defined in section • AddOn - Aggregation Hierarchy on page 152. At reporting time t , the aggregate add-on is calculated as:

1. Trade-level adjusted notional for trade i :

   $d_{t,i}^{(CM)} = N_i \cdot FX_{ccy_i, ccy_d}$

   where N_i is the notional amount of the derivative (defined as the product of the current price of one unit of commodity and the number of units referenced by the trade) and ccy_d is the domestic currency.
2. Trade-level effective notional for trade i

   $D_{t,i}^{(CM)} = \delta_{t,i} \cdot d_{t,i}^{(CM)} \cdot MF_{t,i}$
3. Hedging-bucket-level effective notional for hedging bucket (commodity type) $Type_k$ of hedging set HS_j :

   $D_{t,jk}^{(CM)} = \sum_{i \in \{HS_j, Type_k\}} D_{t,i}^{(CM)}$
4. Hedging-bucket-level add-on:

   $AddOn_{t,jk}^{(CM)} = SF_k^{(CM)} \cdot D_{t,jk}^{(CM)}$
5. Hedging-set-level add-on for hedging set HS_j :

   $AddOn_{t,j}^{(CM)} = \left[\left(\rho_k^{(CM)} \cdot \sum_k AddOn_{t,jk}^{(CM)}\right)^2 + \left(1 - \left(\rho_k^{(CM)}\right)^2\right) \cdot \sum_k \left(AddOn_{t,jk}^{(CM)}\right)^2\right]^{1/2}$
6. Trade-type-level add-on for trade type TT_x

   $AddOn_{t,x}^{(CM)} = \sum_{j \in TT_x} AddOn_{t,j}^{(CM)}$
7. Asset-class-level add-on at time t :

   $AddOn_t^{(CM)} = \sum_x AddOn_{t,x}^{(CM)}$

6.32 SACCR Profile Potential Future Exposure

The function `SACCR Profile Potential Future Exposure` computes the potential future exposure of the portfolio over time based on the Basel SA-CCR methodology.

- **Input Parameters**

  - ReportingTimeline: A list of reporting time steps relative to processing date.
  - IRTenorOffsetting: A boolean attribute, defaults to True. Setting it to False disallows offsetting benefit between maturity buckets for Interest Rate derivatives.
  - FXOffsetting: A boolean attribute, defaults to False. Setting it to True enables offsetting between FX currency pairs for Foreign Exchange derivatives.
  - Alpha1: A parameter to scale the portfolio RC, defaults to 1.4
  - Alpha2: A parameter to scale the portfolio AddOn, defaults to 1.4
  - MultiplierFloor: Sets the floor for AddOn multiplier, defaults to 0.05.
  - MultiplierFactor: Sets the factor for AddOn multiplier, defaults to 2.
  - MaturityFactorMultiplier: Sets the multiplier for margined maturity factor, defaults to 1.5.
  - NettingRule: accepts NO_NET, AGREEMENT and LEGAL_OPINION.
  - SupervisoryDurationDF: Sets the discount rate in supervisory duration calculation, defaults to 0.05.

- IRShift: accepts PER_CURRENCY or PER_TRADE. It specifies whether the IR shift amount is currency dependent or trade dependent.
- EarlyExit: A boolean attribute, defaults to False. When set to True, trades with nextEarlyTerminationDate before the report date are removed from the exposure profile.
- CollateralModel: Sets the collateral model for the expression: NONE, NonSim_CollateralModel, etc.
- IncludeVMInRCUnmargined: An optional collateral model parameter, defaults to 1 (True).
- TableColumn: A lookup key that is associated with the following table inputs: SACCRSupervisoryFactor, SACCROptionVolatility, SACCRCorrelations, SACCRIRCorrelations, SACCRFXCorrelations, SACCRFXHaircut, SACCRCCYPFEHaircuts and VolatilityAdjustments.

Potential Future Exposure Profile

The Potential Future Exposure (PFE) for a given reporting date t consists of two terms: the replacement cost and the add-on times a multiplier. The replacement cost captures the current exposure at the processing date or at the beginning of margin period of risk. The add-on times the multiplier captures an increase in exposure up to the reporting date corresponding to a quantile. The quantile is captured through the stressed parameters applied in the add-on calculation. The lookup key for the stressed inputs are defined through the expression parameter `TableColumn`.

The PFE Profile follows the same methodology as function EE Profile. Please refer to topic 6.31 SACCR Profile Expected Exposure on page 135 for modelling details. The only difference lies in the multiplier calculation, as explained below.

Multiplier

The multiplier reduces the add-on when the portfolio is over-collateralized or out-of-the-money. The add-on is first calculated for all MAs with no CSAs, CSAs, NCs, and trades under NN. If X is such a node, then the corresponding multiplier is defined as:

Case 1: X consists of a single SA-CCR netting set

At a future time step $t > 0$, the multiplier is:

- If the node X has `SettledtoMarketAgreement` = False,

  Unmargined set:

  - If `CollateralModel` ≠ NONE:

    $$m_t = \min\left(1, \max\left(m\_floor, 1 + \frac{a_1 \cdot \left(V_{t,NS} - cpIM(t,ccy) + uIM_{unseg}(t,ccy) - VM_t\right)}{a_2 \cdot AddOn_t}\right)\right)$$

  - If `CollateralModel` = NONE:

    $$m_t = \min\left(1, \max\left(m\_floor, 1 + \frac{a_1 \cdot V_{t,NS}}{a_2 \cdot AddOn_t}\right)\right)$$

  Margined set:

  $$m_t = \min\left(1, \max\left(m\_floor, 1 + \frac{a_1 \cdot \left(TH_{t,ctpy} + MTA_{t,ctpy} - cpIA(t) + \delta_{user} \cdot uIA(t) - cpIM(t,ccy) + uIM_{unseg}(t,ccy)\right)}{a_2 \cdot AddOn_t}\right)\right)$$

- If the node X is a VMCSA (Legacy, VM or IMVMCSA) with `SettledtoMarketAgreement` = True,

  $$m_t^{STM} = \begin{cases} min\{1, max\{m\_floor, 1 + \dfrac{\alpha_1 \cdot (-(cpIM(t,ccy) - uIM_{unseg}(t_0,ccy)))}{\alpha_2 \cdot Addon_{t,Aggregate}^{STM}}\}\} & \text{if CollateralModel } \neq NONE \\ & \text{if CollateralModel } = NONE \end{cases}$$

where $cpIM(t, ccy)$ and $uIM_{unseg}(t, ccy)$ are defined in the section • Formula: Trade-level IM and IA on page 148.

Case 2: X consists of multiple SA-CCR netting sets

The multiplier is calculated at netting set level individually. Since it is problematic to allocate common collateral to individual netting sets, the multiplier is activated only when the current value of the derivative transactions in a netting set is negative.

$$m_{t,NS} = \min\left(1, \max\left(m\_floor, 1 + \frac{a_1 \cdot CV_{t,NS}}{a_2 \cdot AddOn_{t,NS}}\right)\right)$$

The parameters are defined in topic 6.31 SACCR Profile Expected Exposure on page 135, section • Multiplier on page 151.

6.33 SACCR Capital Attribution

• Overview

It is often useful to attribute the risk of a portfolio to a subset of trades within it. Such an attribution can be used, for example, to identify the riskiest trades in the portfolio, to attribute the risk to different businesses within the bank, or to determine the capital consumption (and hence the profitability) of individual trading desks.

A desirable property of an attribution of a risk measure is that it be additive, that is, the contribution of the trades to the measure should add up to the measure itself. When the measure is homogeneous and differentiable in the portfolio's positions, a common technique to obtain an additive attribution is to use Euler's Homogeneous Function Theorem.

In this chapter, whenever possible, Euler's Theorem is used to define an additive attribution for 6.27 SACCR Capital Exposure At Default on page 107.

• Assumptions

The collateral attribution described in this chapter is based on the assumption that the variation margin is either posted by the counterparty or posted by the user, that is, either VM_{ctpy} or VM_{user} is zero.

• Notation and Definitions

We use the notation and definitions from chapters 6.27 SACCR Capital Exposure At Default on page 107, 6.28 SACCR Capital Replacement Cost on page 111 and 6.29 SACCR Capital Potential Future Exposure on page 121.

Define:

$$x^+ = \max(x, 0)$$

$$\mathbf{1}_{x>0} = \begin{cases} 1 & x > 0 \\ 0 & x \le 0 \end{cases}$$

$$\mathbf{1}_{x<0} = \begin{cases} 1 & x < 0 \\ 0 & x \ge 0 \end{cases}$$

Let X denote a legal hierarchy node or a single trade in the NN node, and i a trade in X . In the following sections we define the attribution (or contribution) of trade i to SACCR Capital EAD on X .

• Exposure at Default Attribution

Recall that if X is:

- a CSA node (VM CSA or VMIM CSA or Legacy CSA)
- an MA node with no CSA
- an NC node
- a trade in the NN node

the unmargined and margined exposure at default are defined by:

$$EAD^{\text{Unmargined}} = a_1 \cdot RC^{\text{Unmargined}} + a_2 \cdot PFE^{\text{Unmargined}}$$

$$EAD^{\text{Margined}} = a_1 \cdot RC^{\text{Margined}} + a_2 \cdot PFE^{\text{Margined}}$$

A node X with `SettledtoMarketAgreement` = False falls under one of the two cases below:

- Case 1: X consists of a single SA-CCR netting set
- Case 2: X consists of multiple SA-CCR netting sets

and the exposure at default of X is:

$$EAD = \begin{cases} EAD^{\text{Unmargined}} & \text{if X is an unmargined set} \\ \min\left(EAD^{\text{Unmargined}}, EAD^{\text{Margined}}\right) & \text{if X is a margined set and X follows Case 1} \\ EAD^{\text{Margined}} & \text{if X is a margined set and X follows Case 2} \end{cases}$$

If the node X is a VMCSA (Legacy, VM or IMVMCSA) with `SettledtoMarketAgreement` = True, the exposure at default of X is

$$EAD = EAD^{STM} .$$

For any other legal hierarchy node X , the EAD is the sum of EADs of all children node of X .

As exposure at default is a linear function of replacement cost and potential future exposure, we only need to define the contribution of a trade to these two components.

The contribution of i to unmargined and margined EAD on X with `SettledtoMarketAgreement` = False is defined by:

$$EAD_i^{\text{Unmargined}} = a_1 \cdot RC_i^{\text{Unmargined}} + a_2 \cdot PFE_i^{\text{Unmargined}}$$

$$EAD_i^{\text{Margined}} = a_1 \cdot RC_i^{\text{Margined}} + a_2 \cdot PFE_i^{\text{Margined}}$$

where $RC_i^{\text{Unmargined/Margined}}$ is defined in section • Replacement Cost Attribution on page 172, $PFE_i^{\text{Unmargined/Margined}}$ is defined in section • PFE Attribution on page 173.

The contribution of i to EAD on a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True is defined by:

$$EAD_i^{STM} = \alpha_1 \cdot RC_i^{STM} + \alpha_2 \cdot PFE_i^{STM}$$

where RC_i^{STM} is defined in section • Replacement Cost Attribution on page 172, PFE_i^{STM} is defined in section • PFE Attribution on page 173.

• AddOn Attribution

The aggregate AddOn is a homogeneous function with respect to the weights of the trades in a portfolio, and hence we can apply Euler's Homogeneous Function Theorem (see • Appendix:

Euler's Homogeneous Function Theorem on page 175). In this section, we calculate the derivative of the AddOn function for each asset class to obtain the contribution of a trade to the aggregate addon.

The formula below follows the same notation and definition used in chapter 6.29 SACCR Capital Potential Future Exposure on page 121. In particular:

- $D_i^{(AC)}$ denotes the (margined or unmargined) effective notional of trade i in asset class $AC \in \{IR, FX, CR, EQ, CM\}$
- $D_{jk}^{(IR)}$ denotes the sum of the effective notional of all IR trades in hedging set j and hedging (maturity) bucket k
- $D_{j1}^{(FX)}$ denotes the sum of the effective notional of all FX trades in hedging set j (j has a unique hedging bucket)
- $D_{jk}^{(CR)}$ denotes the sum of the effective notional of all CR trades in hedging set j and hedging (entity) bucket k
- $D_{jk}^{(EQ)}$ denotes the sum of the effective notional of all EQ trades in hedging set j and hedging (entity) bucket k
- $D_{jk}^{(CM)}$ denotes the sum of the effective notional of all CM trades in hedging set j and hedging (commodity) bucket k
- $AddOn_{jk}^{(CR)} = SF_k^{(CR)} \cdot D_{jk}^{(CR)}$, where $SF_k^{(CR)}$ is the supervisory factor for entity k
- $AddOn_{jk}^{(EQ)} = SF_k^{(EQ)} \cdot D_{jk}^{(EQ)}$, where $SF_k^{(EQ)}$ is the supervisory factor for entity k
- $AddOn_{jk}^{(CM)} = SF_k^{(CM)} \cdot D_{jk}^{(CM)}$, where $SF_k^{(CM)}$ is the supervisory factor for commodity k
- $AddOn_j^{(AC)}$ denotes the addon for hedging set j in asset class $AC \in \{IR, FX, CR, EQ, CM\}$
- $AddOn_{\text{Unmargined}}^{(AC)}$ is the sum of (unmargined) $AddOn_j^{(AC)}$ over all hedging sets j in asset class AC
- $AddOn_{\text{Margined}}^{(AC)}$ is the sum of (margined) $AddOn_j^{(AC)}$ over all hedging sets j in asset class AC

The aggregate AddOn is:

$$AddOn_{Aggregate}^{Unmargined} = AddOn_{Unmargined}^{(IR)} + AddOn_{Unmargined}^{(FX)} + AddOn_{Unmargined}^{(CR)} + AddOn_{Unmargined}^{(EQ)} + AddOn_{Unmargined}^{(CM)}$$

$$AddOn_{Aggregate}^{Margined} = AddOn_{Margined}^{(IR)} + AddOn_{Margined}^{(FX)} + AddOn_{Margined}^{(CR)} + AddOn_{Margined}^{(EQ)} + AddOn_{Margined}^{(CM)}$$

Interest Rate

For a hedging set j , the addon is:

$$AddOn_j^{(IR)} = SF_j^{(IR)} \cdot EffectiveNotional_j^{(IR)}$$

If `IRTenorOffsetting` = TRUE,

$$EffectiveNotional_j^{(IR)} = \left[\left(D_{j1}^{(IR)}\right)^2 + \left(D_{j2}^{(IR)}\right)^2 + \left(D_{j3}^{(IR)}\right)^2 + 2 \cdot \rho_{1,2} \cdot D_{j1}^{(IR)} \cdot D_{j2}^{(IR)} + 2 \cdot \rho_{2,3} \cdot D_{j2}^{(IR)} \cdot D_{j3}^{(IR)} + 2 \cdot \rho_{1,3} \cdot D_{j1}^{(IR)} \cdot D_{j3}^{(IR)}\right]^{1/2}$$

If `IRTenorOffsetting` = FALSE,

$EffectiveNotional_j^{(IR)} = \left|D_{j1}^{(IR)}\right| + \left|D_{j2}^{(IR)}\right| + \left|D_{j3}^{(IR)}\right|$

For attribution, let j be a hedging set and $k \in \{1, 2, 3\}$ be a hedging (maturity) bucket of j . Let l and m be the other two hedging maturity buckets, that is, $\{l, m\} \in \{1, 2, 3\} \backslash \{k\}$.

If `IRTenorOffsetting` = TRUE,

$$AddOn_j^{(IR)}\left[k, i\right] = SF_j^{(IR)} \cdot \begin{cases} \frac{\left(D_{jk}^{(IR)} + \rho_{k,l} \cdot D_{jl}^{(IR)} + \rho_{k,m} \cdot D_{jm}^{(IR)}\right) \cdot D_i^{(IR)}}{EffectiveNotional_j^{(IR)}} & \text{if } i \text{ is in } k \text{ and } EffectiveNotional_j^{(IR)} \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

If `IRTenorOffsetting` = FALSE,

$$AddOn_j^{(IR)}\left[k, i\right] = SF_j^{(IR)} \cdot \begin{cases} -D_i^{(IR)} & \text{if } i \text{ is in } k \text{ and } D_{jk}^{(IR)} < 0 \\ D_i^{(IR)} & \text{if } i \text{ is in } k \text{ and } D_{jk}^{(IR)} \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Define the contribution of trade i to the hedging set j by

$AddOn_j^{(IR)}\left[i\right] = \sum_k AddOn_j^{(IR)}\left[k, i\right]$

Note

[*] If i has multiple SACCR containers, $D_i^{(IR)}$ in the definition of $AddOn_j^{(IR)}\left[k, i\right]$ is the sum of the effective notionals of the SACCR containers of i belonging to bucket k .

Foreign Exchange

For a hedging set j , the addon is:

$AddOn_j^{(FX)} = SF_j^{(FX)} \cdot \left|EffectiveNotional_j^{(FX)}\right|$

where $EffectiveNotional_j^{(FX)} = D_{j1}^{(FX)}$ and $SF_j^{(FX)}$ is a constant.

For attribution, assume trade i is in hedging set j .

If $EffectiveNotional_j^{(FX)} \neq 0$, define

$$AddOn_j^{(FX)}\left[k, i\right] = SF_j^{(FX)} \cdot \begin{cases} -D_i^{(FX)} & \text{if } EffectiveNotional_j^{(FX)} < 0 \\ D_i^{(FX)} & \text{if } EffectiveNotional_j^{(FX)} \geq 0 \end{cases}$$

Credit

For a hedging set j , the addon is:

$$AddOn_j^{(CR)} = \left[\left(\sum_l \rho_l^{(CR)} \cdot AddOn_{jl}^{(CR)}\right)^2 + \sum_l \left(1 - \left(\rho_l^{(CR)}\right)^2\right) \cdot \left(AddOn_{jl}^{(CR)}\right)^2\right]^{1/2}$$

For attribution, let j be a hedging set and k a hedging bucket (entity bucket of regular trades) of j .

If $AddOn_j^{(CR)} \neq 0$, define

$$AddOn_j^{(CR)}\left[k,i\right] = \begin{cases} \frac{1}{AddOn_j^{(CR)}} \cdot \left[\rho_k^{(CR)} \cdot \sum_l \rho_l^{(CR)} \cdot AddOn_{jl}^{(CR)} + \left(1-\left(\rho_k^{(CR)}\right)^2\right) \cdot AddOn_{jk}^{(CR)}\right] \cdot SF_k^{(CR)} \cdot D_i^{(CR)} & \text{if } i \text{ is in } k \\ 0 & \text{otherwise} \end{cases}$$

Note

[**] When there is a single bucket (as is always the case for basis/volatility hedging sets), the formula reduces to:

$$AddOn_j^{(CR)}\left[k,i\right] = SF_k^{(CR)} \cdot \begin{cases} -D_i^{(CR)} & \text{if } D_{jk}^{(CR)} < 0 \\ D_i^{(CR)} & \text{if } D_{jk}^{(CR)} \geq 0 \end{cases}$$

$$AddOn_j^{(CR)}\left[i\right] = \sum_k AddOn_j^{(CR)}\left[k,i\right]$$

If $AddOn_j^{(CR)} = 0$, define $AddOn_j^{(CR)}\left[i\right] = 0$.

Note

A similar note as in [*] applies here for $D_i^{(CR)}$

Equity

For a hedging set j , the addon is:

$$AddOn_j^{(EQ)} = \left[\left(\sum_l \rho_l^{(EQ)} \cdot AddOn_{jl}^{(EQ)}\right)^2 + \sum_l \left(1-\left(\rho_l^{(EQ)}\right)^2\right) \cdot \left(AddOn_{jl}^{(EQ)}\right)^2\right]^{1/2}$$

For attribution, let j be a hedging set and k a hedging bucket (entity bucket of regular trades) of j .

If $AddOn_j^{(EQ)} \neq 0$, define

$$AddOn_j^{(EQ)}\left[k,i\right] = \begin{cases} \frac{1}{AddOn_j^{(EQ)}} \cdot \left[\rho_k^{(EQ)} \cdot \sum_l \rho_l^{(EQ)} \cdot AddOn_{jl}^{(EQ)} + \left(1-\left(\rho_k^{(EQ)}\right)^2\right) \cdot AddOn_{jk}^{(EQ)}\right] \cdot SF_k^{(EQ)} \cdot D_i^{(EQ)} & \text{if } i \text{ is in } k \\ 0 & \text{otherwise} \end{cases}$$

$$AddOn_j^{(EQ)}\left[i\right] = \sum_k AddOn_j^{(EQ)}\left[k,i\right]$$

If $AddOn_j^{(EQ)} = 0$, define $AddOn_j^{(EQ)}\left[i\right] = 0$.

Note

A similar note as in [*] applies here for $D_i^{(EQ)}$. A similar note to [**] applies when there is a unique bucket.

Commodity

For a hedging set j , the addon is:

$$AddOn_j^{(CM)} = \left[\left(\sum_l \rho_l^{(CM)} \cdot AddOn_{jl}^{(CM)}\right)^2 + \sum_l \left(1-\left(\rho_l^{(CM)}\right)^2\right) \cdot \left(AddOn_{jl}^{(CM)}\right)^2\right]^{1/2}$$

For attribution, let j be a hedging set and k a hedging bucket (commodity bucket of regular trades) of j .

If $AddOn_j^{(CM)} \neq 0$, define

$$AddOn_j^{(CM)}\left[k,i\right] =$$

$$\begin{cases} \frac{1}{AddOn_j^{(CM)}} \cdot \left[\rho_k^{(CM)} \cdot \sum_l \rho_l^{(CM)} \cdot AddOn_{jl}^{(CM)} + \left(1 - \left(\rho_k^{(CM)}\right)^2\right) \cdot AddOn_{jk}^{(CM)}\right] \cdot SF_k^{(CM)} \cdot D_i^{(CM)} & \text{if } i \text{ is in } k \\ 0 & \text{otherwise} \end{cases}$$

$$AddOn_j^{(CM)}\left[i\right] = \sum_k AddOn_j^{(CM)}\left[k,i\right]$$

If $AddOn_j^{(CM)} = 0$, define $AddOn_j^{(CM)}\left[i\right] = 0$.

Note

A similar note as in [*] applies here for $D_i^{(CM)}$. A similar note to [**] applies when there is a unique bucket.

Aggregate AddOn

The contribution of trade i to $AddOn_{Aggregate}^{Unmargined/Margined}$ is the sum of $AddOn_j^{(AC)}\left[i\right]$ over all asset classes AC and all hedging sets j to which i belongs. We denote this sum in both the unmargined and margined cases simply by $AddOn_i$.

$$AddOn_i = \sum_{AC \in \{IR, FX, CR, EQ, CM\}} \sum_{j \in AC} AddOn_j^{(AC)}\left[i\right]$$

"Netted" AddOn

The "netted" addon is used in the following cases:

- when calculating the attribution of initial margin, we will need to calculate the addon assuming all trades on the node to which initial margin is attached net;
- when calculating the attribution of variation margin in the case of a node with multiple netting sets (Case 2), we will also need to calculate the addon assuming all trades net.

For a node X , define the "netted" addon to the aggregate addon calculate as if all trades net. Denote the contribution of trade i to this "netted" addon on X by:

$AddOn_{X,\,net,\,i}$

• Collateral Attribution

To determine the contribution of a trade to SACCR EAD, we must allocate collateral (variation margin, initial margin and independent amounts) to the trade. To do this, we make the following assumptions:

- the difference between the variation margin (which in our model includes independent amounts) and the mark-to-market of the portfolio, and
- the initial margin

are functions of the volatility of the trades.

With these assumptions, we allocate collateral as follows:

- If a variation margin agreement, CSA , is in place, we first assume a trade i is fully collateralized with variation margin (that is, we allocate MTM_i variation margin to the trade), and then split the excess/deficit variation margin, $VM_{ctpy} - VM_{user,unseg} - V_{CSA}$, among all the trades using the AddOn's as proxies for the volatility of the trades.

- The initial margin $IM_{ctpy} - IM_{user,unseg}$ is also split using AddOn's as proxies for the volatility of the trades.

Recall that:

$C^{Unmargined} = IM_{ctpy} + \sum_{i \in X} cpIM(i, t_0, ccy) - IM_{user,unseg} - \sum_{i \in X} cpIM_{unseg}(i, t_0, ccy) + VM^{Unmargined}$

$C^{Margined} = IM_{ctpy} + \sum_{i \in X} cpIM(i, t_0, ccy) - IM_{user,unseg} - \sum_{i \in X} cpIM_{unseg}(i, t_0, ccy) + VM_{ctpy} - VM_{user,unseg}$

$NICA = IM_{ctpy} + \sum_{i \in X} cpIM(i, t_0, ccy) - IM_{user,unse} - \sum_{i \in X} cpIM_{unseg}(i, t_0, ccy) + IA^{CSA}_{ctpy} + \sum_{i \in CSA} cpIA(i, t_0) - \delta_{user} \cdot (IA^{CSA}_{user} + \sum_{i \in CSA} uIA(i, t_0))$

where:

- $VM^{Unmargined} = \begin{cases} -VM_{user,unseg} & \text{if IncludeVMinRCUnmargined=True} \\ \delta_{ctpy} \cdot cpIA(t_0) - \delta_{user} \cdot uIA(t_0) & \text{if IncludeVMinRCUnmargined=False} \end{cases}$
- δ_{user} = 1 if `UserVMPostStyle` ≠ None and `UserVMSegregated` = False; 0 otherwise.
- δ_{ctpy} = 1 if `CPVMPostStyle` ≠ None; 0 otherwise.
- If `CPVMPostStyle` = None, the netting set is considered unmargined and all counterparty VM should be set to zero. If `UserVMPostStyle` = None, all user VM should be set to zero.
- If `CollateralModel` = None, the netting set is considered unmargined and all counterparty and user VM and IM should be set to zero.

Define:

- $VM_{user} = VM_{user,unseg} + VM_{user,seg}$
- $VM = \begin{cases} VM^{Unmargined} & \text{in unmargined case and single netting set} \\ -VM_{user,unseg} & \text{in unmargined case and multiple netting sets} \\ VM_{ctpy} - VM_{user,unseg} & \text{in margined case} \end{cases}$
- $IM = IM_{ctpy} + \sum_{i \in X} cpIM(i, t_0, ccy) - IM_{user,unseg} - \sum_{i \in X} cpIM_{unseg}(i, t_0, ccy)$
- $IA = IA^{CSA}_{ctpy} + \sum_{i \in CSA} cpIA(i, t_0) - \delta_{user} \cdot (IA^{CSA}_{user} + \sum_{i \in CSA} uIA(i, t_0))$
- V_X = the net mark-to-market of all trades in .

Note

If $X \neq CSA$, then $VM = VM_{ctpy} = VM_{user} = 0$

Note

The attribution formulas below are based on the assumption that either VM_{ctpy} or VM_{user} is zero.

Initial Margin Attribution

Define IM_X to be the initial margin attached to a node X . A trade only contributes to IM_X if it belongs to X .

The contribution of a trade $i \in X$ to IM_X is defined by

$IM_{X,i} = \frac{AddOn_{X,net,i}}{AddOn_{X,net}} \cdot IM_X + cpIM(i, t_0, ccy) - uIM_{unseg}(i, t_0, ccy)$

where $AddOn_{X,net,i}$ is defined in • AddOn Attribution on page 163, $AddOn_{X,net} = \sum_{j \in X} AddOn_{X,net,j}$, $cpIM(i, t_0, ccy)$ and $uIM_{unseg}(i, t_0, ccy)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.

If $AddOn_{X,net} = 0$, replace $\frac{AddOn_{X,net,i}}{AddOn_{X,net}}$ by $\frac{1}{N_X}$, where N_X is the number of trades in X .

In both the margined and unmargined case, the contribution of trade $i \in X$ to IM is

$IM_i = IM_{X,i}$

Variation Margin Attribution

Case 1: Single SA-CCR netting set

- Unmargined:

  The contribution of trade i to VM is defined by

  $$VM_i = \begin{cases} MtM_i + \frac{AddOn_i}{AddOn} \cdot \left(VM - IA_{Trades} - V_{NS}\right) + IA(i, t_0) & \text{if } VM \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

  - MtM_i is the mark-to-market of trade i
  - $AddOn_i$ is defined in • AddOn Attribution on page 163.
  - $AddOn = \sum_{j \in X} AddOn_j$, V_{NS} is defined in • Formula: V on page 114.
  - VM is defined in • Collateral Attribution on page 167.
  - $IA_{Trades} =$

    $$\begin{cases} \sum_{i \in CSA} cpIA(i, t_0) - \sum_{i \in CSA} uIA(i, t_0) & \text{if IncludeVMInRCUnmargined = True} \\ \delta_{ctpy} \cdot \sum_{i \in CSA} cpIA(i, t_0) - \delta_{user} \cdot \sum_{i \in CSA} uIA(i, t_0) & \text{if IncludeVMInRCUnmargined = False} \end{cases}$$

  - $IA(i, t_0) = \begin{cases} cpIA(i, t_0) - uIA(i, t_0) & \text{if IncludeVMInRCUnmargined = True} \\ \delta_{ctpy} \cdot cpIA(i, t_0) - \delta_{user} \cdot uIA(i, t_0) & \text{if IncludeVMInRCUnmargined = False} \end{cases}$
  - $cpIM(i, t_0, ccy)$, $uIM_{unseg}(i, t_0, ccy)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.
  - $cpIA(i, t_0)$, $uIA(i, t_0)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.

- Margined:

  Define

  $\bar{VM} = \min(VM, V_{NS} + IA - (TH + MTA))$

  where VM , V_{NS} and IA are defined in • Collateral Attribution on page 167.

  The contribution of trade i to $\bar{VM}$ is defined by

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Functions

$$\bar{VM}_i = \begin{cases} MtM_i + \frac{AddOn_i}{AddOn} \cdot \left(\bar{VM} - IA_{Trades} - V_{NS}\right) + IA(i, t_0) & \text{if } \bar{VM} = V_{NS} + IA - (TH + MTA) \text{ or } VM_{user, seg} = 0 \\ 0 & \text{otherwise} \end{cases}$$

where

- MtM_i is the mark-to-market of trade i .
- $AddOn_i$ is defined in • AddOn Attribution on page 163.
- $AddOn = \sum_{j \in X} AddOn_j$
- V_{NS} is defined in • Collateral Attribution on page 167.
- $IA_{Trades} =$

  $$\begin{cases} \delta_{ctpy} \cdot \sum_{i \in CSA} cpIA(i, t_0) - \delta_{user} \cdot \sum_{i \in CSA} uIA(i, t_0)) & If\ \ \bar{VM} = V_{NS} + IA - (TH + MTA) \\ \sum_{i \in CSA} cpIA(i, t_0) - \sum_{i \in CSA} uIA(i, t_0)) & Otherwise \end{cases}$$
- $IA(i, t_0) = \begin{cases} \delta_{ctpy} \cdot cpIA(i, t_0) - \delta_{user} \cdot uIA(i, t_0) & If\ \ \bar{VM} = V_{NS} + IA - (TH + MTA) \\ cpIA(i, t_0) - uIA(i, t_0) & Otherwise \end{cases}$
- $cpIM(i, t_0, ccy)$, $uIM_{unseg}(i, t_0, ccy)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.
- $cpIA(i, t_0)$, $uIA(i, t_0)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.

Note

If $AddOn = 0$, replace $\frac{AddOn_i}{AddOn}$ by $\frac{1}{N}$, where N is the number of trades in X .

Case 2: Multiple SA-CCR netting sets

- Unmargined:

  The contribution of trade i to VM is defined by

  $$VM_i = \begin{cases} MtM_i + \frac{AddOn_{X, net, i}}{AddOn} \cdot \left(VM - IA_{Trades} - V_X\right) + IA(i, t_0) & \text{if } VM_{user, unseg} > 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

  - MtM_i is the mark-to-market of trade i
  - $AddOn_{X, net, i}$ is defined in • AddOn Attribution on page 163
  - $AddOn = \sum_{j \in X} AddOn_{X, net, i}$
  - V_X is defined in • Collateral Attribution on page 167.
  - $IA_{Trades} = \sum_{i \in CSA} cpIA(i, t_0) - \sum_{i \in CSA} uIA(i, t_0)$ where $cpIA(i, t_0)$, $uIA(i, t_0)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.
  - $IA(i, t_0) = cpIA(i, t_0) - uIA(i, t_0)$ where $cpIA(i, t_0)$, $uIA(i, t_0)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.

- Margined:

  The contribution of trade i to VM is defined by

  $$VM_i = \begin{cases} MtM_i + \frac{AddOn_{X,net,i}}{AddOn} \cdot \left(VM - IA_{Trades} - V_X\right) + IA(i, t_0) & \text{if } VM_{user,seg} = 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

  - MtM_i is the mark-to-market of trade i
  - $AddOn_{X,net,i}$ is defined in • AddOn Attribution on page 163
  - $AddOn = \sum_{j \in X} AddOn_{X,net,i}$
  - V_X is defined in • Collateral Attribution on page 167.
  - $IA_{Trades} = \sum_{i \in CSA} cpIA(i, t_0) - \sum_{i \in CSA} uIA(i, t_0)$ where $cpIA(i, t_0)$ and $uIA(i, t_0)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.
  - $IA(i, t_0) = cpIA(i, t_0) - uIA(i, t_0)$ where $cpIA(i, t_0)$ and $uIA(i, t_0)$ are defined in • Formula: Initial Margin and Independent Amount on page 118.

Collateral Attribution

Case 1: Single SA-CCR netting set

- Unmargined

  The contribution of trade i to $C^{Unmargined}$ is defined by

  $C_i^{Unmargined} = VM_i + IM_i$

  where VM_i and IM_i are defined in • Collateral Attribution on page 167, respectively.

- Margined

  Define

  $C = \min\left(C^{Margined}, V_{NS} - (TH + MTA - NICA)\right) = \bar{VM} + IM$

  The contribution of trade i to C is defined by

  $C_i = \bar{VM}_i + IM_i$

  where $\bar{VM}_i$ and IM_i are defined in • Collateral Attribution on page 167, respectively.

Case 2: Multiple SA-CCR netting sets

Define

$C_1 = \left(C^{Unmargined/Margined}\right)^+$

$C_2 = \left(C^{Unmargined/Margined}\right)^-$

where $C^{Unmargined/Margined}$ is defined in • Collateral Attribution on page 167.

The contribution of trade i to C_1 and C_2 are defined by

$C_{1,i} = 1_{C_1 > 0} \cdot (VM_i + IM_i)$

$C_{2,i} = 1_{C_2 < 0} \cdot (VM_i + IM_i)$

where VM_i (for margined and unmargined case) and IM_i are defined in • Collateral Attribution on page 167.

- **Replacement Cost Attribution**

- For a node X with `SettledtoMarketAgreement` = False,

  Case 1: Single SA-CCR netting set

  - Unmargined:

    Recall that the replacement cost of X is

    $$RC^{Unmargined} = \max\left(V_{NS} - C^{Unmargined}, 0\right)$$

    where V_{NS} is defined in • Formula: V on page 114 and $C^{Unmargined}$ is defined in • Formula: C Unmargined on page 114.

    The contribution of a trade $i \in X$ to $RC^{Unmargined}$ is defined by

    $$RC_i^{Unmargined} = \begin{cases} 0 & \text{if } RC^{Unmargined} = 0 \\ MtM_i - C_i^{Unmargined} & \text{if } RC^{Unmargined} > 0 \end{cases}$$

    where MtM_i is the mark-to-market of trade i and $C_i^{Unmargined}$ is defined in • Formula: C Unmargined on page 114. Hence,

    $$RC_i^{Unmargined} = \begin{cases} 0 & \text{if } RC^{Unmargined} = 0 \\ \frac{AddOn_i}{AddOn} \cdot \left(V_{NS} + IA_{Trades} - VM\right) - IM_i - IA(i, t_0) & \text{if } RC^{Unmargined} > 0 \text{ and } VM \neq 0 \\ MtM_i - IM_i & \text{if } RC^{Unmargined} > 0 \text{ and } VM = 0 \end{cases}$$

    where IM_i is defined in • Collateral Attribution on page 167.

  - Margined:

    Recall that the replacement cost of X is

    $$RC^{Margined} = \max\left(V_{NS} - C^{Margined}, TH + MTA - NICA, 0\right)$$

    where V_{NS} , $C^{Margined}$ and $NICA$ are defined in • Collateral Attribution on page 167.

    Rewrite $RC^{Margined}$ as

    $$RC^{Margined} = \max(V_{NS} - C, 0)$$

    where C is defined in • Collateral Attribution on page 167.

    The contribution of trade i to $RC^{Margined}$ is defined by

    $$RC_i^{Margined} = \begin{cases} 0 & \text{if } RC^{Margined} = 0 \\ MtM_i - C_i^{Margined} & \text{if } RC^{Margined} > 0 \end{cases}$$

    where MtM_i is the mark-to-market of trade i and C_i is defined in • Collateral Attribution on page 167. Hence,

$$RC_i^{Margined} = \begin{cases} 0 & \text{if } RC^{Margined} = 0 \\ \frac{AddOn_i}{AddOn} \cdot \left(V_{NS} + IA_{Trades} - \bar{VM}\right) - IM_i - IA(i, t_0) & \text{if } RC^{Margined} > 0 \text{ and } \left(C = V_{NS} - (TH + MTA - NICA) \text{ or } VM_{user, seg} = 0\right) \\ MtM_i - IM_i & \text{otherwise} \end{cases}$$

  Case 2: Multiple SA-CCR netting sets

  - Unmargined/Margined

In this case, recall that the replacement cost is

$$RC = RC^{Unmargined/Margined} = RC_1^{Unmargined/Margined} + RC_2^{Unmargined/Margined}$$

$$RC_1 = RC_1^{Unmargined/Margined} = \left(\sum_{NS \in X} (V_{NS})^+ - C_1\right)^+$$

$$RC_2 = RC_2^{Unmargined/Margined} = \left(\sum_{NS \in X} (V_{NS})^- - C_2\right)^+$$

where V_{NS} , C_1 and C_2 are defined in • Collateral Attribution on page 167.

The contribution of trade i to RC is defined by

$$RC_i = RC_{1,i} + RC_{2,i}$$

where RC_1 and RC_2 are defined above.

Note that the trade i belongs to a unique SACCR netting set, denote this set by $NS(i)$

The contribution of trade i to (unmargined/margined) RC_1 and RC_2 are defined by:

$$RC_{1,i} = \begin{cases} 0 & \text{if } RC_1 = 0 \\ \mathbf{1}_{V_{NS(i)} > 0} \cdot MtM_i - C_{1,i} & \text{if } RC_1 > 0 \end{cases}$$

$$RC_{2,i} = \begin{cases} 0 & \text{if } RC_2 = 0 \\ \mathbf{1}_{V_{NS(i)} < 0} \cdot MtM_i - C_{2,i} & \text{if } RC_2 > 0 \end{cases}$$

where MtM_i is the mark-to-market of trade i , and $NS(i)$, V_{NS} , $C_{1,i}$ and $C_{2,i}$ are defined in • Collateral Attribution on page 167.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True, $RC_i^{STM} = \begin{cases} -IM_i & if\ RC^{STM} \neq 0 \\ 0 & if\ RC^{STM} = 0 \end{cases}$ where IM_i is defined in • Collateral Attribution on page 167..

• PFE Attribution

The potential future exposure for an SACCR netting set NS of a node X with `SettledtoMarketAgreement` = False is :

$$PFE^{Unmargined} = \mu^U \cdot AddOn_{AG}^U$$

$$PFE^{Margined} = \mu^M \cdot AddOn_{AG}^M$$

where

- $\mu^U = multiplier^{Unmargined} = \min\left(1, f + \left(1 - f\right) \cdot \exp\left(\frac{CV_{NS}^U}{m \cdot \left(1 - f\right) \cdot AddOn_{AG}^U}\right)\right)$
- $\mu^M = multiplier^{Margined} = \min\left(1, f + \left(1 - f\right) \cdot \exp\left(\frac{CV_{NS}^M}{m \cdot \left(1 - f\right) \cdot AddOn_{AG}^M}\right)\right)$
- $AddOn_{AG}^U = AddOn_{Aggregate}^{Unmargined}$, the unmargined aggregate addon of
- $AddOn_{AG}^M = AddOn_{Aggregate}^{Margined}$, the margined aggregate addon of
- $f = m\_floor$, multiplier floor
- $m = m\_factor$, multiplier factor

- $CV_{NS}^{U} = \begin{cases} V_{NS} - C^{Unmargined} & X = NS \\ V_{NS} - IM_{NS}^{Trade} & X \text{ has multiple SA-CCR netting sets} \end{cases}$
- $CV_{NS}^{M} = \begin{cases} V_{NS} - C^{Margined} & X = NS \\ V_{NS} - IM_{NS}^{Trade} & X \text{ has multiple SA-CCR netting sets} \end{cases}$
- $IM_{NS}^{Trade} = \sum_{i \in NS} cpIM(i, t_0, ccy) - \sum_{i \in NS} cpIM_{uns}(i, t_0, ccy)$ is the sum of trade-level initial margin over all trades in the netting set. See section • Formula: Initial Margin and Independent Amount on page 118 for details.

The potential future exposure for an SACCR netting set NS of a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True is:

$PFE^{STM} = \mu^{STM} \cdot Addon_{Aggregate}^{STM}$

where

- $\mu^{STM} = multiplier^{STM} = \begin{cases} min\left\{1, f + (1-f)exp\left(\frac{-IM_{NS}^{Trade}}{m(1-f)Addon_{Aggregate}^{STM}}\right)\right\} & \text{if CollateralModel} \neq \text{NONE} \\ 1 & \text{if CollateralModel} = \text{NONE} \end{cases}$
- $f = m\_floor$, multiplier floor
- $m = m\_factor$, multiplier factor
- $IM_{NS}^{Trade} = \sum_{i \in NS} cpIM(i, t_0, ccy) - \sum_{i \in NS} cpIM_{uns}(i, t_0, ccy)$ is the sum of trade-level initial margin over all trades in the netting set. See section • Formula: Initial Margin and Independent Amount on page 118 for details.

Multiplier = 1

- For a node X with `SettledtoMarketAgreement` = False,
  - If $\mu^{U} = 1$, the contribution of trade i to $PFE^{Unmargined}$ is

    $PFE_{i}^{U} = AddOn_{i}$
  - If $\mu^{M} = 1$, the contribution of trade i to $PFE^{Margined}$ is

    $PFE_{i}^{M} = AddOn_{i}$
- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True,

  If $\mu^{STM} = multiplier^{STM} = 1$, the contribution of trade i to $PFE^{STM} = \mu^{STM} * Addon_{Aggregate}^{STM}$ is $PFE_{i}^{STM} = [Addon_{Aggregate}^{STM}]_{i}$.

Multiplier < 1

- For a node X with `SettledtoMarketAgreement` = False,
  - **Case 1: Single SA-CCR netting set**

    The contribution of trade i to $PFE^{Unmargined}$ is

$$PFE_i^U = \mu^U \cdot AddOn_i + \exp\left(\frac{V_{NS} - C^{Unmargined}}{m \cdot \left(1 - f\right) \cdot AddOn_{AG}^U}\right) \cdot \frac{1}{m} \cdot$$

$$\left[-IM_i + \frac{AddOn_i}{AddOn_{AG}^U} \cdot IM + \begin{cases} 0 & \text{if } VM \neq 0 \\ MtM_i - \frac{AddOn_i}{AddOn_{AG}^U} \cdot V_{NS} & \text{otherwise} \end{cases}\right]$$

The contribution of trade i to $PFE^{Margined}$ is

$$PFE_i^M = \mu^M \cdot AddOn_i + \exp\left(\frac{V_{NS} - C^{Margined}}{m \cdot \left(1 - f\right) \cdot AddOn_{AG}^M}\right) \cdot \frac{1}{m} \cdot$$

$$\left[-IM_i + \frac{AddOn_i}{AddOn_{AG}^M} \cdot IM + \begin{cases} 0 & \text{if } VM_{user,\,seg} = 0 \\ MtM_i - \frac{AddOn_i}{AddOn_{AG}^M} \cdot V_{NS} & \text{otherwise} \end{cases}\right]$$

- **Case 2: Multiple SA-CCR netting sets**

  The contribution of trade i to $PFE^{Unmargined}$ is

  $$PFE_i^{Unmargined} = \mu^U \cdot AddOn_i + \exp\left(\frac{CV_{NS}}{m \cdot \left(1 - f\right) \cdot AddOn_{AG}^U}\right) \cdot \frac{1}{m} \cdot \left(MtM_i - IM_{NS,i} - \frac{AddOn_i}{AddOn_{AG}^U} \cdot CV_{NS}\right)$$

  where .

Note

In all cases above, if $AddOn_{AG}^{U/M} = 0$, define $PFE_i^{Unmargined/Margined} = 0$

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True,

  If $\mu^{STM} = multiplier^{STM} < 1$, the contribution of trade i to $PFE^{STM} = \mu^{STM} * Addon_{Aggregate}^{STM}$ is

  $PFE_i^{STM} = \mu^{STM} \cdot [Addon_{Aggregate}^{STM}]_i + exp\left(\frac{-IM}{m\_factor \cdot (1 - m\_floor) \cdot Addon_{Aggregate}^{STM}}\right) \cdot \frac{1}{m\_factor}$ where

  $\cdot \left(- IM_i + \frac{[Addon_{Aggregate}^{STM}]_i}{Addon_{Aggregate}^{STM}} \cdot IM\right)$

  $IM = cpIM(t_0, ccy) - uIM_{unseg}(t_0, ccy)$, see section • Formula: Initial Margin and Independent Amount on page 118 for details and IM_i is defined in • Collateral Attribution on page 167.

• Sold Option Exception

If node X is subject to sold option exception, the Initial Margin is assumed to be 0. Hence, the contribution of trade i to EAD of such node is 0.

• Appendix: Euler's Homogeneous Function Theorem

A function $f: R^N \to R$ is said to be homogeneous of degree k , if for all $c \in R$ such that $c > 0$,

$f(cx_1, cx_2, ...cx_N) = c^k f\left(x_1, x_2, ...x_N\right)$

Euler's homogeneous Function Theorem

If the function $f: R^N \to R$ is homogeneous of degree k and differentiable, then

$kf(x_1, x_2, ...x_n) = \sum_{j=1}^{N} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} x_i$

In the special case when $k = 1$, we get

$f(x_1, x_2, ...x_n) = \sum_{j=1}^{N} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} x_i$

Application of Euler's Theorem to Risk Measures

Any risk measure M , such as PFE , $SACCR$, etc., is a function of its trades. By changing the weights of the trades within a portfolio, we can define a function $f: R^N \to R$, where the domain R^N represents the set of weights of the trades and such that $f(1, 1, ...1) = M$.

If the function f is homogeneous of degree 1 and differentiable in the weights, we can decompose the risk measure M by evaluating equation of Euler's homogeneous Function Theorem at $\mathbf{1} = (1, 1, ...1)$:

$M = f(1, 1, ...1) = \sum_{j=1}^{N} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} \bigg|_{\mathbf{x} = (x_1, x_2, ...x_N) = \mathbf{1}}$

The value $\frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} \bigg|_{\mathbf{x} = \mathbf{1}}$ is the contribution of trade i to the measure M .

This is the general idea that is applied to the measures RC , $AddOn$ and PFE to determine the contribution of a trade to SACCR Capital EAD.

6.34 SACCR Profile Attribution

- **Overview**

It is often useful to attribute the risk of a portfolio to a subset of trades within it. Such an attribution can be used, for example, to identify the riskiest trades in the portfolio, to attribute the risk to different businesses within the bank, or to determine the capital consumption (and hence the profitability) of individual trading desks.

A desirable property of an attribution of a risk measure is that it be additive, that is, the contribution of the trades to the measure should add up to the measure itself. When the measure is homogeneous and differentiable in the portfolio's positions, a common technique to obtain an additive attribution is to use Euler's Homogeneous Function Theorem.

In this chapter, whenever possible, Euler's Theorem is used to define an additive attribution for the 6.31 SACCR Profile Expected Exposure on page 135 and 6.32 SACCR Profile Potential Future Exposure on page 160 measures.

- **Assumptions**

The collateral attribution described in this chapter is based on the assumption that the variation margin is either posted by the counterparty or posted by the user, that is, either $VM_{t, ctpy}$ or $VM_{t, user}$ is zero.

- **Notation and Definitions**

We use the notation and definitions from chapters 6.31 SACCR Profile Expected Exposure on page 135 and 6.32 SACCR Profile Potential Future Exposure on page 160.

Define:

$x^+ = \max(x, 0)$

$\mathbf{1}_{x > y} = \begin{cases} 1 & x > y \\ 0 & x \le y \end{cases}$

$$\mathbf{1}_{x \geq y} = \begin{cases} 1 & x \geq y \\ 0 & x < y \end{cases}$$

$$\mathbf{1}_{x < y} = \begin{cases} 1 & x < y \\ 0 & x \geq y \end{cases}$$

Let X denote a legal hierarchy node or a single trade in the NN node, i a trade in X , and t a reporting date. In the following sections we define the contribution of trade i to SACCR profile measures, PFE and EE..

• Expected Exposure and Potential Future Exposure Attribution

Let X be:

- a CSA node (VM CSA or VMIM CSA or Legacy CSA)
- an MA node with no CSA
- an NC node, or
- a trade in the NN node

A node X with `SettledtoMarketAgreement` = False falls under one of the two cases below:

Case 1: Single SA-CCR netting set: $X = NS$

$EE_t = a_1 \cdot RC_t + a_2 \cdot m_{t,X} \cdot AddOn_{t,X}$

$PFE_t = a_1 \cdot RC_t + a_2 \cdot m_{t,X} \cdot AddOn_{t,X}$

where RC_t is defined in section • Replacement Cost Attribution on page 186.

Case 2: Multiple SA-CCR netting sets:

$EE_t = a_1 \cdot RC_t + a_2 \cdot \sum_{NS \in X} m_{t,NS} \cdot AddOn_{t,NS}$

$PFE_t = a_1 \cdot RC_t + a_2 \cdot \sum_{NS \in X} m_{t,NS} \cdot AddOn_{t,NS}$

where RC_t is defined in • Replacement Cost Attribution on page 186.

In both cases, we will refer to $m_{t,NS}$ and $AddOn_{t,NS}$ by m_t and $AddOn_t$.

The contribution of i to EE and PFE on X is defined as follows:

Case 1: Single SA-CCR netting set

$EE_{t,i} = a_1 \cdot RC_{t,i} + a_2 \cdot [m_{t,X} \cdot AddOn_{t,X}]_i$

$PFE_{t,i} = a_1 \cdot RC_{t,i} + a_2 \cdot [m_{t,X} \cdot AddOn_{t,X}]_i$

Case 2: Multiple SA-CCR netting sets

$EE_{t,i} = a_1 \cdot RC_{t,i} + a_2 \cdot \sum_{NS \in X} [m_{t,NS} \cdot AddOn_{t,NS}]_i$

$PFE_{t,i} = a_1 \cdot RC_{t,i} + a_2 \cdot \sum_{NS \in X} [m_{t,NS} \cdot AddOn_{t,NS}]_i$

where $RC_{t,i}$, $[m_{t,X} \cdot AddOn_{t,X}]_i$ and $[m_{t,NS} \cdot AddOn_{t,NS}]_i$ are defined in section • Replacement Cost Attribution on page 186 and • Multiplier times AddOn Attribution on page 188.

If the node X is a VMCSA (Legacy, VM or IMVMCSA) with `SettledtoMarketAgreement` = True, the Expected Exposure and Potential Future Exporsure of X is

$EE_t^{STM} = \alpha_1 \cdot RC_t^{STM} + \alpha_2 \cdot m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}$

$PFE_t^{STM} = \alpha_1 \cdot RC_t^{STM} + \alpha_2 \cdot m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}$.

6

Functions

The contribution of i to EE and PFE on X is defined as follows:

$EE_{t,i}^{STM} = \alpha_1 \cdot RC_{t,i}^{STM} + \alpha_2 \cdot [m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}]_i$

$PFE_{t,i}^{STM} = \alpha_1 \cdot RC_{t,i}^{STM} + \alpha_2 \cdot [m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}]_i$.

- **AddOn Attribution**

The aggregate AddOn is a homogeneous function with respect to the weights of the trades in a portfolio, and hence we can apply Euler's Homogeneous Function Theorem. In this section, we calculate the derivative of the AddOn function for each asset class to obtain the contribution of a trade to the aggregate addon.

The formula below follows the same notation and definition used in chapter 6.29 SACCR Capital Potential Future Exposure on page 121. For a netting set NS and a reporting time t :

- $D_{t,i}^{(AC)}$ denotes the (margined or unmargined) effective notional of trade i in asset class $AC \in \{IR, FX, CR, EQ, CM\}$
- $D_{t,jk}^{(IR)}$ denotes the sum of the effective notional of all IR trades in hedging set j and hedging (maturity) bucket k
- $D_{t,jk}^{(FX)}$ denotes the sum of the effective notional of all FX trades in hedging set j (If `FXOffsetting` = False, j has a unique hedging bucket; if `FXOffsetting` = True, there is a unique j)
- $D_{t,jk}^{(CR)}$ denotes the sum of the effective notional of all CR trades in hedging set j and hedging (entity) bucket k
- $D_{t,jk}^{(EQ)}$ denotes the sum of the effective notional of all EQ trades in hedging set j and hedging (entity) bucket k
- $D_{t,jk}^{(CM)}$ denotes the sum of the effective notional of all CM trades in hedging set j and hedging (commodity) bucket k
- $AddOn_{t,jk}^{(CR)} = SF_k^{(CR)} \cdot D_{t,jk}^{(CR)}$, where $SF_k^{(CR)}$ is the supervisory factor for entity k
- $AddOn_{t,jk}^{(EQ)} = SF_k^{(EQ)} \cdot D_{t,jk}^{(EQ)}$, where $SF_k^{(EQ)}$ is the supervisory factor for entity k
- $AddOn_{t,jk}^{(CM)} = SF_k^{(CM)} \cdot D_{t,jk}^{(CM)}$, where $SF_k^{(CM)}$ is the supervisory factor for commodity k
- $AddOn_{t,j}^{(AC)}$ denotes the addon for hedging set j in asset class $AC \in \{IR, FX, CR, EQ, CM\}$
- $AddOn_t^{(AC)}$ is the sum of $AddOn_{t,j}^{(AC)}$ over all hedging sets j in asset class AC

The aggregate AddOn is:

$AddOn_t = AddOn_t^{(IR)} + AddOn_t^{(FX)} + AddOn_t^{(CR)} + AddOn_t^{(EQ)} + AddOn_t^{(CM)}$

Interest Rate

For a hedging set j , the addon is:

$AddOn_{t,j}^{(IR)} = SF_j^{(IR)} \cdot EN_{t,j}^{(IR)}$

If `IRTenorOffsetting` = TRUE,

$$EN_{t,j}^{(IR)} = \left[\left(D_{t,j1}^{(IR)}\right)^2 + \left(D_{t,j2}^{(IR)}\right)^2 + \left(D_{t,j3}^{(IR)}\right)^2 + 2 \cdot \rho_{1,2} \cdot D_{t,j1}^{(IR)} \cdot D_{t,j2}^{(IR)} + 2 \cdot \rho_{2,3} \cdot D_{t,j2}^{(IR)} \cdot D_{t,j3}^{(IR)} + 2 \cdot \rho_{1,3} \cdot D_{t,j1}^{(IR)} \cdot D_{t,j3}^{(IR)}\right]^{1/2}$$

If `IRTenorOffsetting` = FALSE,

$$EN_{t,j}^{(IR)} = \left|D_{t,j1}^{(IR)}\right| + \left|D_{t,j2}^{(IR)}\right| + \left|D_{t,j3}^{(IR)}\right|$$

For attribution, let j be a hedging set and $k \in \{1,2,3\}$ be a hedging (maturity) bucket of j . Let l and m be the other two hedging maturity buckets, that is, $\{l,m\} \in \{1,2,3\} \backslash \{k\}$.

If `IRTenorOffsetting` = TRUE,

$$AddOn_{t,j}^{(IR)}\left[k,i\right] = SF_j^{(IR)} \cdot \begin{cases} \dfrac{\left(D_{t,jk}^{(IR)} + \rho_{k,l} \cdot D_{t,jl}^{(IR)} + \rho_{k,m} \cdot D_{t,jm}^{(IR)}\right) \cdot D_{t,i}^{(IR)}}{EN_{t,j}^{(IR)}} & \text{if } i \text{ is in } k \text{ and } EN_{t,j}^{(IR)} \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

If `IRTenorOffsetting` = FALSE,

$$AddOn_{t,j}^{(IR)}\left[k,i\right] = SF_j^{(IR)} \cdot \begin{cases} -D_{t,i}^{(IR)} & \text{if } i \text{ is in } k \text{ and } D_{t,jk}^{(IR)} < 0 \\ D_{t,i}^{(IR)} & \text{if } i \text{ is in } k \text{ and } D_{t,jk}^{(IR)} \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Define the contribution of trade i to the hedging set j by

$$AddOn_{t,j}^{(IR)}\left[i\right] = \sum_k AddOn_{t,j}^{(IR)}\left[k,i\right]$$

Note

[*] If i has multiple SACCR containers, $D_{t,i}^{(IR)}$ in the definition of $AddOn_{t,j}^{(IR)}\left[k,i\right]$ is the sum of the effective notionals of the SACCR containers of i belonging to bucket k .

Foreign Exchange

For a hedging set j , the addon is:

$$AddOn_{t,j}^{(FX)} = SF_j^{(FX)} \cdot \left|EN_{t,j}^{(FX)}\right|$$

If `FXOffsetting` = False,

$$EN_{t,j}^{(FX)} = D_{t,j1}^{(FX)}$$

(In this case, there is a unique hedging bucket for each hedging set j or currency pair.)

If `FXOffsetting` = True,

$$EN_{t,j}^{(FX)} = \sqrt{\sum_m \sum_n \rho_{m,n} \left(SF_m^{(FX)} \cdot \left|D_{t,jm}^{(FX)}\right|\right) \cdot \left(SF_n^{(FX)} \cdot \left|D_{t,jn}^{(FX)}\right|\right)}$$

(In this case, there is a unique hedging set $j = 1$ and the hedging buckets are the currency pairs.)

For attribution, assume trade i is in hedging set j and hedging bucket k .

If `FXOffsetting` = False (there is only one hedging bucket, $k = 1$):

$$AddOn_{t,j}^{(FX)}\left[k,i\right] = SF_j^{(FX)} \cdot \begin{cases} -D_{t,i}^{(FX)} & \text{if } EN_{t,j}^{(FX)} < 0 \\ D_{t,i}^{(FX)} & \text{if } EN_{t,j}^{(FX)} \geq 0 \end{cases}$$

If `FXOffsetting` = True (there is only one hedging set, $j = 1$):

$$AddOn_{t,j}^{(FX)}\left[k,i\right] = \begin{cases} -\frac{\sum_m \rho_{m,k} \cdot \left(SF_m^{(FX)} \cdot \left|D_{t,jm}^{(FX)}\right|\right) \cdot \left(SF_k^{(FX)} \cdot D_{t,i}^{(FX)}\right)}{EN_{t,j}^{(FX)}} & \text{if } EN_{t,j}^{(FX)} \neq 0 \text{ and } D_{t,jk}^{(FX)} < 0 \\ \frac{\sum_m \rho_{m,k} \cdot \left(SF_m^{(FX)} \cdot \left|D_{t,jm}^{(FX)}\right|\right) \cdot \left(SF_k^{(FX)} \cdot D_{t,i}^{(FX)}\right)}{EN_{t,j}^{(FX)}} & \text{if } EN_{t,j}^{(FX)} \neq 0 \text{ and } D_{t,jk}^{(FX)} \geq 0 \\ 0 & \text{if } EN_{t,j}^{(FX)} = 0 \end{cases}$$

Note

A similar note as in [*] applies here for $D_{t,i}^{(FX)}$.

Credit

For a hedging set j , the addon is:

$$AddOn_{t,j}^{(CR)} = \left[\left(\sum_l \rho_l^{(CR)} \cdot AddOn_{t,jl}^{(CR)}\right)^2 + \sum_l \left(1 - \left(\rho_l^{(CR)}\right)^2\right) \cdot \left(AddOn_{t,jl}^{(CR)}\right)^2\right]^{1/2}$$

For attribution, let j be a hedging set and k a hedging bucket (entity bucket of regular trades) of j .

If $AddOn_{t,j}^{(CR)} \neq 0$, define

$$AddOn_{t,j}^{(CR)}\left[k,i\right] =$$
$$\begin{cases} \frac{1}{AddOn_{t,j}^{(CR)}} \cdot \left[\rho_k^{(CR)} \cdot \sum_l \rho_l^{(CR)} \cdot AddOn_{t,jl}^{(CR)} + \left(1 - \left(\rho_k^{(CR)}\right)^2\right) \cdot AddOn_{t,jk}^{(CR)}\right] \cdot SF_k^{(CR)} \cdot D_{t,i}^{(CR)} & \text{if } i \text{ is in } k \\ 0 & \text{otherwise} \end{cases}$$

$$AddOn_{t,j}^{(CR)}\left[i\right] = \sum_k AddOn_{t,j}^{(CR)}\left[k,i\right]$$

If $AddOn_j^{(CR)} = 0$, define $AddOn_{t,j}^{(CR)}\left[i\right] = 0$.

Note

A similar note as in [*] applies here for $D_{t,i}^{(CR)}$.

Equity

For a hedging set j , the addon is:

$$AddOn_{t,j}^{(EQ)} = \left[\left(\sum_l \rho_l^{(EQ)} \cdot AddOn_{t,jl}^{(EQ)}\right)^2 + \sum_l \left(1 - \left(\rho_l^{(EQ)}\right)^2\right) \cdot \left(AddOn_{t,jl}^{(EQ)}\right)^2\right]^{1/2}$$

For attribution, let j be a hedging set and k a hedging bucket (entity bucket of regular trades) of j .

If $AddOn_{t,j}^{(EQ)} \neq 0$, define

$$AddOn_{t,j}^{(EQ)}\left[k,i\right] =$$
$$\begin{cases} \frac{1}{AddOn_{t,j}^{(EQ)}} \cdot \left[\rho_k^{(EQ)} \cdot \sum_l \rho_l^{(EQ)} \cdot AddOn_{t,jl}^{(EQ)} + \left(1 - \left(\rho_k^{(EQ)}\right)^2\right) \cdot AddOn_{t,jk}^{(EQ)}\right] \cdot SF_k^{(EQ)} \cdot D_{t,i}^{(EQ)} & \text{if } i \text{ is in } k \\ 0 & \text{otherwise} \end{cases}$$

$$AddOn_{t,j}^{(EQ)}\left[i\right] = \sum_k AddOn_{t,j}^{(EQ)}\left[k,i\right]$$

If $AddOn_{t,j}^{(EQ)} = 0$, define $AddOn_{t,j}^{(EQ)}\left[i\right] = 0$.

Note

A similar note as in [*] applies here for $D_{t,i}^{(EQ)}$.

Commodity

For a hedging set j , the addon is:

$$AddOn_{t,j}^{(CM)} = \left[\left(\sum_l \rho_l^{(CM)} \cdot AddOn_{t,jl}^{(CM)}\right)^2 + \sum_l \left(1 - \left(\rho_l^{(CM)}\right)^2\right) \cdot \left(AddOn_{t,jl}^{(CM)}\right)^2\right]^{1/2}$$

For attribution, let j be a hedging set and k a hedging bucket (commodity bucket of regular trades) of j .

If $AddOn_{t,j}^{(CM)} \neq 0$, define

$$AddOn_{t,j}^{(CM)}\left[k,i\right] =$$

$$\begin{cases} \frac{1}{AddOn_{t,j}^{(CM)}} \cdot \left[\rho_k^{(CM)} \cdot \sum_l \rho_l^{(CM)} \cdot AddOn_{t,jl}^{(CM)} + \left(1 - \left(\rho_k^{(CM)}\right)^2\right) \cdot AddOn_{t,jk}^{(CM)}\right] \cdot SF_k^{(CM)} \cdot D_{t,i}^{(CM)} & \text{if } i \text{ is in } k \\ 0 & \text{otherwise} \end{cases}$$

$$AddOn_{t,j}^{(CM)}\left[i\right] = \sum_k AddOn_{t,j}^{(CM)}\left[k,i\right]$$

If $AddOn_{t,j}^{(CM)} = 0$, define $AddOn_{t,j}^{(CM)}\left[i\right] = 0$.

Note

A similar note as in [*] applies here for $D_i^{(CM)}$.

Aggregate AddOn

If a trade i matures on or after time t , its contribution to $AddOn_t$ is the sum of $AddOn_{t,j}^{(AC)}\left[i\right]$ over all asset classes AC and all hedging sets j to which i belongs. We denote this sum in both the unmargined and margined cases simply by $AddOn_{t,i}$.

$$AddOn_{t,i} = \sum_{AC \in \{IR, FX, CR, EQ, CM\}} \sum_{j \in AC} AddOn_{t,j}^{(AC)}\left[i\right]$$

"Netted" AddOn

The "netted" addon is used in the following cases:

- when calculating the attribution of initial margin, we will need to calculate the addon assuming all trades on the node to which initial margin is attached net;
- when calculating the attribution of variation margin in the case of a node with multiple netting sets, we will also need to calculate the addon assuming all trades net.

For a node X , define the "netted" addon to the aggregate addon calculate as if all trades net. Denote the contribution of trade i to this "netted" addon on X by:

$$AddOn_{t,X,net,i}$$

Note

At time zero, this addon is calculated using maturity factors equal to 1.

- **Collateral Attribution**

To determine the contribution of a trade to EE and PFE, we must allocate collateral (variation margin, initial margin and independent amounts) to the trade. To do this, we make the following assumptions:

- the difference between the variation margin (which in our model includes independent amounts) and the mark-to-market of the portfolio, and
- the initial margin

are functions of the volatility of the trades.

With these assumptions, we allocate collateral as follows:

- If a variation margin agreement is in place, we first assume a trade i is fully collateralized with variation margin (that is, we allocate $MTM_{t,i}$ variation margin to the trade), and then split the excess/deficit variation margin, $VM_t - V_{t,NS}$, among all the trades using the AddOn's as proxies for the volatility of the trades.
- The initial margin IM_t is also split using AddOn's as proxies for the volatility of the trades.

We use the notation and definitions in 6.31 SACCR Profile Expected Exposure on page 135, as well as the following definitions:

Define:

- $IM_t = IM_{t,ctpy} + \sum_{i \in X_t} cpIM(i,t,ccy) - IM_{t,user\_unseg} - \sum_{i \in X_t} uIM_{unseg}(i,t,ccy)$
- $IA_t = IA_{t,ctpy} + \sum_{i \in X_t} cpIA(i,t) - \delta_{user} \cdot (IA_{t,user} + \sum_{i \in X_t} uIA(i,t))$
- $NICA_t = IM_t + IA_t$
- $V_{t,X} = \sum_{i \in X_t} MtM_{t,i}$
- $MtM_{t,i}$ is the mark-to-market of trade at time , as defined in • Formula: V on page 143
- mat_i is the maturity of trade
- $X_t = \{i \in X_t \mid mat_i \geq t\}$
- $\delta_{user} = \begin{cases} 1 & \text{if UserVMPostStyle} \neq \text{NONE and UserVMSegregated} = \text{FALSE} \\ 0 & \text{otherwise} \end{cases}$
- $\delta_{ctpy} = \begin{cases} 1 & \text{if CPVMPostStyle} \neq \text{NONE} \\ 0 & \text{otherwise} \end{cases}$

Case 1: Single SA-CCR netting set

- Unmargined:

  $C_t^{Unmargined} = \begin{cases} IM_t + VM_t & \text{if CollateralModel} \neq \text{NONE} \\ 0 & \text{if CollateralModel=NONE} \end{cases}$

  where

  - $VM_t = \begin{cases} -VM_{t,user\_unseg} & \text{if IncludeVMinRCUnmargined=TRUE} \\ \delta_{ctpy} \cdot cpIA(t) - \delta_{user} \cdot uIA(t) & \text{if IncludeVMinRCUnmargined=FALSE} \end{cases}$
  - $VM_{t,user\_unseg}$ is defined in • Formula: VM on page 147

- IM_t is defined in section Initial Margin Attribution

- Margined:

$$C_t^{Margined} = \begin{cases} IM_t + VM_t & \text{if CollateralModel} \neq \text{NONE} \\ 0 & \text{if CollateralModel=NONE} \end{cases}$$

where

- $VM_t = \begin{cases} VM_{0,ctpy} - VM_{0,user\_unseg} & \text{if } t = 0 \\ V_{t,X} + IA_t - TH_{t,ctpy} - MTA_{t,ctpy} & \text{if } t > 0 \end{cases}$

- IM_t , $V_{t,X}$ and IA_t are defined in • Collateral Attribution on page 182.

Define:

$$VM_{t,user\_seg} = \begin{cases} -VM_t^{all} & \text{if } t = 0, VM_t^{all} < 0, \text{UserVMPostStyle} \neq \text{NONE and UserVMSegregated} = \text{TRUE} \\ 0 & \text{otherwise} \end{cases}$$

where

$$VM_t^{all} = \begin{cases} VM_{0,ctpy} - VM_{0,user} & \text{if } t = 0 \\ V_{t,X} + IA_t - TH_{t,ctpy} - MTA_{t,ctpy} & \text{if } t > 0 \end{cases}$$

Case 2: Multiple SA-CCR netting sets

VM_t , is defined in • Replacement Cost on page 140 if `CollateralModel`NONE and 0 otherwise.

$$C_t^{Unmargined/Margined} = \begin{cases} VM_t + IM_t & \text{if CollateralModel} \neq \text{NONE} \\ 0 & \text{if CollateralModel=NONE} \end{cases}$$

$$C_t^{Unmargined/Margined,\,+} = \left(C_t^{Unmargined/Margined}\right)^+$$

$$C_t^{Unmargined/Margined,\,-} = \left(C_t^{Unmargined/Margined}\right)^-$$

$$VM_{t,user\_unseg} = \begin{cases} -VM_t & \text{if } VM_t < 0 \text{ UserVMSegregated} = \text{FALSE} \\ 0 & \text{otherwise} \end{cases}$$

$$VM_{t,user\_seg} = \begin{cases} -VM_t & \text{if } VM_t < 0 \text{ UserVMSegregated} = \text{TRUE} \\ 0 & \text{otherwise} \end{cases}$$

Note

The attribution formulas below are based on the assumption that either $VM_{t,ctpy}$ or $VM_{t,user}$ is zero.

Initial Margin Attribution

Define:

$IM_{t,X}$, as initial margin on at time

The contribution of a trade $i \in X$ to $IM_{t,X}$ is defined by

$$IM_{t,X,i} = \frac{AddOn_{t,X,net,i}}{AddOn_{t,X,net}} \cdot IM_{t,X} + cpIM(i,t,ccy) - uIM_{unseg}(i,t,ccy)$$

where

- $AddOn_{t,X,net,i}$ is defined in • AddOn Attribution on page 178, and $AddOn_{t,X,net} = \sum_{j \in X} AddOn_{t,X,net,j}$

- $cpIM(i,t,ccy)$: the haircut adjusted value of trade-level segregated and unsegregated initial margins on deal i at time t posted by the counterparty. See • Formula: Trade-level IM and IA on page 148 for details.
- $uIM_{unseg}(i,t,ccy)$: the haircut adjusted value of trade-level unsegregated initial margin on deal i at time t posted by the user. See • Formula: Trade-level IM and IA on page 148 for details.

If $AddOn_{t,X,net} = 0$, replace $\frac{AddOn_{t,X,net,i}}{AddOn_{t,X,net}}$ by $\frac{1}{N_X}$, where N_X is the number of trades in X .

In both the margined and unmargined case, the contribution of trade $i \in X$ to $IM_{t,X}$ is

$IM_{t,i} = IM_{t,X,i}$

Variation Margin Attribution

Case 1: Single SA-CCR netting set

- Unmargined:

  The contribution of trade i to VM_t is defined by

$$VM_{t,i} = \mathbf{1}_{\mathrm{mat}_i \geq t} \cdot \begin{cases} MtM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot \left(VM_t - IA_{t,Trades} - V_{t,NS} \right) + IA(i,t) & \text{if } VM_t \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

  - VM_t , $V_{t,NS}$, $MtM_{t,i}$, mat_i , δ_{ctpy} , δ_{user} and $AddOn_{t,i}$ are defined in • Collateral Attribution on page 182 and • AddOn Attribution on page 178, and $AddOn_t = \sum_{j \in X} AddOn_{t,j}$.
  - $IA_{t,Trades} =$

$$\begin{cases} \sum_{i \in CSA} cpIA(i,t) - \sum_{i \in CSA} uIA(i,t) & if \text{ IncludeVMinRCunmargined} = \text{True} \\ \delta_{ctpy} \cdot \sum_{i \in CSA} cpIA(i,t) - \delta_{user} \cdot \sum_{i \in CSA} uIA(i,t) & if \text{ IncludeVMinRCunmargined} = \text{False} \end{cases}$$

  - $IA(i,t) = \begin{cases} cpIA(i,t) - uIA(i,t) & if \text{ IncludeVMinRCunmargined} = \text{True} \\ \delta_{ctpy} \cdot cpIA(i,t) - \delta_{user} \cdot uIA(i,t) & if \text{ IncludeVMinRCunmargined} = \text{False} \end{cases}$
  - $cpIA(i,t)$: the trade-level independent amount on deal i at time t posted by the counterparty. See section • Formula: Trade-level IM and IA on page 148 for details.
  - $uIA(i,t)$: the trade-level independent amount on deal i at time t posted by the user. See • Formula: Trade-level IM and IA on page 148 for details.

Note

$VM_t = 0$ if `CPPostStyle` = `UserPostStyle` = NONE or `CollateralModel` = NONE

- Margined:

  The contribution of trade i to VM_t is defined by

$$VM_{t,i} = \mathbf{1}_{\mathrm{mat}_i \geq t} \cdot \begin{cases} MtM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot \left(VM_t - IA^*_{t,Trades} - V_{t,NS} \right) + IA^*(i,t) & \text{if } VM_{t,user\_seg} = 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

- VM_t , $VM_{t,user\_seg}$, $V_{t,NS}$, $MtM_{t,i}$, mat_i , δ_{ctpy} , δ_{user} and $AddOn_{t,i}$ are defined in • Collateral Attribution on page 182 and • AddOn Attribution on page 178, and $AddOn_t = \sum_{j \in X} AddOn_{t,j}$.
- $IA^*_{t,Trades} = \delta_{ctpy} \cdot \sum_{i \in CSA} cpIA(i,t) - \delta_{user} \cdot \sum_{i \in CSA} uIA(i,t)$
- $IA^*(i,t) = cpIA(i,t) - \delta_{user} \cdot uIA(i,t)$
- $cpIA(i,t)$: the trade-level independent amount on deal i at time t posted by the counterparty. See section • Formula: Trade-level IM and IA on page 148 for details.
- $uIA(i,t)$: the trade-level independent amount on deal i at time t posted by the user. See • Formula: Trade-level IM and IA on page 148 for details.

Note

If $AddOn_t = 0$, replace $\frac{AddOn_{t,i}}{AddOn_t}$ by $\frac{1}{N}$, where N is the number of trades in X .

Case 2: Multiple SA-CCR netting sets

- Unmargined:

  The contribution of trade i to VM_t is defined by

  $$VM_{t,i} = \mathbf{1}_{\text{mat}_i \geq t} \cdot \begin{cases} MtM_{t,i} + \frac{AddOn_{t,X,net,i}}{AddOn_{t,X,net}} \cdot \left(VM_t - IA_{t,Trades} - V_{t,X} \right) + cpIA(i,t) - uIA(i,t) & \text{if } VM_{t,user\_unseg} > 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

  - VM_t , $V_{t,X}$, $MtM_{t,i}$, mat_i and $AddOn_{t,X,net,i}$ are defined in • Collateral Attribution on page 182 and • AddOn Attribution on page 178, and $AddOn_{t,X,net} = \sum_{j \in X} AddOn_{t,X,net,i}$
  - $IA_{t,Trades} = \sum_{i \in CSA} cpIA(i,t) - \sum_{i \in CSA} uIA(i,t)$
  - $cpIA(i,t)$: the trade-level independent amount on deal i at time t posted by the counterparty. See section • Formula: Trade-level IM and IA on page 148 for details.
  - $uIA(i,t)$: the trade-level independent amount on deal i at time t posted by the user. See • Formula: Trade-level IM and IA on page 148 for details.

- Margined:

  The contribution of trade i to VM_t is defined by

  $$VM_{t,i} = \mathbf{1}_{\text{mat}_i \geq t} \cdot \begin{cases} MtM_{t,i} + \frac{AddOn_{t,X,net,i}}{AddOn_{t,X,net}} \cdot \left(VM_t - IA_{t,Trades} - V_{t,X} \right) + cpIA(i,t) - uIA(i,t) & \text{if } VM_{t,user\_seg} = 0 \\ 0 & \text{otherwise} \end{cases}$$

  where

  - VM_t , $V_{t,X}$, $MtM_{t,i}$, mat_i and $AddOn_{t,X,net,i}$ are defined in • Collateral Attribution on page 182 and • AddOn Attribution on page 178, and $AddOn_{t,X,net} = \sum_{j \in X} AddOn_{t,X,net,i}$
  - $IA_{t,Trades} = \sum_{i \in CSA} cpIA(i,t) - \sum_{i \in CSA} uIA(i,t)$
  - $cpIA(i,t)$: the trade-level independent amount on deal i at time t posted by the counterparty. See section • Formula: Trade-level IM and IA on page 148 for details.

- $uIA(i, t)$: the trade-level independent amount on deal i at time t posted by the user. See ● Formula: Trade-level IM and IA on page 148 for details.

Collateral Attribution

Case 1: Single SA-CCR netting set

The contribution of trade i to $C_t^{Unmargined/Margined}$ is defined by

$$C_{t,i}^{Unmargined/Margined} = VM_{t,i} + IM_{t,i}$$

where $VM_{t,i}$ and $IM_{t,i}$ are defined in ● Collateral Attribution on page 182.

Case 2: Multiple SA-CCR netting sets

The contribution of trade i to $C_t^{Unmargined/Margined,\,+}$ and $C_t^{Unmargined/Margined,\,-}$ are defined by

$$C_{t,i}^{Unmargined/Margined,\,+} = \mathbf{1}_{C_t^{Unmargined/Margined,\,+} > 0} \cdot \left(VM_{t,i} + IM_{t,i}\right)$$

$$C_{t,i}^{Unmargined/Margined,\,-} = \mathbf{1}_{C_t^{Unmargined/Margined,\,-} < 0} \cdot \left(VM_{t,i} + IM_{t,i}\right)$$

where VM_i (for margined and unmargined case) and IM_i are where $VM_{t,i}$ and $IM_{t,i}$ are defined in ● Collateral Attribution on page 182.

● Replacement Cost Attribution

- For a node X with `SettledtoMarketAgreement` = False:

  Case 1: Single SA-CCR netting set

  - Unmargined:

    Recall that the replacement cost of $X = NS$ at time t is

    $$RC_t = \max\left(V_{t,NS} - C_t^{Unmargined}, 0\right)$$

    where $V_{t,NS}$ and $C_t^{Unmargined}$ is defined in ● Collateral Attribution on page 182.

    The contribution of a trade $i \in X$ to RC_t is defined by

    $$RC_{t,i} = \begin{cases} 0 & \text{if } RC_t = 0 \\ \mathbf{1}_{\mathrm{mat}_i \geq t} \cdot MtM_{t,i} - C_{t,i}^{Unmargined} & \text{if } RC_t > 0 \end{cases}$$

    where $MtM_{t,i}$ and $C_{t,i}^{Unmargined}$ are defined in ● Collateral Attribution on page 182. Hence,

    $$RC_{t,i} = \mathbf{1}_{\mathrm{mat}_i \geq t} \cdot \begin{cases} 0 & \text{if } RC_t = 0 \\ \frac{AddOn_{t,i}}{AddOn_t} \cdot \left(V_{t,NS} - VM_t + IA_{t,Trades}\right) - IA(i,t) - IM_{t,i} & \text{if } RC_t > 0 \text{ and } VM_t \neq 0 \\ MtM_{t,i} - IM_{t,i} & \text{if } RC_t > 0 \text{ and } VM_t = 0 \end{cases}$$

    where $IM_{t,i}$ is defined in section Initial Margin Attribution.

  - Margined:

    Recall that the replacement cost of X is

    $$RC_t = \max\left(V_{t,NS} - C_t^{Margined}, 0\right)$$

    where $V_{t,NS}$, $C_t^{Margined}$ are defined in ● Collateral Attribution on page 182.

    The contribution of trade i to RC_t is defined by

$$RC_{t,i} = \begin{cases} 0 & \text{if } RC_t = 0 \\ \mathbf{1}_{\text{mat}_i \geq t} \cdot MtM_{t,i} - C_{t,i}^{Margined} & \text{if } RC_t > 0 \end{cases}$$

where $MtM_{t,i}$ and $C_{t,i}^{Margined}$ are defined in • Collateral Attribution on page 182. Hence,

$$RC_{t,i} = \mathbf{1}_{\text{mat}_i \geq t} \cdot$$
$$\begin{cases} 0 & \text{if } RC_t = 0 \\ \frac{AddOn_{t,i}}{AddOn_t} \cdot \left(V_{t,NS} - VM_t + IA^*_{t,Trades}\right) - IA^*(i,t) - IM_{t,i} & \text{if } RC_t > 0 \text{ and } VM_{t,user\_seg} = 0 \\ MtM_{t,i} - IM_{t,i} & \text{if } RC_t > 0 \text{ and } VM_{t,user\_seg} \neq 0 \end{cases}$$

where $IM_{t,i}$ is defined in section Initial Margin Attribution.

Case 2: Multiple SA-CCR netting sets

- Unmargined/Margined

  Recall that the replacement cost of X is

  $RC_t = RC_t^{(+)} + RC_t^{(-)}$

  $RC_t^{(+)} = \left(\sum_{NS \in X} (V_{t,NS})^+ - C_t^{Unmargined/Margined,\,+}\right)^+$

  $RC_t^{(-)} = \left(\sum_{NS \in X} (V_{t,NS})^- - C_t^{Unmargined/Margined,\,-}\right)^-$

  where $V_{t,NS}$ is the net mark-to-market of all the trades in NS with maturities on or after t , and $C_t^{Unmargined/Margined,\,+}$, $C_t^{Unmargined/Margined,\,-}$ are defined in • Collateral Attribution on page 182.

  The contribution of trade i to RC_t is defined by

  $RC_{t,i} = RC_{t,i}^{(+)} + RC_{t,i}^{(-)}$

$$RC_{1,i}^{(+)} = \begin{cases} 0 & \text{if } RC_t^{(+)} = 0 \\ \mathbf{1}_{V_{t,NS(i)} > 0} \cdot \mathbf{1}_{\text{mat}_i \geq t} \cdot MtM_{t,i} - C_{t,i}^{Unmargined/Margined,\,+} & \text{if } RC_t^{(+)} > 0 \end{cases}$$

$$RC_{1,i}^{(-)} = \begin{cases} 0 & \text{if } RC_t^{(-)} = 0 \\ \mathbf{1}_{V_{t,NS(i)} < 0} \cdot \mathbf{1}_{\text{mat}_i \geq t} \cdot MtM_{t,i} - C_{t,i}^{Unmargined/Margined,\,-} & \text{if } RC_t^{(-)} > 0 \end{cases}$$

  where $NS(i)$ is the unique SACCR netting set to which i belongs, and $V_{t,NS(i)}$, $C_{t,i}^{Unmargined/Margined,\,+}$ and $C_{t,i}^{Unmargined/Margined,\,-}$ are defined in • Collateral Attribution on page 182.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True:

$$RC_{t,i}^{STM} = \begin{cases} -IM_{t,i} & if\ RC_t^{STM} = 0 \\ 0 & if\ RC_t^{STM} \neq 0 \end{cases}$$

where

  - $IM_{t,i}$ is defined in Initial Margin Attribution.
  - $RC_t^{STM} = \begin{cases} max\{-(cpIM(t,ccy) - uIM_{unseg}(t,ccy), 0\} & \text{if CollateralModel} \neq \text{NONE} \\ 0 & \text{if CollateralModel} = \text{NONE} \end{cases}$, see section • Formula: Trade-level IM and IA on page 148 for details.

- **Multiplier times AddOn Attribution**

For an SACCR netting set NS of a node X (equal to an MA with no CSA, CSA, NC or NN trade) with `SettledtoMarketAgreement` = False, and time $t > 0$, the multiplier is defined as follows:

For EE_t :

$$m_{t,NS} = m_t = \min\left(1, f + \left(1 - f\right) \cdot \exp\left(\frac{CV_t}{m \cdot (1-f) \cdot AddOn_t}\right)\right)$$

For PFE_t :

$$m_{t,NS} = m_t = \min\left(1, \max\left(f, 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t}\right)\right)$$

$$CV_t = \begin{cases} V_{t,NS} - C_t^{Unmargined/Margined} & X = NS \\ V_{t,NS} - IM_{t,NS}^{Trade} & X \text{ has multiple SACCR netting sets} \end{cases}$$

where:

- $AddOn_t$ is the (unmargined or margined) aggregate addon of NS
- $f = m\_floor$ is the multiplier floor
- $m = m\_factor$ is the multiplier factor
- $C_t^{Unmargined/Margined}$ is defined in • Collateral Attribution on page 182
- $IM_{t,NS}^{Trade} = \sum_{i \in NS} cpIM(i,t,ccy) - \sum_{i \in NS} uIM_{unseg}(i,t,ccy)$
- $cpIM(i,t,ccy)$: the haircut adjusted value of trade-level segregated and unsegregated initial margins on deal i at time t posted by the counterparty. See • Formula: Trade-level IM and IA on page 148 for details.
- $uIM_{unseg}(i,t,ccy)$: the haircut adjusted value of trade-level unsegregated initial margin on deal i at time t posted by the user. See • Formula: Trade-level IM and IA on page 148 for details.

For an SACCR netting set NS of a NS of a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = False is:

For EE_t^{STM} :

$$m_{t,NS}^{STM} = m_t^{STM} = min\left\{1, f + (1-f) \cdot exp\left(\frac{-IM_{t,NS}^{Trade}}{m \cdot (1-f) \cdot Addon_{t,Aggregate}^{STM}}\right)\right\}$$

For PFE_t^{STM} :

$$m_{t,NS}^{STM} = m_t^{STM} = min\left\{1, max\left\{f, 1 + \frac{\alpha_1 \cdot (-IM_{t,NS}^{Trade})}{\alpha_2 \cdot Addon_{t,Aggregate}^{STM}}\right\}\right\}$$

where:

- $f = m\_floor$ is the multiplier floor
- $m = m\_factor$ is the multiplier factor
- $IM_{t,NS}^{Trade} = \sum_{i \in NS} cpIM(i,t,ccy) - \sum_{i \in NS} uIM_{unseg}(i,t,ccy)$
- $cpIM(i,t,ccy)$: the haircut adjusted value of trade-level segregated and unsegregated initial margins on deal i at time t posted by the counterparty. See • Formula: Trade-level IM and IA on page 148 for details.

- $uIM_{unseg}(i, t, ccy)$: the haircut adjusted value of trade-level unsegregated initial margin on deal i at time t posted by the user. See • Formula: Trade-level IM and IA on page 148 for details.

Expected Exposure: Multiplier = 1

- For a node X with `SettledtoMarketAgreement` = False:

  If $m_t = 1$, the contribution of trade i to $m_{t,NS} \cdot AddOn_{t,NS} = m_t \cdot AddOn_t$ is

  $[m_t \cdot AddOn_t]_i = AddOn_{t,i}$ See • AddOn Attribution on page 178 for the definition of

  in both the margined and unmargined cases.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True:

  If $m_t^{STM} = 1$, the contribution of trade i to $m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}$ is
  $[m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}]_i = [Addon_{t,Aggregate}^{STM}]_i$.

Expected Exposure: Multiplier < 1

- For a node X with `SettledtoMarketAgreement` = False:
  - **Case 1: Single SA-CCR netting set**

    The contribution of trade i to the unmargined $m_t \cdot AddOn_t$ is

    $$[m_t \cdot AddOn_t]_i = m_t \cdot AddOn_{t,i} + \exp\left(\frac{CV_{t,NS}}{m \cdot (1-f) \cdot AddOn_t}\right) \cdot \frac{1}{m}$$

    $$\cdot \left[-IM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot IM_t + \begin{cases} 0 & \text{if } VM_t \neq 0 \\ MtM_{t,i} - \frac{AddOn_{t,i}}{AddOn_t} \cdot V_{t,NS} & \text{otherwise} \end{cases}\right]$$

    The contribution of trade i to the margined $m_t \cdot AddOn_t$ is

    $$[m_t \cdot AddOn_t]_i = m_t \cdot AddOn_{t,i} + \exp\left(\frac{CV_{t,NS}}{m \cdot (1-f) \cdot AddOn_t}\right) \cdot \frac{1}{m}$$

    $$\cdot \left[-IM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot IM_t + \begin{cases} 0 & \text{if } VM_{t,user\_seg} = 0 \\ MtM_{t,i} - \frac{AddOn_{t,i}}{AddOn_t} \cdot V_{t,NS} & \text{otherwise} \end{cases}\right]$$

  - **Case 2: Multiple SA-CCR netting sets**

    The contribution of trade i to the unmargined $m_t \cdot AddOn_t$ is

    $$[m_t \cdot AddOn_t]_i = m_t \cdot AddOn_{t,i} + \exp\left(\frac{CV_{t,NS}}{m \cdot (1-f) \cdot AddOn_t}\right) \cdot \frac{1}{m} \cdot \left(MtM_{t,i} - IM_{t,NS,i} - \frac{AddOn_{t,i}}{AddOn_t} \cdot CV_{t,NS}\right)$$

    where $IM_{t,NS,i} = cpIM(i, t, ccy) - uIM_{unseg}(i, t, ccy)$.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True, the contribution of trade i to $m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}$ is

  $$[m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}]_i = m_t^{STM}[AddOn_{t,Aggregate}^{STM}]_i + exp\left(\frac{-IM_t}{m \cdot (1-f) \cdot Addon_{t,Aggregate}^{STM}}\right) \cdot \left(\frac{1}{m}\right)$$

  $$\cdot \left[-IM_{t,i} + \frac{[AddOn_{t,Aggregate}^{STM}]_i}{AddOn_{t,Aggregate}^{STM}} IM_t\right]$$

  where $IM_t = cpIM(t, ccy) - uIM_{unseg}(t, ccy)$, see section • Formula: Trade-level IM and IA on page 148 for details and $IM_{t,i}$ is defined in section Initial Margin Attribution.

Note

In all cases above, if $AddOn_t = 0$, define $[m_t \cdot AddOn_t]_i = 0$.

Potential Future Exposure: Multiplier = 1

- For a node X with `SettledtoMarketAgreement` = False:

  If $m_t = 1$, the contribution of trade i to $m_{t,NS} \cdot AddOn_{t,NS} = m_t \cdot AddOn_t$ is $[m_t \cdot AddOn_t]_i = AddOn_{t,i}$ See • AddOn Attribution on page 178 for the definition of in both the margined and unmargined cases.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True:

  If $m_t^{STM} = 1$, the contribution of trade i to $m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}$ is $[m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}]_i = [AddOn_{t,Aggregate}^{STM}]_i$.

Potential Future Exposure: Multiplier < 1

- For a node X with `SettledtoMarketAgreement` = False:
  - **Case 1: Single SA-CCR netting set**

    The contribution of trade i to the unmargined $m_t \cdot AddOn_t$ is

$$[m_t \cdot AddOn_t]_i = m_t \cdot AddOn_{t,i} + \frac{a_1}{a_2} \cdot$$

$$\left\{\begin{array}{ll} -IM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot IM_t & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} \geq f \text{ and } VM_t \neq 0 \\ MtM_{t,i} - \frac{AddOn_{t,i}}{AddOn_t} \cdot V_{t,NS} - IM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot IM_t & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} \geq f \text{ and } VM_t = 0 \\ 0 & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} < f \end{array}\right]$$

    The contribution of trade i to the margined $m_t \cdot AddOn_t$ is

$$[m_t \cdot AddOn_t]_i = m_t \cdot AddOn_{t,i} + \frac{a_1}{a_2} \cdot$$

$$\left\{\begin{array}{ll} -IM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot IM_t & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} \geq f \text{ and } VM_{t,user\_seg} \neq 0 \\ MtM_{t,i} - \frac{AddOn_{t,i}}{AddOn_t} \cdot V_{t,NS} - IM_{t,i} + \frac{AddOn_{t,i}}{AddOn_t} \cdot IM_t & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} \geq f \text{ and } VM_{t,user\_seg} = 0 \\ 0 & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} < f \end{array}\right]$$

  - **Case 2: Multiple SA-CCR netting sets**

    The contribution of trade i to the unmargined $m_t \cdot AddOn_t$ is

$$[m_t \cdot AddOn_t]_i = m_t \cdot AddOn_{t,i} + \begin{cases} \frac{a_1}{a_2} \cdot \left(MtM_{t,i} - IM_{t,NS,i} - \frac{AddOn_{t,i}}{AddOn_t} \cdot CV_{t,NS}\right) & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} \geq f \\ 0 & \text{if } 1 + \frac{a_1 \cdot CV_t}{a_2 \cdot AddOn_t} < f \end{cases}$$

where $IM_{t,NS,i} = cpIM(i,t,ccy) - uIM_{unseg}(i,t,ccy)$.

- For a VMCSA (Legacy, VM or IMVMCSA) node X with `SettledtoMarketAgreement` = True, the contribution of trade i to $m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}$ is

  $$[m_t^{STM} \cdot Addon_{t,Aggregate}^{STM}]_i = m_t^{STM} \cdot [AddOn_{t,Aggregate}^{STM}]_i + \frac{\alpha_1}{\alpha_2} \cdot \begin{cases} -IM_{t,i} + \frac{[AddOn_{t,Aggregate}^{STM}]_i}{AddOn_{t,Aggregate}^{STM}} IM_t & if\ f \leq 1 + \frac{\alpha_1 \cdot (-IM_t)}{\alpha_2 \cdot AddOn_{t,Aggregate}^{STM}} \\ 0 & if\ f > 1 + \frac{\alpha_1 \cdot (-IM_t)}{\alpha_2 \cdot AddOn_{t,Aggregate}^{STM}} \end{cases}$$

  where $IM_t = cpIM(t,ccy) - uIM_{unseg}(t,ccy)$, see section ● Formula: Trade-level IM and IA on page 148 for details and

  $IM_{t,i}$ is defined in section Initial Margin Attribution.

Note

In all cases above, if $AddOn_t = 0$, define $[m_t \cdot AddOn_t]_i = 0$.

- **Sold Option Exception**

If node X is subject to sold option exception, the Initial Margin is assumed to be 0. Hence, the contribution of trade i to EE/PFE of such node is 0.

6.35 SACCR Notional Weighted Effective Maturity

This function is applied to implement Effective Maturity calculation in the SACCR engine. The measure can be calculated at various nodes of the legal hierachy including the OTC nodes, as well as on trades. Effective maturity is expressed in years.

- **Input Parameters**

  - **Function Parameters**

    The funtion parameters used in SACCR Effective Maturity function and their notations used in this section are defined as follows:

Function Paramters	Default	Notation
`Capped`	1 (True)	
`maturityCapTermValue`	5	b
`maturityCapTermUnit`	Y (Year)	
`maturityFloorTermValue`	1	c
`maturityFloorTermUnit`	Y (Year)	
`shortTermExpMaturityFloorTermValue`	1	d
`shortTermExpMaturityFloorTermUnit`	D (Day)	
`weightedAverageRemainingMaturity`	1 (True)	f

  - **Other Inputs**

    - `RegulatoryApproach`: a standard attribute defined at the own-counterparty or counterparty view level, can be set to Foundation or Advanced.
    - `FloorOverride`: an optional custom attribute at any node level, in years.

- `regulatoryshorttermexposure`: a boolean attribute at the trade level.
- `regulatoryEffectiveMaturity`: an optional numerical standard attribute at the trade level, in years.

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- $Notional_i$ = the standard trade attribute `recnotional` of trade i , converted to domestic currency
- $ShortTermExp_i$ = the standard trade attribute `regulatoryshorttermexposure` of trade i
- $Maturity_i = \begin{cases} \text{the standard trade attribute } maturitydate \text{ of trade i} & \text{if EarlyExit} = True \\ \text{the standard trade attribute } nextearlyterminationdate \text{ of trade i} & \text{if EarlyExit} = False \end{cases}$
- $RegEffectiveMaturity_i$ = the custom attribute `regulatoryEffectiveMaturity` of trade i
- $\mu_i = \begin{cases} Maturity_i - Processing\ Date & \text{if} f = True \text{ and } Processing\ Date \leq Maturity_i \\ 0 & \text{if} f = True \text{ and } Processing\ Date > Maturity_i \\ RegEffectiveMaturity_i & \text{if} f = False \end{cases}$

Note

1. If $f = False$ and $RegEffectiveMaturity_i$ is not provided, μ_i is defined as when $f = True$.
2. For structured deals, the custom attribute `regulatoryEffectiveMaturity` is read at the componenent-deal level only.
3. For structured deals, the standard attribute `regulatoryshorttermexposure` is read at the structured-deal level and applies to all its component deals.

Node and Trade Level Effective Maturity

Effective Matuirty can be calculated on the following nodes of the legal hierachy.

- OTC Node
- MA/NN Node
- CSA/NC Node

Effective Maturity will also be calculated at the trade level for trades in an NN node.

- **Effective Maturity at Node Level**

  Let X denote a CSA, NC, MA, NN or OTC node and μ denote the effective maturity of the node before any caps/floors are applied.

  If `RegulatoryApproach` = Foundation: $\mu = 2.5\,Years$.

  If `RegulatoryApproach` = Advanced: $\mu = \frac{\sum_{i \in X} Notional_i \times \mu_i}{\sum_{i \in X} Notional_i}$.

  Note

  Each component of a structured deal contributes to μ separately.

  The Effecitve Maturity at node-level with cap/floor is defined as

  $EM = \begin{cases} max\{min\{\mu, b\}, floor\} & if\ Capped = True \\ max\{\mu, floor\} & if\ Capped = False \end{cases}$

  where $floor$ is defined as follows:

```
If (FloorOverride is populated)
    floor = FloorOverride
Else
    If (the node is a CSA node)
        floor is calculated using the table below
    Else if (there are subnodes under the node)
        floor = maximum of the floors of all subnodes
    Else
        floor = 1 year
```

The *floor* value at CSA level is defined as follows:

Fully Collateralized	Call Period = 1 AND Liquid Collateral	CSA Maturity Floor
True	True	1 day
True	False	MPOR + CallPeriod - 1
False	*	1 year

6

Functions

Note

1. In the above table, Fully Collateralized = True if the `RegulatoryFullyCollateralized` attribute of a CSA node is set to True, and `CallPeriod` and `LiquidCollateral` are CSA level attributes. For the definition and calculation of MPOR, refer to • Formula: MPOR on page 116
2. The netting structure of the expression is irrelevant in the above calculation.
3. Node-level Effective Maturity $EM = 0$ if the node is empty.

- **Effective Maturity at Trade Level**

  Let $\bar{\mu}_i$ denote the effective maturity of trade i before any caps/floors are applied.

  If `RegulatoryApproach` = Foundation: $\bar{\mu}_i = 2.5\,Years$.

  If `RegulatoryApproach` = Advanced:

$$\bar{\mu}_i = \begin{cases} \mu_i & \text{if } i \text{ is not a structured deal} \\ \frac{\sum_{j \in \text{components of the SD}} Notional_j \times \mu_j}{\sum_{j \in \text{components of the SD}} Notional_j} & \text{if } i \text{ is a structured deal} \end{cases} .$$

  The Effecitve Maturity of trade i with cap/floor is defined as

  If trade $i \notin$ an NN node: $EM_i = 0$.

  If trade $i \in$ an NN node:

$$EM_i = \begin{cases} max\{min\{\bar{\mu}_i, b\}, d\} & if\ Capped = True \text{ and } ShortTermExp_i = True \\ max\{\bar{\mu}_i, d\} & if\ Capped = False \text{ and } ShortTermExp_i = True \\ max\{min\{\bar{\mu}_i, b\}, c\} & if\ Capped = True \text{ and } ShortTermExp_i = False \\ max\{\bar{\mu}_i, c\} & if\ Capped = False \text{ and } ShortTermExp_i = False \end{cases} .$$

Note

MAG will only generate trade-level Effective Maturity results for trades that belong to an NN node.

6.36 SACCR Intermediate Results

This document summarizes the intermediate results available for capital and profile SACCR expressions. Intermediate results are produced based on the `subresultSpecifier` attribute value, which can be set to VALIDATION, ANALYSIS, VALIDATION_ABOVE_DEALS, ANALYSIS_ABOVE_DEALS.

- **Capital Intermediate Results**

This section summarizes the intermediate results available for SACCR capital expressions: 6.27 SACCR Capital Exposure At Default on page 107, 6.28 SACCR Capital Replacement Cost on page 111 and 6.29 SACCR Capital Potential Future Exposure on page 121.

The SACCR Capital is calculated for node X under the two cases below:

- **Case 1**: X consists of a single SA-CCR netting set
- **Case 2**: X consists of multiple SA-CCR netting sets

Under each node X , the node, deal and mitigant level intermediate results are reported. Please refer to the corresponding financial documents for definition of X and other symbols referenced in the tables below.

Note

DealResult and MitigantResult are not available in ABOVE_DEALS mode.

Table 6.36-1 Capital subresults reporting for the NodeResult result type

Result ID	Symbol	Description
Alpha1	a_1	a scalar for RC **Note** Not available in Analysis mode.
Alpha2	a_2	a scalar for AddOn **Note** Not available in Analysis mode.
Sold Options Exception		A boolean that indicates whether the netting set meets the sold options exception criteria. Reported only if True. When True, PFE is not calculated and reported as 0.
MTM	V_{NS}	net mark-to-market value of all derivative transactions in the netting set.
Positive Exposure	$\sum_{NS \in X} \max(V_{NS}, 0)$	sum of positive exposures; reported under case 2 only.
Negative Exposure	$\sum_{NS \in X} \min(V_{NS}, 0)$	sum of negative exposures, reported under case 2 only.
TH_ctpy	TH	counterparty threshold; before adjusted by the allocation factors in the case of synthetic netting set expressions
MTA_ctpy	MTA	minimum transfer amount; before adjusted by the allocation factors in the case of synthetic netting set expressions
IA_ctpy	IA_{ctpy}^{CSA}	counterparty independent amount; before adjusted by the allocation factors in the case of synthetic netting set expressions
IA_usr	IA_{user}^{CSA}	user independent amounts; before adjusted by the allocation factors in the case of synthetic netting set expressions

Result ID	Symbol	Description
VM_ctpy	VM_{ctpy}	variation margin posted by the counterparty, used in t = 0 only.
VM_usr_unseg	VM_{user_unseg}	variation margin posted by the user
IM_ctpy	IM_{ctpy}	initial margin posted by the counterparty
IM_usr_unseg	IM_{user_unseg}	initial margin posted by the user
VM	VM	haircut adjusted value of net variation margin received; reported under case 2 only.
C_net_plus	$\max\left(C^{Unmargined}, 0\right)$; $\max\left(C^{Margined}, 0\right)$	net collateral received; reported under case 2 only.
C_net_minus	$\min\left(C^{Unmargined}, 0\right)$; $\min\left(C^{Margined}, 0\right)$	net collateral paid; reported under case 2 only.
RC	RC	replacement cost
FX Offsetting		(not used for capital expressions)
IR Tenor Offsetting		a boolean output that indicates whether offsetting benefit between maturity buckets are allowed for Interest Rate derivatives.
Long Currency		indicates the long currency of the hedging set currency pair; reported for Foreign Exchange asset class.
Short Currency		indicates the short currency of the hedging set currency pair; reported for Foreign Exchange asset class.
Correlation		correlation between hedging bucket; reported for Credit, Equity and Commodity asset classes.
< 1 Y AddOn; 1-5 Y AddOn; >5Y AddOn; Hedging Subset AddOn		hedging bucket level add-on
Regular Hedging Set AddOn; Basis Hedging Set AddOn; Volatility Hedging Set AddOn	$AddOn_j^{(a)}$	hedging set level add-on
FX, IR, Credit, Equity, Commodity	$AddOn^{(a)}$	asset class level add-on
AddOn	$AddOn_{Aggregate}^{Unmargined}$; $AddOn_{Aggregate}^{Margined}$	aggregated add-on
Multiplier	$multiplier^{Unmargined}$; $multiplier^{Margined}$	multiplier for add-on **Note** Drilldown not available in Analysis mode.
AllocationFactorCSA	$1 - \rho_0^{CSA parameters}$	allocation factor of the CSA parameters as specified by the splitting rule of the CSA parameters; reported for synthetic netting set expressions only.
AllocationFactorVM	$1 - \rho_0^{VM}$	allocation factor of the current excess variation margin as specified by the splitting rule of the

Result ID	Symbol	Description
		variation margin; reported for synthetic netting set expressions only.
AllocationFactorIM	$1 - \rho_0^{IM}$	allocation factor of the current initial margin as specified by the splitting rule of intial margin; reported for synthetic netting set expressions only. **Note** In case of a VMCSA node that has IMCSA sub-node, the AllocationFactorIM sub-results will include the `path` tag with the path to the IMCSA node.
AddOn Non-sim	$\lvert Addon_{VM,0,n} \rvert$ for AllocationVM and AllocationFactorCSA; $\lvert Addon_{IM,0,n} \rvert$ for AllocationIM	For AllocationFactorCSA and AllocationFactorVM: SACCR AddOn using the set of trades in VMCSA/Legacy CSA/IMVMCSA using any non-zero maturity factor. For AllocationFactorIM: SACCR AddOn using the set of trades in IMCSA/IMVMCSA using any non-zero maturity factor.
AddOn Sim	$\lvert Addon_{VM,0,s} \rvert$ for AllocationVM and AllocationFactorCSA; $\lvert Addon_{IM,0,s} \rvert$ for AllocationIM	For AllocationFactorCSA and AllocationFactorVM: SACCR AddOn using the set of trades in VMCSA/Legacy CSA/IMVMCSA using any non-zero maturity factor. For AllocationFactorIM: SACCR AddOn using the set of trades in IMCSA/IMVMCSA using any non-zero maturity factor.
EAD_Unmargined		expected exposure or potential future exposure under the unmargined calculation; reported for margined node only. **Note** Not available in Analysis mode.
PFE		potential future exposure
EAD_Unmargined		expected exposure or potential future exposure under the unmargined calculation; reported for margined node only. **Note** Not available in Analysis mode.
EAD_Margined		expected exposure or potential future exposure under the margined calculation; reported for margined node only. **Note** Not available in Analysis mode.
EAD	EAD	final expected exposure or potential future exposure

Table 6.36-2 Capital subresults reporting for the DealResult result type

Result ID	Symbol	Description
Deal MtM		deal MtM
MtM Currency		deal MtM currency
Trade Date		standard attribute std.tradedate

Result ID	Symbol	Description
Maturity Date		standard attribute std.maturitydate
Deal-level MtM		deal MtM. 0 is reported on time steps that deal has matured.
Delta	δ_i	supervisory delta
Adjusted Notional	$d_i^{(a)}$	adjusted notional
Maturity Factor	$MF_i^{Margined}$; $MF_i^{Unmargined}$	maturity factor
Supervisory Factor	$SF_k^{(a)}$	supervisory factor
Supervisory Factor Multiplier		supervisory factor multiplier for basis and volatility trades
Effective Notional	$D_{t,i}^{(a)}$	effective notional
Ancillary Risk Factor		the ancillary risk factor for basis transactions
Direction		direction of the trade. For regular/volatility non-FX trades, it is inferred from the long/short indicator, as read from saccr.longentity for EQ/CS, saccr.longhedgingset for IR, saccr.longcommodity for CM. For basis trades, it is inferred from the long/short indicator and the order of hedging set risk factor pair vs. trade risk factor pair. For regular/volatility FX trades, it is inferred from the order of hedging set currency pair vs. trade currency pair. With the exception of delta override being populated, in general positive delta has Long direction and negative delta has Short direction.
Long Currency		saccr.longnotionalcurrency of FX trades
Entity		saccr.entity of Credit and Equity trades
Commodity Hedging Set		saccr.hedgingset of Commodity trades
Commodity		saccr.commodity of Commodity trades
Deal-level AddOn		deal-level addon

Table 6.36-3 Capital subresults reporting for the MitigantResult results type

Result ID	Symbol	Description
Margin Type		margin type: Initial or Variation
Poster		poster of the mitigant: User or Counterparty
Mitigant Type		mitigant type: Cash or Physical
Amount Pre-HC		mitigant value before haircut
Amount Post-HC	$MtM_{t,i}$	mitigant value after haircut. 0 is reported for time steps that are before the delivery date or after the return date of the mitigant.
Mitigant Currency		mitigant currency
Node Currency		node currency is the CSA Agreement Currency for CSA netting sets and MA Termination Currency for MA or NC netting set.
Delivery Date		delivery date

Result ID	Symbol	Description
Return Date		return date
Cash Haircut	HC_i	annualized haircut for cash mitigant, read from table `SACCRCCYPFEHaircuts`
PFE Haircut	HC_i	annualized haircut for physical mitigant, read from table `VolatilityAdjustments`
FX Haircut	HC_{FX}	annualized FX haircut, read from table `SACCRFXHaircut`
Haircut	H_i	final volatility haircut amount (Cash or PFE plus FX) for the mitigant
Haircut At Default	$H_{default}$	haircut at default for the mitigant
Mitigant Amount		final mitigant value for each time step

- **Profile Intermediate Results**

This section summarizes the intermediate results available for SACCR profile expressions: 6.31 SACCR Profile Expected Exposure on page 135 and 6.32 SACCR Profile Potential Future Exposure on page 160.

The SACCR Profile EE and PFE are calculated for node X under the two cases below:

- **Case 1**: X consists of a single SA-CCR netting set
- **Case 2**: X consists of multiple SA-CCR netting sets

Under each node X , the node, deal and mitigant-level intermediate results are reported. Please refer to the corresponding financial documents for definition of X and other symbols referenced in the tables below.

Note

DealResult and MitigantResult are not available in ABOVE_DEALS mode.

Table 6.36-4 Profile subresults reporting for the NodeResult result type

Result ID	Symbol	Description
Alpha1	a_1	a scalar for RC **Note** Not available in Analysis mode.
Alpha2	a_2	a scalar for AddOn **Note** Not available in Analysis mode.
Sold Options Exception		A boolean that indicates whether the netting set meets the sold options exception criteria. Reported only if True.
Result Type		reports either Margined or Unmargined for each time step that indicates whether the final EE comes from margined or unmargined calculation.
CPPostNone		True if CPVMPostStyle=None, False otherwise.

Result ID	Symbol	Description
UserPostNone		True if UserVMPostStyle=None, False otherwise
MTM	$V_{t,NS}$	net mark-to-market value of all derivative transactions in the netting set.
Positive Exposure	$\sum_{NS \in X} \max(V_{t,NS}, 0)$	sum of positive exposures; reported under case 2 only.
Negative Exposure	$\sum_{NS \in X} \min(V_{t,NS}, 0)$	sum of negative exposures, reported under case 2 only.
TH_ctpy	$TH_{t,ctpy}$	counterparty threshold; before adjusted by the allocation factors in the case of synthetic netting set expressions
TH_usr	$TH_{t,user}$	user threshold; reported under case 2 only; before adjusted by the allocation factors in the case of synthetic netting set expressions.
MTA_ctpy	$MTA_{t,ctpy}$	counterparty minimum transfer amount; before adjusted by the allocation factors in the case of synthetic netting set expressions
MTA_usr	$MTA_{t,user}$	user minimum transfer amount; reported under case 2 only; before adjusted by the allocation factors in the case of synthetic netting set expressions.
IA_ctpy	$IA_{t,ctpy}$	counterparty independent amount; before adjusted by the allocation factors in the case of synthetic netting set expressions
IA_usr	$IA_{t,user}$	user independent amounts; before adjusted by the allocation factors in the case of synthetic netting set expressions
VM_ctpy	$VM_{0,ctpy}$	variation margin posted by the counterparty, used in t =0 only
VM_usr_unseg	$VM_{t,user\_unseg}$	variation margin posted by the user
IM_ctpy	$IM_{t,ctpy}$	initial margin posted by the counterparty
IM_usr_unseg	$IM_{t,user\_unseg}$	initial margin posted by the user
VM	VM_t	haircut adjusted value of net variation margin received; reported under case 2 only
C_net_plus	$\max(VM_t + IM_{t,ctpy} - IM_{t,user\_unseg}, 0)$	net collateral received; reported under case 2 only
C_net_minus	$\min(VM_t + IM_{t,ctpy} - IM_{t,user\_unseg}, 0)$	net collateral paid; reported under case 2 only
RC	RC_t	replacement cost
FX Offsetting		a boolean output that indicates whether offsetting between FX currency pairs are allowed for Foreign Exchange derivatives.
IR Tenor Offsetting		a boolean output that indicates whether offsetting benefit between maturity buckets are allowed for Interest Rate derivatives

Result ID	Symbol	Description
Long Currency		indicates the long currency of the hedging set currency pair; reported for Foreign Exchange asset class
Short Currency		indicates the short currency of the hedging set currency pair; reported for Foreign Exchange asset class
Correlation		correlation between hedging bucket; reported for Credit, Equity and Commodity asset classes
< 1 Y AddOn; 1-5 Y AddOn; >5Y AddOn; Hedging Subset AddOn		hedging bucket level add-on
Regular Hedging Set AddOn; Basis Hedging Set AddOn; Volatility Hedging Set AddOn		hedging set level add-on
FX, IR, Credit, Equity, Commodity		asset class level add-on
AddOn	$AddOn_t$	aggregated add-on
Multiplier	m_t	multiplier for add-on **Note** Drilldown not available in Analysis mode.
Multiplier * AddOn		netting set level multiplier * addon; reported under case 2 only
AllocationFactorCSA	$1 - \rho_t^{CSAparameters}$	allocation factor of the CSA parameters as specified by the splitting rule of the CSA parameters; reported for synthetic netting set expressions only.
AllocationFactorVM	$1 - \rho_0^{VM}$	allocation factor of the current excess variation margin as specified by the splitting rule of the variation margin; reported for synthetic netting set expressions only.
AllocationFactorIM	$1 - \rho_t^{IM}$	allocation factor of the initial margin as specified by the splitting rule of intial margin; reported for synthetic netting set expressions only. **Note** In case of a VMCSA node that has IMCSA sub-node, the AllocationFactorIM sub-results will include the `path` tag with the path to the IMCSA node.
AddOn Non-sim	$\lvert Addon_{VM,t,n} \rvert$ for AllocationVM and AllocationFactorCSA; $\lvert Addon_{IM,t,n} \rvert$ for AllocationIM	For AllocationFactorCSA and AllocationFactorVM: SACCR AddOn using the set of trades in VMCSA/Legacy CSA/IMVMCSA using any non-zero maturity factor. For AllocationFactorIM: SACCR AddOn using the set of trades in IMCSA/IMVMCSA using any non-zero maturity factor.

Result ID	Symbol	Description
AddOn Sim	$\|Addon_{VM,t,s}\|$ for AllocationVM and AllocationFactorCSA; $\|Addon_{IM,t,s}\|$ for AllocationIM	For AllocationFactorCSA and AllocationFactorVM: SACCR AddOn using the set of trades in VMCSA/Legacy CSA/IMVMCSA using any non-zero maturity factor. For AllocationFactorIM: SACCR AddOn using the set of trades in IMCSA/IMVMCSA using any non-zero maturity factor.
EE_Unmargined; PFE_Unmargined		expected exposure or potential future exposure under the unmargined calculation; reported for margined node only **Note** Not available in Analysis mode.
EE_Margined; PFE_Margined		expected exposure or potential future exposure under the margined calculation; reported for margined node only **Note** Not available in Analysis mode.
EE; PFE	EE_t ; PFE_t	final expected exposure or potential future exposure

Table 6.36-5 Profile subresults reporting for the DealResult result type

Result ID	Symbol	Description
Deal MtM		deal MtM
MtM Currency		deal MtM currency
Trade Date		standard attribute std.tradedate
Maturity Date		standard attribute std.maturitydate
Deal-level MtM		deal MtM. 0 is reported on time steps that deal has matured.
Delta	$\delta_{t,i}$	supervisory delta
Adjusted Notional	$d_{t,i}^{(a)}$	adjusted notional
Maturity Factor	$MF_{t,i}$	maturity factor
Supervisory Factor	$SF_k^{(a)}$	supervisory factor
Supervisory Factor Multiplier		supervisory factor multiplier for basis and volatility trades
Effective Notional	$D_{t,i}^{(a)}$	effective notional
Ancillary Risk Factor		the ancillary risk factor for basis transactions
Direction		direction of the trade. For regular/volatility non-FX trades, it is read from saccr.longentity for CM/CS, saccr.longhedgingset for IR, saccr.longcommodity for CM. For regular/volatility FX trades, it is inferred from the currency pair. For basis trades, it is inferred from the hedging set risk factor pair. With

6 Functions

Result ID	Symbol	Description
		the exception of delta override being populated, in general positive delta has Long direction and negative delta has Short direction.
Long Currency		saccr.longnotionalcurrency of FX trades
Entity		saccr.entity of Credit and Equity trades
Commodity Hedging Set		saccr.hedgingset of Commodity trades
Commodity		saccr.commodity of Commodity trades
Deal-level AddOn		deal-level addon

Table 6.36-6 Profile subresults reporting for the MitigantResult result type

Result ID	Symbol	Description
Margin Type		margin type: Initial or Variation
Poster		poster of the mitigant: User or Counterparty
Mitigant Type		mitigant type: Cash or Physical
Amount Pre-HC		mitigant value before haircut
Amount Post-HC	$MtM_{t,i}$	mitigant value after haircut. 0 is reported for time steps that are before the delivery date or after the return date of the mitigant.
Mitigant Currency		mitigant currency
Node Currency		node currency is the CSA Agreement Currency for CSA netting sets and MA Termination Currency for MA or NC netting set.
Delivery Date		delivery date
Return Date		return date
Cash Haircut	HC	annualized haircut for cash mitigant, read from table `SACCRCCYPFEHaircuts`
PFE Haircut	HC	annualized haircut for physical mitigant, read from table `VolatilityAdjustments`
FX Haircut	HC_{FX}	annualized FX haircut, read from table `SACCRFXHaircut`
Haircut		final volatility haircut amount (Cash or PFE plus FX) for each time step
Haircut At Default	$H_{default}$	haircut at default for the mitigant
Mitigant Amount		final mitigant value for each time step

6.37 SACCR Current Exposure

There are 4 functions relevant and defined for CSA, NC, MA, NN and OTC nodes:

- SACCR Current Value
- SACCR Collateral Balance
- SACCR Collateralized Value

- SACCR Current Exposure

Note

The netting allowed under SACCR applies to these functions.

Note

These functions are defined for both the forward and reverse views.

Note

These functions are not defined at the deal level.

Preliminary Definitions

Mark-to-Market

If all the trades in a set *X* net, define

$$\mathrm{MtM}_{0,X} := \begin{cases} \sum_{i \in X} \mathrm{MtM}_{0,\mathrm{i}}, & \text{if forward view} \\ -\sum_{i \in X} \mathrm{MtM}_{0,\mathrm{i}}, & \text{if reverse view} \end{cases}$$

If none of the trades in a set *X* net, define

$$P_{0,X} := \begin{cases} \sum_{i \in X} \max\left(\mathrm{MtM}_{0,\mathrm{i}}, 0\right), & \text{if forward view} \\ \sum_{i \in X} \max\left(-\mathrm{MtM}_{0,\mathrm{i}}, 0\right), & \text{if reverse view} \end{cases}$$

$$N_{0,X} := \begin{cases} \sum_{i \in X} \min\left(\mathrm{MtM}_{0,\mathrm{i}}, 0\right), & \text{if forward view} \\ \sum_{i \in X} \min\left(-\mathrm{MtM}_{0,\mathrm{i}}, 0\right), & \text{if reverse view} \end{cases}$$

where $MtM_{0,i}$ is the current mark-to-market of trade *i*.

Collateral

For a node *X*, define

- In forward view

$$C_{0,X} := \begin{cases} \mathrm{VM}_{0,\mathrm{cpty}} - \mathrm{VM}_{0,\mathrm{user\_unseg}} + \mathrm{IM}_{0,\mathrm{cpty}} - \mathrm{IM}_{0,\mathrm{user\_unseg}}, & \text{if X = CSA and CollateralModel} \neq \text{NONE} \\ 0, & \text{otherwise} \end{cases}$$

- In reverse view

$$C_{0,X} := \begin{cases} \mathrm{VM}_{0,\mathrm{user}} - \mathrm{VM}_{0,\mathrm{cpty\_unseg}} + \mathrm{IM}_{0,\mathrm{user}} - \mathrm{IM}_{0,\mathrm{cpty\_unseg}}, & \text{if X = CSA and CollateralModel} \neq \text{NONE} \\ 0, & \text{otherwise} \end{cases}$$

where

$VM_{0,cpty}$ = the un-haircut variation margin posted by the counterparty at t_0 on $X = CSA$

$VM_{0,user\_unseg}$ = the un-haircut unsegregated variation margin posted by the user at t_0 on $X = CSA$

$IM_{0,cpty}$ = the un-haircut initial margin posted by the counterparty at t_0 on $X = CSA$

$IM_{0,user\_unseg}$ = the un-haircut unsegregated initial margin posted by the user at t_0 on $X = CSA$

Note

All variation and initial margin values in this section are assumed to be positive.

- **Current Value, Collateral Balance, Collateralized Value, and Current Exposure**

Note

Closeout Netting Pools are only supported if the closeout netting pools are empty, that is if none of the trades in an MA net. If non-empty closeout netting pools are present, trades in the closeout netting pools don't contribute to the current value thus to the current exposure defined in this section.

CSA/NC Node

- Single Netting Set

  When all trades in a node (*X = CSA or NC*) net, Current Value (V), Collateral Balance (CB), Collateralized Value (CV) and Current Exposure (CE) are defined as:

  $$V_{0,X} := \mathrm{MtM}_{0,X}$$

  $$\mathrm{CB}_{0,X} := C_{0,X}$$

  $$\mathrm{CV}_{0,X} := V_{0,X} - \mathrm{CB}_{0,X}$$

  $$\mathrm{CE}_{0,X} := \max\{0, \mathrm{CV}_{0,X}\}$$

- No Netting

  If none of the trades in a node (*X = CSA or NC*) net, Current Value (V), Collateral Balance (CB), Collateralized Value (CV) and Current Exposure (CE) are defined as:

  $$V_{0,X} := P_{0,X}$$

  $$\mathrm{CB}_{0,X} := \max\{0, C_{0,X}\} - \max\{0, N_{0,X} - \min(0, C_{0,X})\}$$

  $$\mathrm{CV}_{0,X} := V_{0,X} - \mathrm{CB}_{0,X}$$

  $$\mathrm{CE}_{0,X} := \max\{0, \mathrm{CV}_{0,X}\}$$

MA Node

$$V_{0,\mathrm{MA}} := \sum_{\mathrm{CSA} \subseteq \mathrm{MA}} V_{0,\mathrm{CSA}} + V_{0,\mathrm{NC}}$$

$$\mathrm{CB}_{0,\mathrm{MA}} := \sum_{\mathrm{CSA} \subseteq \mathrm{MA}} \mathrm{CB}_{0,\mathrm{CSA}} + \mathrm{CB}_{0,\mathrm{NC}}$$

$$\mathrm{CV}_{0,\mathrm{MA}} := \sum_{\mathrm{CSA} \subseteq \mathrm{MA}} \mathrm{CV}_{0,\mathrm{CSA}} + \mathrm{CV}_{0,\mathrm{NC}}$$

$$\mathrm{CE}_{0,\mathrm{MA}} := \max\{0, \mathrm{CV}_{0,\mathrm{MA}}\}$$

NN Node

$$V_{0,\mathrm{NN}} := P_{0,\mathrm{NN}}$$

$$\mathrm{CB}_{0,\mathrm{NN}} := 0$$

$$\mathrm{CV}_{0,\mathrm{NN}} := V_{0,\mathrm{NN}}$$

$$\mathrm{CE}_{0,\mathrm{NN}} := \max\{0, \mathrm{CV}_{0,\mathrm{NN}}\}$$

Note

Since there is no netting on the NN node, the collateralized value is just the exposure.

OTC Node

$$V_{0,\mathrm{OTC}} := \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \max\{0, V_{0,\mathrm{MA}}\} + V_{0,\mathrm{NN}} = \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \max\{0, V_{0,\mathrm{MA}}\} + P_{0,\mathrm{NN}}$$

$$\mathrm{CB}_{0,\mathrm{OTC}} := \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \mathrm{CB}_{0,\mathrm{MA}} + \mathrm{CB}_{0,\mathrm{NN}} = \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \mathrm{CB}_{0,\mathrm{MA}}$$

$$\mathrm{CV}_{0,\mathrm{OTC}} := \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \max\{0, \mathrm{CV}_{0,\mathrm{MA}}\} + \mathrm{CV}_{0,\mathrm{NN}} = \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \mathrm{CE}_{0,\mathrm{MA}} + \mathrm{CE}_{0,\mathrm{NN}}$$

$$\mathrm{CE}_{0,\mathrm{OTC}} := \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \max\{0, \mathrm{CV}_{0,\mathrm{MA}}\} + \mathrm{CV}_{0,\mathrm{NN}} = \sum_{\mathrm{MA} \subseteq \mathrm{OTC}} \mathrm{CE}_{0,\mathrm{MA}} + \mathrm{CE}_{0,\mathrm{NN}}$$

Note

Since there is no netting above the MA/NN nodes, the OTC current value is the current exposure without collateral, and the OTC collateralized value is just the current exposure.

- **Stressed Results**

The four functions defined in this section can be combined with function Simulated SACCR. The Current Value, Collateral Balance, Collateralized Value and Current Exposure measures defined above for each scenario are thus stressed by using the stressed MtM and stressed FX from the cube. The stressed FX is also applied to the mark-to-market of the collateral.

6.38 Simulated SACCR

This function is combined with an SACCR function to generate stressed SACCR measure.

A cube of stressed mark-to-market values of the trades for a single timestep is provided, as well as a vector of FX rates. For each scenario, the Stressed SACCR measures are defined as in the regular SACCR function document except that the stressed MtM values are used instead of the book of record MtM. Meanwhile, the stressed FX rates are applied to both the MtM of trades and mitigants.

6.39 Exposure backtesting

The functions to perform exposure backtesting are introduced in this section.

- **Preliminary background**

Backtesting counterparty credit risk (CCR) is becoming increasingly important in the financial industry but it is still a challenging task. A quantitative methodology has been developed and implemented as IMCR functionality. With this methodology, exposure envelopes can be constructed using Monte Carlo scenarios. The envelope boundaries are used to compute the probability distribution of τ , which is the fraction of time the exposure spent outside of the envelope. The quantiles of the distribution of τ are compared with the computed fraction $\tilde{\tau}$ for the realized exposure to map the model to the color bands. Another element of the exposure backtesting framework is EPE and EEPE backtesting based on the concept of excess exposure. The two elements are complementary to each other and the combination of the two methods give users a complete exposure backtesting framework.

Tau

- **Function Parameters**

  - **Envelope Option**: Two Sided, Upper (default), Lower
  - **Tau Method**: Count (default), Distance
  - **Envelope Probability**: 0.95 (default)

This MAG function returns a sheet (time and scenario) of $\psi_{t_k}\left(s_j\right)$, as defined below.

Chaining the **Tau** function and the MAG function **Time Weight** with **start day** of 0 and **end day**, being the last time step, produces tau statistic per scenario. Applying time weight would also make sure that the time average stops after the longest dated transaction expires to avoid underestimating tau.

For hypothesis testing, the right tail of tau distribution would be needed, which can be calculated by chaining the non-parametric right tail function to the result of the time-weighted tau function.

- **Formulas**

The tau statistic per scenario is defined as average time spent outside of an "envelope", with upper and lower bounds, or only upper, or only lower:

$\tau\left(s_j\right) = \frac{1}{(t_M - t_0)} \sum_{k=1}^{M} \psi_{t_k}\left(s_j\right) \cdot \left(t_k - t_{k-1}\right)$

where

Upper and Lower:

- Tau Method = Count

$\psi_{t_k}(s_j) = I(E_{t_k}(s_j) < L_{t_k}) + I(E_{t_k}(s_j) > U_{t_k})$

- Tau Method = Distance

$\psi_{t_k}(s_j) = -(E_{t_k}(s_j) - L_{t_k}) \times I(E_{t_k}(s_j) < L_{t_k}) + (E_{t_k}(s_j) - U_{t_k}) \times I(E_{t_k}(s_j) > U_{t_k})$

$P\left(E_{t_k}\left(s_j\right) > U_{t_k}\right) = \frac{1-p}{2}, P\left(E_{t_k}\left(s_j\right) < L_{t_k}\right) = \frac{1-p}{2}$

Upper:

- Tau Method = Count

$\psi_{t_k}(s_j) = I(E_{t_k}(s_j) > U_{t_k})$

- Tau Method = Distance

$\psi_{t_k}(s_j) = (E_{t_k}(s_j) - U_{t_k}) \times I(E_{t_k}(s_j) > U_{t_k})$

$P\left(E_{t_k}\left(s_j\right) > U_{t_k}\right) = 1 - p$

Lower:

- Tau Method = Count

$\psi_{t_k}(s_j) = I(E_{t_k}(s_j) < L_{t_k})$

- Tau Method = Distance

$\psi_{t_k}(s_j) = -(E_{t_k}(s_j) - L_{t_k}) \times I(E_{t_k}(s_j) < L_{t_k})$

$P\left(E_{t_k}\left(s_j\right) < L_{t_k}\right) = 1 - p$

L_{t_k} and U_{t_k} are the lower and upper envelope corresponding to the envelope probability computed. The logic of the non-parametric right and left tail MAG functions can be used for the estimation of upper and lower envelopes respectively (for every time step) inside the **Tau** function using envelope probability. For a two-sided envelope, right and left tail functions should use two-sided as the confidence interval option.

$E_{t_k}(s_j)$ are the given exposures on Monte Carlo scenarios, $s_j,\ j=1,\ 2,\ ...,\ N,$ at the time step $t_k,\ k=0,\ 1,\ ...,\ M.$

$I(\cdot)$ is an index function that returns 1 if inside logic is true, or 0 otherwise.

■ Realized Tau

● Function Parameters

- **Expression Definition Id**: ID of the expression that generated the tau envelope to be used

This MAG function returns a sheet (time) of realized $\widetilde{\psi}_{t_k}$, as defined below.

The function chain of **Realized Tau** with the MAG function **Time Weight** produces tau statistic.

● Formulas

The tau statistic is defined as average time spent outside of an "envelope", with upper and lower bounds, or only upper, or only lower:

$$\widetilde{\tau}=\frac{1}{(t_M-t_0)}\sum_{k=1}^{M}\widetilde{\psi}_{t_k}\cdot\left(t_k-t_{k-1}\right)$$

where

Upper and Lower:

- Tau Method = Count

$$\widetilde{\psi}_{t_k}=I(\widetilde{E}_{t_k}<L_{t_k})+I(\widetilde{E}_{t_k}>U_{t_k})$$

- Tau Method = Distance

$$\widetilde{\psi}_{t_k}=-(\widetilde{E}_{t_k}-L_{t_k})\times I(\widetilde{E}_{t_k}<L_{t_k})+(\widetilde{E}_{t_k}-U_{t_k})\times I(\widetilde{E}_{t_k}>U_{t_k})$$

Upper:

- Tau Method = Count

$$\widetilde{\psi}_{t_k}=I(\widetilde{E}_{t_k}>U_{t_k})$$

- Tau Method = Distance

$$\widetilde{\psi}_{t_k}=(\widetilde{E}_{t_k}-U_{t_k})\times I(\widetilde{E}_{t_k}>U_{t_k})$$

Lower:

- Tau Method = Count

$$\widetilde{\psi}_{t_k}=I(\widetilde{E}_{t_k}<L_{t_k})$$

- Tau Method = Distance

$$\widetilde{\psi}_{t_k}=-(\widetilde{E}_{t_k}-L_{t_k})\times I(\widetilde{E}_{t_k}<L_{t_k})$$

L_{t_k} and U_{t_k} are the same lower and upper envelope generated by **Tau** MAG function.

$\widetilde{E}_{t_k}$ are the realized exposures on the realized scenario based on historical data.

$I(\cdot)$ is an index function that returns 1 if inside logic is true, or 0 otherwise.

Excess Exposure

Function Parameters

- **Excess Exposure Option**: EPE (default), EEPE

This MAG function returns a sheet (time and scenario) of $E^{+}_{t_k}(s_j)$, as defined below.

The result of this function chained with the function **Time Weight** with start day of 0 and end day of 365 would produce excess exposure per scenario. For hypothesis testing, the right tail of the excess exposure distribution would be needed, which can be calculated by chaining the non-parametric right tail function to the result of the time-weighted excess exposure function.

Formulas

$$E^{+}_{t_k}(s_j) = \left(E_{t_k}(s_j) - M\right)^{+}$$

where the parameter M is the benchmark for defining excess exposure for a given scenario and time. EPE and EEPE are used as choices of parameter M. EPE is computed based on EE (expected exposure) then applying **Time Weight**, whereas EEPE is computed based on applying Ratchet to EE then applying **Time Weight**. Start and end day of this time weight is (0, 365).

Realized Excess Exposure

Function Parameters

- **Expression Definition Id**: ID of the expression that generated the corresponding M

This MAG function returns a sheet (time) of realized excess exposure $\widetilde{E}^{+}_{t_k}$ as defined below.

The chaining of this function with the function **Time Weight** produces the realized value for hypothesis testing.

Formulas

$$\widetilde{E}^{+}_{t_k} = \left(\widetilde{E}_{t_k} - M\right)^{+}$$

where $\widetilde{E}_{t_k}$ are the realized exposures on the realized scenario based on historical data.

The file, which is created by the MAG function Excess Exposure, contains information on M such that the realized excess exposure can be calculated consistent with Monte Carlo that generated the corresponding excess exposure sheet.

Excess Exposure Buffer

Function Parameters

- **Excess Exposure Option**: EPE (default), EEPE
- **Lower Confidence Interval**: 0.95 (default)
- **Higher Confidence Interval**: 0.99 (default)
- **Ratio**: True, False (default)

This MAG function returns a single value as defined below. It can be used to assess extra capital buffer to be held in case backtesting results show model failure.

- **Formulas**

$\zeta\left(s_j\right) = \frac{1}{\left(t_M - t_0\right)} \sum_{k=1}^{M} E_{t_k}^{+}\left(s_j\right) \cdot \left(t_k - t_{k-1}\right)$

$R = \frac{E\left[\zeta \mid \chi_\zeta\left(p^-\right) \le \zeta \le \chi_\zeta\left(p^+\right)\right]}{M}$

where

$\chi_\zeta\left(p^-\right)$: the right tail of ζ distribution with lower confidence level

$\chi_\zeta\left(p^+\right)$: the right tail of ζ distribution with higher confidence level

The logic of the MAG function non-parametric right tail should be used to get the tails above with one-sided confidence interval option.

The conditional expectation $E\left[\zeta \mid \chi_\zeta\left(p^-\right) \le \zeta \le \chi_\zeta\left(p^+\right)\right]$ of ζ is a simple average of ζ on all scenarios (all s_j) that satisfy the inequality $\chi_\zeta\left(p^-\right) \le \zeta \le \chi_\zeta\left(p^+\right)$. When Ratio is false, this conditional expectation is returned, otherwise, ratio R is returned.

6

Functions

6.40 PFE Attribution

This function calculates additive trade level attributions for PFE[*1] at entity level when path dependent collateral model or settlement adjusted path dependent collateral model with parameter `UseSettlementFlows = FALSE` is used. The trade level contributions then need to be aggregated across various dimensions like currency, desk, type, etc., to produce PFE of the portfolio of choice.

To determine the trade level attribution for PFE, the exposure needs to be allocated additively to each position first for every time step and scenario. Allocation of exposure sheet happens at the MA level since collateralized value is calculated there.[*2] Refer to 4.1.1.5 Collateralized Exposure on page 21 for the calculation of collateralized value. Collateralized value (CV) is an additive function of portfolio and collateral. Thus, the allocation CV is actually the allocation of portfolio and collateral value separately to every position.

Note

Node X used throughout this document denotes the MA (master agreement) node.

Note

Even though this document describes allocation of exposure at MA level, NoNet (NN) node can use the same methodology, given that NN node is equivalent to the MA node with no CSA node under it and `NettingRule = NONET`.

- **Portfolio Value Attribution**

Let $V_k^X(t_i, s_j)$ denote the portfolio value $V^X(t_i, s_j)$ allocated to position *k*.

Depending on the netting features specified by `NettingRule`, `CloseoutNetting` and the existence of complex structure, portfolio value is allocated as follows:

- Case 1. Forward view with all positions netted (the portfolio value at Node X is calculated as $V^X(t_i, s_j) = MtM^X(t_i, s_j) = \sum_{k=1}^{K} v_k(t_i, s_j)$)

The portfolio value allocated to position k is calculated as:

$V_k^X(t_i, s_j) = \upsilon_k(t_i, s_j)$

- Case 2. Forward view with sum of positions that have positive value to user (the portfolio value at Node X is calculated as $V^X(t_i, s_j) = P^X(t_i, s_j) = \sum_{k=1}^{K} \upsilon_k^+(t_i, s_j)$)

  The portfolio value allocated to position k is calculated as:

  $V_k^X(t_i, s_j) = \upsilon_k^+(t_i, s_j)$

- Case 3. Forward view with `NettingRule = LEGAL_OPINION` and complex structure (i.e. closeout netting pool) present (the portfolio value at Node X is calculated as $V^X(t_i, s_j) = \sum_{l=1}^{L} \max\left(MtM_{CNP_l}^X(t_i, s_j), 0\right) + P_{N\_CNP}^X(t_i, s_j)$, where CNP_l denotes the closeout netting pools on Node X and N_CNP the pool that doesn't net.)

  The portfolio value allocated to position k is calculated as:

  If position k is in one of the closeout netting pool (denoted by CNP_l):

  $$V_k^X(t_i, s_j) = \upsilon_k(t_i, s_j) \times 1_{MtM_{CNP_l}^X(t_i, s_j) > 0}$$

  Else (i.e. position k is in N_CNP)

  $$V_k^X(t_i, s_j) = \upsilon_k^+(t_i, s_j)$$

  where, $\upsilon_k(t_i, s_j) \times 1_{MtM_{CNP_l}^X(t_i, s_j) > 0}$ is the contribution of position k to $\sum_{l=1}^{L} \max\left(MtM_{CNP_l}^X(t_i, s_j), 0\right)$, and $1_{MtM_{CNP_l}^X(t_i, s_j) > 0}$ is an indicator function whose value is either 1 if the condition in the subscript is satisfied or 0 otherwise.

- Case 4. Reverse view with all positions netted (the portfolio value at Node X is calculated as $V^X(t_i, s_j) = -MtM^X(t_i, s_j) = -\sum_{k=1}^{K} \upsilon_k(t_i, s_j)$)

  The portfolio value allocated to position k is calculated as:

  $V_k^X(t_i, s_j) = -\upsilon_k(t_i, s_j)$

- Case 5. Reverse view with sum of positions that have positive value to counterparty (the portfolio value at Node X is calculated as $V^X(t_i, s_j) = -N^X(t_i, s_j) = -\sum_{k=1}^{K} \upsilon_k^-(t_i, s_j)$)

  The portfolio value allocated to position k is calculated as:

  $V_k^X(t_i, s_j) = -\upsilon_k^-(t_i, s_j)$

- **Collateral Attribution**

The collateral at MA level includes the Variation Margin (VM) associated with the CSA nodes under it if any and the Initial Margin (IM) assigned. VM is allocated at CSA level, while IM is allocated at MA level.

VM Attribution

Note

In current implementation, `CPPostStyle/UserPostStyle = Gross` is not supported.

Let c denote the relevant call date for reporting date t_i. c is determined as follows.

```
Cfirst = first call date on or before ti-lagmax
Clast = first call date on or before ti-lagmin
If lagcp > laguser
      If ∑CSA vk(cfirst, sj) < ∑CSA vk(clast, sj)

            c = Cfirst
      Else
            c = Clast
Else If lagcp < laguser
      If ∑CSA vk(cfirst, sj) > ∑CSA vk(clast, sj)

            c = Cfirst
      Else
            c = Clast
Else
      c = Cfirst = Clast
```

For a CSA, denote $VM^{CSA}(t_i, s_j)$ to be the counterparty variation margin minus the unsegregated user variation margin at report time t_i and scenario s_j.

Note

$VM^{CSA}(t_i, s_j) = 0$, if `CPVMPostStyle` = `UserVMPostStyle` = `NONE` or `CollateralModel` = `NONE`.

Define the total independent amount of all trades in a CSA as

$$IA_{Trades}(c, s_j) := \sum_{x \in CSA} cpIA(x, c, s_j) - \sum_{x \in CSA} uIA(x, c, s_j)$$

For definitaion of $cpIA(x, c, s_j)$ and $uIA(x, c, s_j)$, refer to Step 4: Compute the amount of variation margin required at each call date in the document of "Variation Margin Balance and Margin Call Algorithm when CPVMPostStyle = Net and UserVMPostStyle = Net".

Let $VM_k^{CSA}(t_i, s_j)$ denote the amount from $VM^{CSA}(t_i, s_j)$ allocated to position *k*, position *k* is in one of the CSA nodes.

Note

The VM attribution would be zero if the position is not in one of the CSA nodes.

$VM_k^{CSA}(t_i, s_j)$ is calculated as follows.

- Case 1. Isolate Collateral Spikes = False

If $VM^{CSA}(t_i, s_j) \neq 0$

$$VM_k^{CSA}(t_i, s_j) := \frac{v_k(c, s_j)}{\sum_{x \in CSA} v_x(c, s_j)} \times [VM^{CSA}(t_i, s_j) - IA_{Trades}(c, s_j)] + [cpIA(k, c, s_j) - uIA(k, c, s_j)]$$

Else

$$VM_k^{CSA}(t_i, s_j) = 0$$

where, $v_k(c,s_j)$ is the value of position *k* on *c* approximated by using linear interpolation as follows:

```
If c > t0
```
$$v_k(c, s_j) = v_k(t_{m-1}, s_j) + \frac{v_k(t_m, s_j) - v_k(t_{m-1}, s_j)}{t_m - t_{m-1}} \times (c - t_{m-1})$$
```
Else
```
$$v_k(c, s_j) = v_k(t_0)$$

Here, t_{m-1} is the simulation date immediately before c; and t_m is the simulation dates immediately after c. The interpolation should be done based on values in `AgreementCurrency` and then converted to simulation currency on t_i.

- Case 2. Isolate Collateral Spikes = True

Note

This case is only supported when the the path dependent collateral model is used.

```
If
```
$$VM^{CSA}(t_i, s_j) \neq 0$$

$$VM_k^{CSA}(t_i, s_j) := \frac{v_k^{nm}(c, s_j)}{\sum_{x \in CSA} v_x^{nm}(c, s_j)} \times [VM^{CSA}(t_i, s_j) - IA_{Trades}(c, s_j)] + [cpIA(k, c, s_j) - uIA(k, c, s_j)]$$
```
Else
```
$$VM_k^{CSA}(t_i, s_j) = 0$$

where, $v_k^{nm}(c, s_j)$ is the value of position k on c if it has not matured before reporting date t_i. If position k matures before t_i, $v_k^{nm}(c, s_j) = 0$. Otherwise, it is approximated by using linear interpolation as follows:

```
If c > t0
```
$$v_k(c, s_j) = v_k^{nm}(t_{m-1}, s_j) + \frac{v_k(t_m, s_j) - v_k^{nm}(t_{m-1}, s_j)}{t_m - t_{m-1}} \times (c - t_{m-1})$$
```
Else
```
$$v_k(c, s_j) = v_k(t_0)$$

IM Attribution

MA level IM is allocated to all simulated trades, not just the ones in one of the CSA nodes.

The amounts of node-level initial margins allocated to trade k are calculated as

$$cpIM_{X,k}(t_i, s_j, ccy) = \frac{v_k(t_i, s_j)}{MtM^X(t_i, s_j)} \times cpIM_X(t_i, s_j, cc)$$

$$uIM_{X,k}(t_i, s_j, ccy) = \frac{v_k(t_i, s_j)}{MtM^X(t_i, s_j)} \times uIM_X(t_i, s_j, ccy)$$

where $cpIM_X(t_i, s_j, ccy)$ is the node-level counterparty initial margin posted on node X (coverted to currency ccy) and $uIM_X(t_i, s_j, ccy)$ is the the node-level user initial margin posted on node X (coverted to currency ccy).

Note

The same logic of IM attribution applies to both segregated and unsegregated IM.

For a trade k in a node X, define the attribution of the initial margins to the trade by

$cpIM_k(t_i, s_j, ccy) := cpIM_{X,k}(t_i, s_j, ccy) + cpIM(k, t_i, s_j, ccy)$

$uIM_k(t_i, s_j, ccy) := uIM_{X,k}(t_i, s_j, ccy) + uIM(k, t_i, s_j, ccy)$

$cpIM_{unseg,k}(t_i, s_j, ccy) := cpIM_{unseg,X,k}(t_i, s_j, ccy) + cpIM_{unseg}(k, t_i, s_j, ccy)$

$uIM_{unseg,k}(t_i, s_j, ccy) := uIM_{unseg,X,k}(t_i, s_j, ccy) + uIM_{unseg}(k, t_i, s_j, ccy)$

For the definition of $cpIM(k, t_i, s_j, ccy)$, $uIM(k, t_i, s_j, ccy)$, $cpIM_{unseg}(k, t_i, s_j, ccy)$, and $uIM_{unseg}(k, t_i, s_j, ccy)$, refer to the Initial Margins section of the document 4.1.2.4 Collateralized Exposure on page 43.

Collateral Offset and Allocation

In the case of `NettingRule` = `LEGAL_OPINION`, `User Collateral Offset` = `Netting Pools` and a complex structure is specified, collateral posted by user can get offset by closeout netting pools when calculating the collateralized value from the perspective of user. This impacts the collateral attribution described above for position *k* summed to the MA level as follows:

If $C^X_{user,unseg}(t_i, s_j) + N^X_{CNP}(t_i, s_j) \le 0$

$C^X_{user,unseg,k}(t_i, s_j) = 0$

Else If position *k* is in one of the closeout netting pools (*CNP*$_l$)

$C^X_{user,unseg,k}(t_i, s_j) = C^X_{user,unseg,k}(t_i, s_j) + v_k(t_i, s_j) \times 1_{MtM^X_{CNP_l}(t_i, s_j) \le 0}$

Else

No change (i.e. $C^X_{user,unseg,k}(t_i, s_j) = C^X_{user,unseg,k}(t_i, s_j)$

)

where

$N^X_{CNP}(t_i, s_j) = \sum_{l=1}^{L} \min\left(MtM^X_{CNP_l}(t_i, s_j), 0\right)$, CNP_l is the closeout netting pool on Node X; and

$v_k(t_i, s_j) \times 1_{MtM^X_{CNP_l}(t_i, s_j) \le 0}$ is the contribution of position *k* to $N^X_{CNP}(t_i, s_j)$.

- **Exposure Attribution**

Given the allocation for both portfolio value and collateral, the collateralized value allocated to position *k* can be calculated by following the description in 4.1.1.5 Collateralized Exposure on page 21.

Then the allocated exposure to each simulated trade is computed as:

$E^X_k(t_i, s_j) = CV^X_k(t_i, s_j) \times 1_{CV^X(t_i, s_j) > 0}$

Non-simulated trades are added to $E^X(t_i, s_j)$ separately as Add-on. Therefore, we only need to calculate their contribution to Add-on. This Add-on piece is calculated as sum of values of non-simulated trades that are positive, i.e. $E^X_k(t_i, s_j) = v^+_k(t_i, s_j)$.

- **PFE Attribution**

PFE can be allocated once the scenario that produces the quantile of interest is identified.

Let s_{α, t_i} denote the legal entity level scenario corresponding to the confidence level α for time step t_i.

PFE attribution to each simulated trade is:

$$E_k^X(t_i, s_{\alpha,t_i}) = CV_k^X(t_i, s_{\alpha,t_i}) \times 1_{CV^X(t_i, s_{\alpha,t_i}) > 0}$$

If non-simulated trades were added as Add-on to PFE, the add-on term is allocated to those trades using trade value floored to zero.

*1 In current implementation, we only support the combination of PFE attribution and RightTail function.

*2 The collateralized value used here refers to the one used for exposure calculation (as opposed to the one used for FVA).

6.41 CVA Attribution

This function calculates additive trade level attributions for CVA*1 at entity level when path dependent collateral model is used. The trade level contributions then need to be aggregated across various dimensions like currency, desk, type, etc. to produce CVA of the portfolio of choice.

- **Exposure Attribution**

Refer to 6.40 PFE Attribution on page 209 for the calculation of Exposure Attribution.

- **CVA Attribution**

Given the exposure allocation to each position for every time step and scenario, CVA can be allocated as follows.

For simulated trades, the attributed CVA is:

$$CVA_k(m, t_0) = Mean\left[\sum_i E_k^X(t_i, s_j) \times d(t_0, t_i, s_j) \times q(m, t_{i-1}, t_i, s_j) \times (1 - r(m))\right]$$

For non-simulated trades, the allocated Add-on is used instead of $E_k^X(t_i, s_j)$ in the above formulae to calculate the allocated CVA for those trades. Refer to 6.40 PFE Attribution on page 209 for the calculation of Add-on.

The results are then additive across all the trades.

*1 This function needs to be combined with CVA Amount function.

6.42 Comprehensive Approach for SFT

MAG implements the following two functions to calculate risk management measures using the comprehensive approach:

- SFT CA Capital E Star
- SFT CA Profile E Star

These two functions are applicable to Securities Financing Transactions (SFTs). They are not applicable to OTCs.

SFTs cover the following financial products:

- Repos
- Basket repos

- Tri-party repos
- Securities lending

An SFT trade is represented by at least one leg (*loan or collateral*).

These two functions are applicable on any of the following nodes of the legal hierarchy:

- SFT
- NN_SFT, and
- MA

It is assumed that an MA in the SFT legal hierarchy either

- has no CSA/NC sub-nodes, or
- equals to a unique CSA.

Note

In current implementation, it is assumed that initial margin is not posted for SFTs. If any initial margin is entered at an SFT node, it will be ingored.

Preliminary Definitions

The definitions introdued in this section is relevant to the caluclation of both SFT capital and SFT profile.

Data Inputs

Attributes	Source	Comments
NettingRule	SFTExpressions.csv	Used to define netting sets. It can be set to NO_NET, AGREEMENT, LEGAL_OPINION.
AgreementEnforceable	Counterparty.csv	Used to define netting sets.
DomesticCurrency	Counterparty.csv	The same domestic currency used in SACCR.
CloseoutNetting	MasterAgreement.csv	Used to define netting sets.
TerminationCurrency	MasterAgreement.csv	Used as the settlement currency if the NS is a subset of an MA without a CSA.

Netting Set

An MA is a netting set if all trades in the MA net. All trades in an MA net when the following conditions are satisfied:

- The expression attribute `NettingRule` = `AGREEMENT` and the MA attribute `CloseoutNetting` = `TRUE`; OR
- The expression attribute `NettingRule` = `LEGAL_OPINION` and
  - no closeout netting pools are present and the counterparty attribute `AgreementEnforceable` = `TRUE` and the MA attribute `CloseoutNetting` = `TRUE`, or
  - all trades in the MA belong to a unique non-empty closeout netting pool, or
  - there is a single trade in the MA; OR
- The expression attribute `NettingRule` = `NONET` and there is a single trade in the MA.

Other than the MA netting set defined above, a netting set can also be determined using the following algorithm:

```
If the node is an MA node AND not all trades in the MA net
     If closeout netting pools are present
```

```
          each closeout netting pool on the node forms a netting set
          any trade on the node not in any closeout netting pool forms a netting s
et
     Else
          each trade on the node forms a netting set
Else if the node is an NN_SFT node
     each trade on the node forms a netting set
```

Settlement, Domestic and Reporting Currency

All calculations are done in the domestic currency and final results are converted to the reporting currency of the expression using spot exchange rate.

The settlement currency of a netting set (NS) is defined as

- the termination currency of the MA if the NS is a subset of an MA without a CSA,
- the CSA currency if the NS is a subset of an MA = CSA,
- the trade's `terminationCurrency` if the NS is a set containing a single trade of the NN_SFT node.

- **General Notation**

S = {*sec* | *sec* is a valid `securityID`}

C = {*ccy* | *ccy* is a valid currency code}

$S_{Eligible}$ = {$sec \in S$ | *sec* is in an eligible `securityClass` for the given expression}

$FX_{CCY1 \rightarrow CCY2}$ = spot foreign exchange from *CCY1* to *CCY2*

For a CSA, define the following sets:

VM_{CSA} = {*m* | *m* is a variation margin mitigant of the CSA} (excluding mitigants posted by the counterparty if `CPVMPostStyle` = `NONE` and mitigants posted by the user if `UserVMPostStyle` = `NONE`)

VM_{CSA}^{CPTY} = { $m \in VM_{CSA}$ | *m* is posted by counterparty}

$VM_{CSA}^{USERUnseg}$ = { $m \in VM_{CSA}$ | *m* is posted by user and is not segregated}

$VM_{CSA}^{SECURITY}$ = { $m \in VM_{CSA}$ | *m* is a security mitigant}

VM_{CSA}^{sec} = { $m \in VM_{CSA}$ | *m* has `securityID` = *sec*} for $sec \in S$

VM_{CSA}^{CASH} = { $m \in VM_{CSA}$ | *m* is a cash mitigant}

VM_{CSA}^{ccy} = { $m \in VM_{CSA}$ | *m* has `currency` = *ccy*} for $ccy \in C$

6.42.1 Comprehensive Approach for SFT - Capital

This function calculates SFTs exposure (E^*) using the comprehensive approach for the purpose of determining the capital requirement of SFTs.

- **Preliminary Definitions and Notation**

Data Inputs

Various inputs are used in the calculation of E^* , as listed in the following table.

Attributes	Source	Comments
MarginLag	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
SettlementLag	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
CurePeriod	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
CallPeriod	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
startDate	standardTradeAttributes	tradeDate is on or before startDate.
maturityDate	standardTradeAttributes	Used to calculate time to maturity of a trade.
terminationCurrency	standardTradeAttributes	Used as settlement currency when the trade is in an NN node.
legType	SFTLegs	The value can be LOAN or COLLATERAL.
assetType	SFTLegs	The value can be SECURITY or CASH.
direction	SFTLegs	The value can be TO_COUNTERPARTY or TO_USER.
securityId	SFTLegs	A valid security ID.
quantity	SFTLegs	Number of units of the security; amount of cash on trade start date.
securityPrice	SFTLegs	MtM of the security
cashMtM	SFTLegs	Value of cash as-of processing date, including accrued interest since start date.
currency	SFTLegs	The currency of the security or cash.
PFEHaircutLookup	standardSecurityAttributes	Used to look up the volatilityadjustment in user-defined table.
MTM	nodeMitigantCash.csv / nodeMitigantSecurity.csv	Total value of the mitigant.
MTMCcy	nodeMitigantCash.csv / nodeMitigantSecurity.csv	The currency of the mitigant.

Margin Period of Risk

For a netting set $NS \subseteq CSA$, define

$MPoR^{NS}$ = *max(MarginLag, SettlementLag) + CurePeriod + CallPeriod -1*

For a netting set $NS \subseteq MA$ without a CSA or $NS \subseteq NN$, define

$MPoR^{NS}$ = *max(7, longest time to maturity of the trades in NS),*

where *time to maturity of a trade = maturityDate - processingDate*.

Note

For the defintion of Netting Set, please refer to 6.42 Comprehensive Approach for SFT on page 214.

Notation

For a netting set *NS*, define the following sets:

L^{NS} = {L | L is a loan or collateral leg of a trade $\in NS$}

$L^{NS}_{SECURITY}$ = { $L \in L^{NS}$ | L has `assetType` = SECURITY}

L^{NS}_{CASH} = { $L \in L^{NS}$ | L has `assetType` = CASH}

L^{NS}_{ccy} = { $L \in L^{NS}$ | L has `currency` = *ccy*} for $ccy \in C$

L^{NS}_{sec} = { $L \in L^{NS}$ | L has `securityID` = *sec*} for $\mathrm{sec} \in S$

- **Mark-to-Market**

Mark-to-Market of a Trade Leg

For $L \in L^{NS}$, define

$$MtM_L = Sign \times FX_{currency \to DomesticCurrency} \times \begin{cases} quantity \times security\mathrm{Pr}\,ice, & L \in L^{NS}_{SECURITY} \\ cashMtM, & L \in L^{NS}_{CASH} \end{cases} \tag{(1)}$$

where

$$Sign = \begin{cases} +1, & if \ \ the \ \ direction \ \ of \ \ L = TO\_COUNTERPARTY \\ -1, & if \ \ the \ \ direction \ \ of \ \ L = TO\_USER \end{cases}$$

Mark-to-Market of a Mitigant

For $m \in VM_{CSA}$, define

$$MtM_m = Sign \times MTM \times FX_{MTMCcy \to DomesticCurrency} \tag{(2)}$$

where

$$Sign = \begin{cases} -1, & if \ \ the \ \ PostDirection \ \ of \ \ m = TO\_COUNTERPARTY \\ +1, & if \ \ the \ \ PostDirection \ \ of \ \ m = TO\_USER \end{cases}$$

Mark-to-Market of Cash Position

The cash position for a netting set *NS* is the sum of the mark-to-market of all the cash legs of its trades, and if NS = CSA, minus the mark-to-market of the CSA cash mitigants. Thus, define the cash position mark-to-market for the netting set *NS* as:

$$MtM^{NS}_{CASH} = \sum_{L \in L^{NS}_{CASH}} MtM_L - \begin{cases} \sum_{m \in VM^{CASH}_{CSA}} MtM_m, & if \ \ NS = CSA \\ 0, & otherwise \end{cases} \tag{(3)}$$

where MtM_L is defined in (1) and MtM_m is defined in (2).

The mark-to-market of the positive and negative cash positions for the netting set *NS* are defined as:

$(MtM^{NS}_{CASH})^+ = \max(0, MtM^{NS}_{CASH})$ and $(MtM^{NS}_{CASH})^- = \min(0, MtM^{NS}_{CASH})$, respectively.

For a currency $ccy \in C$, define the cash position mark-to-market in that currency as:

$$MtM^{NS}_{CASH,\,ccy} = \sum_{L \in L^{NS}_{CASH} \cap L^{NS}_{ccy}} MtM_L - \begin{cases} \sum_{m \in VM^{CASH}_{CSA} \cap VM^{ccy}_{CSA}} MtM_m, & if \ \ NS = CSA \\ 0, & otherwise \end{cases} \tag{(4)}$$

Mark-to-Market of a Security Position

For a given security, the security position for a netting set *NS* in that security is the sum of all the legs of the trades in *NS* minus all the variation margin mitigants (if any) with that security as underlying. Thus, for a security $\mathrm{sec} \in S$, define

$$MtM_{\sec}^{NS} = \sum_{L \in L_{\sec}^{NS}} MtM_L - \begin{cases} \sum_{m \in VM_{CSA}^{\sec}} MtM_m, & if \quad NS = CSA \\ 0, & otherwise \end{cases} \qquad ((5))$$

where MtM_L is defined in (1) and MtM_m is defined in (2).

- **Eligible Positions**

Not all securities are eligible to mitigate exposure. There are two reasons that negative security positions are excluded from the exposure calculation. The first is that the security belongs to a list of ineligible collateral. The second is that the haircut on the security (or set of securities) is not above a specified floor. This second condition only applies to some type of trading with some counterparties.*1

The set of eligible security positions excludes all negative positions of the security which is not eligible as specified by user. Thus, for a netting set *NS*, define the set of eligible securities in which we have positions as:

$$S_{ELIGIBLE\_POS}^{NS} = \left\{ \sec \in S \;\middle|\; MtM_{\sec}^{NS} > 0 \;\; or \;\; (MtM_{\sec}^{NS} < 0 \;\; and \;\; \sec \in S_{Eligible}) \right\} \qquad ((6))$$

Define the set of eligible securities in which we have positions in currency $ccy \in C$ as

$$S_{ELIGIBLE\_POS, ccy}^{NS} = \left\{ \sec \in S_{ELIBIGLE\_POS}^{NS} \;\middle|\; the \;\; currency \;\; of \;\; \sec \;\; = ccy \right\} \qquad ((7))$$

Mark-to-Market of Eligible Security Positions

Define

$$MtM_{SECURITIES}^{NS} = \sum_{\sec \in S_{ELIGIBLE\_POS}^{NS}} MtM_{\sec}^{NS} \qquad ((8))$$

$$MtM_{SECURITIES, ccy}^{NS} = \sum_{\sec \in S_{ELIGIBLE\_POS, ccy}^{NS}} MtM_{\sec}^{NS} \qquad ((9))$$

where $MtM_{\sec}^{NS}$, $S_{ELIGIBLE\_POS}^{NS}$, and $S_{ELIGIBLE\_POS, ccy}^{NS}$ are defined in (5), (6) and (7) respectively.

Number of Material Security Positions

Let

$$MtM_{\max} = \max_{\sec \in S_{ELIGIBLE\_POS}^{NS}} |MtM_{\sec}|$$

Let $N = \#\left\{ \sec \in S_{ELIGIBLE\_POS}^{NS} \;\middle|\; |MtM_{\sec}| \geq 10\,\% \times MtM_{\max} \right\}$, that is, *N* is the number of material (eligible) positions in a security.

Mark-to-Market of Eligible Variation Margin (VM)

For a CSA node, define

$$VM_{cpty}^{CSA} = \sum_{m \in VM_{CSA}^{CPTY} \cap VM_{CSA}^{CASH}} MtM_m^{HC} + \sum_{m \in VM_{CSA}^{CPTY} \cap VM_{CSA}^{ELIGIBLE\_SECURITY}} MtM_m^{HC} \qquad ((10))$$

$$VM_{user\_unseg}^{CSA} = \sum_{m \in VM_{CSA}^{USERUnseg}} MtM_m^{HC} \qquad ((11))$$

where

$VM_{CSA}^{ELIGIBLE\_SECURITY}$ = { $m \in VM_{CSA}^{SECURITY}$ | *m*'s underlying security is in $S_{Eligible}$},

and

MtM_m^{HC} is the mitigant's mark-to-market, MtM_m, adjusted by the haircuts of volatility and FX:

- The FX haircut is only applied if the mitigant's currency (either underlying security or cash) is different from the settlement currency of the MA = CSA.
- The haircuts are scaled by the $MPoR^{CSA}$

- The value of a mitigant posted by the counterparty (which is assumed to be positive) becomes less positive, $MtM_m^{HC} = \min\left(1, MtM_m \times (1 - HC)\right)$.
- The value of a mitigant posted by the user (which is assumed to be negative) becomes more negative, $MtM_m^{HC} = MtM_m \times (1 + HC)$.

Note

VM_{cpty}^{CSA} and $VM_{user\_unseg}^{CSA}$ are only used in equation (14). That is, they are only used when a master agreement is made up of multiple netting sets.

Exposure Measure

Here we define the exposure measure E^* at each of the following nodes of the legal hierarchy:

- SFT
- NN_SFT and
- MA

At an SFT node,

$$E_{SFT}^* = \sum_{MA \ \ under \ \ SFT} E_{MA}^* + E_{NN\_SFT}^* \qquad ((12))$$

At X = NN_SFT or an MA with no CSA underneath it, or an MA = CSA with a unique netting set,

$$E_X^* = \sum_{NS \ \ under \ \ X} E_{NS}^+ \qquad ((13))$$

At an MA = CSA with multiple netting sets,

$$E_{MA}^* = \max\left\{0, \sum_{NS \ \ under \ \ MA} E_{NS}^+ - VM_{cpty}^{CSA}\right\} + \max\left\{0, \sum_{NS \ \ under \ \ MA} E_{NS}^- - VM_{user\_unseg}^{CSA}\right\} \qquad ((14))$$

In the above equations, E_{NS}^+ and E_{NS}^- are defined in equations in (15) and (16), and VM_{cpty}^{CSA} and $VM_{user\_unseg}^{CSA}$ are defined in (10) and (11).

Netting Set Exposure

The netting set *NS* positive and negative exposures are defined as follows:

$$E_{NS}^+ = \max\left\{0, V_{NS} + AddOn_{NS}\right\} \qquad ((15))$$

$$E_{NS}^- = \min\{0, V_{NS} + AddOn_{NS}\} \qquad ((16))$$

where V_{NS} and $AddOn_{NS}$ are defined in equations (17) and (18).

Netting Set Value

For a netting set *NS*, define

$$V_{NS} = MtM_{CASH}^{NS} + MtM_{SECURITIES}^{NS} \qquad ((17))$$

where MtM_{CASH}^{NS} and $MtM_{SECURITIES}^{NS}$ are defined in (3) and (8).

Netting Set AddOn

The AddOn of a netting set *NS* is:

$$AddOn_{NS} = AddOn_{Securities} + AddOn_{FX} \qquad ((18))$$

where the securities and FX addons are defined in equations (19) and (20).

- Securities AddOn

Define the addon for the eligible security positions as

$AddOn^{GROSS}_{securities} = \sum_{sec \in S^{NS}_{ELIGIBLE\_POS}} |MtM_{sec}| \times HCF_{sec}$

$AddOn^{NET}_{securities} = \left|\sum_{sec \in S^{NS}_{ELIGIBLE\_POS}} MtM_{sec} \times HCF_{sec}\right|$

where

$HCF_{sec} = HC_{sec} \times \sqrt{\frac{MPoR^{NS}}{LH_{sec}}}$, HC_{sec} and LH_{sec} are the haircut and liquidity horizon of security *sec* defined by user in csv file; and MtM_{sec} and $S^{NS}_{ELIGIBLE\_POS}$ are defined in (5) and (6), respectively.

Finally,

$$\mathrm{AddOn_{sec}} = \begin{cases} \mathrm{AddOn^{GROSS}_{securities}}, & \text{if AddOnOffsetting} = \text{NONE} \\ 0.4 \times \mathrm{AddOn^{NET}_{securities}} + \frac{0.6}{\sqrt{N}} \mathrm{AddOn^{GROSS}_{securities}}, & \text{if AddOnOffsetting} = \text{REGULATORY} \end{cases} \quad ((19))$$

Note

If AddOnOffsetting is set to INTERNAL, it will be defaulted to REGULATORY.

- FX AddOn

For each currency *ccy* other than the settlement currency, define

$MtM_{ccy} = MtM^{NS}_{CASH,ccy} + MtM^{NS}_{SECURITIES,ccy}$

$AddOn_{ccy} = |MtM_{ccy}| \times HCF_{FX}$

where

$HCF_{FX} = HC_{FX} \times \sqrt{\frac{MPoR^{NS}}{LH_{FX}}}$, HC_{FX} and LH_{FX} are the FX haircut and liquidity horizon defined by user in csv file; and $MtM^{NS}_{CASH,ccy}$ and $MtM^{NS}_{SECURITIES,ccy}$ are defined in (4) and (9), respectively.

Finally, the FX addon is defined by:

$$AddOn_{FX} = \sum_{ccy \in C \setminus \{settCcy\}} AddOn_{ccy} \quad ((20))$$

*1 The second condition is not implemented in current MAG.

6.42.2 Comprehensive Approach for SFT - Profile

This function calculates a risk management profile measure ($\{E^*_t\}$) based on the non-simulated capital measure for SFTs.

- **Preliminary Definitions and Notation**

Data Inputs

Various inputs are used in the calculation of $\{E^*_t\}$, as listed in the following table.

Attributes	Source	Comments
MarginLag	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
SettlementLag	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.

Attributes	Source	Comments
CurePeriod	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
CallPeriod	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Used to calculate the MPoR.
CPVMThreshold	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Assumed to be a positive value.
UserVMThreshold	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Assumed to be a negative value.
CPVMMinTransfer	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Assumed to be a positive value.
UserVMMinTransfer	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Assumed to be a negative value.
marginMethodology	CollateralAgreementVM.csv / CollateralAgreementIMVM.csv	Apply margin ratio to trade leg LOAN or haircut to trade leg COLLATERAL to determine a trade's exposure for the purpose of calculating addition collateral. It can be set to RATIO or HAIRCUT.
startDate	standardTradeAttributes	tradeDate is on or before startDate.
maturityDate	standardTradeAttributes	Used to calculate time to maturity of a trade.
terminationCurrency	standardTradeAttributes	Used as settlement currency when the trade is in an NN node.
legType	SFTLegs	The value can be LOAN or COLLATERAL.
assetType	SFTLegs	The value can be SECURITY or CASH.
direction	SFTLegs	The value can be TO_COUNTERPARTY or TO_USER.
securityId	SFTLegs	A valid security ID.
quantity	SFTLegs	Number of units of the security; amount of cash on trade start date.
securityPrice	SFTLegs	MtM of the security
cashMtM	SFTLegs	Value of cash as-of processing date, including accrued interest since start date.
currency	SFTLegs	The currency of the security or cash.
haircut	SFTLegs	For legs of type = COLLATERAL, this is a value between 0 and 100; for legs of type = LOAN, this is a value greater than or equal to 1.
PFEHaircutLookup	standardSecurityAttributes	Used to look up the volatilityadjustment in user-defined table.
MTM	nodeMitigantCash.csv / nodeMitigantSecurity.csv	Total value of the mitigant.
MTMCcy	nodeMitigantCash.csv / nodeMitigantSecurity.csv	The currency of the mitigant.

Attributes	Source	Comments
AddOnOffsetting	SFTExpressions.csv	Used to determine how Securities Addon is calculated.

Margin Period of Risk

For a netting set $NS \subseteq CSA$, define

$MPoR^{NS}$ = *max(MarginLag, SettlementLag) + CurePeriod + CallPeriod -1*

Note

For the definition of netting set, please refer to 6.42 Comprehensive Approach for SFT on page 214.

Notation

$x^{+} = \max\{x, 0\}$ and $x^{-} = \min\{x, 0\}$

For a set *X* of SFT transactions, define the following sets:

L_X = {*L* | *L* is a leg of a trade ∈ *X*}

$L_X^{SECURITY}$ = { $L \in L_X$ | *L* has `assetType` = SECURITY}

L_X^{CASH} = { $L \in L_X$ | *L* has `assetType` = CASH}

L_X^{LOAN} = { $L \in L_X$ | *L* is of `legType` = LOAN}

$L_X^{COLLATERAL}$ = { $L \in L_X$ | *L* is of `legType` = COLLATERAL}

For a set *X* of SFT transactions and a time *t*, define the following sets:

$L_{t,X}$ = { $L \in L_X$ | `startDate` ≤ *t* ≤ `processingDate`

$L_{t,X}^{SECURITY}$ = { $L \in L_{t,X}$ | *L* has `assetType` = SECURITY}

$L_{t,X}^{CASH}$ = { $L \in L_{t,X}$ | *L* has `assetType` = CASH}

$L_{t,X}^{LOAN}$ = { $L \in L_{t,X}$ | *L* is of `legType` = LOAN}

$L_{t,X}^{COLLATERAL}$ = { $L \in L_{t,X}$ | *L* is of `legType` = COLLATERAL}

$L_{t,X}^{ELIGIBLE\_SECURITY}$ = { $L \in L_{t,X}^{SECURITY}$ | *L* has `securityID` ∈ $S_{Eligible}$}

$L_{t,X}^{ccy}$ = { $L \in L_{t,X}$ | *L* has `currency` = *ccy*} for $ccy \in C$

$L_{t,X}^{\sec}$ = { $L \in L_{t,X}^{SECURITY}$ | *L* has `securityID` = *sec*} for $\sec \in S$

Time *t* is a reporting date set by the attribute `ReportingTimeline` in AggregatorInstances.csv and is measured with respect to the processing date so that *t* = 0 refers to the processing date.

- ## Mark-to-Market

Mark-to-Market of a Trade Leg

For $L \in L_{NS}$, define

$$MtM_L = Sign \times FX_{currency \to DomesticCurrency} \times \begin{cases} quantity \times securityPrice, & L \in L_{NS}^{SECURITY} \\ cashMtM, & L \in L_{NS}^{CASH} \end{cases} \quad ((1))$$

where

$$Sign = \begin{cases} +1, & if \;\; the \;\; direction \;\; of \;\; L = TO\_COUNTERPARTY \\ -1, & if \;\; the \;\; direction \;\; of \;\; L = TO\_USER \end{cases}$$

Mark-to-Market of a Mitigant

For $m \in VM_{CSA}$, define

$$MtM_m = Sign \times MTM \times FX_{MTMCcy \rightarrow DomesticCurrency} \tag{(2)}$$

where

$$Sign = \begin{cases} -1, & if \;\; the \;\; PostDirection \;\; of \;\; m = TO\_COUNTERPARTY \\ +1, & if \;\; the \;\; PostDirection \;\; of \;\; m = TO\_USER \end{cases}$$

Mark-to-Market of Cash Position

For a netting set *NS*, a currency $ccy \in C$, and time *t*, define:

$$MtM_{t,NS}^{TRADES,CASH} = \sum_{L \in L_{t,NS}^{CASH}} MtM_L \tag{(3)}$$

$$MtM_{t,NS}^{TRADES,CASH,ccy} = \sum_{L \in L_{t,NS}^{CASH} \cap L_{t,NS}^{ccy}} MtM_L \tag{(4)}$$

$$MtM_{t,NS}^{CASH} = MtM_{t,NS}^{TRADES,CASH} - \begin{cases} \sum_{m \in VM_{CSA}^{CASH}} MtM_m, & if \;\; t = 0 \;\; and \;\; NS = CSA \\ 0, & otherwise \end{cases} \tag{(5)}$$

$$MtM_{t,NS}^{CASH,ccy} = MtM_{t,NS}^{TRADES,CASH,ccy} - \begin{cases} \sum_{m \in VM_{CSA}^{CASH} \cap VM_{CSA}^{ccy}} MtM_m, & if \;\; t = 0 \;\; and \;\; NS = CSA \\ 0, & otherwise \end{cases} \tag{(6)}$$

where MtM_L is defined in (1) and MtM_m is defined in (2).

Note

$MtM_{t,NS}^{TRADES,CASH}$ only differs from $MtM_{t,NS}^{CASH}$ when *t = 0* and *NS = CSA*.

Mark-to-Market of a Security Position

For a netting set *NS*, a security $\sec \in S$, and time *t*, define:

$$MtM_{t,NS}^{TRADES,\sec} = \sum_{L \in L_{t,NS}^{\sec}} MtM_L \tag{(7)}$$

and

$$MtM_{t,NS}^{\sec} = MtM_{t,NS}^{TRADES,\sec} - \begin{cases} \sum_{m \in VM_{CSA}^{\sec}} MtM_m, & if \;\; t = 0 \;\; and \;\; NS = CSA \\ 0, & otherwise \end{cases} \tag{(8)}$$

where MtM_L is defined in (1) and MtM_m is defined in (2).

Note

$MtM_{t,NS}^{TRADES,\sec}$ only differs from $MtM_{t,NS}^{\sec}$ when *t = 0* and *NS = CSA*.

- **Eligible Positions**

For a netting set *NS*, a currency $ccy \in C$, and time *t*, define:

$$S_{t,NS}^{TRADES,ELIGIBLE\_POS} = \left\{\sec \in S \;\middle|\; MtM_{t,NS}^{TRADES,\sec} > 0 \;\; or \;\; (MtM_{t,NS}^{TRADES,\sec} < 0 \;\; and \;\; \sec \in S_{Eligible})\right\} \tag{(9)}$$

$$S_{t,NS}^{TRADES,ELIGIBLE\_POS,ccy} = \left\{\sec \in S_{t,NS}^{TRADES,ELIBIGLE\_POS} \;\middle|\; the \;\; currency \;\; of \;\; \sec \;\; = ccy\right\} \tag{(10)}$$

$$S_{t,NS}^{ELIGIBLE\_POS} = \left\{\sec \in S \;\middle|\; MtM_{t,NS}^{\sec} > 0 \;\; or \;\; (MtM_{t,NS}^{\sec} < 0 \;\; and \;\; \sec \in S_{Eligible})\right\} \tag{(11)}$$

$$S_{t,NS}^{ELIGIBLE\_POS,ccy} = \left\{\sec \in S_{t,NS}^{ELIBIGLE\_POS} \;\middle|\; the \;\; currency \;\; of \;\; \sec \;\; = ccy\right\} \tag{(12)}$$

where $S_{Eligible}$ is defined in 6.42 Comprehensive Approach for SFT on page 214.

Note

$S_{t,NS}^{TRADES,ELIGIBLE\_POS}$ and $S_{t,NS}^{TRADES,ELIGIBLE\_POS,ccy}$ only differ from $S_{t,NS}^{ELIGIBLE\_POS}$ and $S_{t,NS}^{ELIGIBLE\_POS,ccy}$ when *t* = *0* and *NS* = *CSA*.

Mark-to-Market of Eligible Security Positions

For a netting set *NS*, a currency $ccy \in C$, and time *t*, define:

$$MtM_{t,NS}^{TRADES,SECURITIES} = \sum_{\sec \in S_{t,NS}^{TRADES,ELIGIBLE\_POS}} MtM_{t,NS}^{TRADES,\sec} \quad ((13))$$

$$MtM_{t,NS}^{TRADES,SECURITIES,ccy} = \sum_{\sec \in S_{t,NS}^{TRADES,ELIGIBLE\_POS,ccy}} MtM_{t,NS}^{TRADES,\sec} \quad ((14))$$

$$MtM_{t,NS}^{SECURITIES} = \sum_{\sec \in S_{t,NS}^{ELIGIBLE\_POS}} MtM_{t,NS}^{\sec} \quad ((15))$$

$$MtM_{t,NS}^{SECURITIES,ccy} = \sum_{\sec \in S_{t,NS}^{ELIGIBLE\_POS,ccy}} MtM_{t,NS}^{\sec} \quad ((16))$$

where $MtM_{t,NS}^{TRADES,\sec}$, $MtM_{t,NS}^{\sec}$, $S_{t,NS}^{TRADES,ELIGIBLE\_POS}$, $S_{t,NS}^{TRADES,ELIGIBLE\_POS,ccy}$, $S_{t,NS}^{ELIGIBLE\_POS}$ and $S_{t,NS}^{ELIGIBLE\_POS,ccy}$ are defined in (7), (8), (9), (10), (11) and (12), respectively.

Note

$MtM_{t,NS}^{TRADES,SECURITIES}$ and $MtM_{t,NS}^{TRADES,SECURITIES,ccy}$ only differ from $MtM_{t,NS}^{SECURITIES}$ and $S_{t,NS}^{SECURITIES,ccy}$ when *t* = *0* and *NS* = *CSA*.

Number of Material Security Positions

Let

$$\mathrm{MtM}_{t,\max} = \max_{\sec \in S_{t,\mathrm{NS}}^{\mathrm{ELIGIBLE\_POS}}} |\mathrm{MtM}_{t,\mathrm{NS}}^{\sec}|$$

Define the number of material (eligible) security positions at time *t*, for the netting set *NS* as

$$n_{t,\mathrm{NS}} := \#\{\sec \in S_{t,\mathrm{NS}}^{\mathrm{ELIGIBLE\_POS}} \mid |\mathrm{MtM}_{t,\mathrm{NS}}^{\sec}| \geq 10\% \times \mathrm{MtM}_{t,\max}\}$$

- ## Variation Margin

Current Eligible Variation Margin

For a CSA node, define

$$VM_{t,cpty}^{CSA} = \sum_{m \in VM_{CSA}^{CPTY} \cap VM_{CSA}^{CASH}} MtM_{t,m}^{HC} + \sum_{m \in VM_{CSA}^{CPTY} \cap VM_{CSA}^{ELIGIBLE\_SECURITY}} MtM_{t,m}^{HC} \quad ((17))$$

$$VM_{t,user\_unseg}^{CSA} = \sum_{m \in VM_{CSA}^{USERUnseg} \cap VM_{CSA}^{CASH}} MtM_{t,m}^{HC} + \sum_{m \in VM_{CSA}^{USERUnseg} \cap VM_{CSA}^{SECURITY}} MtM_{t,m}^{HC} \quad ((18))$$

where

$VM_{CSA}^{ELIGIBLE\_SECURITY}$ = { $m \in VM_{CSA}^{SECURITY}$ | *m*'s underlying security is in $S_{Eligible}$} and

$$MtM_{t,m}^{HC} = MtM_m \times \begin{cases} 1 - HC_{t,m}, & if \;\; m \;\; is \;\; posted \;\; by \;\; the \;\; counterparty \\ 1 + HC_{t,m}, & if \;\; m \;\; is \;\; posted \;\; by \;\; the \;\; user \end{cases}$$

MtM_m is the mitigant's mark-to-market (which is positive/negative if posted by the counterparty/user).

$$HC_{t,m} = \left(\frac{HC_m}{\sqrt{LH_m}} + \frac{HC_{m,FX}}{\sqrt{LH_{FX}}}\right) \times \sqrt{t}$$

$$HC_m = \begin{cases} HC_{\text{sec}}, \ if \ \ m \ \ is \ \ a \ \ \text{security} \ \ mitigant \ \ with \ \ underlying \ \ security \ \ \text{sec} \\ 0, \qquad\qquad\qquad\qquad\qquad if \ \ m \ \ is \ \ a \ \ cash \ \ mitigant \end{cases}$$

$$HC_{m,FX} = \begin{cases} HC_{FX}, \ if \ \ m's \ \ currency \ \ is \ \ not \ \ equal \ \ to \ \ the \ \ settlement \ \ currency \ \ of \ \ the \ \ MA(=CSA) \\ 0, \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad otherwise \end{cases}$$

$LH_m = LH_{sec}$

HC_{sec} and LH_{sec} are the haircut and liquidity horizon of security *sec* provided by user.

HC_{FX} and LH_{FX} are the FX haircut and liquidity horizon provided by user.

For $t < MPoR$, define

$$VM_{t,CSA} = VM^{cpty}_{t,CSA} + VM^{user\_unseg}_{t,CSA} \qquad ((19))$$

Note

Only one of the summands should be zero as only one party should be posting variation margin at any give point of time.

Note

$VM_{t,CSA}$ where $t < MPoR$ is only used in equation (23). That is, it is only used when a CSA is made up of multiple netting sets.

Variation Margin at Future Timesteps

To estimate the variation margin posted on a CSA at future timesteps, we assume that for $t \geq MPoR$, the variation margin in the calculation of E^*_t is obtained from the variation margin called up to the beginning of the MPoR, i.e., at *t - MPoR*. We also assume that this collateral is cash in the settlement currency so that a haircut is not applied.

More precisely, for $t \geq MPoR$,

$$VM_{t,CSA} = \delta_{CptyPost} \times \left[\left(MtM^{MCalls}_{t-MPoR,CSA}\right)^+ - CPVMThreshold_{t-MPoR} - CPVMMinTransfer_{t-MPoR}\right]^+$$
$$+\delta_{UserPost} \times \delta_{unseg} \times \left[\left(MtM^{MCalls}_{t-MPoR,CSA}\right)^- - UserVMThreshold_{t-MPoR} - UserVMMinTransfer_{t-MPoR}\right]^-$$

((20))

where:

If `marginMethodology` = Ratio,

$$MtM^{MCalls}_{t-MPoR,CSA} = \sum_{L \in L^{LOAN}_{t-MPoR,CSA}} MtM^{\text{MarginCalls}}_L + \sum_{L \in L^{COLLATERAL}_{t-MPoR,CSA}} MtM_L$$

If `marginMethodology` = Haircut,

$$MtM^{MCalls}_{t-MPoR,CSA} = \sum_{L \in L^{LOAN}_{t-MPoR,CSA}} MtM_L + \sum_{L \in L^{COLLATERAL}_{t-MPoR,CSA}} MtM^{\text{MarginCalls}}_L$$

$$MtM^{\text{MarginCalls}}_L = MtM_L \times \begin{cases} haircut, \qquad L \in L^{LOAN}_{CSA} \\ 1 - haircut, \ L \in L^{COLLATERAL}_{CSA} \end{cases}$$

MtM_L is defined in (1)

$$\delta_{unseg} = \begin{cases} 1, \ if \ \ UserVMSegregated \ \ = \ \ FALSE \\ 0, \ if \ \ UserVMSegregated \ \ = \ \ TRUE \end{cases}$$

$$\delta_{CptyPost} = \begin{cases} 1, \ if \ \ CPVMPostStyle \ \ = \ \ NET \ \ or \ \ GROSS \\ 0, \qquad if \ \ CPVMPostStyle \ \ = \ \ NONE \end{cases}$$

$$\delta_{UserPost} = \begin{cases} 1, & if \quad UserVMPostStyle = NET \ or \ GROSS \\ 0, & if \quad UserVMPostStyle = NONE \end{cases}$$

Note

To calculate the variation margin called up to the beginning of the MPoR, we do not exclude the ineligible security positions.

- ### Exposure Measure

Here we define the exposure measure E_t^* , at time *t*, at each of the following nodes of the legal hierarchy:

- SFT
- NN_SFT and
- MA

At an SFT node,

$$E_{t,SFT}^* = \sum_{MA \ under \ SFT} E_{t,MA}^* + E_{t,NN\_SFT}^* \quad ((21))$$

At X = NN_SFT or an MA with no CSA underneath it, or an MA = CSA with a unique netting set,

$$E_{t,X}^* = \sum_{NS \ under \ X} E_{t,NS}^+ \quad ((22))$$

At an MA = CSA with multiple netting sets,

$$E_{t,MA}^* = \max\left\{0, \sum_{NS \ under \ MA} E_{t,NS}^+ - (VM_{t,CSA})^+\right\} + \max\left\{0, \sum_{NS \ under \ MA} E_{t,NS}^- - (VM_{t,CSA})^-\right\} \quad ((23))$$

In the above equations, NS stands for a netting set as defined in 6.42 Comprehensive Approach for SFT on page 214, $E_{t,NS}^+$ and $E_{t,NS}^-$ are defined in equations in (24) and (25), and $VM_{t,CSA}$ is defined in (19) and (20).

- ### Netting Set Exposure

The netting set *NS* positive and negative exposures are defined as follows:

$$E_{t,NS}^+ = \max\{0, CV_{t,NS} + AddOn_{t,NS}\} \quad ((24))$$

$$E_{t,NS}^- = \min\{0, CV_{t,NS} + AddOn_{t,NS}\} \quad ((25))$$

where $CV_{t,NS}$ and $AddOn_{t,NS}$ are defined in equations (26) or (27), and (28).

Netting Set Collateralized Value

For a netting set *NS* which is not a subset of a CSA, at time *t*, define

$$CV_{t,NS} = MtM_{t,NS}^{CASH} + MtM_{t,NS}^{SECURITIES} \quad ((26))$$

For a netting set $NS \subseteq CSA$ at time $t < MPoR$, define

$$CV_{t,NS} = V_{0,NS} = MtM_{0,NS}^{CASH} + MtM_{0,NS}^{SECURITIES}$$

In the equations above, $MtM_{t,NS}^{CASH}$ and $MtM_{t,NS}^{SECURITIES}$ are defined in (5) and (15).

Note

If $NS \subseteq CSA$, $V_{0,NS}$ already has variation margin incorporated in it, but if $NS \nsubseteq CSA$, $V_{0,NS}$ does not have any variation margin incorporated in it. In the latter case, variation margin is incorporated in the exposure formula for a *MA* = *CSA* with multiple netting sets, $E_{t,MA}^*$.

For a netting set $NS \subseteq CSA$ at time $t \geq MPoR$, define

$$CV_{t,NS} = MtM_{t-MPoR,NS}^{TRADES,CASH} + MtM_{t-MPoR,NS}^{TRADES,SECURITIES} - \begin{cases} VM_{t,NS} & if \quad NS = CSA \\ 0 & otherwise \end{cases} \qquad ((27))$$

where $VM_{t,CSA}$ is defined in (20).

Note

When $NS \subseteq CSA$, we are using the mark-to-market of the netting set at *t-MPoR* instead of its value at *t*. The underlying assumption is that no exchange of securities/cash occurs during the MPoR; that is, if a trade matures between *t-MPoR* (exclusive) and *t* (inclusive), the legs of the trade are not exchanged. We are doing this to mimic the behavior in the internal model for the smooth exposure in which spikes caused by maturing trades are excluded.

Netting Set AddOn

The AddOn of a netting set *NS* is:

$$AddOn_{t,NS} = AddOn_{t,NS}^{SECURITIES} + AddOn_{t,NS}^{FX} \qquad ((18))$$

where the securities and FX addons are defined in equations (29) and (30).

- Securities AddOn

Define the addon for the eligible security positions as

$$\mathrm{AddOn}_{t,\mathrm{NS}}^{\mathrm{SECURITIES}} =$$

$$\begin{cases} \mathrm{AddOn}_{t,\,\mathrm{securities}}^{\mathrm{GROSS}}, & \text{if AddOnOffsetting} = \text{NONE} \\ 0.4 \times \mathrm{AddOn}_{t,\,\mathrm{securities}}^{\mathrm{NET}} + \frac{0.6}{\sqrt{N_{t,\,\mathrm{NS}}}} \mathrm{AddOn}_{t,\,\mathrm{securities}}^{\mathrm{GROSS}}, & \text{if AddOnOffsetting} = \text{REGULATORY} \\ \sqrt{\sum_{\mathrm{sec}_i \in S_{t,\,\mathrm{NS}}} \sum_{\mathrm{sec}_j \in S_{t,\,\mathrm{NS}}} \rho_{\mathrm{sec}_i,\,\mathrm{sec}_j} \times \mathrm{AddOn}_{t,\,\mathrm{sec}_i} \times \mathrm{AddOn}_{t,\,\mathrm{sec}_j}}, & \text{if AddOnOffsetting} = \text{INTERNAL} \end{cases} \qquad ((29))$$

where

$$\mathrm{AddOn}_{t,\,\mathrm{securities}}^{\mathrm{GROSS}} = \sum_{\mathrm{sec} \in S_{t,\,\mathrm{NS}}} |\mathrm{AddOn}_{t,\,\mathrm{sec}}|$$

$$\mathrm{AddOn}_{t,\,\mathrm{securities}}^{\mathrm{NET}} = \left| \sum_{\mathrm{sec} \in S_{t,\,\mathrm{NS}}} \mathrm{AddOn}_{t,\,\mathrm{sec}} \right|$$

$$S_{t,NS} = \begin{cases} S_{0,NS}^{ELIGIBLE\_POS}, & if \quad NS \subseteq CSA \quad and \quad t < MPoR \\ S_{t-MPoR,NS}^{TRADES,ELIGIBLE\_POS}, & if \quad NS \subseteq CSA \quad and \quad t \geq MPoR \\ S_{t,NS}^{ELIGIBLE\_POS}, & otherwise \end{cases}$$

$$AddOn_{t,\mathrm{sec}} = HCF_{t,NS}^{\mathrm{sec}} \times \begin{cases} MtM_{0,NS}^{\mathrm{sec}}, & if \quad NS \subseteq CSA \quad and \quad t < MPoR \\ MtM_{t-MPoR,NS}^{TRADES,\mathrm{sec}}, & if \quad NS \subseteq CSA \quad and \quad t \geq MPoR \\ MtM_{t,NS}^{\mathrm{sec}}, & otherwise \end{cases}$$

$$HCF_{\mathrm{sec}} = \frac{HC_{\mathrm{sec}}}{\sqrt{LH_{\mathrm{sec}}}} \times \sqrt{RH_{t,NS}}$$

$$RH_{t,NS} = \begin{cases} \min(MPoR^{NS}, t), & if \quad NS = CSA \\ t, & otherwise \end{cases}$$

HC_{sec} and LH_{sec} are the haircut and liquidity horizon of security *sec* provided by user.

$MtM_{t,NS}^{\sec}$, $MtM_{t,NS}^{TRADES,\sec}$, $S_{t,NS}^{TRADES,ELIGIBLE\_POS}$ and $S_{t,NS}^{ELIGIBLE\_POS}$ are defined in (8), (7), (9) and (11), respectively;

$$N_{t,\mathrm{NS}} = \begin{cases} n_{0,\mathrm{NS}} & \text{if NS} \subseteq \text{CSA and t} < \text{MPOR} \\ n_{t-\mathrm{MPOR},\mathrm{NS}} & \text{if NS} \subseteq \text{CSA and t} \geq \text{MPOR} \\ n_{t,\mathrm{NS}} & \text{otherwise} \end{cases}$$

$n_{u,NS}$ (for *u = 0, t-MPOR, and t*) is defined in Number of Material Security Positions.

$\rho_{\sec_i,\sec_j}$ is the correlation provided by user.

Note

$\rho_{\sec_i,\sec_j} = 1$ if $sec_i = sec_j$.

- FX AddOn

For each currency *ccy* other than the settlement currency, define

$AddOn_{t,NS}^{ccy} = \left|MtM_{t,NS}^{ccy}\right| \times HCF_{t,NS}^{FX}$

where

$$MtM_{t,NS}^{ccy} = \begin{cases} MtM_{0,NS}^{CASH,ccy} + MtM_{0,NS}^{SECURITIES,ccy}, & if \quad NS \subseteq CSA \quad and \quad t < MPoR \\ MtM_{t-MPoR,NS}^{TRADES,CASH,ccy} + MtM_{t-MPoR,NS}^{TRADES,SECURITIES,ccy}, & if \quad NS \subseteq CSA \quad and \quad t \geq MPoR \\ MtM_{t,NS}^{CASH,ccy} + MtM_{t,NS}^{SECURITIES}, & otherwise \end{cases}$$

$HCF_{t,NS}^{FX} = \frac{HC_{FX}}{\sqrt{LH_{FX}}} \times \sqrt{RH_{t,NS}}$,

$$RH_{t,NS} = \begin{cases} \min(MPoR^{NS}, t), & if \quad NS = CSA \\ t, & otherwise \end{cases}$$

HC_{FX} and LH_{FX} are the FX haircut and liquidity horizon provided by user.

$MtM_{t,NS}^{CASH,ccy}$, $MtM_{t,NS}^{SECURITIES,ccy}$, $MtM_{t,NS}^{TRADES,CASH,ccy}$ and $MtM_{t,NS}^{TRADES,SECURITIES,ccy}$ are defined in (6), (16), (4) and (14), respectively.

Finally, the FX addon is defined by:

$$AddOn_{t,NS}^{FX} = \sum\nolimits_{ccy \in C \backslash \{settCcy\}} AddOn_{t,NS}^{ccy} \qquad ((30))$$

*1 The second condition is not implemented in current MAG.

6.43 Synthetic Netting Sets

Regulatory approval to use internal models is sometimes restricted to a subset of products. When this is the case, the CCR capital measure is calculated using both an internal model and a standardized methodology (CEM or SACCR for OTC transactions and the comprehensive approach for SFT transactions).

As well, some transactions for which the pricing models don't perform well when compared to benchmarks must be excluded from the internal model calculations.

There are also operational reasons for excluding some trades from the internal model, such as when the trade fails to simulate properly.

When a legally binding netting set has some trades that are included in the internal model and others in the standard model, the contractual netting of these trades is artificially broken. In such a case, various inputs into the calculation of risk measures must be split between the internal and standard model in such a way that risk is not under-estimated. To fulfill this purpose, MAG provides 2 functions:

- Synthetic Netting Set SACCR
- Synthetic Netting Set CEM

Note

Currently, these two functions only cover the synthetic netting sets calculation for OTC transactions. For the simulated SFT transactions, internal models are used. For the non-simulated SFT and direct lending transactions, addons are used.

Definitions and Inputs

Synthetic Netting Sets

When a netting set contains both simulated and non-simulated trades, the set is split into two subsets called "synthetic netting sets". Trades are allocated to synthetic netting sets by standard trade attribute `IMMApproved`, which is defined by user:

- `IMMApproved` = TRUE
  - In case of a regular trade, it is allocated to the simulated synthetic netting set.
  - In case of a structured deal, all component deals are allocated to the simulated synthetic netting set after MAG ensures that all component deals are with `IMMApproved` = TRUE and simulated successfully.
- `IMMApproved` = FALSE
  - In case of a regular trade, it is allocated to the non-simulated synthetic netting set.
  - In case of a structured deal, all component deals are allocated to the non-simulated synthetic netting set.
- `IMMApproved` is empty
  - In case of a regular trade, it is treated as with `IMMApproved` = FALSE.
  - In case of a structured deal, it is treated as with `IMMApproved` = TRUE. If `IMMApproved` of the component deal is empty, it is treated as with `IMMApproved` = FALSE.

Note

Trades that fail to simulate are treated as with `IMMApproved` = FALSE.

Note

In current implementation, there are some limitations in structured deal real time operations. Please refer to "Synthetic Netting Set - Structured Deals realtime limitations" in Integratd Market and Credit Risk Solution Guide.

Note, if a netting set contains only a non-empty synthetic netting set (i.e. all trades have the same `IMMApproved` value), there is no need to split the CSA parameters, current variation margin and initial margin specified as below. All the CSA parameters, current variation margin and initial margin are allocated to this non-empty synthetic netting set.

Inputs

Synthetic netting set functions have three parameters to allow alternatives in splitting the CSA parameters, current variation margin, and current initial margin. Here, the assumption is that there is no dependency between these parameters and value chosen for one parameter is fully independent of the value chosen for the others.

- **`Allocation CSA Parameter`**

  This parameter is used to specify the splitting rule for the CSA parameters, that is, user/counterparty threshold, user/counterparty MTA and user/counterparty IA between synthetic netting sets over time. Four values are allowed for this parameter:

  - `SimulatedSet`

    Thresholds, MTAs and IAs are first fully allocated to the simulated synthetic netting set. Then they are reallocated to the non-simulated synthetic netting set after the maturity of all simulated trades.

  - `NonSimulatedSet`

    Thresholds, MTAs and IAs are first fully allocated to the non-simulated synthetic netting set. Then they are reallocated to the simulated synthetic netting set after the maturity of all non-simulated trades.

  - `ProportionateToMtM`

    Thresholds, MTAs and IAs are split between the simulated and non-simulated synthetic netting sets in proportionate to the absolute value of MtM for each set.

  - `ProportionateToSACCRAddon`

    Thresholds, MTAs and IAs are split between the simulated and non-simulated synthetic netting sets in proportionate to the absolute value of modifed SACCR Addon for each set. See section Definition: Addon_{VM} for details.

- **`Allocation VM`**

  This parameter is used to specify the splitting rule for the current variation margin between synthetic netting sets. Four values are allowed for this parameter:

  - `SimulatedSet`

    The current excess variation margin after fully collateralizing both synthetic netting sets above respective threshold plus MTA is fully allocated to the simulated synthetic netting set.

  - `NonSimulatedSet`

    The current excess variation margin after fully collateralizing both synthetic netting sets above respective threshold plus MTA is fully allocated to the non-simulated synthetic netting set.

  - `ProportionateToMtM`

    The current excess variation margin after fully collateralizing both synthetic netting sets above respective threshold plus MTA is split in proportion to the absolute value of MtM for each set.

  - `ProportionateToSACCRAddon`

    The current excess variation margin after fully collateralizing both synthetic netting sets above respective threshold plus MTA is split in proportion to the absolute value of modified SACCR Addon for each set. See section Definition: Addon_{VM} for details.

- **`Allocation IM`**

Note

In current version of MAG, any static initial margin is ignored for synthetic netting set measures.

This parameter is used to specify the splitting rule for the node mitigant initial margin between synthetic netting sets. Only node mitigant initial margin is split between synthetic netting sets. Four choices are allowed for this parameter:

- `SimulatedSet`

  The current initial margin is first fully allocated to the simulated synthetic netting set, and then reallocated to the non-simulated synthetic netting set after the maturity of all the simulated trades.

- `NonSimulatedSet`

  The current initial margin is first fully allocated to the non-simulated synthetic netting set, and then reallocated to the simulated synthetic netting set after the maturity of all the non-simulated trades.

- `ProportionateToMtM`

  The current initial margin is split in proportion to the absolute value of MtM for each synthetic netting set.

- `ProportionateToSACCRAddon`

  The current initial margin is split in proportion to the absolute value of modified SACCR Addon for each set. See section Definition: $\mathrm{Addon}_{\mathrm{IM}}$ for details.

- **`Allocation MtM`**

  This parameter is used to define the mark-to-market value of a trade that belongs to the simulated synthetic netting set. Three values are allowed for this parameter:

  - `BoR`

    This is the default value of function parameter `Allocation MtM`. When this option is selected, the current book of record mark-to-market is used in below Allocation Algorithm.

  - `Modelled`

    When this option is selected, the simulated mark-to-market is used.

  - `ModelledAsBackup`

    When this option is selected, the simulated mark-to-market is used if the current book of record mark-to-market is not provided by the user.

AddOn

- **`Initial Margin`**

  Let $T_{IM,t}$ be the set of trades in an IMCSA or IMVMCSA that have not matured by time *t* and $T_{IM,t,s}$ and $T_{IM,t,n}$ be the trades in $T_{IM,t}$ which belong to the simulated and non-simulated synthetic netting set, respectively.

  We define $AddOn_{IM,t,s}$, $AddOn_{IM,t,n}$ as the SACCR AddOn at time *t* using the trades in $T_{IM,t,s}$ and $T_{IM,t,n}$ respectively and using $\sqrt{\frac{IMMarginHorizon}{365}}$ as the maturity factor.

  Note

  Any non-zero maturity factor can be used, since they cancel out in the formula calculating the allocation factor.

Note

These addons should use the SACCR formula for a single netting set and are always positive.

These addons can be used to allocate the Initial Margin at time t_0 and counterparty/user IM Thresholds at time *t*.

- **`Variation Margin`**

  Let $T_{VM,t}$ be the set of trades in an VMCSA/Legacy CSA/IMVMCSA that have not matured by time *t* and $T_{VM,t,s}$ and $T_{VM,t,n}$ be the trades in $T_{VM,t}$ which belong to the simulated and non-simulated synthetic netting set, respectively.

  We define $AddOn_{VM,t,s}$, $AddOn_{VM,t,n}$ as the SACCR AddOn at time *t* using the trades in $T_{VM,t,s}$ and $T_{VM,t,n}$ respectively and using $\sqrt{\frac{MPoR}{365}}$ as the maturity factor, where $MPoR$ is the margin period of risk. See section • Formula: MPOR on page 116 for details.

  Note

  Any non-zero maturity factor can be used, since they cancel out in the formula calculating the allocation factor.

  Note

  These addons should use the SACCR formula for a single netting set and are always positive.

  These addons can be used to allocate the excess Variation Margin at time t_0 and the CSA parameters at time *t*.

Mark-to-Market of a Set

For a set *T*, we define the mark-to-market of this set at time *t* by

$MtM_t = \sum_{i \in T_t} MtM_{0,i}$

where T_t is the set of trades in *T* that have not matured by time *t*; and $MtM_{0,i}$ is defined as follows:

```
If trade i ∈ the non-simulated synthetic netting set
     MtM_{0,i} = the current book of record mark-to-market of trade i
Else
     MtM_{0,i} is determined by the function parameter Allocation MtM
```

This mark-to-market is split into the mark-to-market of the simulated and non-simulated trades as follows:

$MtM_{t,s} = \sum_{i \in T_{t,s}} MtM_{0,i}$

$MtM_{t,n} = \sum_{i \in T_{t,n}} MtM_{0,i}$

where $T_{t,s}$ and $T_{t,n}$ are the sets of simulated and non-simulated trades in T_t, respectively.

Note

In case of allocating IM, T_t is the set of trades in an IMCSA or IMVMCSA that have not matured by time *t*. In case of allocating VM and CSA Parameters, T_t is the set of trades in an VMCSA/Legacy CSA/IMVMCSA that have not matured by time *t*.

- **Allocation Algorithm**

Allocation Factor (ρ_t^X)

Let t denote the reporting date, and X CSA Parameter, VM, or IM.

The allocation factor ρ_t^X used in this section is defined as follows:

- `Allocation_X` = `SimulatedSet`

  ρ_t^X = 1 up to maturity of simulated trades and 0 afterwards;

- `Allocation_X` = `NonSimulatedSet`

  ρ_t^X = 0 up to maturity of non- simulated trades and 1 afterwards;

- `Allocation_X` = `ProportionateToMtM`

$$\rho_t^X = \begin{cases} \frac{|MtM_{t,s}|}{|MtM_{t,s}| + |MtM_{t,n}|}, \; if \;\; MtM_{t,s} \neq 0 \;\; or \;\; MtM_{t,n} \neq 0 \\ \frac{1}{2}, \qquad\qquad otherwise \end{cases}$$

- `Allocation_X` = `ProportionateToSACCRAddon`
  - For `X = CSA parameters`:

$$\rho_t^X = \begin{cases} \frac{|Addon_{VM,t,s}|}{|Addon_{VM,t,s}| + |Addon_{VM,t,n}|} & \text{if } |Addon_{VM,t,s}| \neq 0 \text{ or } |Addon_{VM,t,n}| \neq 0 \\ \frac{1}{2} & \text{otherwise} \end{cases}$$

  - For `X = VM`:

$$\rho_t^X = \begin{cases} \frac{|Addon_{VM,t_0,s}|}{|Addon_{VM,t_0,s}| + |Addon_{VM,t_0,n}|} & \text{if } |Addon_{VM,t_0,s}| \neq 0 \text{ or } |Addon_{VM,t_0,n}| \neq 0 \\ \frac{1}{2} & \text{otherwise} \end{cases}$$

  - For `X = IM`:

$$\rho_t^X = \begin{cases} \frac{|Addon_{IM,t,s}|}{|Addon_{IM,t,s}| + |Addon_{IM,t,n}|} & \text{if } |Addon_{IM,t,s}| \neq 0 \text{ or } |Addon_{IM,t,n}| \neq 0 \\ \frac{1}{2} & \text{otherwise} \end{cases}$$

Note

There is a separate ρ_t^X for each value X takes, and ρ_t^X doesn't need to be the same for different X.

In the special case where there are no trades in a CSA but variation margin or/and initial margin or/and independent amount for the CSA, the allocation factor ρ_t^X is calculated as:

$$\rho_t^X = \begin{cases} 1 \;, \qquad if \;\; Allocation\_X = \; SimulatedSet \\ 0 \;, \qquad if \;\; Allocation\_X = \; Non - SimulatedSet \\ 1 \;, \; if \;\; Allocation\_X = \; ProportionateToMtM \; or \; ProportionatetoSACCRAddon \end{cases}$$

CSA Parameters

To ensure that future exposure is not overestimated, CSA parameters need to be split between synthetic netting sets.

For the computation of SACCR/IMM or CEM/IMM capital measures, the threshold and MTA are only allocated at t_0. For IMM capital calculation, this means that the EEPE is computed only using threshold and MTA allocated at t_0. When SACCR/IMM capital measure is requested, the parameter `Nonsimulated Type` of the function Synthetic Netting Set SACCR should be set to `Capital`.

Note

Although Synthetic Netting Set CEM function also has the parameter `Nonsimulated Type` in defineFunctions tool, MAG always uses the option of `Capital` in the calculation regarless of what value is set.

For the computation of SACCR/IMM profile measures, we allocate threshold and MTA over time between the IMM and SACCR synthetic netting sets because these parameters are used in both the internal model and SACCR methodologies and the threshold and MTA can change over time as a result of matured trades. In the case of changing threshold and MTA, if a relevant call date is not a reporting date, the margin call on the call date should be computed using the allcoated threshold and MTA on the reporting date that is closest to the call date; if the closest reporting date is not unique, the allocated threshold and MTA on the reporting date right after the call date should be used. When SACCR/IMM profile measure is requested, the parameter `Nonsimulated Type` of the function Synthetic Netting Set SACCR should be set to `Profile`.

The Independent Amount also need to be allocated to the simulated/non-simulated trades.

Let:

- *cptyTh*, *userTh*, *cptyMTA*, *userMTA*, *cptyIA*, *userIA* *cptyIMTh* and *userIMTh*be the counterparty threshold, user threshold, counterparty MTA, user MTA, independent amount posted by counterparty, independent amount posted by user, counterparty IM threshold and user IM threshold as specified in the CSA;
- $cptyTh_{t,s}$, $userTh_{t,s}$, $cptyMTA_{t,s}$, $userMTA_{t,s}$, $cptyIA_{t,s}$, $userIA_{t,s}$ $cptyIMTh_{t,s}$ and $userIMTh_{t,s}$ be the portions of the counterparty threshold, user threshold, counterparty MTA, user MTA, counterparty independent amount, user independent amount, counterparty IM threshold and user IM threshold allocated to the simulated trades;
- $cptyTh_{t,n}$, $userTh_{t,n}$, $cptyMTA_{t,n}$, $userMTA_{t,n}$, $cptyIA_{t,n}$, $userIA_{t,n}$, $cptyIMTh_{t,n}$ and $userIMTh_{t,n}$ be the portions of the counterparty threshold, user threshold, counterparty MTA, user MTA, counterparty independent amount, user independent amount, counterparty IM threshold and user IM threshold allocated to the non-simulated trades.
- $cpIA_{t,n}$ and $uIA_{t,n}$ be the sum of trade-level counterparty independent amounts of non-simulated trades and the sum of trade-level user independent amounts of non-simulated trades, see section Trade-level IM and IA for more details.
- $cpIA_{t,s}$ and $uIA_{t,s}$ be the sum of trade-level counterparty independent amounts of simulated trades and the sum of trade-level user independent amounts of simulated trades.

Following table illustrates the threshold and MTA allocated to simulated and non-simulated trades respectively.

Threshold and MTA allocated to simulated trades	Threshold and MTA allocated to non-simulated trades
$cptyTh_{t,s} = \rho_t^{CSAParameter} \times cptyTh$	$cptyTh_{t,n} = (1 - \rho_t^{CSAParameter}) \times cptyTh$
$userTh_{t,s} = \rho_t^{CSAParameter} \times userTh$	$userTh_{t,n} = (1 - \rho_t^{CSAParameter}) \times userTh$
$cptyMTA_{t,s} = \rho_t^{CSAParameter} \times cptyMTA$	$cptyMTA_{t,n} = (1 - \rho_t^{CSAParameter}) \times cptyMTA$
$userMTA_{t,s} = \rho_t^{CSAParameter} \times userMTA$	$userMTA_{t,n} = (1 - \rho_t^{CSAParameter}) \times userMTA$
$cptyIA_{t,s} = \rho_t^{CSAParameter} \times cptyIA + cpIA_{t,s}$	$cptyIA_{t,n} = (1 - \rho_t^{CSAParameter}) \times cptyIA + cpIA_{t,n}$

Threshold and MTA allocated to simulated trades	Threshold and MTA allocated to non-simulated trades
$userIA_{t,s} = \rho_t^{CSAParameter} \times userIA + uIA_{t,s}$	$userIA_{t,n} = (1 - \rho_t^{CSAParameter}) \times userIA + cpIA_{t,n}$
$cptyIMTh_{t,s} = \rho_t^{IM} \times cptyIMTh$	$cptyIMTh_{t,n} = (1 - \rho_t^{IM}) \times cptyIMTh$
$userIMTh_{t,s} = \rho_t^{IM} \times userIMTh$	$userIMTh_{t,n} = (1 - \rho_t^{IM}) \times userIMTh$

Note

$\rho_t^{CSAParameter}$ and ρ_t^{IM} are defined in section Allocation Factor (ρ_t^X).

Current Variation Margin

The ECB considers that variation margin should be split between synthetic netting sets in a way that considers the sets current exposures, or the bank must justify their methodology.

When `SettledToMarketAgreement = TRUE`, the current variation margin of the simulated synthetic netting set is calculated as the settled amount. The settled amount of the simulated trades is defined as:

$$\mathrm{SA}_0 := \mathrm{SA}_{0,\,\mathrm{CSA},\,\mathrm{sim}} := \sum_{i \in T_{\mathrm{sim}}} \mathrm{MtM}_{0,i}$$

where T_{sim} := the trades in the simulated synthetic netting set.

When `SettledToMarketAgreement = FALSE`, to allocate variation margin, it is assumed that each synthetic netting set is first fully collateralized above the counterparty threshold and MTA or below the user threshold and MTA, and the excess variation margin is allocated according to one of the three alternatives specified by `Allocation VM`.

Note

It is assumed that the allocation of variation margin is based on the current variation margin before any haircut.

Note

When CEM measures are calculated for non-simulated set at MA level, the current variation margin allocated to non-simulated set at CSA level is aggregated.

1. Define the current variation margin amount that is required to fully collateralize the simulated and non-simulated synthetic netting sets at t_0

$$VM_{0,s}^{fully\_collateralized} =$$

$$\begin{cases} C_{0,s} \;, if \quad C_{0,s} > 0 \\ D_{0,s} \;, if \quad D_{0,s} < 0 \\ 0 \quad , \quad otherwise \end{cases}$$

where

$$C_{0,s} = MtM_{0,s} + cptyIA_{0,s} + userIA_{0,s} - (cptyTh_{0,s} + cptyMTA_{0,s})$$

$$D_{0,s} = MtM_{0,s} + cptyIA_{0,s} + userIA_{0,s} - (userTh_{0,s} + userMTA_{0,s})$$

$$VM_{0,n}^{fully\_collateralized} =$$

$$\begin{cases} C_{0,n} \;, if \quad C_{0,n} > 0 \\ D_{0,n} \;, if \quad D_{0,n} < 0 \\ 0 \quad , \quad otherwise \end{cases}$$

where

$C_{0,n} = MtM_{0,n} + cptyIA_{0,n} + userIA_{0,n} - (cptyTh_{0,n} + cptyMTA_{0,n})$

$D_{0,n} = MtM_{0,n} + cptyIA_{0,n} + userIA_{0,n} - (userTh_{0,n} + userMTA_{0,n})$

2. Compute the excess variation margin at t_0

$$VM_0^{Excess} = VM_0 - VM_{0,s}^{fully\_collateralized} - VM_{0,n}^{fully\_collateralized}$$

where VM_0 is the current variation margin to be allocated between synthetic netting sets.

3. Allocate the VM_0^{Excess} between simulated and non-simulated synthetic netting sets

$$VM_{0,s}^{Excess} = \rho_0^{VM} \times VM_0^{Excess}$$

$$VM_{0,n}^{Excess} = (1 - \rho_0^{VM}) \times VM_0^{Excess}$$

where ρ_0^{VM} is defined in section Allocation Factor (ρ_t^X)

4. Compute the current variation margin allocated to simulated and non-simulated synthetic netting sets

$$VM_{0,s} = VM_{0,s}^{Excess} + VM_{0,s}^{full\_collateralized}$$

$$VM_{0,n} = VM_{0,n}^{Excess} + VM_{0,n}^{full\_collateralized}$$

5. Compute the unit of physical collateral allocated to simulated and non-simulated synthetic netting sets and update the `PostDirection` attribute value.

Note

This step is only needed when there is physical collateral in the current variation margin.

Note

The physical collateral unit allocation and post direction override in this step should be done before checking the user and counterparty variation margin segregation rule.

Let PU_0^k be the position unit of physical collateral mitigant *k* at t_0.

The unit of *k* allocated to the simulated synthetic netting set is:

$$PU_{0,s}^k = PU_0^k \times \left|\frac{VM_{0,s}}{VM_0}\right|$$

The unit of *k* allocated to the non-simulated synthetic netting set is:

$$PU_{0,n}^k = PU_0^k \times \left|\frac{VM_{0,n}}{VM_0}\right|$$

Note

If $VM_0 = 0$, $PU_{0,s}^k = PU_{0,n}^k = 0$.

After allocating the physical collateral mitigant *k* between synthetic netting sets, the `PostDirection` associated with $PU_{0,s}^k$ and $PU_{0,n}^k$ is determined by the sign of $\frac{VM_{0,s}}{VM_0}$ and $\frac{VM_{0,n}}{VM_0}$ as follows:

- If $\frac{VM_{0,s}}{VM_0} \geq 0$, keep `PostDirection` for $PU_{0,s}^k$ the same as the `PostDirection` in the mitigant file for PU_0^k ; otherwise, change `PostDirection` for $PU_{0,s}^k$ to the opposite of the `PostDirection` in the mitigant file for PU_0^k .

- If $\frac{VM_{0,n}}{VM_0} \geq 0$, keep `PostDirection` for $PU^k_{0,n}$ the same as the `PostDirection` in the mitigant file for PU^k_0 ; otherwise, change `PostDirection` for $PU^k_{0,n}$ to the opposite of the `PostDirection` in the mitigant file for PU^k_0 .

  Note

  The above relationship between the sign of $\frac{VM_{0,s}}{VM_0}$, $\frac{VM_{0,n}}{VM_0}$ and the `PostDirection` attribute value means that the `PostDirection` in the user input mitigant file might need to be overriden. This follows the convention in the existing mitigant file and model that the Position Unit is always a positive number and the `PostDirection` attribute is used to reflect the direction/sign of a mitigant.

Initial Margin

We will assume that initial margin is entered as node mitigants.

Note

For the purpose of Synthetic Netting Set calculation, IM allocation only happens when DIM is applied.

- Allocating node level initial margin mitigants

  Let $IM_{0,ctpy}$ be the initial margin posted by the counterparty at t_0 and $IM_{0,user\_unseg}$ be the unsegregated initial margin posted by the user at t_0 . The $IM_{0,ctpy}$, $IM_{0,user\_unseg}$, $IM^{calibration}_{0,ctpy}$ and $IM^{calibration}_{0,user}$ allocated to the simulated synthetic netting set is:

  $IM_{0,ctpy,s} = \rho^{IM}_0 \times IM_{0,ctpy}$

  $IM_{0,user\_unseg,s} = \rho^{IM}_0 \times IM_{0,user\_unseg,}$

  $IM^{calibration}_{0,ctpy,s} = \rho^{IM}_0 \times IM^{calibration}_{0,ctpy}$

  $IM^{calibration}_{0,user,s} = \rho^{IM}_0 \times IM^{calibration}_{0,user}$

  The $IM_{0,ctpy}$, $IM_{0,user\_unseg}$, $IM^{calibration}_{0,ctpy}$ and $IM^{calibration}_{0,user}$ allocated to the non-simulated synthetic netting set is:

  $IM_{0,ctpy,n} = (1 - \rho^{IM}_0) \times IM_{0,ctpy}$

  $IM_{0,user\_unseg,n} = (1 - \rho^{IM}_0) \times IM_{0,user\_unseg}$

  $IM^{calibration}_{0,ctpy,n} = (1 - \rho^{IM}_0) \times IM^{calibration}_{0,ctpy}$

  $IM^{calibration}_{0,user,n} = (1 - \rho^{IM}_0) \times IM^{calibration}_{0,user}$

  where ρ^{IM}_0 is defined in section Allocation Factor (ρ^X_t). The posting direction of the mitigant is unchanged.

- Allocating trade-level initial margin mitigants

  First node-level IMs are allocated as usual to the simulated and non-simulated synthetic netting sets. Then the trade-level IMs are added to the synthetic netting set amount to which the trade belongs. In particular, trades that fail simulation should have their trade-level IMs allocated to the non-simulated set.

• Reallocation Switch for Real-Time Trades

When there are trades coming in in real time, we use the function parameter `Reallocation for Real Time Trades` to control whether the CSA parameters, current variation margin, and independent amount are reallocated, or not, in real time.

- `Reallocation for Real Time Trades` = TRUE (default value)

  The CSA parameters, current variation margin and independent amount are reallocated to reflect the real-time trades.

- `Reallocation for Real-Time Trades` = FALSE

  The CSA parameters, current variation margin and independent amount are not reallocated. The real time trades will not affect the allocation until the next batch run.

  Note

  If the CSA is empty before any real time trades come in and `Reallocation for Real Time Trades` = FALSE, `Reallocation for Real Time Trades` is treated as TRUE when the first real time trade(s) are added. The allocation factors resulted from the first non-whatif type real-time request are used in the subsequent real-time operations.

Synthetic Netting Sets and Closeout Netting Pools

When Closeout Netting Pools (CNPs) are present under a master agreement (MA), the MA is split into sub-netting sets. In this case, we will:

- First allocate collateral between synthetic netting sets
- Then apply the multiple netting sets methodologies

Let CSA_k *(k = 1, ..., K)* be the VM CSAs on node *X*.

By following the Allocation Algorithm specified in the previous section, the current variation margin for each CSA_k, $VM_0^{CSA_k}$, can be allocated to simulated and non-simulated trades as $VM_{0,s}^{CSA_k}$ and $VM_{0,n}^{CSA_k}$.

Combined Measures

The allocated VM, threshold and MTA for each synthetic netting set are used as inputs to the computation of exposures according to IMM or SA-CCR/CEM for OTC transactions.

Capital and risk profile (PFE, EE) computed from different synthetic netting sets are then combined by simple sum on the higher level in the aggregation hierarchy.

Note

User can access the measure at synthetic netting set level by using function Synthetic Netting Set SACCR Detail or function Synthetic Netting Set CEM Detail. The function parameter `OutputType` of Synthetic Netting Set SACCR Detail can be set to either `Simulated` or `SACCR`. The function parameter `OutputType` of Synthetic Netting Set CEM Detail can be set to either `Simulated` or `CEM`.

Note

We do not add up EAD from synthetic netting sets. EAD is used to compute capital for each synthetic netting set and the capital is then added up to the higher level in the hierarchy.

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Notices

- **Algo One® version 5.2.0.6**

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SS&C Technologies, Inc.
80 Lamberton Road Windsor, CT 06095-0760
USA
Phone: 800-234-0556
860-298-5400
Fax: 860-298-4900
Internet: http://www.ssctech.com